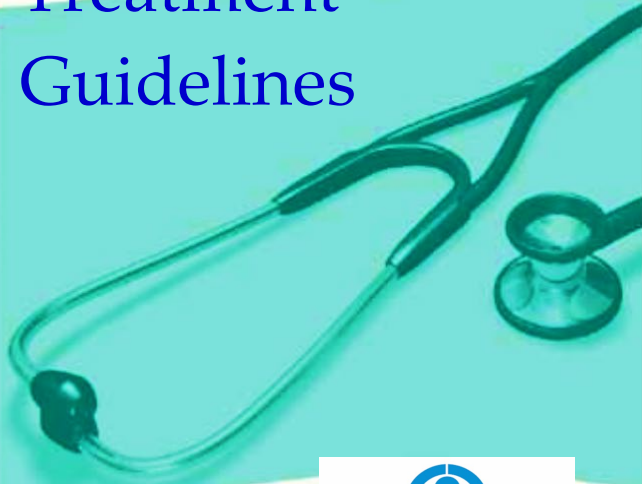


6th Edition

January 2023

Gertrude's Standard Treatment Guidelines



Gertrude's Standard Treatment Guidelines

6th Edition

Disclaimer

Whereas all care has been taken in the compilation on this document, it is advised that any user of this book must exercise their clinical judgement in the management of each patient. Gertrude's Children's Hospital will not take any responsibility for the use of information that is herein contained.

Review

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APPROVAL:

This protocol has been approved for use in consultation with the Medicines and Therapeutics Committee by the Head of Clinical Services who is herein identified as the protocol owner

AUTHOR:

Whereas the Medicines and Therapeutics Committee appears as the protocol author, the actual development was done by several people as acknowledged in the document, the Medicines and Therapeutics Committee Coordinated its development. The Medicines and Therapeutics Committee is a sub-committee of the Medical Advisory Committee

ACCOUNTABILITY:

The Head of Clinical Services is responsible for the implementation of this protocol

DESCRIPTION:

This protocol contains standard treatment guidelines to be used in the management of the described diseases/conditions as will be encountered at Gertrude's Children's Hospital

WARNING

We have endeavoured to ensure that the information in this Standard Treatment Guidelines is correct at the time of publication, but errors may have occurred thereafter, and the diagnostic and therapeutic regimes may have been altered. Where there is any doubt, information should be checked against published literature, manufacturers' summary of product characteristics (data sheets), or other specialist sources.

The Standard Treatment Guideline is intended for rapid reference and cannot always contain all the information necessary for diagnosis, investigations, prescribing and dispensing

ABBREVIATIONS AND ACRONYMS

ADH	:	Antidiuretic Hormone
ALI	:	Acute Lung Injury
ARDS	:	Acute Respiratory Distress Syndrome
CCU	:	Critical Care Unit
CDC	:	Child Development Centre
CGP	:	Clinical Governance Protocol
CNS	:	Central Nervous System
CO₂	:	Carbon dioxide
CPR	:	Cardiopulmonary Resuscitation
CRP	:	C-Reactive Protein
CVP	:	Central Venous Pressure
CVVH	:	Continuous veno-venous Haemodialysis
DDAVP	:	Desmopressin
DVT	:	Deep venous Thrombosis
ECG	:	Electrocardiogram
ENT	:	Ear Nose and Throat
ESR	:	Erythrocyte Sedimentation Rate
ET	:	Endotracheal
ETCO₂	:	End Tidal Carbon dioxide level
GABHS	:	Group A beta haemolytic streptococci
HAE	:	Hereditary Angioedema
HCO₃	:	Bicarbonate
HIV	:	Human Immunodeficiency Virus
ICS	:	Inhaled Corticosteroid
IO	:	Intraosseous
IV	:	Intravenous
Kg	:	Kilogram
LMWH	:	Low molecular weight Heparin
ORS	:	Oral Rehydration Solution
PCA	:	Patient Controlled Analgesia
PCP	:	Pneumocystis jirovecii Pneumonia
PEEP	:	Positive End Expiratory Pressure
SAM	:	Severe Acute Malnutrition
SVT	:	Supraventricular tachycardia
TRH	:	Thyroid Releasing Hormone

TSH	:	Thyroid Stimulating Hormone
UFH	:	Ultrafiltration
UTI	:	Urinary Tract Infection
VF	:	Ventricular Fibrillation
VT	:	Ventricular Tachycardia

CHANGES IN THIS EDITION

- Anaphylaxis
 - Doses of adrenaline adjusted
 - a section of refractory anaphylaxis added
 - Use of Antihistamines and steroids discouraged
- Cardioversion energy dose adjusted
- Fluid bolus for most conditions changed to 10 mL/Kg to be given over 10 – 15 minutes
- Maintenance fluid requirements (Holliday Segar Formula) have been presented in a table for children weighing up to 40 Kg
- Management of severe sepsis and septic shock shortened
- The fluid management in Diabetic Ketoacidosis has been updated to use less fluid
- Antihistamine doses for allergy adjusted
- A section of malignant Hyperthermia has been added
- A section of otitis externa has been added
- Reference ranges for Vitamin D levels adjusted
- Daily vitamin D routine supplementation for all infants
- A chapter on animal bites has been added
- Acyclovir dosage has been simplified for most conditions
- Page numbers and table of contents added

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1. RESUSCITATION GUIDELINES

1.1 CARDIORESPIRATORY ARREST

Signs of shock, cyanosis, bradycardia/tachycardia, apnoea or increasing tachypnoea are warning signs and an indication for urgent resuscitation.

Most arrests in children are due to hypoxia, hypotension and acidosis. The most common dysrhythmias are:

- severe bradycardia
- pulseless electrical activity
- asystole.

Ventricular arrhythmias - Ventricular fibrillation (VF) and pulseless ventricular tachycardia (VT) - are seen with:

- pre-existing cardiac disease (cardiomyopathies, hereditary prolongation of QT interval, congenital heart disease)
- poisoning (e.g., tricyclic antidepressants)
- low voltage electrocution (less than 1000volts)
- may occur during resuscitation.

Supraventricular Tachycardia (SVT) may cause shock in newborn infants.

1.1.1 Assessment

The order of assessment and intervention for any seriously ill or injured child follows the **ABCDE** principles.

- **A** indicates airway (Ac for airway and cervical spine stabilisation for the injured child)
- **B** indicates breathing
- **C** indicates circulation (with haemorrhage control in injured child)
- **D** indicates disability (level of consciousness and neurological status)
- **E** indicates exposure to ensure full examination (whilst respecting dignity and temperature conservation)

A - Airway

- Position the head - neutral position (<1-year-old) or sniffing position (≥ 1 year of age).
- Open airway – head tilt or chin lift or jaw thrust.
- Use Oropharyngeal airway if required.

B - Breathing

- Assess for signs of respiratory inadequacy.
- If respiration is adequate, administer Oxygen by Non rebreathing mask at 10-15L/min. Do not use self-inflating bags in spontaneously breathing patients. They are designed to deliver Oxygen only if squeezed.
- If the child is not breathing, commence artificial ventilation.

C - Circulation

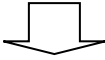
- If there are no signs of circulation, i.e., no pulse, slow pulse (<60/min) or you are not sure, commence external cardiac compression and determine the cardiac rhythm by attaching an ECG monitor to display the ECG.

1.1.2 Initial Management

Assessment and Management of the airway and breathing are the initial priorities.

Table 1: Initial Assessment and Management

Observe, Stimulate and Assess Response

Call for HELP if not alert

Assess the Airway (mouth and nose) and Assess Breathing

Bag-Valve and Mask Ventilation

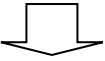
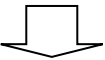
Assess Large Pulse and Adequacy of Circulation and manage appropriately

Assess Disability and manage appropriately

Assess Exposure and manage appropriately

Table 2: Selection of the correct endotracheal tube size

Age	Weight (Kg)	Tube size (mm)	Length at lip (cm)
Newborn	3.5	3.0	8.5
2 months	5	3.5	9
6 months	8	4.0	10
1 year	10	4.0	11

1.1.2.1 Artificial Ventilation

- Select the appropriately sized resuscitator bag
 - Child up to 2 years – 500 mL bag.
 - Child above 2 years – 1000 mL bag.
- Select an appropriately sized mask.
- Obtain an airtight seal.
- Oxygen flow rate of 10 – 15L/min and attachment of a reservoir assembly will give nearly 100% oxygen.
- An oropharyngeal airway will facilitate maintenance of the airway.
- Brief suction of the mouth and pharynx if needed, is done using a wide bore sucker under direct vision.
- Ventilate to have normal chest rise and fall. Do not over ventilate. The target respiratory rate is based on the child's age and should not exceed 30 breaths/minute even for a newborn.
- Intubation should be attempted by those skilled to do so.

1.1.2.2 Endotracheal intubation

- Selection of the appropriate endotracheal tube in a child older than 1 year:

Tube size (mm) = (Age in years)/4 + 4

Oral Length at lip (cm) = (Age in years)/2 +12

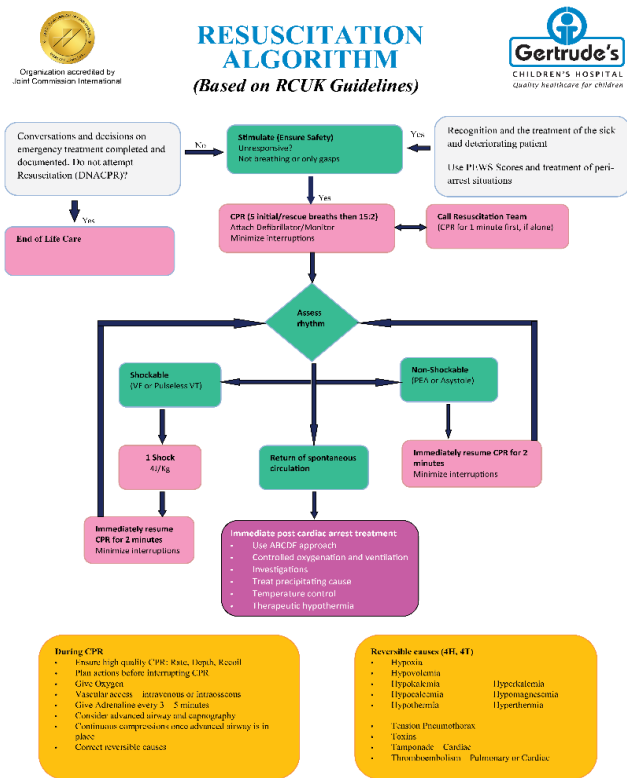
Nasal Length at nares (cm) = (Age in yr)/2 +15

- Cuffed tracheal tubes are as safe as uncuffed tubes for infants (except neonates) and children.
- Monitoring end tidal CO₂ (ETCO₂) reliably confirms tracheal tube placement in a patient with a perfusing rhythm, and its use is strongly recommended after intubation, and during transport of an intubated child.
- Hyperoxemia can be detrimental in neonates and there are some advantages in using room air while monitoring the oxygen saturation during initial newborn resuscitation.

1.1.2.3 External cardiac compression

- DO NOT interrupt except for defibrillation.
- Place the child on a firm surface. If on a bed, place the cardiac massage board under the patient, not under the mattress.
- Apply massage to the lower half of the sternum in all patients including newborns.
 - Compress sternum one third the antero-posterior diameter of the chest.
 - Use the hand technique that allows you to achieve this.
 - With large children use the “heel” of one hand with the other superimposed.
 - For small children use the heel of one hand.
 - For infants use two fingers.
 - For newborn infants the best technique is a two-handed hold (encirclage) in which both thumbs compress the sternum.
- Gain IV or IO access as soon as possible - at least the second dose of adrenaline should be given via this route
- Frequent change of personnel (every few minutes) is desirable
- During resuscitation do not stop to check for a pulse unless for defibrillation or the ECG shows an organised rhythm
 - After DC shock, continue CPR for 2 minutes prior to checking rhythm

Diagram 1: Resuscitation Algorithm Chart



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1.1.3 During resuscitation

Correct treatable causes (Hs, Ts)

- Hypoxaemia
- Hypovolaemia
- Hypothermia/Hyperthermia
- Hypokalemia/Hyperkalemia
- Cardiac Tamponade
- Tension pneumothorax
- Toxins/Poisons
- Cardiac or Pulmonary Thromboembolism

1.1.4 Blood gas analysis

- It is not a priority in initial resuscitation attempts and obtaining a sample should not distract from other resuscitation manoeuvres.
- Arterial (and to some extent venous) blood gas analysis can help determine degree of hypoxaemia, adequacy of ventilation, degree of acidosis and presence of electrolyte abnormalities such as of sodium and potassium.

1.1.5 Post resuscitation care

- Ensure airway and breathing are managed effectively including intubation if not already performed. Do not extubate. Use adequate sedation and analgesia.
- Ventilate to normocarbida.
- Circulation - maintain adequate blood pressure with use of inotropes as needed. Monitor for further arrhythmia.
- Aim for core temperature of 35°C (do not actively warm if core temp < 32°C).
- Ongoing antiarrhythmic drugs.

- Ensure normoglycemia.

Table 3: Vital signs and Equipment size

Age	Length or Height (cm)	Weight (Kg)	Pulse	Resp Rate	Systolic BP mmHg	Diastolic BP mmHg	Temp (°C)	ET tube (mm)	ET depth (cm tip to tip)	Laryngeal Blade	Supraglottic Airway (i-gel)	NG Tube	Suction Catheter
Pre-term	45	1 - 2	100-180	40 - 60	50 - 60	35 - 45	34.0 - 38.0	2.5	6 + WT (Kg)	0	1	5	5 - 6
Term Newborn	50	3 - 4	100-180	40 - 60	60 - 90	40 - 45	34.0 - 38.0	3.0	6 + WT (Kg)	1	1	5 - 8	6 - 8
6months	67	7	100-160	30 - 60	83 - 105	40 - 45	34.0 - 38.0	3.5	11	1	1.5	8	8
1 year	75	10	80 - 110	26 - 34	95 - 105	50 - 65	34.0 - 38.0	4.0	11 - 12	1	1.5	10	8 - 10
3 years	95	15	70 - 110	24 - 26	96 - 110	55 - 75	36.1 - 37.8	4.5	13 - 14	2	2	10	10
5 years	110	18	70 - 110	20 - 24	96 - 110	55 - 75	36.1 - 37.8	5.0	14 - 15	2	2	12	10
6 years	115	20	65 - 110	20 - 24	97 - 112	65 - 80	36.1 - 37.8	5.5	15 - 16	2	2	12	10
8 years	127	25	65 - 110	20 - 24	97 - 112	65 - 80	36.1 - 37.8	6.0	17 - 18	2	2.5	14	10
12 years	150	40	60 - 100	12 - 20	112 - 128	70 - 85	35.9 - 37.6	6.5	19 - 20	3	3	14	12
16 years	160	<50	60 - 100	12 - 20	112 - 128	70 - 85	35.9 - 37.6	7.0	20 - 24	3	3	18	12
Adult Female	160 - 165	50 - 75	60 - 100	12 - 20	112 - 130	70 - 90	35.9 - 37.6	7.5	22 - 24	3 - 4	4	18	14
Adult Male	160 - 175	75 - 100	60 - 100	12 - 20	112 - 128	70 - 85	35.9 - 37.6	8.0	22 - 24	3 - 4	5	18	14

Appropriate internal diameter (mm) of ET tube = (Age in years/4) + 4; NB: For ET tube size, you choose a size larger and a size smaller in addition to the indicated size.
Appropriate length of Oral tube (cm) = Newborn = 6 + weight (Kg); in infant and child = (Age in years/2) + 12, or three times internal tube diameter
Appropriate length of Nasal tube (cm) = (Age in years/2) + 15

Table 4: Adrenaline Resuscitation Doses

Age	Endotracheal	Intravenous
	1:1 000	1:10 000
Pre-term	0.5 mL (1:10 000)	0.3 mL
Term Newborn	1mL (1:10 000)	0.5 – 1 mL
6 months	0.7mL	0.7mL
1 year	1mL	1 mL
3 years	1.5 mL	1.5 mL
5 years	1.8 mL	1.8 mL
6 years	2 mL	2 mL
8 years	2.5 mL	2.5mL
12 years	2.5 mL	4 mL
16 years	2.5 mL	5 mL
Adult	2 – 2.5 mL	10 mL

Table 5: Adrenaline and Hydrocortisone doses in Anaphylaxis

Age	Adrenaline IM	Adrenaline Infusion (1mg in 100 mL Normal Saline)	Hydrocortisone (IV)
< 6 mo.	100 - 150mcg	0.5 – 1 mL/Kg/hr	25 mg
6 mo. – 6 yr	150mcg	Consider other vasopressors if refractory to Adrenaline	50 mg
6 –12 yr	300mcg		100 mg
>12 yr	500mcg		200 mg

Other Resuscitation Medication and Electricity Doses

Adenosine: 0.1 mg/Kg rapid IV/IO push; increase to 0.2 mg/Kg if needed; maximum single dose 12 mg

Amiodarone: 5 mg/Kg IV/IO; rapid bolus for pulseless VF/VT; over 20 – 60 min for perfusion

Atracurium: IV 0.2 - 0.4mg/Kg for children aged ≤ 2 years; for aged > 2 years the dose is 0.4 - 0.5 mg/Kg given over 1 minute onset 2 – 3 minutes and effect last for 30 - 45 minutes. The dose is on ideal body weight whenever obesity is present.

Atropine: 0.02 mg/Kg IV/OI/ET maximum single dose 0.6 mg

Cardioversion: 1 Joule/Kg; increase to 2 Joule/Kg if needed Adult: 50 Joules for SVT or atrial flutter; 100 Joules for monomorphic VT or A-fib; 200 joules for Polymorphic VT; (increase to 300 and 360 joules if needed)

Cisatracurium: IV 0.15- 0.2mg/kg. Onset of action in 2-3 minutes and duration of action 20-40 minutes.

Defibrillation: 4 Joules/Kg (monophasic or biphasic waveform) should be used for all shocks. The MAXIMUM DOSE for the first shock is 360 J (monophasic) or 150 – 200 J biphasic) and subsequent shocks is 360 J (monophasic) or 200 J (biphasic) (Adult: 200 joules; then 200 – 300 joules; 360 joules thereafter)

Dopamine: (Central line) IV 5-10 mcg/kg/min, titrated to effect

Dopamine: (Peripheral line) IV 1 mcg/kg/min, titrated to effect

Fentanyl: IV 1-4 ug/kg. Onset of action in 2-3 minutes, Peak 3-4 minutes and duration of action up to 60 minutes

Fluids in circulatory compromise: Use an ISOTONIC, non-glucose containing solution (Normal Saline or Ringer's lactate) 10 mL/Kg bolus over 10 – 15 minutes

Ketamine: (To be used with Atropine to counter hypersalivation) 1 – 2 mg/Kg IV Effects lasts 5 – 15 minutes and is contraindicated where there is suspected or proven increased intracranial pressure

Lignocaine: 1 - 1.5 mg/Kg IV/IO or ET; followed with an infusion. (Adult: 1.0 - 1.5 mg/Kg IV/IO; repeat 0.5 – 0.75 mg/Kg every 5 - 10min if needed MAX 3 mg/Kg. ET dose 2 - 4 mg/Kg). Never give Lignocaine after Amiodarone since they have similar effect

Lorazepam: 0.1 mg/Kg IV at 2 mg/min to maximum of 4 mg. May repeat at 10-minute intervals twice.

Midazolam: IV 0.1 – 0.2 mg/Kg; onset of action 3 – 5 minutes and lasts up to 60 minutes

Morphine IV: Add 0.5 mg/Kg of Morphine to 50 mL of Normal Saline. Administer an initial bolus of 5 mL (50 microgram/Kg) of the solution over 3 – 5 minutes, then infuse 1 – 4 mL/hour (10 – 40 microgram/Kg/hr.) or boluses of 1-2 mL (10-20mcg/Kg) over 2 – 3 minutes at intervals not less than 10 minutes (Reassess before each dose to determine the pain level)

Naloxone: Prepare a Naloxone solution by adding 0.4 mg of Naloxone to 20 mL of Normal Saline (20 microgram/mL) and infuse 1 – 10 microgram/Kg according to indication (Opioid induced pruritus (1 mcg/Kg), excess sedation (2 mcg/Kg) and opioid induced respiratory depression; apnea (10 mcg/Kg)

Noradrenaline: (Central line) IV 0.02-0.1 mcg/kg/min, titrated to effect

Rocuronium: IV 0.6-1.2mg/kg. Onset of action in 30-60 seconds and duration of action 30-40 minutes.

Tachycardia: MAXIMUM SINGLE DOSE: 300 mg. May repeat to MAX DOSE= 15 mg/Kg/day (2.2g/day)

Note: Neuromuscular blocking agents like Atracurium, Cisatracurium and Rocuronium are to be used only when the airway can be secured

2. EMERGENCIES

2.1 ACUTE UPPER AIRWAY OBSTRUCTION

The signs of partial acute upper airway obstruction are:

- Stridor.
- Increased work of breathing as evidenced by suprasternal, intercostal and subcostal retraction along with an increased use of accessory muscles of respiration.

2.1.1 Management

- Allow child to settle quietly on parent's lap in the position the child feels most comfortable.
- Observe closely with minimal interference.
- Treat specific cause.
- Admit to CCU if worsening or severe obstruction.
- Oxygen may be given while awaiting definitive treatment.

This can be falsely reassuring because a child with quite severe obstruction may look pink in oxygen.

2.1.2 Notes

- Intravenous access should be deferred – upsetting the child can cause increasing obstruction.
- Lateral cervical soft tissue x-rays do not assist in management.
- In severe airways obstruction imaging cause undue delay in definitive treatment and may be dangerous (positioning may precipitate respiratory arrest).

2.2 ANAPHYLAXIS

Anaphylaxis is an acute potentially life-threatening multi-systemic allergic reaction characterized by at least one of the following:

1. Respiratory compromise [e.g., dyspnea, hoarseness, central cyanosis, wheeze/bronchospasm, stridor, reduced peak expiratory flow, hypoxemia)

or

2. Cardiovascular decompensation [reduced blood pressure (BP) or associated symptoms and signs of end-organ under perfusion (e.g., hypotonia [collapse], syncope, incontinence)

And

3. Gastrointestinal symptoms and signs (e.g., crampy abdominal pain, vomiting)

or

4. Skin-mucosal tissue involvement (e.g., generalized hives, itch-flush, swollen lips, tongue or uvula).

For reactions which do not fulfil this definition, see Urticaria guidelines and seek expert advice early.

2.2.1 Management

- Remove the likely allergen [Blood product/Antibiotic], Call for Help
- Airway - Potentially at risk of being obstructed. Open and maintain airway.

- Breathing - Supplement with 100% Oxygen [with BVM if required]. Seek expert assistance for early intubation if airway involvement is seen.
- Circulation - Give intramuscular adrenaline 1:1,000 into the upper lateral thigh. Repeat in 5 minutes if there is no clinical improvement.
 - Child aged > 12years/Adult: 500 micrograms [0.5 mL]
 - Child 6 – 12 years: 300 micrograms [0.3mL]
 - Child < 6years: 100 - 150 micrograms [0.1 – 0.15 mL]
- An adrenaline infusion in PICU should be considered if repeated doses of IM adrenaline are required – 0.05 – 0.1 mcg/Kg/min
- In addition to adrenaline, repeated 10 mL/Kg boluses of IV Normal Saline may be required for shock.
- If airway oedema is not responding to parenteral and nebulised adrenaline, early intubation is indicated
- Give supplemental oxygen.
- Admit to PCCU if:
 - Child requires more than 2 adrenaline doses or 2 fluid boluses for the management of anaphylaxis
 - If ongoing airway concerns persist

2.2.2 Other therapies to consider

- Nebulised salbutamol is recommended if the patient has respiratory distress with wheezing.
- Antihistamines may be given for symptomatic relief of pruritus after the patient has stabilized – non sedating oral antihistamines are preferred e.g., Cetirizine.

- Routine use of corticosteroids is not recommended. They may be given for refractory reactions or ongoing Asthma/Shock.

2.2.3 Admission

All children with anaphylaxis should be admitted. At least 6-12 hours observation is recommended

Overnight admission should be considered in the following circumstances:

- Greater than one dose of adrenaline (including nebulised) is required
- A fluid bolus is required
- The child lives a long distance from medical services

2.2.4 Discharge Plan and Follow up

- Outpatient follow up in 24 hours is recommended.
- Consider Adrenaline autoinjector for patients with suspected anaphylaxis to insect sting or food which cannot be easily avoided

2.3 URTICARIA

Urticaria is a condition characterized by wheals (hives) or angioedema, or both. It is more common in atopic patients. It usually occurs due to release of chemical mediators (including histamine, bradykinin, platelet-activating factor and cytokines) from tissue mast cells and circulating basophils. These activate sensory nerves and cause dilation of blood vessels with third spacing of fluid into surrounding tissues. These mediators may be released by allergic or non-allergic mechanisms.

- Wheal: superficial swelling, usually surrounded by erythema, may have itchy or burning sensation.

- Angioedema: deeper swelling within skin or mucous membranes, may be erythematous, itchy or painful, but often asymptomatic and localized. Commonly affects the face, hands, feet and genitalia.
- Acute urticaria: less than 6 weeks duration, often gone within hours to days.
- Chronic urticaria: more than 6 weeks duration, with daily or episodic symptoms. May be inducible.

Severe allergic urticaria may lead to anaphylactic shock and should be managed as an emergency (see section on Anaphylaxis).

Triggers include infections (bacterial or viral), allergies to food or medication, vaccination, insect bites or stings, blood transfusion, extremes of temperature or contact with allergens in atopic patients. Chronic spontaneous urticaria may be associated with chronic autoimmune disease. Urticaria in response to a physical stimulus is inducible urticaria e.g., dermographism.

2.3.1 Management

There are no routine tests, but one may investigate for an underlying infection or chronic disorder if the history is suggestive.

If any signs of anaphylaxis (at least one respiratory or cardiovascular feature AND at least one gastrointestinal or skin feature), manage as an emergency. See Anaphylaxis guidelines.

The main treatment of all forms of urticaria is:

1. Oral antihistamines. In children above 2 years, second generation antihistamines (such as cetirizine, desloratadine) are preferable since they are longer acting, less sedating and with fewer anticholinergic side effects.
2. Eliminate trigger where this can be identified.
3. If antihistamines are not effective, consider a short course of oral prednisone (1mg/Kg daily for 5 days). Tapering of the corticosteroid dose is not necessary after this course.

4. Intramuscular injection of adrenaline is reserved for life-threatening anaphylaxis.

Refer to allergologist – skin prick tests and radioallergosorbent tests (RAST) may be helpful if drug or food allergy suspected, and for specialist management of chronic urticaria.

2.4 HEREDITARY ANGIOEDEMA

(C1 Esterase Inhibitor Deficiency)

- Hereditary angioedema (HAE) causes recurrent episodes of angioedema in the upper respiratory tract, gastrointestinal tract or in subcutaneous tissues.
- Acute episodes of angioedema may be triggered by infection, stress, menstruation, surgery, dental work, trauma and some medicines (including oestrogen containing contraceptives and ACE inhibitors) or may have no clear trigger.
- HAE is a rare autosomal dominant condition in which C1 esterase inhibitor levels are reduced (HAE type I) or poorly functional (HAE type II). HAE is diagnosed by the finding of low C1 esterase inhibitor level or function. C4 level is also low during episodes of angioedema.

2.4.1 Management

2.4.1.1 *Mild/moderate angioedema episodes*

Treatment is conservative:

- Hospital admission is not usually required
- Other causes of abdominal pain may need to be excluded and abdominal ultrasound may help show intestinal wall oedema or ascites in HAE-related angioedema.

- A 3-day course of tranexamic acid may be considered to shorten the duration of symptoms (15 - 25 mg/Kg (max. 1.5g) 2 - 3 times daily).

2.4.1.2 Severe angioedema episodes

Severe angioedema episodes can be fatal. Management includes:

- Hospital admission for all severe angioedema episodes. If upper airway obstruction is present, notify a colleague skilled in endotracheal intubation.
- Intravenous C1 esterase inhibitor concentrate should be administered at 25 mg/Kg infusion over 1 hour.

2.4.1.3 Planned Surgery or Oropharyngeal Procedures

Surgery or any traumatic procedure of the oropharyngeal area such as dental work should be carefully planned. The use of a prophylactic agent prior to such procedures reduces the risk of precipitating angioedema.

- Consult with an intensivist before the procedure.
- For planned procedures, Danazol is the first choice of prophylactic agent, (10 mg/Kg/day for 5-10 days before and 2-5days after).
- The concentrate (25units/Kg infusion given 1 hour prior to the procedure).
- Due to the risk of precipitating laryngeal oedema, oropharyngeal procedures should usually involve general anaesthesia with endotracheal intubation.

Note:

- Antihistamines and corticosteroids have no role in the management of HAE
- The role of adrenaline in the treatment of HAE is not well established

- There are anecdotal reports of efficacy using nebulised or intramuscular adrenaline to treat upper airway angioedema, however C1 esterase inhibitor is the treatment of choice for airway angioedema caused by HAE.

2.5 DROWNING/IMMERSION

All children should receive full resuscitative efforts after an episode of immersion or near drowning.

2.4.1 Assessment and management

2.5.1.1 Airway and Breathing

- Rapidly assess airway, breathing, circulation and level of consciousness
- If child is in cardiorespiratory arrest proceed immediately with cardiopulmonary resuscitation
- Airway and breathing (protect cervical spine if any possibility of injury)
- If spontaneously breathing, administer 100% Oxygen by face mask
- Intubate and ventilate if:
 - Inadequate respiration
 - Falling arterial oxygen concentration (PaO_2) despite increased fractional inspired oxygen (FiO_2)
- Monitor oxygen saturation
- Perform an arterial blood gas
- Do a chest x-ray

2.5.1.2 Circulation

- Assess pulse rate and volume, blood pressure and capillary refill.

- Insert intravenous line.
- Perform haemogram, serum glucose, electrolytes and Creatinine.
- If circulation is inadequate give fluid bolus of 10 mL/Kg Normal Saline.
- Consider early inotropic support.

2.5.1.3 Cerebral support

- Avoid any further episodes of hypoxia and hypercarbia.
- Optimise circulation as best possible.
- Once shock is reversed restrict fluids to 75% of maintenance.
- Monitoring and control of raised intracranial pressure is occasionally required.

2.5.1.4 Temperature

- Actively rewarm children slowly to a core temperature of 34 °C.
- Passively rewarm if temperature over 34 °C.

2.5.2 General

- Admit to appropriate inpatient unit.
- Corticosteroids are not recommended.

2.5.3 Adverse Prognostic Factors

- Immersion time > 10 minutes
- Rectal temperature < 30°C
- Absence of any initial resuscitative efforts
- Arrival in hospital with CPR in progress or in coma.
- Requirement of cardiopulmonary resuscitation
- Initial serum PH < 7.0

2.6 FLUID MANAGEMENT

Table 6: Maintenance Fluid Volume Requirements (Holliday Segar)

Body Weight (Kg)	Fluid Therapy per Day (mL)	Fluid Infusion Rate per hour
Day of Life	mL/Kg/day	mL/Kg/hr
1	65 – 80	3
2	80	3 - 4
3	100	4
4	120	5
5 - 28	135 – 150	5 - 6
Older than 28 days	Use weight ranges below	
Body weight (Kg)	Total fluid required in 24 hours (mL)	Fluid Infusion rate per hour (mL/hr)
4 – 10Kg	100 mL /Kg	4 x wt.
11 – 20Kg	1000 mL + 50 mL/Kg for each Kg greater than 10 Kg	40 + 2 x (wt-10)
> 20 Kg	1500 mL+ 20 mL/Kg for each Kg greater than 20 Kg	60 plus 2 x (wt-20)
The requirements and infusion rates have been simplified in Table 8		

2.6.1 Maintenance Electrolyte Requirements

- Sodium: 2 - 3 mmol/Kg/24hr
- Potassium 1 - 2 mmol/Kg/24hr
- Calcium 1 - 2 mmol/Kg/24hr
- Glucose 4 – 6 mg/Kg/min (5.7 – 8.6g/Kg/day)

Table 7: Maintenance Fluid Requirements for 24 hours

Weight	Maintenance Fluid Requirements for 24 hours	Infusion rate mL/hr.
4	400	16.7
5	500	20.8
6	600	25.0
7	700	29.2
8	800	33.3
9	900	37.5
10	1000	41.7
11	1050	43.8
12	1100	45.8
13	1150	47.9
14	1200	50.0
15	1250	52.1
16	1300	54.2
17	1350	56.3
18	1400	58.3
19	1450	60.4
20	1500	62.5
22	1540	64.2
24	1580	65.8
26	1620	67.5
28	1660	69.2
30	1700	70.8
32	1740	72.5
34	1780	74.2
36	1820	75.8

Table 8: Composition of Common Intravenous Fluids

Fluid	Na	Cl	K	Ca	Lactate	Osmolality
0.9% Normal Saline	154	154	–	–	–	308
½ Strength Normal Saline	77	77	–	–	–	154
¼ Strength Normal Saline	38.5	38.5	–	–	–	77
Ringers Lactate	130	109	4	3	28	275
Half strength Darrows	62	17	–	–	25	170
5%Dextrose	-	-	-	-	-	253

2.6.2 Hypernatremic dehydration

Restore intravascular volume

- If in shock, administer Normal Saline 10 mL/Kg over 10 – 15 minutes. Repeat until out of shock

Determine time for correction based on the initial sodium concentration, maximum drop of 12 mmol/L/24hours

- 145 – 157 mmol/L: 24 hours
- 158 – 170 mmol/L: 48 hours
- 171 – 183 mmol/L: 72 hours
- 184 – 196 mmol/L: 84 hours

Administer fluid at constant rate over the time for correction

- Typical fluids 5% dextrose + $\frac{1}{4}$ normal saline OR 5% dextrose + $\frac{1}{2}$ normal saline (both with 20mmol/L potassium chloride unless contraindicated).
- Typical rate: 1.25 -1.5 times maintenance.
- Monitor serum sodium concentration 4hourly.

Adjust fluid based on clinical status and serum sodium concentration:

- Signs of volume depletion: Administer normal saline (10 mL/Kg over 10 – 15 minutes)

If Sodium decreases too rapidly,

- Increase sodium concentration of IV fluids.
- Decrease IV fluids infusion rate.

If Sodium decreases too slowly,

- Decrease sodium concentration of IV fluids.
- Increase IV fluids infusion rate.
- Replace ongoing losses as they occur.

2.6.3 Fluid replacement in shock

- Fix an IO line.
- Draw blood for analysis including a blood gas.
- Infuse 10 mL/Kg of Normal saline or Ringer's lactate over 10 – 15 minutes.
- Reassess after first infusion: if no improvement, repeat 10 mL/Kg over 10 – 15 minutes.
- Reassess after second infusion: if no improvement, repeat 10 mL/Kg over 10 – 15 minutes.
- Reassess after third infusion: if no improvement, repeat 10 mL/Kg over 10 – 15 minutes.

- If there is no improvement after 40 – 60 mL/Kg, ongoing losses must be suspected e.g., diarrhoea or bleeding. The aim is to prevent onset of decompensated circulatory failure which may lead to irreversible circulatory failure and death.
- Identify and treat the cause.
- Consider inotropic support – start adrenaline infusion.
- In septic shock, up to 60 – 80 mL/Kg is often required in the first hour of resuscitation. Inotropes and intubation may be required and should be considered at an early stage (40 mL/Kg)
- Circulatory failure without hypovolemia may be due to other causes like cardiac failure. The effect of each small bolus should be assessed to prevent worsening of the circulation – crackles or enlarging liver.
- In trauma, especially if there is bleeding, consider blood and blood products for third and subsequent boluses.
- Where there is massive bleeding at presentation, massive transfusion protocol should be activated – pRBC, FFP, Platelet ratio 1:1:1.

2.7 MANAGEMENT OF SEPSIS AND SEPTIC SHOCK

2.7.1 Definitions

Systemic Inflammatory Response Syndrome (SIRS):

Presence of at least two of:

- Core temperature < 38.5°C or < 36°C

- Tachycardia
- Bradycardia if age < 1 year
- Tachypnoea
- Leukocytosis or leucopenia
- < 10% immature neutrophils

Sepsis:

Infection + SIRS

Septic Shock:

Sepsis and hypotension

Severe Sepsis:

Sepsis and multiorgan dysfunction of the following:

- Lactic acidosis
- Acute lung injury
- Oliguria with elevated creatinine
- Elevated liver enzymes
- Coagulopathy

2.7.2 Recognition within 5 minutes

A, B assessment (recognize respiratory failure)

- Airway, RR, work of breathing, Oxygen saturation, breath sounds
- Open airways, start high flow oxygen by non-rebreather mask with reservoir or BMV as appropriate

C assessment (recognize rhythm, or circulatory failure)

- HR, CRT, BP, peripheral and central perfusion, rhythm recognition, recognition of circulatory failure/shock
- Establish IO/IV access (take samples: Blood culture, full blood count, glucose, urea, electrolytes, lactate, blood gas, coagulation profile, liver function, HIV)
Give fluid resuscitation
Continuous monitoring

D assessment (recognize altered mentation)

- AVPU level
- Get blood glucose level and intervene

E assessment (recognize evidence of infection)

- Skin rash, Fever
- Give broad spectrum antibiotic

2.7.3 Initial Resuscitation 5 – 15 minutes

- If no fluid overload (hepatomegaly, basal lung crepitations), administer 10 mL/Kg Ringers Lactate or Normal saline over 5 – 10 minutes and reassess after each bolus up to 40 – 60 mL/Kg or until perfusion normalizes (CRT <2 secs, Normal BP for age, Urine output ≥ 1 mL/Kg/hr., normal pulses, alert). Maximum fluid bolus 500 mL
- If fluid overload detected, start inotropic support
- Recognize and treat reversible causes (hypovolemia, hypoglycemia, hypokalemia, hyperkalemia, hypocalcemia, hypomagnesemia)
- Start broad spectrum antibiotic (Ceftriaxone at 50 mg/Kg unless in Neonate or young infant with jaundice)

- Call PCCU for assistance and patient transfer
- Stabilize the cardiovascular system before endotracheal intubation

2.7.4 Fluid refractory shock 15 – 60 minutes

- Start IO/IV inotrope infusion preferably centrally but can be started peripherally
 - Adrenaline 0.05 – 0.3 mcg/Kg/min (use more dilute infusion peripherally)
 - And/or Noradrenaline 0.05 mcg/Kg/min
- Titrate with clinical response and haemodynamic monitoring
- Use Ketamine with or without Atropine to gain central access and airway placement
- Give Hydrocortisone 2 – 4 mg/Kg/day and intermittent dosing or continuous infusion maximum 200 mg/day

2.7.5 Fluid and Catecholamine refractory shock over 60 minutes

- Seek PCCU advise (Consider baseline serum cortisol or adrenocorticotrophic hormone stimulation testing) prior to corticosteroid administration
- Goal is Normal MAP, CVP, CI 3.3 - 6.0L/min/m²; SCVO₂>70%
- Hydrocortisone in stress doses (2 to 4 mg/Kg/day, intermittent or continuous infusion, maximum dose 200 mg/day)
- Monitor fluid, Inotrope, Vasopressor, vasodilator therapy/response

2.7.6 Diagnosis

Obtain appropriate cultures before starting antibiotics provided this does not significantly delay antimicrobial administration.

- Obtain two or more blood cultures.
- One or more blood cultures should be percutaneous.
- One blood culture from each vascular access device in place < 48 hours.
- Culture other sites as clinically indicated.

Perform imaging studies promptly to confirm and sample any source of infection if safe to do so.

2.7.7 Antibiotic therapy

Begin intravenous antibiotics as early as possible, and always **within the first hour** of recognising severe sepsis and septic shock.

- Broad spectrum: one or more agents active against likely bacterial/fungal pathogens and with good penetration into presumed source.
- Reassess antimicrobial regimen daily to optimise efficacy, prevent resistance, avoid toxicity and minimise costs.
- Consider combination therapy in *Pseudomonas* infections.
- Consider combination empiric therapy in neutropenic patients.

Combination therapy no more than 3 - 5 days and de-escalation following confirmation of susceptibilities.

Duration of therapy typically limited to 7 - 10 days; longer if response is slow or undrainable foci of infection or immunologic deficiencies.

Stop antimicrobial therapy as soon as they are not required.

2.7.8 Source identification and control

A specific anatomic site of infection should be established as rapidly as possible and within the first 6 hours of presentation.

- Formally evaluate patient for a focus of infection amenable to source control measures.
- Implement source control measures as soon as possible following successful initial resuscitation.
- Choose source control measure with maximum efficacy and minimal physiologic upset.
- Remove intravascular access devices if potentially infected.

2.7.8 Steroids

- Consider intravenous hydrocortisone at 2 mg/Kg/6hr (maximum of 200 mg/day) for septic shock when hypotension remains poorly responsive to adequate fluid resuscitation and vasopressors.
- Steroid therapy may be weaned once vasopressors are no longer required.
- Do not use corticosteroids to treat sepsis in the absence of shock unless the patient's endocrine or corticosteroid history warrants it.

2.7.9 Blood product administration

Give red blood cells when haemoglobin decreases to <7.0 g/dL

Do not use erythropoietin to treat sepsis related anemia.

Do not use fresh frozen plasma to correct laboratory clotting abnormalities unless there is bleeding or planned invasive procedures.

Do not use antithrombin therapy.

Administer platelets when:

- Counts are $<5 \times 10^9/L$ regardless of bleeding
- Counts are $5 - 30 \times 10^9/L$ and there is significant bleeding risk

- Higher platelet counts $\geq 50\,000/\text{mm}^3$ ($50 \times 10^9/\text{L}$) are typically required for surgery or invasive procedures

2.7.10 Mechanical ventilation of sepsis-induced acute lung injury (ALI) or acute respiratory distress syndrome (ARDS)

- Target a tidal volume of 6 mL/Kg body weight in patients with ALI/ARDS.
- Target initial upper limit plateau pressure $\leq 30\text{cmH}_2\text{O}$. Consider chest wall compliance when assessing plateau pressure.
- Allow CO_2 to increase above normal if needed to minimize plateau pressures and tidal volumes.
- Positive end expiratory pressure (PEEP) should be set to avoid extensive lung collapse at end expiration.
- Maintain mechanically ventilated patients in a semi recumbent 30 – 45 degrees position unless contraindicated.
- Non-invasive ventilation may be considered in the minority of ALI/ARDS patients with mild-moderate hypoxemic respiratory failure. The patients need to be haemodynamically stable, comfortable, easily arousable, able to protect/clear their airway and expected to recover rapidly.
- Use a weaning protocol and a spontaneous breathing trial (SBT) regularly to evaluate the potential for discontinuing mechanical ventilation. SBT options include a low level of pressure support with continuous positive airway pressure of 5 cmH_2O or a T-piece.
- Before the SBT, patients should:
 - be arousable.
 - be haemodynamically stable without vasopressors.

- have no new potentially serious conditions.
- have low ventilatory and end-expiratory pressure requirement.
- require FIO₂ levels that can be safely delivered with the face mask or nasal cannula.
- Do not use a pulmonary artery catheter for the routine monitoring of patients with ALI/ARDS.
- Use a conservative fluid strategy for patients with established ALI who do not have evidence of tissue hypoperfusion.

2.7.11 Sedation, analgesia, and neuromuscular blockade in sepsis

Use sedation protocols with a sedation goal for critically ill mechanically ventilated patients.

Use either intermittent bolus sedation or continuous infusion sedation to predetermined end points (sedation scales), with daily interruption/ lightening to produce awakening.

Avoid neuromuscular blockers where possible. Monitor depth of block with train of four when using continuous infusions.

2.7.12 Glucose control

- Use IV insulin to control hyperglycaemia in patients with severe sepsis following stabilisation in the CCU. Aim to keep blood glucose < 8.3mmol/L using a validated protocol for insulin dose adjustment.
- Provide a glucose calorie source and monitor blood glucose values every 1-2 hours (4 hours when stable) in patients receiving intravenous insulin.

- Interpret with caution low glucose levels obtained with point of care testing, as these techniques may overestimate arterial blood or plasma glucose values.

2.7.13 Renal replacement therapy

- Intermittent haemodialysis and continuous venovenous hemofiltration (CVVH) are considered equivalent. CVVH offers easier management in haemodynamically unstable patients.
- In younger children, peritoneal dialysis may be a better option.

2.7.14 Bicarbonate therapy

Do not use bicarbonate therapy for the purpose of improving haemodynamic or reducing vasopressor requirements when treating hypo perfusion-induced lactic acidaemia with $\text{pH} \geq 7.15$.

2.7.15 Deep vein thrombosis (DVT) prophylaxis

Use either low-dose unfractionated heparin (UFH) or low molecular weight heparin (LMWH), unless contraindicated.

Use a mechanical prophylactic device, such as compression stockings or an intermittent compression device, when heparin is contra indicated.

Use a combination of pharmacologic and mechanical therapy for patients who are at very high risk for DVT.

In patients at very high risk LMWH should be used rather than UFH.

2.7.16 Stress ulcer prophylaxis

Provide stress ulcer prophylaxis using proton pump inhibitor. Benefits of prevention of upper GI bleed must be weighed against the potential for development of ventilator- acquired pneumonia.

2.7.17 Consideration for limitation of support

As the patient is managed, discuss at all stages the care plan and care objectives.

Once a decision has been made by a multidisciplinary team to change the care objectives from treatment to terminal support, the parents/care givers must be given an opportunity to give their guided input.

Once they have been informed, they need to sign the Treatment plan for a patient with life limiting illness, which will be executed thereafter.

2.8 DIABETIC KETOACIDOSIS

2.8.1 Diagnosis

- Hyperglycaemia Glucose >11.1 mmol/L
- Blood PH <7.3 or HCO_3^- <15 mmol/L
- Raised Ketones in the blood and urine

2.8.2 Categorization of DKA severity

- Mild DKA – venous pH 7.2 - 7.29 or bicarbonate < 15 mmol/L. No clinical signs. Assume 5% dehydration
- Moderate DKA – venous pH 7.1 - 7.19 or bicarbonate < 10 mmol/L. easily detectable loss of skin turgor, reduced capillary return. Assume 5% dehydration

- Severe DKA – venous pH less than 7.1 or serum bicarbonate < 5 mmol/L. Poor perfusion, rapid pulse, hypotension. Assume 10% dehydration

Due to risk of overestimation, do not use more than 7% dehydration to calculate fluid deficit for replacement over 48 hours

2.8.3 Investigations

- Blood glucose, urea, and electrolytes, finger prick Ketones
- Arterial or capillary acid/base status
- Urine - Ketones, microscopy, culture
- Check for precipitating cause e.g., Infection (Urine, Full Haemogram, blood cultures; consider Chest x-ray)

2.8.4 Management

- Where feasible, weigh the patient to reduce the chances of errors in fluid calculations. For overweight or obese patients, or those heavier than 65 Kg, use the 97th percentile weight for age
- Airway: Ensure that the airway is patent and maintain with an oropharyngeal airway if patient is not alert and insert a nasogastric tube if vomiting and leave on free drainage
- Breathing: Give high flow oxygen
- Circulation: Insert an intravenous line and take blood samples, take heart rate and blood pressure and attach a vital signs/cardiac monitor

2.8.4.1 Fluid Requirements

- If in shock, give Normal Saline 10 mL/Kg over 10 - 15 minutes. Reassess circulatory status after the bolus and repeat until perfusion is re-established (alert, warm peripheries, capillary

refill <2 secs, normal volume pulses), maximum 40 mL/Kg, beyond which inotropes may be required

- If not in shock but dehydrated, give one bolus of Normal Saline 10 mL/Kg over 20 – 30 minutes
- The Holliday Segar formula for 24-hour maintenance fluid requirements is 100 mL/Kg for the first 10 Kg of weight, 50 mL/Kg for the second 10 Kg and 20 mL/Kg for the remaining weight, to a maximum of 2400 mL per day.
- The fluid requirement over 48 hours is calculated using the Holliday and Segar formula plus 70 mL/Kg (assuming 7% dehydration). The maximum fluid volume should not exceed twice the maintenance requirements

This has been simplified in Table 9 below:

- The patient should be kept nil per oral for the first 24 hours, thereafter, the patient can be allowed oral fluids or feeds if metabolically stable and is alert
- Use (0.9%) normal saline for the first 12 hours, and thereafter, switch to (0.45%) half normal saline, depending on the serum sodium concentrations with guidance of the endocrinologist
- Once the blood sugar levels fall below 15 mmol/L, the infusion fluid is changed to 5% dextrose-saline (DNS)
- If the blood sugar falls very quickly, i.e., within the first few hours, change to 5% dextrose-saline. When the blood sugar reaches 12 - 15 mmol/L, use 0.45% saline with 5% dextrose. Aim to keep the blood sugar at 10 - 12 mmol/L

Table 9: Fluid maintenance and replacement infusion

Weight	Maintenance Fluid 48 hours	Estimated Fluid loss 7%	Total Fluid for 48 hours	Infusion rate
4	800	280	1080	23
5	1000	350	1350	28
6	1200	420	1620	34
7	1400	490	1890	39
8	1600	560	2160	45
9	1800	630	2430	51
10	2000	700	2700	56
11	2100	770	2870	60
12	2200	840	3040	63
13	2300	910	3210	67
14	2400	980	3380	70
15	2500	1050	3550	74
16	2600	1120	3720	78
17	2700	1190	3890	81
18	2800	1260	4060	85
19	2900	1330	4230	88
20	3000	1400	4400	92
22	3080	1540	4620	96
24	3160	1680	4840	101
26	3240	1820	5060	105
28	3320	1960	5280	110
30	3400	2100	5500	115
32	3480	2240	5720	119
34	3560	2380	5940	124
36	3640	2520	6160	128
38	3720	2660	6380	133
40	3800	2800	6600	138
45	4000	3150	7150	149
50	4200	3500	7700	160
55	4400	3850	8250	172
60	4600	4200	8800	183
65	4800	4550	9350	195
>65	4800	4800	9600	200

- If the blood glucose falls below 10 mmol/L, and the patient is still sick and acidotic, increase the dextrose in the infusion to 7.5% or higher. **Do not turn down insulin infusion**
- Hartmann solution may be used in the maintenance fluid after the initial 12 hours with less risk of hyperchloremic metabolic acidosis but higher chances of hyperkalemia.

2.8.4.2 Admission criteria to PCCU (otherwise ward)

Admission criteria to PCCU

- Initial pH < 7.1
- Marked respiratory distress
- Shock requiring fluid resuscitation
- Age < 2 years
- Coma
- Seizures

2.8.4.3 Management on/after admission

Special Instructions

- Strict fluid balance (input and output charting)
- Check all urine drained for Ketones
- Hourly observations: Temperature pulse, Blood pressure, respiratory rate, level of consciousness and pupillary size and reaction to light
- Hourly glucose (Glucometer) while on insulin infusion; other biochemistry as clinically indicated

2.8.4.3.1 Insulin

Note: Beware the rare entity of hyperglycaemic-hyperosmolar non-ketotic coma: Insulin should ONLY be used after discussion with endocrinologist

Add 50 units of clear/short-acting insulin (Actrapid HM or Humulin R) to 50 mL Normal Saline (1 unit/mL solution). Ensure that the insulin is clearly labelled

- Start insulin infusion after 1 – 2 hours of intravenous fluid therapy at 0.05 units/Kg/hr. and adjust the concentration of dextrose to keep blood glucose 10 - 12 mmol/L (range 0.05 – 0.1 Units/Kg/hr.). Adequate insulin must be continued to clear acidosis (Ketonemia).
- The insulin infusion can be discontinued when the child is alert and is metabolically stable
 - blood glucose <10 -12 mmol/L
 - PH > 7.30 and HCO₃ > 15 mmol/L
 - normal Ketones in blood or none in urine).
- The best time to change to Subcutaneous insulin (SC) is just before mealtime. The insulin infusion should only be stopped 30 minutes after the first SC injection of insulin or as guided by the endocrinologist.

2.8.4.3.2 Potassium

- Start Potassium Chloride infusion after the initial resuscitation fluid has been given at a concentration of 40mmol/L unless there is evidence of acute kidney injury. Measure levels 2 hours after starting therapy and every 2 to 4 hours thereafter.
- **Give no Potassium if the serum level is >5.5 mmol/L or if the patient is anuric**
- If Potassium is low at presentation (<3.0 mmol/L), then insulin administration should be deferred until Potassium is >3.0 mmol/L. This may require high concentration of potassium chloride requiring a central line insertion.

2.8.4.3.3 Bicarbonate

- This is usually not necessary if shock has been adequately corrected. Continuing acidosis usually means insufficient

resuscitation. In patients with severe DKA (PH <6.9 or HCO₃ <5mmol/L, small amounts may be given after discussion with the intensivist.

- Only consider bicarbonate if there is life threatening hyperkalemia or in severe acidosis with impaired myocardial contractility.
- Bicarbonate dose (mmol) = $0.15 \times \text{body weight (Kg)} \times \text{base deficit}$ below 15mmol/L. Give over 30 min with cardiac monitoring. Reassess acid base status. Remember risk of Hypokalemia

2.8.5 Hazards

2.8.5.1 Hypernatremia

- Measured serum sodium is depressed by the dilutional effect of the hyperglycaemia. To "adjust" sodium concentration, use the following formula:
- Adjusted sodium = measured sodium + $0.3 \times (\text{measured glucose} - 5.5 \text{ mmol/L})$, i.e., 3 mmol/L of sodium to be added for every 10 mmol/L of glucose above 5.5 mmol/L.
- The sodium level should rise as the glucose level falls during treatment. If this does not happen, or if there is hyponatremia, it indicates overzealous volume correction and insufficient electrolyte replacement which may place the patient at risk of cerebral oedema

2.8.5.2 Hypoglycaemia

- If blood glucose is <2.2 mmol/L give IV 10% dextrose 2 mL/Kg and continue with a 10% infusion until stable
- **Do not discontinue the insulin infusion**, but consider reducing the rate to 0.05 Units/Kg/hour if it was higher

2.8.5.3 Cerebral Oedema

Some degree of subclinical brain swelling is present during most episodes of diabetic ketoacidosis. Clinical cerebral oedema occurs suddenly, usually between 6 and 12 hours after starting therapy (range 2 to 24 hours). Mortality or severe morbidity is very high without early treatment

2.8.5.3.1 Prevention

Go slow on the correction of fluid and biochemical abnormalities. Optimally, the rate of fall of blood glucose and serum Osmolality should not exceed 5 mmol/L/hr, but in children there is often a quicker initial fall in glucose. Patients should be nursed head up

2.8.5.3.2 Warning signs

- First presentation
 - long history of poor control
 - young age (<5yr)
 - No rise in sodium level as glucose level falls
 - hyponatremia during therapy
 - initial adjusted Hypernatremia
 - Headache
 - Irritability
 - Lethargy
 - depressed consciousness
 - incontinence, thermal instability
- Late presentation
 - Bradycardia
 - increased systolic blood pressure
 - respiratory impairment
 - Papilloedema

2.8.5.3.3 Treatment of Cerebral oedema

- 20% Mannitol 0.5 - 1 g/Kg IV stat. Give immediately when the clinical diagnosis is made - do not delay for confirmatory brain CT scan
- Severe reduce fluid input
- Nurse Head up – about 30° to 45°
- Transfer immediately to PCCU

2.10 PHAEOCHROMOCYTOMA CRISIS

Phaeochromocytoma crisis is caused by the action of unopposed high circulating levels of catecholamines acting at adrenoreceptors: α -receptors cause a pressor response with increases in blood pressure, while β -receptor activation has positive inotropic and chronotropic effects, any patient presenting with acute hypertension and tachycardia should be considered at risk of Phaeochromocytoma, especially if young

2.10.1 Risk factors

Spontaneous

2.10.2 Associated conditions:

Multiple Endocrine Neoplasia type 2

Neurofibromatosis type 1

Von Hippel-Lindau syndrome

Ataxia telangiectasia

Tuberous sclerosis

Sturge-Weber syndrome

Precipitating factors

Spontaneous

Haemorrhage into Pheochromocytoma

Others

- Exercise
- Pressure on abdomen
- Urination
- Drugs: Guanethidine, Glucagon, Naloxone, Metoclopramide, ACTH, cytotoxics, tricyclic antidepressants

2.10.3 Clinical features

Hypertension, palpitations, sweating, pallor, pounding headache, anxiety and tremulousness, pulmonary oedema, feeling of impending death, hyperhidrosis, nausea and vomiting, abdominal pain (tumour haemorrhage), paralytic ileus, hyperglycaemia, altered consciousness (hypertensive encephalopathy), myocardial infarction, and stroke

The attacks last between 15 to 60 minutes

Signs of end organ hypertensive damage; hypertensive retinopathy and papilledema, left ventricular hypertrophy, renal impairment, and proteinuria

2.10.4 Biochemical diagnosis

Biochemical diagnosis is made by:

- Collecting urine into acidified bottles for estimation of
- 24-hour free catecholamines and metanephrines
- Plasma metanephrines and catecholamines are also increasingly be used for this diagnosis.

2.10.5 Treatment

- Should not wait for biochemical confirmation
 α -adrenergic receptor blockers - Phentolamine and Phenoxybenzamine are the agents of choice
- Intravenous Phentolamine is the agent of choice for the hypertensive crisis secondary to phaeochromocytoma at a dose of 0.1 - 0.2 mg/Kg, it can also be used intramuscularly
- Phenoxybenzamine (alpha blocker) is the drug of choice, with a starting dose of 0.25 – 1 mg/Kg orally three times a day, increasing to a maximum dose of 240 mg/24 hour. Doses above 40 mg three times a day are seldom required. Dose titration is performed every 48 hours until control of blood pressure is achieved
- After the first 48 hours addition of Propranolol 0.5 – 4 mg/Kg/day PO divided every 8 hours; not to exceed 60 mg/day orally three times a day may be added

Important pharmacological issues in treatment

It is vital that 48 hours of α -blockade precede β -blockade to avoid exacerbating a crisis through the unopposed action of catecholamines at α -receptors

Alpha-antagonists such as doxazosin, and calcium channel antagonists, while increasingly used for maintenance therapy before operation for Phaeochromocytoma, are not recommended for management of crisis

Labetalol is not recommended as this has relatively greater β -blocking action compared to its α -blocking action, and hence can even precipitate or worsen Phaeochromocytoma crisis

2.11 ACUTE HYPERCALCEMIA

2.11.1 Causes of hypercalcemia

Endocrine:

- Hyperparathyroidism (adenoma, hyperplasia, carcinoma)
- Multiple endocrine Neoplasia
- PTH related protein production by solid tumours

Neoplastic:

- Carcinoma and bone invasion
- Myeloma
- Granulomatous:
 - Sarcoidosis
 - Tuberculosis
 - Berylliosis
- Iatrogenic:
- Vitamin D toxicity
- Thiazides
- Vitamin A
- Renal failure:
- Tertiary hyperparathyroidism.
- Aluminium toxicity
- Miscellaneous:
- Paget's disease of bone
- Familial hypocalciuric hypercalcemia.
- Hypophosphatemia

2.11.2 History

History of polyuria and polydipsia, and there can be dehydration, bone pain, confusion, anorexia, and constipation. Relevant drug and family histories must be taken

The cornerstone of differential diagnosis is the measurement of serum parathyroid hormone (PTH)

2.11.3 Investigations at presentation should include:

- Parathyroid Hormone, PTH
- Serum total protein with immunoglobulin electrophoresis for myeloma, Albumin, Phosphate, Magnesium
- Erythrocyte sedimentation rate, full blood count
- ECG
- Chest radiography
- Fractional excretion of Calcium

2.11.4 Treatment

- Any precipitating drugs should be stopped
- The mainstay of treatment is adequate volume repletion with intravenous Normal Saline
- Loop diuretics such as Frusemide, which has calciuretic effects, should only be used after initial volume expansion
- Intravenous bisphosphonates, such as sodium pamidronate at a dose of 30–90mg are extremely effective in the treatment of hypercalcemia of malignancy, with duration of action that lasts days to weeks
- It is important to remember that serum PTH will rise after an acute fall in serum calcium induced by such treatment and hence it is vital that the initial sample for PTH is obtained before bisphosphonate treatment
- Cases of primary Hyperparathyroidism should be referred to endocrine surgeons
- Corticosteroids (Prednisolone 1 – 2 mg/Kg/) are the medications of choice, if granulomatous diseases such as Sarcoidosis, or vitamin A or D intoxication are being considered
- In subcutaneous fat necrosis, intravenous fluids, diuretics, and corticosteroids should be use

2.12 THYROID STORM

2.12.1 Precipitating factors

2.12.1.1 *General*

- Infection
- Non-thyroidal trauma or surgery
- Psychosis
- Parturition
- Myocardial infarction or other acute medical problems

2.12.1.2 *Thyroid specific*

- Radioiodine
- High doses of iodine-containing compounds (for example, radiographic contrast media)
- Discontinuation of antithyroid drug treatment
- Thyroid injury (palpation, infarction of an adenoma)
- New institution of Amiodarone therapy

2.12.2 Cardinal features

Include severe tachycardia, fever (usually 38.5°C), gastrointestinal dysfunction (vomiting, diarrhoea, and occasional jaundice), agitation, confusion, delirium, or coma. Congestive heart failure may occur, particularly in the elderly, and most patients have systolic hypertension

2.12.2 Biochemical features

Include hyperglycaemia, leucocytosis, high free T3 and T4, low TSH, mild hypercalcemia, and abnormal liver function tests. Adrenal reserve may be impaired

Treatment

Carbimazole/Methimazole 0.5 – 1 mg/Kg/day orally in 2 or 3 divided doses, or Propylthiouracil in a dose of 5 – 7 mg/Kg/day is given.

They inhibit thyroid hormone synthesis and conversion of thyroxine to triiodothyronine.

One hour after starting any of the above medications, iodide (like, eight drops of Lugol's iodine every six hours) is given to inhibit thyroid hormone release

- High doses of β -blocker should be given, and Propranolol at a dose of 2 - 4 mg/Kg/day every six hours is recommended
- Cholestyramine, 4 g every 6 - 12 hours, binds thyroid hormone in the gut and thus interrupts the modest enterohepatic circulation of thyroid hormone; its use will lead to a more rapid lowering of circulating thyroid hormones
- In exceptional cases, peritoneal dialysis or plasmapheresis may be needed
- Treatment of any underlying illness follows the usual lines
- Supportive treatment includes the use of external cooling, possibly supplemented by chlorpromazine, cautious intravenous fluids, or Oxygen as determined by appropriate assessment and empirical administration of intravenous Hydrocortisone 100mg every eight hours

2.13 MYXOEDEMA COMA

2.13.1 Precipitating factors

Hypothermia

Infections especially pneumonia

Myocardial infarction or congestive heart failure

Cerebrovascular accident

Respiratory depression due to drugs (for example

Anaesthetics, sedatives, tranquillizers)

Trauma or gastrointestinal blood loss

2.13.2 Clinical features

2.13.2.1 The three main features are:

- Altered mental state ranging from poor cognitive function through psychosis to coma
- Hypothermia (as low as 23°C) or absence of fever in spite of severe infection (prognosis worsens as the core temperature falls)
- The presence of a precipitating event.

2.13.2.2 Other features

Most patients with hypothyroidism have typical facies and many of them have respiratory depression secondary to a decreased hypoxic ventilatory drive and an impaired response to hypercapnia: the higher the hypercapnia the higher the likelihood of coma.

Cardiac enlargement, bradycardia, decreased ventricular contractility, hypotension, and ECG changes (low voltage, non-specific ST wave changes and sometimes torsade's des pointes with a long QT interval) are common

Many patients have anorexia, abdominal pain and distension, and constipation and these changes may rarely lead to paralytic ileus and megacolon

2.13.3 Biochemical abnormalities include:

Hyponatremia, normal or increased urine sodium excretion,

Raised creatine phosphokinase and lactate dehydrogenase

Hypoglycaemia

Normocytic or macrocytic anemia

Thyroid stimulating hormone values may only be modestly raised (and will be normal or low in secondary hypothyroidism) but free thyroxine levels are usually very low

2.13.4 Treatment

The three principles of management are:

- Rapid institution of thyroid hormone replacement
- Treatment of the precipitating cause
- Provision of ventilatory and other support

Thyroxine institution may be done in two ways:

- Intravenous thyroxine can be given as a bolus of 300 - 500 mcg, followed by 50 - 100 mcg daily then convert to oral when feasible. If not responding well in 24 hours, intravenous Triiodothyronine 10 mcg can be added to intravenous Thyroxine 200 mcg then can be added, then IV Triiodothyronine 10mcg every 12 hours and IV thyroxine 100 mcg every 24 hours, until the patient resumes normal oral thyroxine. Oral treatment with similar doses is also possible
- Intravenous Thyroxine 200 – 300 mcg plus intravenous Triiodothyronine 10 – 25 mcg, 2.5 – 10 mcg every 8 hours depending on age and presence of cardiovascular risk factors. On clinical improvement, the Triiodothyronine is stopped, and a daily dose of Thyroxine is maintained

Treatment of the underlying precipitant is usually straight forward.

All patients should be admitted to Critical care or the HDU: most patients require ventilator support for 1 - 2 days

Hypothermia should be treated with space blankets, since active rewarming leads to circulatory collapse

Cautious volume expansion using intravenous saline usually suffices, but hypertonic saline may need to be considered if the serum sodium is very low (<120 mmol/L) and intravenous glucose may be required for hypoglycaemia

There is often a degree of temporarily impaired adrenal function, and most authorities advocate routine intravenous administration of 50 - 100 mg Hydrocortisone every eight hours until recovery

2.14 ACUTE PITUITARY APOPLEXY

2.14.1 Causes

Acute pituitary apoplexy occurring in 0.6 – 9.1% of all surgically treated pituitary tumours, but is potentially life threatening

Haemorrhagic infarction of a pituitary tumour

Normal pituitary gland

2.14.2 Risk factors include:

- Spontaneous
- Anticoagulation – pre-existing haemodialysis, cardiac surgery
- Hypertension
- Raised intracranial pressure
- Dynamic pituitary testing - insulin tolerance test, thyroid releasing hormone test, gonadotrophin releasing hormone test, corticotrophin releasing hormone test
- Medications - oestrogens, Bromocriptine, Aspirin
- Pituitary radiotherapy

2.14.3 Treatment

- If pituitary apoplexy is suspected
- Give Hydrocortisone 2 mg/Kg IV/IM stat, then 100 mg/m² in 4 – 6 hourly doses intramuscularly six hourly, or 4 mg/hour intravenously should be administered without delay and continued until the crisis is over
- Before administration bloods should be drawn for cortisol, prolactin, follicle stimulating hormone, luteinising hormone, oestradiol (women), testosterone (men), sex hormone binding globulin, free thyroxine, thyroid stimulating hormone, insulin-like growth factor-1, and preferably
- ACTH (needs immediate cold centrifugation and freezing at negative 220°C), urinary and plasma Osmolality blood glucose should be monitored and treated accordingly
- Standard cardiopulmonary supportive measures are needed
- Early neurosurgical intervention is associated with improved neuro-ophthalmic outcome

2.15 PITUITARY CRISIS

2.15.1 Causes

- Not matching Glucocorticoid dose to stress in known patient with pituitary deficiency
- Undiagnosed Hypopituitarism
 - Pituitary adenomas or other intrasellar and parasellar tumors
 - Inflammatory and infectious destruction
 - Surgical removal and traumatic brain injury (TBI)
 - Radiation-induced destruction of pituitary tissue
- Subarachnoid haemorrhage
- Postpartum agalactasia pituitary necrosis (Sheehan syndrome)

2.15.2 Clinical presentation

Weakness, fatigue, or altered mental status without a clear diagnosis, Hypotonia, bradycardia, decreased skin and nipple pigmentation, muscle weakness, vomiting, nausea, constipation, hypothermia, and hypoventilation

A postpartum galactic is often the first sign of Sheehan's syndrome

2.15.3 Investigations

- TBC, serum electrolytes
- Cortisol level, prolactin level
- Evaluate the hypothalamic-pituitary-adrenal axis - corticotropin stimulation test
- Assess thyroid function – serum thyrotropin (TSH) and thyroxine (T4)
- 24hour fluid balance then a water deprivation test and an aqueous vasopressin stimulation test

2.15.4 Radiological findings

A lateral skull film can delineate contours of the sella turcica

Both MRI and CT scans should be obtained with intravenous contrast to increase sensitivity of the tests.

2.15.5 Treatment

- Standard resuscitation with IV fluids, Vasopressors
- Hypoglycemia must be sought and treated
- Electrolyte imbalance is corrected following the usual guidelines
- Hormone replacements
 - Hydrocortisone – high dose
 - Thyroxine
 - ADH analogues

Note

Do not start on thyroxin therapy before confirmation of normal cortisol levels, or Hydrocortisone replacement

2.16 POISONING GUIDELINES

The management of acute poisoning is discussed in detail in the **Gertrude's Children's Hospital Poisoning Management Guidelines**.

Below is a summary guideline. The actions below are not necessarily outlined in any order of implementation.

Manage patients as per general guidelines below and specific management as per the Gertrude's Children's Hospital Poisoning Management Guidelines.

- Call the Gertrude's Drug and Poison Information unit on 0708 988 400 or
- 020 7206237/ 020 7206239 / 020 7206240 for information support.
- Consult the Paediatrician on call.
- Admit patient.

2.16.1 General management

- Maintain adequate airway.
- Provide adequate oxygenation.

- Treat convulsions if any and make efforts to establish the aetiology of the seizures so that appropriate treatment can be administered.
- Treat coma if patient is in coma. It is important to consider other causes of depressed sensorium, including ruling out structural lesions such as subdural haematoma.
- The following agents can be utilized in the poisoned patient with altered level of consciousness:
 - 100% oxygen in suspected cases of poisoning with carbon monoxide, hydrogen sulphide, cyanide and asphyxiants.
 - Thiamine to prevent Wernicke's encephalopathy in an alcohol intoxicated patient: Dose is 100 mg IV.
 - Glucose to reverse the effects of drug induced hypoglycaemia: Dose is 0.5 - 1g glucose/Kg body weight in children (e.g., 5 – 10 mL/Kg of 10% dextrose).
 - Naloxone in cases of possible opioid toxicity (assess pupils for myosis).
 - Flumazenil in cases of coma known to be caused by benzodiazepines ALONE (because of the risk of provoking convulsions or arrhythmias, flumazenil should not be given when the patient is suspected of having taken other drugs as well).
 - Correct metabolic abnormalities. The following metabolic abnormalities should be corrected:
 - Hypokalemia
 - Hyperkalemia
 - Hypomagnesaemia
 - Hypothermia
 - Hyperthermia
 - Hypoglycaemia
 - Hypocalcemia

- Acid-base abnormalities, particularly metabolic acidosis.
- Prevent absorption of poisons by gut decontamination using single dose activated charcoal or gastric lavage as may be indicated if a patient has taken a potentially toxic amount of a poison up to one hour following ingestion.
- Avoid emesis.
- Wash off skin and/or eye exposure of poisoning if indicated.
- Enhance elimination of poison through multiple activated charcoal doses if indicated.

2.16.2 Supportive management

Provide supportive care as will be indicated:

- Monitor vital signs (blood pressure, heart rate, respiratory rate, temperature).
- Monitor fluid input and output.
- Monitor level of consciousness, pattern of breathing and regularity of the heart rate.
- Monitor oxygen saturation using a pulse oximeter.
- Administer intravenous fluids for maintenance and fluid loss replacement.
- Carry out intensive nursing care to avoid aspiration or the development of bed sores.
- Treat metabolic disturbances such as electrolyte abnormalities, hypoglycaemia and metabolic acidosis.
- Manage underlying illnesses that may be aggravated by existing problem of poisoning.
- Use specific antidotes if indicated.

2.17 MALIGNANT HYPERTHERMIA

Malignant hyperthermia is a severe reaction to certain inhalational anaesthetic agents and succinylcholine that may manifest immediately, during or several hours after exposure.

It has an incidence of 1:5,000 – 50,000 globally. Susceptibility to this condition has an autosomal dominant inheritance thus its occurrence can be obviated by history taking and subsequent use of either regional anaesthetics or total intravenous anaesthesia.

Malignant hyperthermia presents as an initial increase in muscle spasm, jaw rigidity, tachycardia, tachypnea, increase in carbon dioxide production and a very high temperature. these symptoms may vary but the pathogenesis is a hypermetabolic state characterized by increased skeletal muscle activity and resultant often fatal hyperpyrexia, rhabdomyolysis and coagulopathy.

Diagnosis is clinical with a high index of suspicion on the occurrence of the above signs in the perioperative period.

Laboratory tests such as creatine kinase, muscle biopsy and genetic tests are useful in confirming the diagnosis after initiating patient management.

The mainstay of management of an episode of malignant hyperthermia is:

1. Ceasing the offending anaesthetic agent including changing any respiratory circuits and anaesthetic machine in use.
2. Dantrolene IV 2.5 mg/Kg bolus; repeated until normothermic up to 30 mg/Kg then maintenance at 0.25 mg/Kg/hr.
3. Manage metabolic acidosis.
4. Oxygen supplementation due to increased metabolic demand.
5. Ice packs for cooling the patient
6. Diuresis to manage myoglobinuria
7. Symptomatic supportive care.

Possible differential diagnoses of this condition include thyroid storm, pheochromocytoma and sepsis.

3. FEVER AND FEBRILE CONVULSIONS

3.1 FEVER

Fever is an unnaturally high body temperature, which may be accompanied by headache, shivering and in some cases delirium.

Fever is not an illness but an adaptive physiologic response to a threat. During a fever, hypothalamic regulation of body temperature is altered, and a new (higher) temperature set point is created, causing the body to actively retain heat in a bid to raise its core temperature. This is achieved by increased production (e.g., shivering, chills) and decreased loss (e.g., shunting of blood away from cutaneous to deep vascular beds, hence the cold hands and warm body typical of fever).

It is however a common complaint and cause for anxiety amongst both parents and health care providers. It may signal a mild condition or a much more serious bacterial or viral illness.

3.1.1 Approach considerations

In children younger than 2 months, the risk of severe bacterial sepsis is higher, and therefore the presence of fever should trigger a full septic workup.

In otherwise healthy children above the age of 2 months, most fevers are self-limited and benign, provided that the cause is known, and fluid loss is replaced; fever does not cause brain damage. If there are signs of serious illness, a health care provider should be consulted.

There is no evidence that fever makes the illness worse. Initial measures to reduce the child's temperature include provision of extra fluids and reduced activity.

Fever may merit treatment with an antipyretic agent if the child is uncomfortable (as indicated by decreased activity level, decreased fluid intake, etc.). However, decreased temperature after receiving an antipyretic agent does not help to determine whether the child has a bacterial or viral infection.

Children who are receiving treatment for fever do not need to be awakened to receive the antipyretic agent.

3.1.2 Management

Treatment must include identification and management of the underlying cause.

3.1.2.1 Indications of treatment

Indications for the short-term treatment of fever may include:

- Shock
- Underlying neurologic or cardiopulmonary disease, or other condition with increased metabolic rate (e.g., burn, postoperative state)
- Alteration in fluid and electrolyte balance

- High fever $\geq 40^{\circ}\text{C}$, consider use if uncomfortable and $\geq 38.5^{\circ}\text{C}$
- Discomfort
- Major head trauma
- Postcardiac arrest

3.1.2.2 Medications

Children who are receiving antipyretic medications should not be given combination cough and cold preparations, which often contain antipyretic medications; giving both medications may lead to inadvertent overdose.

Antipyretic medications should be dosed according to weight, rather than age. When suggesting the use of antipyretic medications, the risk of under- or overdosing antipyretics can be minimized if the clinician provides written dosing instructions and a measuring device, such as a properly marked syringe (for liquid formulations); the dosing directions and measuring devices that are included with over-the-counter medications are variable and inconsistent. The instructions should include which formulation (or which concentration of a liquid formulation); how to measure the appropriate volume (for liquid formulation); how often to administer; how to monitor the response; when to discontinue; and when to contact the health care provider.

- Rectal or oral Paracetamol for fever 10 - 15 mg/Kg/ or PO Ibuprofen 4 - 10 mg/Kg/6hours

Note:

Diclofenac has a higher risk of gastrointestinal, cardiovascular and other side effects in children, and thus is not indicated for initial management of fever.

Indications for diclofenac use are:

- Short term treatment of post traumatic and post-operative pain, inflammation and swelling.
- As an adjuvant in severe painful inflammatory infections of the ear, nose and throat.

Children should not receive aspirin for the management of fever, due to the risk of Reye's syndrome.

3.2 FEBRILE CONVULSIONS

A febrile convulsion is a seizure occurring with fever in children aged between the ages of 6months and 5 years without evidence of other underlying cause.

- Simple febrile seizure – single episode in 24 hours, lasting less than 15 minutes and generalized tonic clonic activity
- Complex febrile seizure - multiple episodes in 24 hours with localizing signs and lasting more than 15 minutes

3.2.1 Management

3.2.1.1 General measures

- If seizure ended with recovery of consciousness, determine the source of the fever and treat appropriately.
- Expose to lower temperature in conjunction with antipyretic use.
- If seizure continues apply supportive measures with laying on side, protecting from injury, maintaining airway, low flow oxygen.

3.2.2.2 Medications

- Rectal or oral Paracetamol for fever 10 - 15 mg/Kg/ or PO Ibuprofen 10 mg/Kg/6hours

- Anticonvulsants rarely indicated

3.2.2 Considerations for doing a lumbar puncture

- Strongly consider lumbar puncture in children younger than 12 months, because the signs and symptoms of bacterial meningitis may be minimal or absent in this age group.
- Lumbar puncture should be considered in children aged 12-18 months, because clinical signs and symptoms of bacterial meningitis may be subtle in this age group.
- In children older than 18 months, the decision to perform lumbar puncture rests on the clinical suspicion of meningitis.

4. GASTROINTESTINAL DISEASES

4.1 DIARRHOEA

Diarrhoea is defined by the WHO as the passage of three or more loose or liquid stools per day (or more frequent passage than is normal for the individual). Diarrhoea is considered persistent when it lasts more than fourteen days. There is a significant variability in stooling frequency among babies and children. Breastfed babies usually stool more frequently.

Types of diarrhoea presentation include:

- Acute diarrhoea with severe, some and no dehydration
- Severe persistent diarrhoea with and without malnutrition
- Non severe diarrhoea with and without malnutrition

4.1.1 Acute diarrhoea with severe dehydration

Diarrhoea of abrupt onset associated with frequent passage of loose or watery stools, fever, chills, anorexia, vomiting and malaise.

Unable to drink adequately or Altered level of consciousness plus one of Sunken eyes, skin pinch ≥ 2 seconds
Diarrhoea plus being Unable to drink adequately or Altered level of consciousness plus one of:

- Sunken eyes, skin pinch ≥ 2 seconds
- Skin pinch goes back very slowly (> 2 seconds)

4.1.1.1 Management

- Administer oral rehydration fluids while setting up drip.
- Administer IV Hartmann's solution immediately, Normal Saline is an alternative. This should be given at a dose of 100 mL/Kg as follows:

Table 10: Plan C Management for Severe Dehydration

	First give 30 mL/Kg in:	Then, give 70 mL/Kg in:
<12 months	1 hour*	5 hours
≥12 months	30 minutes*	2 ½ hours

**Repeat initial fluid if radial pulse is still weak or not detectable*

Reassess the child after every 15 – 30 minutes. If rehydration status is not improving, give the IV drip more rapidly.

Give ORS at 5 mL/Kg/hour as soon as the child can drink in addition to the IV fluid.

Reassess an infant after 6 hours and a child after 3 hours.

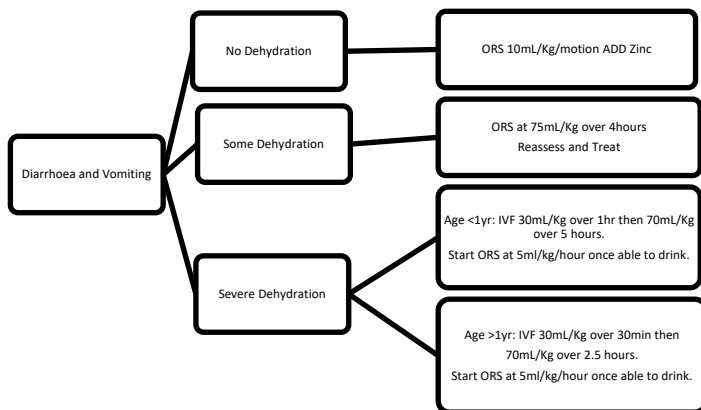
Classify dehydration and treat appropriately.

A normally breastfed child should be breastfed regularly as rehydration continues.

Zinc supplementation when able to tolerate orally:

- <6 months or < weighing 10 Kg – 10 mg elemental Zinc per day for 10 – 14 days
- ≥6 months or weighing ≥10 Kg – 20 mg elemental Zinc per day for 10 – 14 days

Diagram 2: Management of dehydration



4.1.2 Diarrhoea with Some Dehydration

If the child diarrhoea and being Able to drink adequately plus two of:

- Restlessness/irritability
- Sunken eyes, Skin pinch 1- 2 seconds

4.1.2.1 Management

In the first 4 hours, administer ORS at a dose of 75 mL/Kg: If the child wants more ORS, this is given.

- Show the mother how to give the ORS solution, a teaspoonful every 1-2 minutes if the child is under 2 years, frequent sips from a cup for an older child.
- If the child vomits, wait 10 minutes, then resume ORS more slowly.
- If the child becomes puffy, stop ORS and give plain water or breast milk.

Advise breast feeding mothers to continue breastfeeding on demand.

Reassess the child after 4 hours, checking for signs of dehydration.

Discharge on Zinc supplementation.

4.1.2.2 Feeding

In the first 4 hours, do not give any food except breast milk. Breastfed children should continue to breastfeed frequently throughout the episode of diarrhoea.

After 4 hours, if the child still has some dehydration, continue hydration and give food every 3 – 4 hours.

4.1.3 Diarrhoea with no Dehydration

Should be diagnosed if the child is Able to drink adequately plus one of:

- Restlessness/irritability
- Sunken eyes
- Skin pinch 1-2 seconds

4.1.3.1 Management

Management: ORS 10 mL/Kg/motion add Zinc

4.1.4 Severe persistent diarrhoea without malnutrition

Persistent diarrhoea is diarrhoea, with or without blood which begins acutely and lasts for 14 days or longer.

When there is some or severe dehydration, persistent diarrhoea is classified as severe.

4.1.4.1 Management

Assess the child for signs of dehydration and manage with fluids appropriately.

Investigate for intestinal infections and treat appropriately.

Investigate for non – intestinal infections such as pneumonia, sepsis, urinary tract infections, oral thrush and otitis media and treat appropriately.

Give multivitamins plus micronutrients supplementation for 2 weeks.

Treat persistent diarrhoea with blood in the stool with an oral antibiotic effective for Shigella.

4.1.4.2 Feeding

Feeding is essential for all children with persistent diarrhoea.

4.1.5 Non-Severe Persistent Diarrhoea without Malnutrition

Children with diarrhoea lasting 14 days or longer who have no signs of dehydration and no severe malnutrition.

4.1.5.1 Management

- Treat as outpatient.
- Give multivitamins plus micronutrients for 14 days.

4.1.5.2 Prevent dehydration

- Give fluids as for acute diarrhoea with no dehydration (ORS).
- Identify and treat specific infections.

4.1.6 Persistent Diarrhoea with Malnutrition

Refer to section on malnutrition.

4.1.7 Management at point of discharge for all diarrhoea

Counsel the mother on:

- Giving extra fluid (as much as the child can take)
- Give Zinc supplements:
 - <6 months or < weighing 10Kg – 10 mg elemental Zinc per day for 10 – 14 days
 - ≥6 months or weighing ≥10Kg – 20 mg elemental Zinc per day for 10 – 14 days
- Continue feeding
- Follow up – ask the mother to return to clinic immediately if:
 - The child becomes more sick
 - Is unable to drink or to breastfeed
 - Drinks poorly
 - Develops a fever
 - Shows blood in the stool

If the child shows none of these signs but is still not improving, advise the mother to return for follow – up after 5 days.

4.2 ACUTE VOMITING

Nausea and vomiting of abrupt onset.

4.2.1 Management

- Oral rehydration: identify level of dehydration as described in the section on diarrhoea and rehydrate appropriately.
- Use IV fluids if necessary.
- Investigate cause and manage appropriately.
- If vomiting everything, ADMIT and consider a STAT dose of ondansetron to facilitate oral rehydration.

4.3 CHOLERA

An acute infectious disease caused by *vibrio cholera*.

Characteristics include severe diarrhoea with extreme fluid and electrolyte depletion, muscle cramps and prostration. Vomiting may be a feature.

Cholera should be suspected in patients with acute watery diarrhoea and signs of severe dehydration if the disease is in the locality.

4.3.1 Management

Cholera is an easily treatable disease. Most patients can be treated successfully through prompt administration of oral rehydration solution (ORS)

Severely dehydrated patients are at risk of shock and require the rapid administration of intravenous fluids. These patients are also given appropriate antibiotics to diminish the duration of

diarrhoea, reduce the volume of rehydration fluids needed, and shorten the amount and duration of *V. cholerae* excretion in their stool. Give oral Azithromycin 10 mg/Kg

Zinc supplementation when able to tolerate orally

4.3.2 Prevention:

- Hand washing
- Consume clean water or food
- Vaccination - 2 doses separated by 2 - 4 weeks

4.4 DYSENTERY

Usually of infective cause:

Most commonly Bacillary, Amoebic or due to Salmonella or occasionally *E. coli*.

Identify the causative organism by stool examination and/or blood culture.

Maintain hydration.

4.4.1 Treatment:

Amoebic dysentery – Metronidazole 7.5 mg/Kg IV every 8 hours or 5 – 15 mg/Kg PO every 8 hours. The luminal forms should be eradicated by concurrent use of Diloxanide furoate 20 – 25 mg/Kg every 8 hours for children weighing ≥ 25 Kg).

Bacterial dysentery - use Ciprofloxacin 15 mg/Kg PO every 12 hours for three days for those not being admitted. Those admitted use Ceftriaxone 50 mg/Kg IV every 24 hours for 5 days and discharged to complete the course on an oral equivalent like

Cefpodoxime 5 mg/Kg every 12 hours or Cefixime 4 mg/Kg every 12 hours.

4.5 TYPHOID FEVER

This is caused by *Salmonella typhi* and *Salmonella paratyphi*. Consider typhoid fever if a child presents with fever, plus any of the following: diarrhoea or constipation, vomiting, abdominal pain, headache or cough, particularly if the fever has persisted for 7 or more days and malaria has been excluded.

4.5.1 Complications

Complications of typhoid fever include convulsions, confusion or coma, diarrhoea, dehydration, shock, cardiac failure, pneumonia, osteomyelitis and anemia. In young infants, shock and hypothermia can occur.

4.5.2 Treatment

- Supportive care
- If the child has high fever, give Paracetamol.
- Monitor haemoglobin level and, if they are low and falling, consider transfusion.
- Third generation Cephalosporin such as Ceftriaxone 80 mg/Kg IM or IV, once daily or Cefotaxime 50 mg/Kg IV once a day.
- Ciprofloxacin may be used at a dose 15 mg/Kg 12hourly (prescribe with caution in children).
- If there are signs of gastrointestinal perforation, pass a nasogastric tube, keep nil per oral and get surgical opinion.

4.6 ACUTE ABDOMEN

The assessment of the child with acute abdominal pain depends on a good history and careful examination.

Typical features of some important causes of acute abdominal pain in children are described in table 6.

- Acute appendicitis must be considered in any child with severe abdominal pain. In the very young child, in whom the risk of perforation is higher the presenting symptoms are less specific.
- The diagnosis is clinical - no laboratory or radiological tests are required.
- Commonly occurs below the age of 5 years, but the peak age for Intussusception is 4 - 6 months. Plain abdominal x-rays may show signs of bowel obstruction with decreased gas in the right colon. The diagnosis is confirmed by air insufflation or barium enema with reduction usually possible by the same means unless there are signs of peritonitis with the risk of perforation.
- Midgut volvulus is commonest in the newborn period but can occur in later childhood. Predisposing factors include malrotation and abnormal mesentery.
- Some children suffer recurrent non-specific abdominal pain with no organic cause identifiable. Constipation is often an important contributing factor. Psychogenic factors (e.g., family, school issues) need to be considered. These children should be referred for general paediatric assessment.
- Some uncommon diagnoses need to be considered in patients with certain underlying chronic illnesses.
- Hirschsprung's disease can be complicated by enterocolitis, with sudden painful abdominal distension and bloody

diarrhoea. These patients can become rapidly unwell with dehydration, electrolyte disturbances, and systemic toxicity and are at risk of colonic perforation.

- Any neonate presenting with vomiting in the first 72 hours of life – consider small bowel obstruction.
- Primary bacterial peritonitis can occur in children with nephrotic syndrome, post splenectomy and those with ventriculoperitoneal shunts.

Table 11: Some Presentations of Acute Abdomen

Diagnosis	Typical features	
	History	Examination
Acute appendicitis	Abdominal pain becomes increasingly severe, and often localizes to right iliac fossa.	Tenderness, guarding, and rebound. Tenderness usually greatest in right iliac fossa, though may be more diffuse.
Intussusception	Intermittent colicky abdominal pain, vomiting and the passage of blood and or mucus per rectum. There is frequently a preceding respiratory or diarrheal illness.	Pallor, lethargy A sausage-shaped mass is palpable in about two-thirds of cases, crossing the midline in the epigastrium or behind the umbilicus.
Midgut volvulus	Bowel obstruction - abdominal pain, distension; usually bile-stained vomiting.	Distension, tenderness.
Constipation	Can present with quite severe abdominal	Firm stool palpable in lower abdomen

	pain in children; often recurrent.	(sometimes entire colon).
UTI	Infants: fever, vomiting, lethargy. Older children: dysuria, haematuria.	Fever; suprapubic tenderness; loin tenderness if associated pyelonephritis; leukocyte esterase, and nitrites may be positive.
Pneumonia	Fever; may have cough, vomiting.	Fever; tachypnoea, chest indrawing, focal signs at one base.
Gastroenteritis	Vomiting, diarrhoea, fever.	Tenderness, increased bowel sounds; signs of dehydration.

4.7 CONSTIPATION

Constipation is defined variably based on the frequency of defecation, discomfort in passing stools, delayed intestinal transit and weight of the stools. Children complaining of constipation can describe stools that are too small, too big, too infrequent, painful or difficult to expel or that leave a feeling of incomplete evacuation.

4.7.1 Management

4.7.1.1 General measures

- Eliminate medications that may cause or worsen constipation.
- Encourage exercise/activity.
- Eliminate foods identified that may cause constipation.
- Encourage adequate fluid intake.
- Encourage high fibre diet.

- Encourage regular toilet habits, sit on the toilet 5-10min after meals, takes advantage of the gastro-colic reflex.

4.7.1.2 Medicines

- Check for fecal impaction (digital rectal examination): If there is fecal impaction; give enema.

4.7.1.3 Maintenance

- Give osmotic laxatives – Lactulose or magnesium hydroxide, macrogols
- Stimulant laxatives may be used intermittently as rescue therapy to prevent recurrence of impaction but be avoided for long-term, daily use e.g., picosulphate, bisacodyl and senna
- Lubricant laxatives; for those with hard painful/blood-streaked stools; liquid paraffin, castor oil.
- Bulk laxatives; for those with low fibre in diet; bran, ispaghula.
- Drugs acting locally on rectum for anal fissures to relieve pain; Lignocaine creams/ointments.
- If no improvement, refer to a gastroenterologist.
- Children with recurrent constipation should be evaluated for possible hypothyroidism.

4.8 ORAL CANDIDIASIS

An oromucocutaneous disorder (in the oral cavity) caused by infection with various species of *Candida*.

4.8.1 Treatment

First line

- **Nystatin:** Age <30 days: 0.5 mL on each side of the mouth four times a day for ten days. Continue for at least three days after you no longer see any signs of thrush, Age >30 days: 1 mL on each side of the mouth four times a day for ten days.
- **Clotrimazole paint:** 10-20 drops (0.5 – 1 mL) to be applied to affected lesions in the mouth 3-4 times daily. Apply till symptoms disappear completely.
- **Miconazole oral gel:** Neonates, 1 mL to be applied to affected lesions in the mouth 2 - 4 times daily.
Child: 1-2yrs: 2.5 mL to be applied to affected lesions in the mouth three times daily. Retain near lesion for as long as possible before swallowing.
Child: 2 – 6yr: 5 mL to be applied to affected lesions in the mouth three times daily. Retain near lesion for as long as possible, before swallowing.
Child 6 – 12yr: 5 mL 4 times daily. Child 12 – 18yr: 5 – 10 mL 4 times daily.
- **NB:** Miconazole should be given after food or feeding, and treatment should continue for at least 48hrs after lesions have healed. Orthodontic appliances should be removed at night and brushed with the gel.

Second line (severe or refractory)

- Fluconazole Oral/IV:
 - Neonates below 2 weeks: 3 – 6 mg/Kg/day

- Neonates of 2 - 4 weeks 6 mg/Kg stat, then 3 – 6 mg/Kg/day
- Child 1month – 12yrs: 3- 6 mg/Kg/day start (day 1) then 3 mg/Kg (max 100 mg) daily for 7-14days
- Child 12-18yrs: 50mg daily; 100 mg in unusually difficult infections for 7-14 days

Maximum duration is 14 days prolonged in immunosuppression

4.9 GINGIVOSTOMATITIS

Generalized inflammation of the oral mucosa of many possible aetiologies.

4.9.1 Treatment

4.9.1.1 Supportive management

- Oral toilette – Above 6 years: Quarter strength hydrogen peroxide (6% - approx. 20vol): Rinse the mouth for 2 – 3 min. With 15 mL diluted in a tumblerful of warm water 2-3 times daily.
- Topical anaesthetic gel: Mostly combinations of Choline Salicylate, Benzalkonium chloride and Lignocaine hydrochloride. This serves as a local analgesic, antiseptic and surface anaesthetic respectively 1-2 drops or paste is applied to the index finger and gently rubbed on the affected areas when necessary.
- Analgesics: Paracetamol 10 – 15 mg/Kg 6 – 8hourly, Children over 10 years can rinse mouth and gargle with a Chlorhexidine or iodine-based gargle: Chlorhexidine should be used half an hour before or after brushing teeth. Rinse mouth with 10 – 15 mL held in the mouth for about 30 seconds twice daily.

- Povidone-iodine: Age 6 – 18 years; up to 10 mL undiluted or diluted with an equal quantity of warm water held for about 30 seconds up to 4 times daily for up to 14 days.

4.9.1.2 Specific management

Viral: Acyclovir

- Acyclovir oral (tablets/suspension):
 - Child 1 month – 2 years: 100mg 5 times daily usually for 5 days.
 - Child < 2 years: 200mg 5 times daily for 5 days.
 - Duration may be longer if healing incomplete or new lesions appear. Dose may be doubled in immunocompromised or if absorption is impaired
- Oral cream: The cream is to be applied to the infected areas 5 times a day at intervals of 4 hours omitting the night-time application. Treatment should be continued for 5 days, but if healing is not complete, treatment may continue for 5 more days at most
- Severe viral (herpes) infection requires admission and use of parenteral Acyclovir

Septic: Amoxicillin + Metronidazole:

- Amoxicillin oral:
 - Neonate under 7 days, 30 mg/Kg twice daily for 5 – 7 days. Dose doubled in severe infection.
 - Age above 7 days: 30 mg/Kg three times daily for 5 – 7 days; dose doubled in severe infection.
- Metronidazole; Oral:
 - Neonate initially 15 mg/Kg then 7.5 mg/Kg twice daily.
 - Age above 1 month: 7.5mg/Kg (max 400 mg) 8 hourly
- Severe septic infections require admission and use of parenteral CoAmoxiclav

Investigate cause and treat appropriately:

- Micronutrient deficiency – Vitamin C and Vitamin B Complex
 - Vitamin C Oral
 - Child < 4 years; 125 – 250 mg daily in 1 – 2 divided doses
 - Child ≥ 4 – 12years; 250 – 500 mg in 1 -2 divided doses
 - 123 - 18years; 500 mg – 1gm daily in 1-2 divided doses.
 - Vitamin B Complex
 - Child 1 month - 1 year; 5ml 3 times daily
 - Child 1 - 12years; 10 mL3 times daily, 12 – 18 yr 10-15 mL3 times daily
 - The above dosages are for treatment. Prophylaxis is 5mL once daily, twice daily and thrice daily as per the age groups above respectively
 - Iron deficiency: Elemental Iron 6 mg/Kg/day

4.10 GASTRO-OESOPHAGEAL REFLUX DISEASE

Reflux of gastroduodenal contents into the oesophagus, larynx or lungs resulting in symptoms or complications.

4.10.1 Management

Refer to Gastroenterologist.

4.10.1.1 Supportive

- Proper positioning (upright positioning after feeds, elevate head of bed)
- Providing small, frequent feeds
- Weight reduction if obese in the older child

4.10.1.2 Medications

Proton pump inhibitors: Esomeprazole

By mouth

Child 1–11 years (body weight 10–19 kg): 10 mg once daily for 8 weeks.

Child 1–11 years (body weight 20 kg and above): 10 – 20 mg once daily for 8 weeks.

Child 12–17 years: Initially 40 mg once daily for 4 weeks, continued for further 4 weeks if not fully healed or symptoms persist; maintenance 20 mg daily.

By intravenous injection, or by intravenous infusion

Child 1 – 11 years (body weight up to 20 Kg): 10 mg once daily, injection to be given over at least 3 minutes.

Child 1–11 years (body weight 20 Kg and above): 10 – 20 mg once daily, injection to be given over at least 3 minutes.

Child 12–17 years: 40 mg daily, injection to be given over at least 3 minutes.

4.10.1.3 Surgical Options

Fundoplication may be required in a minority of patients.

4.11 PEPTIC ULCER DISEASE

A chronic ulcer in the lining of the gastrointestinal tract

Duodenal ulcer (DU): Most located in the duodenal bulb multiple ulcers and if distal to the bulb raise the possibility of Zollinger-Ellison syndrome.

Gastric ulcer (GU): Much less common than DU (in NSAID absence). Commonly located along lesser curvature of the antrum near the incisura and in the pre-pyloric area.

Oesophageal ulcers: A peptic ulcer in the distal oesophagus may be part of Barrett's epithelial change due to chronic reflux of gastroduodenal contents.

Ectopic gastric mucosal ulceration: May develop in patients with Meckel diverticula or other sites of ectopic gastric mucosa.

4.11.1 Treatment

Aluminium/Magnesium hydroxide: As for gastro-oesophageal reflux.

Refer to gastroenterologist for assessment and follow up.

4.12 HEPATITIS A INFECTION

4.12.1 Treatment

4.12.1.1 Medical Care

- Treatment is supportive because no specific therapy is available.

- Hospitalization is indicated for patients with significant dehydration due to vomiting or those with fulminant hepatitis.
- Medications that have known liver toxicity should be avoided.
- All hepatitis patients should be followed up until their liver function tests normalise.

4.12.1.2 Consultations

- Consultation with a subspecialist for those with prolonged jaundice >1month or with features of hepatic encephalopathy

4.12.1.3 Diet

- No specific dietary changes are needed
- In euvolemic patients with vomiting (but without dehydration that requires intravenous fluid therapy) appropriate intake of oral fluid is recommended, as with other viral illnesses

4.12.1.4 Activity

- Activity can be resumed as tolerated

4.12.1.5 Medication

- No specific medications are indicated

4.12.2 Follow up

4.12.2.1 Further Inpatient Care

- Inpatient care is not needed for most patients with hepatitis A virus (HAV) infection

- Some patients may require hospitalization for intravenous rehydration. Once emesis subsides and the patient can tolerate oral fluids, discharge is appropriate

4.12.2.2 Further Outpatient Care

- Follow up by gastroenterologist for all hepatitis patients.

4.12.3 Prevention

- General prevention measures consist of good personal hygiene, hand washing, ingestion of safe drinking water and proper sanitation.
- Prevention specific to hepatitis A infection includes the use of HAV immune globulin (IG) and HAV vaccine
 - IG is given as an intramuscular injection of 0.02 mL/Kg
 - HAV vaccine is currently licensed for use in children aged 12 months or older

Specific measures to prevent HAV can be divided into 2 groups:

- **Pre-exposure prophylaxis** with HAV vaccine is recommended for persons aged 1 year or older. If the trip is shorter than 2 weeks or if the patient is younger than 1 year, IG should be given. If the trip is longer than 3 months, a larger dose of IG (0.06 mL/Kg) is needed for those who cannot receive the vaccine.
- **Post exposure prophylaxis** consists of the administration of IG to contacts as soon as possible, but not longer than 2 weeks after exposure.

5. EAR NOSE AND TROAT

5.1 TONSILLOPHARYNGITIS

- Tonsillopharyngitis is inflammation of the pharynx and tonsils mostly caused by an acute infection mostly viral.
- Group A streptococcus is a focus of diagnosis due to its potential for preventable rheumatic sequelae.
- Chronic low-grade symptoms usually related to reflux disease or vocal abuse.

5.1.1 Management

5.1.1.1 *Supportive treatment*

- Normal Saline gargle – use 10 to 15 mL every 4 - 6 hours for all children who can gargle usually above 4years
- Children over 10 years can rinse mouth and gargle with a Chlorhexidine or iodine-based gargle: Chlorhexidine should be used half an hour before or after brushing teeth. Rinse mouth with 10 - 15 mL held in the mouth for about 30 seconds twice daily. Povidone iodine: ≥ 6 years up to 10 mL undiluted or diluted with an equal quantity of warm water held for about 30 seconds up to 4 times daily for up to 14 days
- Oral analgesic in pain: Paracetamol 10-15 mg/Kg/dose every 4 – 6 hours as needed

5.1.1.2 *Definitive Treatment*

- Antibiotic treatment is not routinely recommended, due to the prevalent viral aetiology of pharyngitis. It may be indicated

when clinically justified and may be supported by diagnostic tests

- Oral Amoxicillin 25 - 50 mg/Kg/day in divided doses every 8 hours for 5 - 7 days, doubled in severe infection.
- Oral Cefadroxil 30 mg/Kg/day in 2 divided doses for 5 - 7 days

Penicillin Hypersensitivity

- Oral Erythromycin 20 – 40 mg/Kg/day in divided doses every 6 – 8 hours for 5 - 7 days
- Oral Azithromycin 20 mg/Kg once daily for three days
- Oral Clarithromycin 7.5 mg/kg/dose 12 hourly for 10 days

2nd line antibiotics

- Oral Coamoxiclav 90 mg/Kg/day in 2 divided doses for 10 days
- Oral Azithromycin 20 mg/Kg once daily for three days (or other macrolides as above)
- Oral Cefuroxime 30 mg/Kg/day in 2 divided doses for 10 days

3rd line antibiotics

- Intravenous Ceftriaxone 50 - 80 mg/Kg/day as a once daily dose

5.2 OTITIS MEDIA

Otitis media is infection or inflammation of the middle ear

5.2.1 Definitions

- **Acute Otitis Media (AOM):** Usually a bacterial infection accompanied by viral upper respiratory infection; rapid onset of signs and symptoms
- **Recurrent AOM:** 3 or more AOM episodes in 6 months or 4 or more AOM episodes in 1 year
- **Otitis media with effusion (OME):** Persistent inflammation manifested as asymptomatic middle ear fluid that follows treatment of AOM or arises without prior AOM
- **Chronic Otitis Media** with or without cholesteatoma

5.2.2 Acute Otitis Media

This is otitis media described by:

- History of ear pain or pus draining from the ear for a period of less than 2 weeks
- Fever
- Systemic signs may or may not be present
- Ear drum is red, inflamed, bulging and opaque or perforated with discharge on otoscopy

5.2.2.1 Management

Supportive Treatment

- If there is pus draining from the ear, advice the mother to wick the ear repeatedly until the wick is dry and keep doing it several times a day to keep the ear dry

Definitive treatment

- Pain and fever should be managed with oral Paracetamol

10 - 15 mg/Kg/dose every 4 – 6 hours or Ibuprofen 4 – 10 mg/Kg every 6-8 hours

First line antimicrobial therapy

- Oral Amoxicillin 80 - 90 mg/Kg/day in divided doses every 12 hours for 10 days

Penicillin hypersensitivity

- Oral Azithromycin 20 mg/Kg/day once daily for 7 days
or
- Oral Erythromycin 30 – 40 mg/Kg/day in divided doses every 6 – 8 hours for 10 days
or
- Oral Clarithromycin 7.5 mg/kg/dose 12 hourly for 10 days

Second line Antimicrobial Therapy

- Oral Coamoxiclav 80 - 90 mg/Kg/day in divided doses every 12 hours for 10 days
or
- Cefuroxime 30 mg/Kg/day in divided doses every 12 hours for 10 days

Third line Antimicrobial Therapy

- Cefixime 8 mg/Kg/day 12 hourly for 10 days
or
- Cefpodoxime 10 mg/Kg/day given in one or two doses for 10 days

5.2.2.2 Follow up

Refer all patients with ear discharge or recurrent infection to the ENT surgeons

5.2.3 Chronic Suppurative Otitis Media

An Otitis media associated with:

- A history of pus draining from the ear for more than 2 weeks
- Perforated tympanic membrane on examination

Note: Expect gram negative organisms.

5.2.3.1 Management

Supportive treatment

- Advice the mother to wick the ear repeatedly until the wick is dry
- Instil topical antibacterial ear drops: Ciprofloxacin once daily for 2 weeks

Consult an ENT specialist

5.3 Otitis Externa

5.3.1 Definition

Otitis externa, also called swimmer's ear, is an inflammation of the external ear canal or auricle. It often presents with ear pain which may be excruciating, swelling of the ear canal, and sometimes decreased hearing. Typically, there is pain with movement of the ear. Fever is usually not high.

5.3.2 Classification

Otitis externa may be acute (lasting less than six weeks) or chronic. Acute cases are usually due to bacterial infection whereas chronic ones are often due to allergies, otomycosis and autoimmune disorders.

5.3.3 Aetiology

The most common bacteria are staphylococcus aureus followed by streptococcus pneumonia and pseudomonas aeruginosa.

Risk factors:

swimming, minor trauma from cleaning, using hearing aids and ear plugs.

5.3.4 Treatment

1. Pain control - Ibuprofen or paracetamol
2. Antibiotics - local antibiotic, systemic antibiotic
3. Aural wicking or ear toilet

Local antibiotics (may be combined with steroids - betamethasone, hydrocortisone or dexamethasone)

- Ciprofloxacin
- Norfloxacin

Systemic antibiotics only in severe cases

- Coamoxiclav 80 - 90 mg/Kg/day in divided doses every 12 hours for 10 days
- or
- Cefuroxime 30 mg/Kg/day in divided doses every 12 hours for 10 days

Antipseudomonal antibiotics after culture results

5.4 ALLERGIC RHINITIS

Allergic rhinitis is an immediate and/or delayed reaction to airborne allergens associated with:

- Stuffy nose, sneezing, itching, runny nose, cough, halitosis, and repeated throat clearing,
- Sensation of plugged ears and wheezing may occur. Red and itchy eyes suggestive of allergic conjunctivitis
- Seasonal, perennial, or episodic symptoms and exacerbating factor may help identify allergen
- Family history of atopic disease supports diagnosis

5.4.1 Management

5.4.1.1 *Supportive treatment*

- Avoidance therapy – identify and eliminate known or suspected allergens

5.4.1.2 *Definitive treatment*

Selection of the appropriate management strategy depends on the predominant symptoms – rhinitis or congestion.

- Normal Saline nasal drops; 2-3 drops into both nostrils 3-4 times daily when necessary
- Steam inhalation with menthol and eucalyptus drops added into the steaming water twice daily **under supervision**.
- Non-Sedating antihistamines: Cetirizine (use with caution in cardiac patients):
 - Desloratadine; oral
 - Infants – Not recommended

- Child 1 – 2 years; 1.25 mg once daily
 - Child 2 – 5 years; 2.5 mg once daily
 - Child 6 – 11 years; 5 mg once daily
 - Child $\geq$ 12 years; 10 mg once daily
 - Cetirizine; oral
 - Infants – Not recommended
 - Child 12 – 23 months 2.5 mg once daily
 - Child 2 – 5 years 5 mg once daily
 - Child $\geq$ 6 years 5 mg once daily
 - Mast cell Stabiliser – Montelukast once daily dosing:
 - Child $<$ 2 years – Not recommended
 - Child 2 - 5 years; 4 mg,
 - Child 5 - 14 years; 5 mg
 - Child $>$ 14 years; 10 mg
- Corticosteroid Nasal Sprays
 - Fluticasone: 4 years and above; 25mcg in each nostril once daily. Double the dose to 50mcg in severe symptoms and reduce to 25mcg once symptoms are controlled.
 - Budesonide: In children less than 12 years: 64 mcg in each nostril once daily, In children above 12 years, 128 mcg in each nostril once daily for a maximum of 3 months

Recurrence of allergic rhinitis needs follow up by the ENT Surgeons or an Allergist

5.5 SINUSITIS

Sinusitis is a symptomatic inflammation of the mucosal lining of the nasal cavity and paranasal sinuses. It may be caused by a viral or bacterial infection.

Symptoms of acute sinusitis last less than 30 days, subacute sinusitis between 30 - 90 days, and chronic sinusitis beyond 90 days.

5.5.1 Acute Bacterial Sinusitis

5.5.1.1 Definitive Management

- Patients with evidence of invasive infection or any CNS complications should be hospitalized immediately
- Intravenous therapy with third-generation cephalosporin such as Cefotaxime should be initiated until culture results become available
- Less severe cases of acute sinusitis should be treated with oral antibiotics for a minimum of 14 days.

First line

- Coamoxiclav at 80 - 90 mg/Kg/day in 2 divided doses for 10 – 14 days

Second line

- 3rd generation cephalosporins Ceftriaxone or Cefpodoxime.
- Patients with underlying allergic rhinitis may benefit from intranasal corticosteroid nasal spray

- Paracetamol or Ibuprofen if in pain to permit sleep until drainage is achieved
- The application of ice over the sinus may help to relieve pain

5.5.1.2 Supportive treatment:

- Avoidance of allergens
- Paracetamol or Ibuprofen if in pain to permit sleep until drainage is achieved
- The application of ice over the sinus may help to relieve pain

5.5.2 Chronic or recurrent sinusitis

- Recurrent acute bacterial sinusitis is defined as successive episodes of bacterial infections of the sinuses, each lasting less than 30 days and separated by intervals of at least 10 days, during which the patient is asymptomatic
- Chronic sinusitis is defined as episodes of inflammation of the paranasal sinuses lasting more than 90 days.

5.5.2.1 Treatment

- Antibiotic therapy is like that used for acute sinusitis, but the duration is longer.
- Adjuvant therapies such as saline nasal irrigations, antihistamines and topical intranasal steroids may be helpful depending on the underlying cause.
- **Refer to the ENT surgeons**

5.6 EPISTAXIS

Haemorrhage from nostril, nasal cavity or nasopharynx

- Anterior bleed: Originates from anterior nasal cavity, usually Little's area (Kiesselbach's plexus)
- Posterior bleed: Originates from posterior nasal cavity or nasopharynx usually under the posterior half of the inferior turbinate or the roof of the nasal cavity

5.6.1 Management

5.6.1.1 *General measures*

- Resuscitation as indicated
- Sedation, analgesic, antihypertensive or anticoagulant reversal as needed
- Patient should be gowned and sitting, if stable.
- Clear nasal cavity of blood with suction, forceps withdrawal of clot or patient blowing nose

5.6.1.2 *If bleeding has stopped,*

- Rub suspicious areas with wet cotton tipped applicator to identify site. Diffuse ooze or multiple sites suggests systemic cause
- In cases of posterior bleed, identify as either roof or low posterior site since each has different arterial supply (will be important if arterial ligation is necessary).

5.6.1.3 *Anterior bleed*

- Place pledget soaked in vasoconstrictor and local anaesthetic in cavity and pinch nostril for several minutes

to stop bleeding by direct pressure for not more than fifteen minutes.

- If bleeding still not controlled, pack the nostril with gauze soaked in liquid paraffin and refer to ENT surgeon

5.6.1.4 Intractable bleed

- Bilateral packing to achieve adequate compression (admission required)
- Bleeding from roof may be controlled by placing double balloon system with small anterior pack placed above anterior balloon.
- Intractable bleed will require surgical cauterization or arterial ligation (ideally after visual identification of bleeding site to define appropriate arterial supply)

5.6.1.5 Medication Management

- Vasoconstrictor: Cocaine 4%, Phenylephrine 0.25%, Xylometazoline 0.1%, epinephrine 1:1000
- Anaesthetic: Tetracaine 2%, Lidocaine laryngeal spray, Lidocaine jelly 2%, Lidocaine solution 4%, Lidocaine viscous 2%
- Systemic antibiotics and decongestants to prevent sinusitis with packs or balloons
- Consider iron supplementation for patients with considerable blood loss

5.6.2 Prevention/avoidance

- Liberal application of petroleum jelly (Vaseline) to nostril to prevent drying and picking

- Humidification at night
- Cut fingernails

5.7 LUDWIG ANGINA

5.7.1 Definition

Ludwig angina is a life-threatening cellulitis involving the submandibular, sublingual and submental spaces bilaterally. It is a polymicrobial bacterial infection with gram positive bacteria being dominant.

Causes of death in Ludwig angina may include upper airway obstruction, septic shock, mediastinitis and respiratory failure.

Timely diagnosis coupled with prompt management is key in averting untoward effects associated with this condition.

5.7.2 Clinical features

The clinical features include a rapidly progressing firm swelling of the floor of mouth and neck resulting that may result in airway compromise. As a result of progressive obstruction, increased work of breathing including tachypnea, nasal flaring, dyspnea, stridor and head bobbing develop. Fever and tachycardia may be seen.

Predisposing factors include dental caries, upper respiratory tract infections, infection of the submandibular gland, fractures, immunodeficiency states like diabetes and skin and mucosal ulcerations.

5.7.3 Investigations

Full hemogram, blood culture, urea, electrolytes, creatinine, c-reactive proteins, random blood sugar, x-rays (orthopantomogram, chest) and blood gases.

5.7.4 Management

5.6.4.1 Airway management

This is priority to ensure that the airway is secured early

5.7.4.2 Antibiotics

Immediate intravenous antibiotic is required. First line is Amoxicillin + Clavulanic acid combination unless of penicillin allergy. This can be combined with Metronidazole.

5.7.4.3 Antipyretics/Analgesics

Fever and pain control medications are to be used

5.7.4.4 Steroids

There is no consensus of the use of steroids

5.7.4.5 Nutrition and Hydration

Initial fluid resuscitation and maintenance including nutrition intervention is used as required

5.7.4.6 Surgical Intervention

Incision and drainage is required to remove pus and hasten recovery. Where pus may not have formed, the incision done assist

in decompressing the inflammatory process and relieving airway obstruction.

5.7.4.7 Follow up

Follow up is required to establish the cause with and taking appropriate action to prevent subsequent repeat scenario.

6. RESPIRATORY TRACT CONDITIONS

6.1 BACTERIAL PNEUMONIA

An acute bacterial infection of the lung parenchyma

Infection may be community acquired or nosocomial (hospital acquired) by an inpatient admitted for at least 48 hours or inpatient in the previous 3 weeks. In all age groups *Tuberculosis* and *viral infections* can occur.

In the Neonatal period the following organisms are common causes: Group B streptococcus, *Escherichia coli* (and other enteric Gram-negative bacilli), *Listeria monocytogenes*, *Streptococcus pneumoniae*, *Haemophilus influenzae*, *Staphylococcus aureus*, *Neisseria meningitides* and *Salmonella* species

In the age 1 – 3 months, the following organisms are common causes of infection: *Streptococcus pneumoniae*, Group B streptococcus, *Neisseria meningitides*, *Salmonella* species, *Haemophilus influenzae* and *Listeria monocytogenes*.

In the age above 3 months, the following are commoner causes: *Streptococcus*, *Pneumoniae*, *Haemophilus influenzae*, *Neisseria meningitides*, and *Salmonella* species.

Hospital-acquired pneumonia is usually due to gram negative rods such as *Pseudomonas*, *klebsiella*, *Acinetobacter* or at times *Staphylococcus*

In children aged above 8 years and adolescents consider atypical organisms like *mycoplasma* and *pertussis*.

6.1.1 Classification

Patients will commonly present with cough, fever and rapid breathing.

Classification will depend on physical examination findings:

- **Severe pneumonia** - One of the following present: Oxygen Saturation <90%, Central Cyanosis, Inability to drink or Breastfeed, Grunting, AVPU <A
- **Pneumonia** - Tachypnoea, Lower Chest Wall Indrawing
- **No pneumonia** - No feature of Pneumonia or Severe Pneumonia

6.1.1.1 Basic Investigations

Do the following assessments and tests if the child meets the admission criteria or to consider the need for antibiotics if not being admitted:

- Pulse oximetry
- Full blood count and C-reactive protein
- Procalcitonin also useful for the more severe cases at onset to help adjust treatment to response (more sensitive for infectious than for inflammatory states).

6.1.1.2 Further Investigations

If deteriorating or needing admission to the PCCU, do the following additional tests:

- Serum electrolytes - may have hyponatremia owing to syndrome of inappropriate anti-diuretic hormone (SIADH) or dehydration from poor feeding.

- Nasopharyngeal aspirate or nasal swab or sputum expectoration for microscopy, culture and respiratory PCT panel that includes pertussis, SARS-CoV-2, other atypical organisms and viruses.
 - If an effusion is noted, pleural fluid cytology, microscopy, culture and if possible pneumococcal PCR are to be done.
 - If pertussis suspected, nasal swab in charcoal transport medium.
 - Pleural fluid cytology, microscopy and culture, if possible - pneumococcal PCR.
 - Routine chest radiography not advised if pneumonia is not severe or not being admitted.
 - Chest Ultrasonography now useful for emergency point of care diagnosis of pneumothorax, masses, cavitation/cyst, effusions and empyema. Can detect pneumonia too.
 - Do not perform lateral X-ray routinely, useful for suspected TB.

6.1.2 Management

- Supportive – Oxygen, Rest
- Antibiotic therapy

6.1.2.1 Severe Pneumonia (Admit)

- IV Crystalline Penicillin 50 000 – 100 000 IU/Kg/dose every 6 hours or IV Amoxicillin 100 mg/Kg/day in 3 divided doses and IV Gentamicin 5–7 mg/Kg once a day

Second Line Antibiotics

- IV Coamoxiclav 30 mg/Kg/dose (oral 90 mg/Kg/day in two divided doses) 8 hourly for 7 – 10 days

- IV Cefuroxime 90 mg/Kg/day (oral 30 mg/Kg/day in 2 divided doses) in three divided doses for 7 – 10 days
- IV Ceftriaxone 50 mg/Kg/day for 5 days. Use Coamoxiclav or Flucloxacillin if *staph. aureus* is suspected.
- IV Ceftriaxone at the higher dose of 80 - 100mg/Kg/day for 5 days. Sensitive strains of *staph. aureus* responds to Ceftriaxone.
- or Add Flucloxacillin if *staph. aureus* is suspected.
- Pneumonia with influenza; Oseltamivir (Tamiflu oral) + Coamoxiclav.
- Aminoglycosides for gram negative organisms seen in malnourished and other immunosuppressed states.
- Staphylococcal pneumonia may be complicated by effusions, empyema and abscesses or necrosis. This may require high dose Ceftriaxone or flucloxacillin added for extended duration under the guidance of a chest specialist.

For penicillin allergy and atypical cases, use Azithromycin 20 mg/Kg/day once daily or Erythromycin 30 – 50 mg/Kg/day in 3 - 4 divided doses or Clarithromycin 7.5 mg/Kg twice daily maximum 500 mg per dose.

6.1.2.2 Pneumonia

- Consider Admission if not certain about review within 48 hours or those at high risk of deterioration like infants and malnourished or immunocompromised patients.
- Oral Amoxicillin 90 mg/Kg/day in two divided doses
- Review within 48 hours

Note for Pneumonia and Severe Pneumonia

- Consider Acute Respiratory Distress Syndrome (ARDS) if characteristic features present e.g., hypoxemia, acute onset within 7 days, bilateral widespread chest infiltrates and low PaO₂/FiO₂ (PF) ratio below 300.
- Consider tuberculosis if response is poor, or prolonged history e.g., cough over 2 weeks, weight loss and close contact with a household TB case
- Adjust medications to specific organism identified by laboratory tests
- Review in one week
- If there are complications, follow up should be done in consultation with the chest specialist.

Table 12: Pneumonia treatment failure definition and action

Treatment failure definition	Action required
Any time Progression of pneumonia to severe (<i>development of cyanosis or inability to drink in a child with pneumonia without these signs on admission</i>) Cavitation, effusions or increased infiltrates on CXR	Admit the child Treat with Flucloxacillin and Gentamicin IV for <i>Staph. aureus</i> or Gram negative pneumonia
48 hours Severe pneumonia child getting worse, re-assess thoroughly, get chest X ray if not already done (<i>looking for empyema /effusion, cavitation etc.</i>) Pneumonia without improvement in at least one of: • Respiratory rate • Severity of indrawing	Switch to Ceftriaxone unless suspect Staphylococcal pneumonia then use flucloxacillin and <i>Suspect PCP especially if <12mo, a HIV test must be done - treat for Pneumocystis if HIV positive</i>

• <i>Fever</i> • <i>Ability to drink or feed</i>	Change treatment from Amoxicillin to penicillin and Gentamicin
Day 5 At least three of: • <i>Fever, temp > 38°C</i> • <i>Respiratory rate < 60 bpm</i> • <i>Still cyanosed or saturation <90% and no better than admission</i> • <i>Chest in drawing persistent</i> • <i>Worsening CXR</i>	If only on amoxicillin, admit the child and change to penicillin and Gentamicin If on penicillin and Gentamicin change to Ceftriaxone Suspect PCP, an HIV test must be done - treat for Pneumocystis if HIV positive
After 1 week Persistent fever and respiratory distress	Consider TB, perform Mantoux and check TB treatment guidelines. Consult infectious disease or chest specialist

6.1.3 Prevention

- Emphasize well baby/child visits and annual check-up for all children and adolescents, vaccinate and keep up with the schedule even for boosters.
- Practice good cough/respiratory hygiene, this includes covering your mouth and nose with a tissue when coughing or sneezing; the inner elbow can be sneezed into if no tissue available.
- Use the nearest waste receptacle to dispose of the tissue after use.
- Perform hand hygiene (e.g., hand washing with non-antimicrobial soap and water, alcohol-based hand rub, or antiseptic hand wash) after having contact with respiratory secretions and contaminated objects/materials.

- Healthcare facilities should ensure the availability of materials for adhering to Respiratory Hygiene/Cough Etiquette in waiting areas for patients and visitors.
- Provide tissues and no-touch receptacles for used tissue disposal.
- Provide conveniently located dispensers of alcohol-based hand rub; where sinks are available, ensure that supplies for hand washing (i.e., soap, disposable towels) are consistently available.
- Masking and Separation of Persons with Respiratory Symptom especially during periods of increased respiratory infection activity in the community (e.g., when there is increased absenteeism in schools and increased medical office visits by persons complaining of respiratory illness), offer masks to persons who are coughing.
- Procedure masks (i.e., with ear loops) or surgical masks (i.e., with ties) may be used to contain respiratory secretions. N-95 are not necessary for this purpose.
- When space and chair availability permit, encourage coughing persons to sit at least three feet away from others in common waiting areas. Some facilities may find it logistically easier to institute this recommendation year-round.
- Consult infection prevention and control Hospital team for further information.

6.2 BRONCHIOLITIS

Acute viral inflammatory illness of small airways that occurs in cold and rainy season epidemics and affects children aged <2 years, with peak incidence at around 6 months.

Classically caused by RSV and Rhinovirus but influenza, parainfluenza, coronavirus, metapneumovirus, bocavirus and adenovirus infections have been noted.

6.2.1 Presentation

- Respiratory distress and coryza (nasopharyngitis with catarrhal symptoms)
- Fever
- Irritability and poor feeding
- Apnoea in neonates
- Crackles and rhonchi widespread throughout lung fields ± audible wheeze.

6.2.2 Management

- Usually, mild symptoms with good oxygen saturation and able to feed, even though may appear in respiratory distress from the wheeze and tachypnea. Largely supportive therapy and can be treated as an outpatient.
- Ensure Adequate hydration and feeding status.
- Humidified oxygen if required for saturation below 90%.
- Chest physiotherapy (vibration, percussion, passive forced exhalation) is not recommended for routine use.
- Evidence for use of bronchodilators (e.g., ipratropium bromide or salbutamol) not strong, can give a trial stat dose especially if

hypoxic or has underlying atopy and wheeze is from hyperresponsiveness, if child improves then can continue,

- Nebulized hypertonic saline has been shown to reduce hospital stay but minimally, no evidence for routine outpatient use. Works by:
 - reducing airway oedema
 - decreasing mucous plugging
 - improving mucociliary clearance
 - increasing airway surface liquid
- Adrenaline and steroids are not recommended for routine use unless in suspected anaphylaxis or allergic trigger.
- Antibiotics are not recommended unless there is a high index of suspicion concerning a concomitant bacterial infection e.g., pneumonia or otitis media

6.3 ASPIRATION PNEUMONIA

- Aspiration syndromes can be divided into two types: The aspiration of oropharyngeal flora or infected secretions and the aspiration of gastric contents.
- In both cases, the aspiration event may not be witnessed or is accompanied by few, if any acute symptoms

6.3.1 Management

1st line antibiotics

- IV Coamoxiclav 30mg/kg/dose 8hourly
- IV Cefuroxime 20-30mg/kg/dose 8 hourly
- IV Clindamycin 3 - 6 mg/Kg/day 6 hourly, for penicillin hypersensitivity or MRSA infection.

2nd line antibiotics

- Add Gentamicin 5 – 7 mg/Kg once daily to first line antibiotic
- Add IV Metronidazole 7.5 mg/Kg every 8 hourly to first line antibiotic if gastric related
- IV Ceftriaxone 50 - 80mg/Kg once daily.

Supportive care

Oxygen, respiratory support as necessary, chest physiotherapy and airway clearance techniques.

6.4ASTHMA

6.4.1 Definition

Asthma is a heterogeneous disease characterized by chronic airway inflammation with a history of variable symptoms of wheezing, shortness of breath, chest tightness, cough and variable expiratory airflow limitation.

These symptoms are usually relieved by inhaled short acting beta 2 agonist (SABA) and controlled by an inhaled corticosteroid. In children above 6 years when feasible, lung function tests should be done for diagnosis before treatment.

It is estimated that Asthma affects 7.5% of the Kenyan population.

6.4.2 Risk factors for developing Asthma

1. Gender – Childhood male, Adult Female
2. Atopy
3. Airway hyperresponsiveness

6.4.3 Diagnosis of Asthma

- Frequent episodes of wheeze, cough, chest tightness or heaviness, breathlessness particularly experienced at night and/or early morning, or triggered by exercise or during playtime, common irritants like dust and perfumes, emotions like laughter or symptoms that also occur in the absence of a 'common cold'.
- Personal history of atopy/allergy conditions like eczema, allergic rhinitis or conjunctivitis, history of reactions to animal proteins like milk, meat, or eggs.
- Family history of atopy and or Asthma
- Reduced play and easily tires during activity and playtime.
- Widespread poly-phonic wheeze on chest auscultation during a flare-up.
- History of improvement in symptoms or lung function in response to asthma-specific therapy.
- Objective tests of lung function are recommended in those children aged 5 years and above – These include spirometry (Forced Expiratory Volume in 1 second, Forced Vital Capacity) and peak expiratory flow measurements

6.4.4 Asthma severity

Asthma severity is assessed after at least 2 – 3 months of appropriate treatment, from the level of treatment required to control symptoms and exacerbations. Frequency of symptoms, nocturnal symptoms, exercise limitation, acute exacerbations at presentation and FEV1 are assessed to rate severity.

6.4.4.1 Intermittent

- Infrequent symptoms less than once a week or not more than twice a month with peaks during certain seasons such as the rainy/ cold season, when maize plants are flowering or with respiratory infections.
- Between flare-ups there are no symptoms or minimal symptoms that do not interfere with their daily activities.

6.4.4.2 Mild severity:

- Symptoms less than once a week or not more than twice a month
- FEV₁ predicted is between 60% and 80%
- Maintains control with as needed SABA +low dose ICS in children or low dose ICS-formoterol as needed in > 12 years, or with daily low dose ICS, leukotriene receptor antagonists.

6.4.4.3 Moderate severity:

- Symptoms more than once a week but not daily
- Maintains control with low to medium dose daily ICS with or without LABA and a leukotriene receptor antagonist.

6.4.4.4 Severe

- Symptoms almost daily
- Low lung function FEV₁ below 60% of the predicted value and may have comorbid disease contributing to their poor control such as obesity, GERD and chronic rhinosinusitis.

- Have ever been admitted with a severe exacerbation needing oxygen therapy, prolonged hospital stay or intubation and intensive care unit (CCU) management.
- Maintains control with medium to high dose daily ICS with or without LABA and add on therapies such as leukotriene receptor antagonists, oral steroids, tiotropium and biologic agents where available.

Table 13: Classification of Asthma Severity

Components of severity		Classification of Asthma severity (age ≥ 12 years)			
		Intermittent	Persistent		
			Mild	Moderate	Severe
Impairment	Symptoms	≤ 2 days a week	>2 days a week, but not daily	Daily	Throughout the day
	Night awakening	≤ 2 per month	3-4 times per month	< 1 once per week but not daily	Daily
	Short Acting Beta 2 Agonist use	≤ 2 days a week	>2 days a week, but not daily	Daily	Several times a day
	Interference of	None	Minor limitation	Some limitation	Extreme limitation

	normal activity				
	Lung function	Normal FEV ₁ between exacerbations. FEV ₁ more than 80% of predicted. FEV ₁ /FVC normal	FEV ₁ ≥ 80% of predicted. FEV ₁ /FVC normal	FEV ₁ 60-80% of predicted. FEV ₁ /FVC reduced by ≤ 5%	FEV ₁ < 60% of predicted. FEV ₁ /FVC reduced by > 5%
Risk	Exacerbations requiring systemic steroids	0-1 per year	≥ 2 per year		
		Consider severity and interval since last exacerbation. Frequency and severity may fluctuate over time for patients in any severity category			
		Relative annual risk of exacerbations may be related to FEV ₁			

6.4.5 Level of control

Assess the child and find out if in the past 4 weeks, the child has had these symptoms:

- Daytime symptoms for more than a few minutes, more than once a week.
- Any activity limitation due to Asthma (runs/plays less than other children. tires easily during walks/playing).
- Reliever needed more than once a week.

- Any night waking or night coughing due to Asthma.

Well controlled – None of these

Partially controlled – 1 – 2 of these

Uncontrolled – 3 – 4 of these

6.4.6 Management

6.4.6.1 Patient Education

- Basic explanation of Asthma and the factors that influence it.
- Recognise triggers and home allergen avoidance measures for common triggers like dust, mould and certain foods.
- Training in correct inhalation technique and use of inhalers with spacer devices
- The importance of adherence to the prescribed medications

6.4.6.2 Causes of exacerbations

- Aeroallergens (e.g., pollen, pet dander, dust, dust mites, mold spores, cockroach frass)
- Airborne irritants (e.g., cigarette or wood smoke, air pollution, chemical compounds)
- Respiratory infection
- Medications (e.g., NSAIDs including aspirin, β -blockers)
- Psychological or emotional stress
- Cold, dry air (e.g., weather changes)
- Exercise

6.4.6.2 Allergen Control Measures

Allergen avoidance/control strategies should be tailored to the patient. Objective tests such as skin prick tests may be useful in guiding this process.

Food Allergies:

- Avoidance any foods that have been associated with triggering an exacerbation or as may be guided after allergy testing
- Avoid foods and drinks with artificial flavourings, preservatives and colourants “junk food”.

Pollen:

- Keep bedroom windows shut and shut car windows when traveling in a car to prevent pollen blowing in.
- Avoiding sitting outdoors after lawns have been mowed but keeping grass short reduces pollen burden and if mowing is unavoidable wear an industrial mask.

House dust mites:

- Allergy proof bedding creates a barrier from exposure to mites in the bedding.
- Wash beddings, curtains and other soft furnishings at high temperatures and sun-dry regularly at least once a week.
- Removal of carpets/mats/rugs is recommended.
- Clean surfaces with a wet cloth daily.
- Fluffy toys and storage of boxes under the beds and in the wardrobes must be avoided in the bedroom.

Mould:

- Remove mouldy items like stale bread, rubbish bins and bedding
- Limit indoor plants.
- Avoid use of humidifiers in rooms.

- Avoid compost and decaying material, hay vines, granaries and heavy vegetation areas like forest trails.
- Use bleach to clean mould in bathrooms and kitchens.

Pets

- People with allergies to indoor pets must remove them from the home. Those who cannot, must wash the pet regularly, keep out of the bedroom and vacuum clean sofas, curtains and floors daily.

Cockroaches

- Ensure kitchens and stores are kept clean, repair damp and leaking areas, seal food and rubbish containers and regular fumigation of infested areas.

Occupational allergens:

- Certain occupations can expose patients and their household members to allergens from the workplace which can cause airway diseases. Once identified, the doctor can advise the workplace measures which can be taken to minimize exposure to the offending allergens.

Triggers

- Avoid cigarette and tobacco smoke, perfumes, insecticides, cold weather, car and industrial smoke, incense (Udh) and other strong smells.
- Use of charcoal 'jikos' and paraffin stoves should be avoided and done outside away from the house in an open and well-ventilated area.

6.4.7 Asthma Action Plan

A written Asthma Action Plan (see sample plan) should include:

- a description of how the parent or caregiver can recognize when symptoms control is deteriorating
- medications to administer, dosage and method

- when and how to obtain medical care, including emergency telephone numbers.

6.4.8 Treatment

6.4.8.1 Goals

- Achieve and maintain control of symptoms
- Maintain normal activity levels
- Maintain pulmonary function
- Prevent asthma exacerbations
- Minimize side effects of asthma medications
- Prevent asthma mortality

The patient's current level of asthma control and current treatment determine the selection of pharmacologic treatment.

Diagram 3: Asthma Action Plan

Name: Date:
Parent's Name: Review In:
Tel:
PFR: L/min FEV1: Review Date:

PERSONALIZED ACTION PLAN

Take these medicines EVERY DAY for Control and Maintenance

GREEN ZONE:

Doing Well

Child has ALL of these:

- Breathing is good
- No cough or wheeze
- Sleep through the night
- Can work or play

MEDICINE	HOW MUCH TO TAKE	WHEN TO TAKE

YELLOW ZONE:

Caution/Getting Worse

Child has ANY of these:

- First sign of a cold
- Exposure to known trigger
- Cough
- Mild wheeze
- Tight chest
- Coughing at night

CONTINUE WITH GREEN ZONE MEDICINE AND ADD:

MEDICINE	HOW MUCH TO TAKE	WHEN TO TAKE

Go to HOSPITAL if you have been in the YELLOW ZONE for more than 72 hours

RED ZONE:

ALERT

- Medicine is not helping within 30 minutes
- Breathing is hard and fast
- Ribs show
- Lips and/or fingernails blue
- Trouble walking and talking

Take These quick-relief Medicines:

MEDICINE	HOW MUCH TO TAKE	WHEN TO TAKE

GO TO HOSPITAL NOW! CALL 020-7206000 FOR ASSISTANCE

Doctor's Name: Signature:

QMS/OP/REC/45

6.4.8.2 Initiation

Treatment of Asthma follows a stepwise approach with medications adjusted up or down to achieve good symptom control and minimize future risk of flare-ups and medication side effects. The need for controller medications should be assessed regularly.

Children aged below 2 years with recurrent wheezing and those with severe or multisystemic symptoms should be referred to the Chest specialist.

Many children below 5 years will begin at Step 1 - as needed Inhaled bronchodilator only during wheezing episodes for one to a few days until the symptoms disappear. The same is used for isolated viral induced wheezing episodes.

Children who should be put on controller medication (Step 2):

- The symptom pattern suggests a diagnosis of Asthma and are uncontrolled
- Wheezing episodes are frequent - three or more in a season
- Wheezing episodes are frequent but are severe
- The diagnosis of asthma is in doubt and inhaled short acting beta 2 agonist needs to be repeated frequently, i.e., more than 1 - 2 times a week.
- The current Kenya national Asthma guidelines recommend stepping up the ICS dose, if on a low dose, for children aged below 5 years, but consider adding a long acting beta agonist/Inhaled corticosteroids (LABA/ICS) for children aged 6 years and above.
- Leukotriene receptor antagonist (LTRA) are a useful add on therapy for concomitant allergic rhinitis/conjunctivitis patients.

- Low dose ICS with SABA in all age groups and formoterol/low dose ICS above 12 years, are now included as reliever options in the national guideline. Refer to the tables.

Refer patients to chest clinic at Steps 3 and 4.

Table 14: Stepped approach for children aged below 5 years

Steps for Children aged below 5 years	Step 1 Infrequent viral wheezing and no or few interval symptoms	Step 2 Symptoms consistent with Asthma and not well controlled or recurrent wheezing more than 3 exacerbations per year	Step 3 Asthma symptoms not well controlled with low dose ICS	Step 4 Asthma symptoms not well controlled with double low dose ICS
	Check diagnosis, inhaler technique, adherence, comorbidities, exposures, and triggers.			
Preferred Controller Choice		Daily Low dose ICS	Double low dose ICS	Continue controller option and refer to chest specialist
Other controller options		Leukotriene receptor antagonist (LTRA) as add on therapy to ICS – not used as monotherapy* No role of Intermittent ICS	Low dose ICS + LTRA	Add LTRA, Increase ICS frequency to twice if once daily, refer for specialist review.
Reliever	As needed short acting beta 2 agonist (SABA). 2puffs = 1 dose		As needed SABA and low dose ICS, a puff with each SABA dose.	

Table 15: Stepped approach for children aged 6 – 11 years

Steps for Children aged 6 – 11 years	Step 1 Infrequent symptoms and no or few interval symptoms.	Step 2 Symptoms consistent with Asthma and 2 or more exacerbations per month but not daily.	Step 3 Most days or waking with asthma once a week or more.	Step 4 Most days or waking with asthma once a week or more and low lung function.
	Check diagnosis, inhaler technique, adherence, comorbidities, exposures and triggers.			
Preferred Controller Choice	Low dose ICS whenever SABA is used	Daily low dose ICS	Low dose ICS+LABA	Medium dose ICS plus LABA
Other controller options	-	Low dose ICS whenever SABA is taken.	Low dose ICS + LTRA especially for comorbid allergic rhinitis Medium dose ICS.	high dose ICS, high dose ICS plus LABA, add-on tiotropium OR add-on LTRA
Reliever	Low dose ICS whenever SABA is used	As needed SABA. 2puffs = 1 dose		

Table 16: Stepped approach for children aged 12 years and above

Steps for 12 years and above	Step 1 Infrequent symptoms and no or few interval symptoms.	Step 2 Symptoms consistent with Asthma and 2 or more exacerbations per month but not daily.	Step 3 Most days or waking with asthma once a week or more.	Step 4 Most days or waking with asthma once a week or more and low lung function	Step 5 Not controlled by step 4 therapy including OCS.
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	Check diagnosis, inhaler technique, adherence, comorbidities, exposures, and triggers.				
Preferred Controller Choice	low dose ICS +Formoterol is used on an 'as needed' basis.	Daily low dose ICS	Low dose ICS+LABA inhaler of either formoterol salmeterol or vilanterol	medium dose ICS plus LABA	high dose ICS plus LABA
Other controller options				add-on short course of oral corticosteroid (OCS)	add-on tiotropium (inhaled LAMAs), Biologics, OCS,
Reliever	ICS+SABA Preferred as a combination MDI OR each as a single inhaler.	Extra puffs of the Low dose ICS whenever SABA is taken. OR Low dose ICS+ LABA MDI	**Use their daily Low dose ICS+ Formoterol extra puffs as needed. OR Use low dose ICS+SABA MDI if on salmeterol or vilanterol based inhaler.	Low dose ICS+ Formoterol as needed. OR low dose ICS+SABA MDI if on other LABA.	

Table 17: Choosing an inhaler device for children

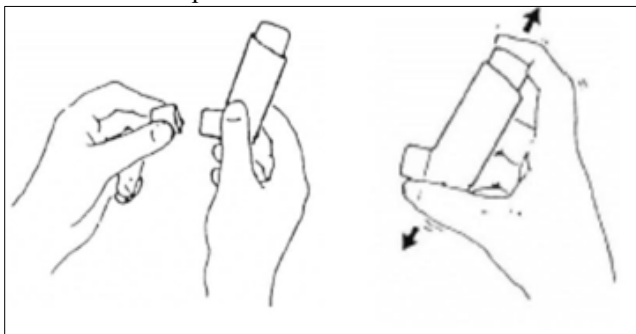
Age group	Preferred device
<1year	Infant spacer with Face mask
1 – 5years	Older child spacer with Face mask
> 5 years	Spacer with mouthpiece

The spacer device should be cleaned with a soft sponge in mild soapy water and drip dried without rinsing every 1 - 2 weeks to reduce hydrostatic charge. Avoid scrubbing the inner surface with rough items like brushes or scrubs.

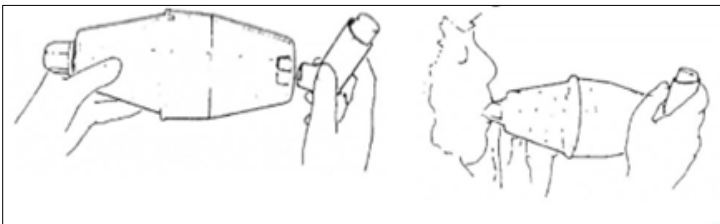
Check the counter or date of issue, frequency of use and number of doses per canister to determine if the inhaler is finished, to avoid using the propellant as the active drug when the doses are over.

Diagram 4: How to use the pressurized MDI with a spacer

1. Remove cap of inhaler



2. Shake inhaler.
3. Insert inhaler into flat end of the spacer device.



4. Place mouthpiece in mouth and press inhaler canister to release a dose of medication. Ask the patient to breathe in slowly and deeply.
5. Hold breaths for at least ten seconds or as long as comfortable.

For children aged below 5 years who need to be assisted:

Step 1-3 remain the same.

Step 4. Place face mask over the nose and mouth of the child and press inhaler canister to release a dose of medication. Ask the patient to breathe in and out (deeply if possible) for 5 – 10 breaths.

Note: Shake inhaler between every puff and NEVER spray two or more puffs at the same time. Give one puff, breath 5-10 times then shake the inhaler and give the second puff.

6.4.8.3 Controller medications

6.4.8.3.1 Inhaled corticosteroids

Most effective and recommended for treatment of children of all ages

Table 18: Equipotent doses of ICS for children

Drug	Low daily	Medium daily	High daily
Beclomethasone	100 – 200	< 200 – 400	< 400
Budesonide*	100 – 200	< 200 – 400	< 400
Budesonide neb	250 – 500	< 500 – 1000	< 1000
Mometasone furoate	100 – 200	< 200 – 400	< 400

Triamcinolone	400 – 800	< 800 – 1200	< 1200
Ciclesonide*	80 – 160	< 160 – 320	< 320
Fluticasone	100 – 200	< 200 – 500	< 500
Flunisolide	500 – 750	< 750 – 1250	< 1250

**Approved for once daily dosing in mild patients. Clinical judgment or response to therapy is required for determining appropriate dosing level*

6.4.8.3.2 Reviewing Treatment Response and Adjusting

Assessment at every visit should include asthma symptom control and risk factors. Because of the potential of steroids affecting growth, the height of these children should be measured at least once a year.

Assess the need for controller medications every 3 - 6 months.

Some children with marked seasonal variability of symptoms may be allowed to stop the controlled medications 4 weeks after their symptom season has passed. In this case, the child should be reviewed after 3 - 6 months to check the need for treatment or otherwise.

Before stepping up medications:

- Check inhaler technique
- Confirm adherence to treatment
- Enquire risk factors - psychosocial stressors, tobacco and indoor fuel exposure, comorbidities like obesity, frequency of exacerbations and previous ICU/Hospital Admissions.
- Confirm symptoms are due to Asthma
- Refer for assessment by the Chest specialist if in doubt, or if there are frequent exacerbations and comorbid allergy disease like rhinitis, conjunctivitis and eczema.

Formoterol is a rapid onset long-acting beta 2 agonist. When combined with an inhaled corticosteroid (ICS) it is recommended for use both as maintenance or controller and reliever (MART) therapy in persons over the age of 12 years.

6.4.9 Asthma Flare ups or Exacerbations

Acute asthma exacerbation (Flare-ups) refers to a change in the patient's usual status characterized by an increase in symptoms of shortness of breath, cough, wheezing or chest tightness which is unresponsive to usual effective therapy, and which may necessitate care in an emergency room or hospital ward. It is important to note that some cases may occur as the initial presentation of asthma in childhood

The patient should be started at onset (at home) on inhaled Salbutamol 200 microgram every 15 to 20 minutes for three doses. The child should be taken to hospital, if:

- More than 4 puffs of inhaled SABA within the first two hours
- The child has not improved in 24 hours.

All patients presenting with acute exacerbation of asthma should have their vital signs, BP and SpO₂ taken and recorded.

Table 19: Initial Assessment of a child with an acute exacerbation

Classification	Clinical features	Measurement
Life threatening Asthma	• Silent chest • Cyanosis • Poor respiratory effort • Hypotension	SpO ₂ < 92%

	• Responds to pain stimulus only or unresponsive • Bradycardia	
Severe Asthma exacerbation	• Unable to complete sentences in one breath • Too breathless to talk/feed • Responds to voice, agitation, or irritability • Has lower chest wall in-drawing • Tachycardia for age • Tachypnoea for age	$\text{SPO}_2 < 92\%$
Moderate Asthma exacerbation	• Not responding to initial treatment of a mild exacerbation within 2 hours • Has expiratory wheezing • Tachypnoea for age • Has no lower chest wall in-drawing • Able to talk but may have episodes of breathlessness • No altered mental state • • May or may not be able to feed	$\text{SPO}_2 \geq 92\%$
Mild Asthma exacerbation	• Has expiratory wheezing • Has no tachypnoea for age • Has no lower chest wall in-drawing • • Able to talk and feed with no breathlessness	$\text{SPO}_2 \geq 92\%$

6.4.9.1 Mild exacerbation

Start treatment with 2 puffs of Salbutamol (200 micrograms) as soon as symptoms begin, even at home. Give 1 puff at a time with each puff inhaled separately in 5 - 10 tidal breaths at 15 - 30-second intervals. Inhaled salbutamol doses can be repeated at 2 puffs every 20 - 30 minutes according to response.

If the child remains unresponsive within 2 hours, escalate the treatment to that recommended for a moderate exacerbation.

If child improves, continue delivering a controller medication and salbutamol every 6 hours for up to 48 hours. Schedule a review within 48 - 72 hours either by phone or at the nearest health centre if feasible.

A spacer and a mask must be used for children under 5 years preferably with a mouthpiece for those aged over 5 years.

6.4.9.2 Moderate exacerbation

Continue inhaled salbutamol every 20 - 30 minutes. Add Ipratropium bromide delivered by nebulization 250 micrograms for those aged 11 years and below and 250 - 500 micrograms for those above every 20 minutes up to 3 times in 1 hour then every 4 - 6 hourly when the patient's condition improves.

Start oral steroids early. Oral Prednisolone should be given at 1 - 2 mg/kg/dose, once daily for 3 - 5 days.

The recommended maximum dosages of prednisolone per age category are 10 mg per dose (daily) for children under 2 years of age,

20 mg for children aged 2 – 5 years and 30 - 40 mg for children older than 5 years.

As an alternative to prednisolone oral dexamethasone at 0.6 mg/Kg/day can be given for up to 2 days with a maximum of 20mg in a day.

Intravenous steroids are **ONLY** recommended for children with severe or life-threatening acute exacerbations who are unable to retain oral medications.

Observe patients in the acute care setting for at least 6 hours. If after this period, vital signs remain stable and the patient's condition has improved with no requirement for oxygen therapy, the patient should be discharged home and be followed-up in 48-72 hours.

Admit those patients who:

- require oxygen therapy
- do not improve after 6 hours of observation in the acute care setting
- cannot use their inhaler medications well at home
- are known to be poorly adherent to treatment
- have poor social factors and may not be able to come for short-term follow-up after discharge.

6.4.9.3 Severe and Life-threatening Acute Exacerbation

- Start oxygen and target SPO₂ of 94 – 98%
- Start Nebulized salbutamol and Nebulized Ipratropium bromide (250 micrograms for those aged 11 years and below and 500 micrograms for those above). The recommended dose

is 0.15mg/ kg for children below 4 years, 2.5mg for children 5 - 11 years and 2.5 - 5mg for those above 12 years 5mg. For those aged 6 years and above, nebulized Levosalbutamol 1.25 mg - 2.5mg for those 6 - 11 years and 2.5mg for those above 12 years. Use these every 20 – 30 minutes for an hour then every 4 - 6 hours as you monitor response.

- A single dose of intravenous (IV) Magnesium Sulphate 50 mg/Kg over 20 minutes can be given in the acute care setting with monitoring of blood pressure to avoid hypotension. Maximum dose 2 grams.
- Intravenous Salbutamol can be given ONLY in the critical care unit where cardiac monitoring is done. Loading dose of salbutamol 5 - 10 µg/Kg/min of for 1 hour, followed by continuous infusion of 1 - 5 µg/Kg/min.
- When oral steroids cannot be used because of the severity of the illness, intravenous steroids are to be used.
- Intravenous Dexamethasone: 0.1 - 0.3mg/Kg/daily in 2 - 3 divided doses with a maximum of 20 mg a day given as a slow IV injection or infusion.
- Intravenous Hydrocortisone may be given at a dose of 4 mg/kg with a maximum of 100 mg repeated 6 hourly for those aged 1 year and below, 25 mg for those aged 2-4 years and 50mg for those 5-17 years.
- Steroids should be continued for about 5 days or until about 2 days after the exacerbation appears to have resolved.
- The benefits of intravenous Aminophylline are not clear and should be used after careful consideration in the Critical care unit. Recommended for those who do not respond to Magnesium Sulphate and Steroids
- Adrenaline is only indicated if the exacerbation is due to anaphylaxis (see under anaphylaxis)

- Mechanical ventilation is rarely indicated in Asthma and should be considered in respiratory failure. If necessary, start with noninvasive positive pressure ventilation (high flow/BiPAP) to avert further deterioration requiring intubation and invasive ventilation which should be avoided, if possible, unless absolutely indicated.

Imaging and evaluation for bacterial infections should be done in patients with severe and life-threatening acute exacerbations.

Diagram 5: Flow chart for mild exacerbation of Asthma in children aged below 6 years

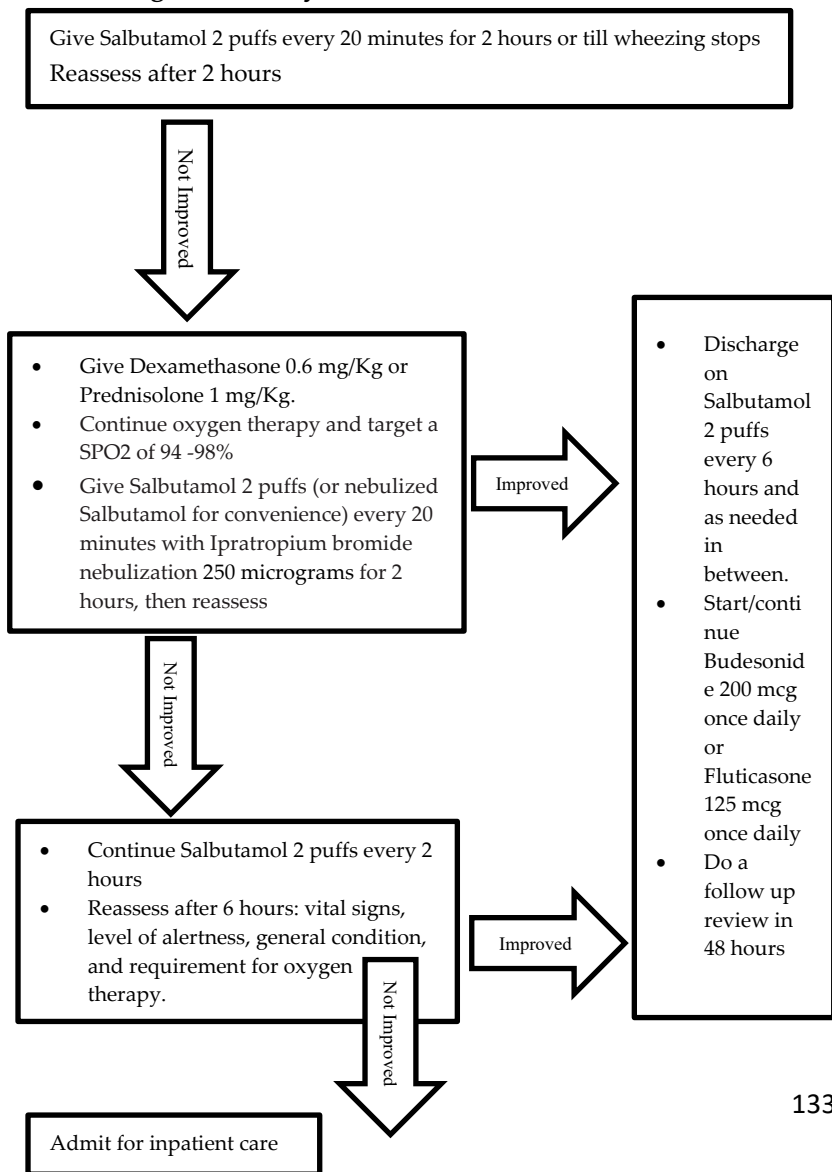


Diagram 6: Flow chart for moderate exacerbation of Asthma in children aged below 6 years

- Give or ensure that a systemic steroid has been given
- Give Salbutamol 2 puffs every 20 minutes for 2 hours or till wheezing stops.
- Add Ipratropium bromide delivered via a pMDI at a dose of 20mcg per actuation.
- Start or continue oxygen therapy and target a SPO₂ of 94 - 98%
(If the patient requires oxygen, or unable to puff, use nebulized Salbutamol 2.5 mg and Nebulized Ipratropium 250 mcg)
- Observe for 6 hours: vital signs, level of alertness, general condition, and requirement for oxygen therapy for at least 6 hours.

No Improvement

Admit for inpatient care

- Continue oral Prednisolone 1 mg/Kg daily for 3 – 5 days
- Continue nebulized Salbutamol every 6 hours till discharge
- Continue nebulized Ipratropium every 6 hours for 24 hours only

Not Improved

Transfer to Critical Care Unit and

Improved

- Discharge after 48 hours, if off oxygen, no wheezing, hydrating well without intravenous fluids, Asthma care plan and education has been given and follow up plan instituted
- Discharge on Salbutamol 2 puffs every 6 hours and as needed in between.
- Start/continue Budesonide 200 mcg once daily or Fluticasone 125 mcg once daily
- Do a follow up review in 48 hours

Diagram 7: Flow chart for severe and life-threatening exacerbation of Asthma in children aged below 6 years

- Give IV Dexamethasone 0.6 mg/Kg or Prednisolone 1 mg/Kg.
- Continue oxygen therapy and target a SPO₂ of 94 - 98%
- Give Nebulized Salbutamol 2.5 mg and Ipratropium bromide 250 micrograms every 20 minutes for 2 hours, then reassess
- Give Magnesium Sulphate 50 mg/Kg (maximum 2 g) infusion over 20 minutes with blood pressure monitoring

No Improvement

Improved

- Give Nebulized Salbutamol 2.5 mg and Ipratropium bromide 250 micrograms every 20 minutes for 2 hours
- Continue oral Prednisolone 1 mg/Kg daily for 3 – 5 days (Or Dexamethasone for 2 days)
- Consider Infusions of Aminophylline and Salbutamol with consultation with the Pulmonologist and/Intensivist

Not Improved

- Complete care in the general wards in 48 hours, then discharge home if off oxygen, alert, no wheezing, hydrating well without intravenous fluids, Asthma care plan and education has been given and follow up plan instituted
- Discharge on Salbutamol 2 puffs every 6 hours and as needed in between.
- Start/continue Budesonide 200 mcg once daily or Fluticasone 125 mcg once daily and
- Do follow up review in 48 hours

- Continue care in the critical care unit in consultation with a pulmonologist
- Consider noninvasive positive pressure ventilation (high flow/BiPAP) to avert further deterioration requiring invasive ventilation
- Adrenaline is only indicated if the exacerbation is due to anaphylaxis

6.5 PNEUMOCYSTIS CARINII PNEUMONIA, PCP

A pneumonia arising in immunosuppressed persons caused by *Pneumocystis jirovecii* (PJP)

This is one of the most common opportunistic infections occurring in patients with human immunodeficiency virus (HIV) infections and in those with other immunodeficiency states like cancer and primary immunodeficiency (PID).

Pneumocystis infection can cause organ involvement and disseminated disease as well as pneumonia

6.5.1 Management

6.5.1.1 Supportive management

- Oxygen – admit to PCCU for high flow oxygen if severe respiratory distress
- Nutrition, intravenous fluids

6.5.1.2 Definitive

IV/PO Cotrimoxazole 100 – 120 mg/Kg every 6 hours, for 2 weeks (3 weeks in immunosuppressed patients, consider addition of steroids)

After treatment, give prophylaxis indefinitely as a single dose of Cotrimoxazole 20 – 25 mg/Kg PO daily

Use Prednisolone 1 – 2 mg/Kg/day until respiratory distress settles

6.6 TUBERCULOSIS (TB)

An infectious disease caused predominantly by mycobacteria tuberculosis

6.6.1 Antituberculosis treatment in children

6.6.1.1 The main objectives of anti-TB treatment are to:

- Cure the patient of TB (by rapidly eliminating most of the bacilli);
- Prevent death from active TB or its late effects
- Prevent relapse of TB (by eliminating the dormant bacilli);
- Prevent the development of drug resistance (by using a combination of drugs);
- Decrease TB transmission to others

6.6.1.2 Recommended treatment regimens as per case definitions

- Antituberculosis treatment is divided into two phases: An intensive phase and Continuation phases
- The number at the front of each phase represents the duration of that phase in months
- Anti TB medications are abbreviated thus: Isoniazid (H), Rifampicin (R), Pyrazinamide (Z) and Ethambutol (E)
- Specimen may include expectorated sputum (child < 5yr), induced sputum, Nasopharyngeal Aspirate and Gastric Aspirate. Attend to obtain specimen in every child
- Do a culture and DST for the following children
 - Rifampicin resistance detected by the Xpert test
 - Refugees and children in contact with anyone who has drug resistant TB

- Those not responding to TB treatment
- Those with indeterminate Xpart results

Notes:

- Steroid therapy should be given for TB meningitis, PTB with respiratory distress, PTB with airway obstruction by hilar lymphnodes, severe military TB or pericardial effusion. Give Prednisolone at 2 mg/Kg (maximum 60 mg) once daily for 4 weeks, taper down over two weeks.

Table 20: Algorithm for Pulmonary Tuberculosis in children

History • Two or more of Cough, Fever, Poor weight gain or weight loss, reduced fullness • Contact with an adolescents or adult with chronic cough or TB in the previous two years
Examination • Temperature > 38 °C (Fever) • Weight – confirm poor gain or loss (see growth chart) • Respiratory rate • Any abnormal respiratory system examination finding • Signs of extrapulmonary Tuberculosis
Investigations • Obtain specimen for Xpart MTB/RIF • Do CXR, Do Mantoux Test, Do HIV test • Do other tests to diagnose extrapulmonary tuberculosis
Diagnosis • Bacteriologically confirmed (Specimen positive for MTB) • Clinically diagnosed – Child has two or more of: Persistent cough, fever, poor weight gain or loss, lethargy PLUS two or more of the following: Positive contact, abnormal respiratory signs, abnormal CXR, Positive Mantoux test
Patients to be started on Treatment • All children with a bacteriologically confirmed TB • All children with a clinical diagnosis of TB
Treatment • TB meningitis, Bone and joint TB – Treat for 12 months- 2RHZE/10RH • All other forms of TB - Treat for 6 months - 2RHZE/4RH

Table 21: Tuberculosis (TB) Medication doses

Medicine	Average daily dose mg/Kg	Range in mg/Kg	Maximum dose in mg
Isoniazid	10	10 - 15	300
Rifampicin	15	10 - 20	600
Pyrazinamide	35	30 - 40	1500
Ethambutol	20	15 - 25	1600

Pyridoxine to be given throughout the whole course of treatment

- 5 - 7Kg – 12.5 mg (1/4 tab)
- 8 – 14Kg – 25 mg (1/2 tab)
- 15 Kg and above – 50 mg (1tab)

6.7 CROUP

A condition classified and characterized as:

Fever, hoarse voice, barking or hacking cough, stridor that is heard only when the child is agitated. The loudness of the stridor is not a good guide to the severity of the obstruction.

- Mild - symptoms on exertion
- Moderate - expiratory wheeze
- Severe - expiratory and inspiratory wheeze

Differential diagnoses include:

- Inhaled foreign body
- Epiglottitis
- Bacterial tracheitis

6.7.1 Assessment:

Children with croup should have minimal examination. Do not examine throat. Do not upset child further.

- barking cough
- inspiratory stridor
- may have associated widespread wheeze
- increased work of breathing
- may have fever, but no signs of toxicity

6.7.2 Risk factors for severe croup

- Pre-existing narrowing of upper airways
- Subglottic stenosis (congenital or secondary to prolonged neonatal ventilation)
- Down Syndrome
- Previous admissions with severe croup
- Uncommon <6 months, rare <3 months of age. Consider differential

Table 22: Assessment of croup severity

	MILD	MODERATE	SEVERE
Behaviour	Normal	Some or intermittent irritability	Increasing irritability and/or lethargy
Stridor	Barking cough Stridor only when active or upset	Some stridor at rest	Stridor present at rest

Respiratory Rate	Normal	Increased Respiratory rate Tracheal Tug Nasal Flaring	Marked increase or decrease Tracheal Tug Nasal Flaring
Accessory Muscle Use	None or minimal	Moderate chest wall retraction	Marked chest wall retraction
Oxygen	No oxygen requirement	No oxygen requirement	Hypoxemia is a late sign of significant upper airways obstruction

6.7.3 Investigations

Investigations including NPA, CXR, blood tests are NOT usually indicated and may cause the child distress and worsening of symptoms

6.7.4 Acute management

- Children with croup need minimal handling. This includes limiting examination, nursing with parents. Supplemental oxygen is not usually required. If needed, consider severe airways obstruction
- Do not forcibly change a child's posture - they will adopt the posture that minimizes airways obstruction
- IV access should be deferred
- Avoid distressing the child further

6.7.5 Treatment

Children with cough only do not require treatment

Steroids have been shown to decrease the length of hospital stay, need for nebulized Adrenaline and other interventions

6.7.5.1 Mild to Moderate Croup

Prednisolone 1 mg/kg, and prescribe a second dose for the next evening, or a single dose of Oral Dexamethasone 0.15 mg/kg.

Observe for half an hour post steroid administration. Discharge once stridor-free at rest.

6.7.5.2 Severe croup

Nebulized adrenaline 0.4 mg/Kg stat and IM/IV Dexamethasone 0.6 mg/kg (max 12 mg)

If good improvement, observe for 4 hours post adrenaline. Consider discharge once stridor free at rest.

If Improvement then deterioration, give further doses of adrenaline and admit.

If no improvement, reconsider diagnosis

Consider admission to PCCU if:

- Severe airways obstruction
- No improvement with nebulized adrenaline
- Child has risk factors

For a child with severe croup who is deteriorating:

- Oxygen should be given.
- If there is incipient airway obstruction tracheostomy should be performed

- Severe indrawing or lower chest wall and restlessness likely indicate the need of tracheostomy or intubation
- Nasal prongs or nasal/nasopharyngeal catheter can upset the child and precipitate obstruction of the airway
- Consider macrolide therapy especially in young infants and those with pneumonia as well

Intubation and tracheostomy:

- Intubation should be done immediately if there are signs of incipient airway obstruction, such as severe indrawing of the lower chest wall and restlessness

6.8 COVID-19 DISEASE

6.8.1 Background

Coronaviruses are a large family of viruses that cause illness ranging from the common cold to more severe diseases such as Middle East Respiratory Syndrome (MERS-CoV) and severe acute respiratory syndrome (SARS CoV)

The novel coronavirus (SARS COV-2) is a new strain that had not been previously identified in humans. Coronaviruses are zoonotic, meaning they are transmitted between animals and people. Common signs of infection include but not limited to fever, muscle aches, dry cough, sore throat, tiredness and breathing difficulty. Other symptoms may include loss of taste and loss of smell. In more severe cases, patients may get severe pneumonia, respiratory failure, kidney failure or may even die

6.8.2 Case Definition

Any patient presenting with:

- Acute respiratory tract infection presenting as sudden onset of fever with either cough or shortness of breath with no other aetiology that fully explains the clinical presentation
- Acute respiratory tract infection AND having been in close contact with a confirmed or probable COVID-19 case in the 14 days preceding onset of illness
- Acute respiratory tract infection presenting as sudden onset of fever with either cough or shortness of breath AND requiring hospitalisation (SARI) and with no other aetiology that fully explains the clinical presentation
- Children may present with atypical symptoms like diarrhoea or altered level of consciousness among others

6.8.3 Management

Any patient with confirmed COVID-19 requiring inpatient care will be admitted under the admitting consultant Paediatrician on call or their primary doctor. The responsible consultant shall invite any other specialists as may be required to manage the patient. Management of the patient shall be guided by the severity of the illness

Mild - Fever, cough, sore throat, malaise, headache, muscle pain
BUT no dyspnoea (shortness of breath) and No abnormalities on chest imaging

Moderate - Clinical signs of non-severe pneumonia (cough or difficulty breathing) AND Fast breathing AND/OR chest indrawing.

Severe - Child with clinical signs of pneumonia (cough or difficulty in breathing) + at least one of the following: Central cyanosis or $\text{SPO}_2 < 90\%$; severe respiratory distress (e.g., fast breathing, grunting, very severe chest indrawing); general danger sign: inability to breastfeed or drink, lethargy or unconsciousness, or convulsions

Fast breathing (breaths/ min): <2months: ≥ 60 ; 2-11months: ≥ 50 ; 1-5years: ≥ 40
Critical - Any of the following: Acute respiratory distress syndrome, Respiratory failure requiring mechanical ventilation, Sepsis/Septic shock, other organ failure requiring ICU care

MIS-C (Multisystem Inflammatory Syndrome in Children) - Children and adolescents – years of age with fever > 3 days AND Two of the following: rash or bilateral non-purulent conjunctivitis or muco-cutaneous inflammation signs (oral, hands or feet); hypotension or shock; features of myocardial dysfunction, pericarditis, valvulitis, or coronary abnormalities; evidence of Coagulopathy, acute gastrointestinal problems; AND No other obvious microbial cause of inflammation AND Evidence of COVID-19 (RT-PCR, antigen test or serology positive), or likely contact with patients with COVID-19.

6.8.4 COVID-19 Treatment Guidelines

Mild illness - Assess for eligibility for home-based isolation and care Counsel caregiver on the following danger signs and when to return (difficulty breathing, fast breathing, grunting, inability to breastfeed/ drink, central cyanosis, confusion, reduced level of consciousness).

Moderate illness – Do Baseline tests – Total blood count, renal and liver function, HIV test, random blood sugar. Do chest x-ray if there are significant respiratory signs

Symptomatic treatment: Fever and sore throat – Paracetamol and gargles as appropriate for age

VTE (venous Thromboembolism) prophylaxis with Enoxaparin 0.75 – 1 mg/Kg 12 hourly (maximum dosing 40 mg): Consider prophylaxis for children older than 5 years with comorbidities and not ambulant

Severe illness - Admit in ward with Oxygen. Do Baseline Tests (Total blood count, renal and liver function, HIV test, random blood sugar). Chest x-ray, sputum M/C/S and PCR to assess for comorbid respiratory pathogens like influenza and pertussis)

Symptomatic treatment - Oxygen supplementation to maintain SPO₂ above 90% (oxygen supplementation can be via nasal prongs, masks, non-rebreather masks or high flow nasal cannula)

Dexamethasone 0.15 mg/Kg IV/PO once daily to a maximum of 6 mg or prednisolone 1 mg/Kg OD maximum 40 mg once daily, Methylprednisolone 0.8 mg/Kg IV once daily; maximum 32mg once daily

VTE (venous Thromboembolism) prophylaxis with Enoxaparin 0.75 – 1 mg/Kg 12 hourly (maximum dosing 40 mg): Consider prophylaxis for children older than 5 years with comorbidities and not ambulant.

Critical illness - Admit to the Critical Care Unit

Baseline Tests (Total blood count, renal and liver function, HIV test, random blood sugar). Chest X-ray, sputum M/C/S and PCR to assess for comorbid respiratory pathogens like influenza and pertussis)

Symptomatic treatment

Conservative fluid management

Mechanical Ventilation if no improvement in oxygenation with maximal oxygen flows with other modalities – start with noninvasive ventilation, consider tracheal intubation and ventilation if noninvasive is not sufficient

Closed suctioning of secretions where available

Dexamethasone 0.15 mg/Kg IV/PO once daily to a maximum of 6 mg or prednisolone 1 mg/Kg OD maximum 40 mg once daily, Methylprednisolone 0.8 mg/Kg IV once daily; maximum 32mg once daily

VTE (venous Thromboembolism) prophylaxis with Enoxaparin 0.75 – 1 mg/Kg 12 hourly (maximum dosing 40 mg): Consider prophylaxis for children older than 5 years with comorbidities and not ambulant.

MIS-C illness - Supportive measures: fluid resuscitation; inotropic support; respiratory support; Intravenous Immunoglobulin (IVIG); Steroids; Enoxaparin

The following agents are still investigational and under emergency use authorization and may have a role in management of COVID-19 – Tocilizumab, Baricitinib plus Remdesivir and

Remdesivir. Specialist input would be required in defining the appropriate patient population, weighing benefit against risk, and cost considerations

6.8.5 Emergency Handling

Any patient with COVID-19 requiring resuscitation and other procedures will be handled with following standard hospital guidelines and using appropriate personal protective equipment as appropriate.

6.8.6 Prevention

Standard recommendations to prevent infection include use face covering, regular hand washing after touching surfaces, covering mouth and nose when coughing and sneezing. One should maintain a distance of at least 1.5m from other people

Vaccines are currently available for the prevention of COVID-19 infection. The vaccine is given in 1 or 2 doses with varying interval between 3 to 8 weeks and a booster dose 6 months later depending on the manufacturer. Vaccines are effective in the prevention of severe disease and death

7. GENITOURINARY DISORDERS

7.1 URINARY TRACT INFECTIONS

Definition: Urinary tract infection (UTI) refers to infection in the urinary bladder (cystitis), kidneys and ureters (pyelonephritis).

- It is difficult to differentiate pyelonephritis from cystitis in infants.
- In older children, cystitis presents with no fever, bacteriuria, dysuria, frequency, urgency or lower abdominal discomfort, no fever, no loin pain or tenderness.
- Pyelonephritis on the other hand would present with bacteriuria and either fever or loin pain or tenderness.
- Severe pyelonephritis additionally presents with tachycardia, nausea, vomiting, or sepsis or septic shock.

7.1.1 Clinical Features

0 to 3 months:

Jaundice, Fever, vomiting, Failure to thrive, Poor feeding, Vomiting, Lethargy, irritability, Hematuria, Offensive urine

3 months to 3 years:

Vomiting, Loin tenderness, Fever, Irritability, Abdominal pain, Hematuria, Offensive urine, Abdominal pain, Strong-smelling urine, Poor feeding, lethargy, irritability, hematuria

Children older than 3 years

Vomiting, Abdominal pain, Fever, Malaise, Cloudy urine, Foul smelling urine, Enuresis, dysuria, urgency, frequency, Flank pain/ back pain, Urine incontinence

Red Flags in history

Congenital genitourinary tract anomalies, spina bifida, phimosis, labial fusion, bladder dysfunction, previous UTI, abnormal urinary stream, surgery of the genitourinary tract.

Examination

May have no specific physical sign, lower abdominal or loin tenderness, dehydration, lethargy, fever, genital anomalies, raised blood pressure

7.1.2 Investigations

- Dipstick testing for leucocyte esterase and nitrites is diagnostically as useful as urine microscopy
- If both are positive, child has most likely has a Urinary Tract Infection and the urine should be sent for culture and antibiotics started immediately.
- If leucocyte esterase is negative and nitrites is positive, do clean catch or midstream urine sample collection and start antibiotics immediately.

- If leucocyte esterase is negative and nitrites is positive send for culture, hold antibiotics unless symptomatic.
- If both are negative, there is no UTI and no need for antibiotics.
- Nitrites may be negative in infants who have a UTI, since they pass urine frequently and there is not much time for the bacteria in the urinary bladder to convert nitrates to nitrites in the urine.

Table 23: Interpretation of Dipstick Urine analysis

Urinalysis result	Implications	Management	Further testing
Leucocyte esterase positive and Nitrite positive	Consistent with UTI	Commence antibiotics	Send urine for microscopy and culture
Leucocyte esterase positive and Nitrite negative	Treat as UTI if clinically indicated	Consider antibiotics	Send urine for microscopy and culture and consider infection outside the urinary tract
Leucocyte esterase negative and Nitrite positive	Treat as UTI if clinically indicated	Consider antibiotics	Send for microscopy and culture
Leucocyte esterase negative and Nitrite negative	Do not treat as UTI	Antibiotics for UTI should not be started	Unlikely to be a UTI, consider other causes of illness. MCS is usually still sent in non-toilet trained children.

Urine culture

- Growth of a single organism at > 100,000 CFU/mL from any collection method suggests infection.

- Growth of a single organism at lower counts of >1,000 CFU/mL from catheter urine suggests infection, and from clean catch or MSU may indicate early infection.
- Growth of any amount from suprapubic aspirate suggests infection.

7.1.3 Management

7.1.3.1 Supportive care

- Encourage the child to drink or breastfeed regularly to maintain a good fluid intake
- Pain should be managed with Paracetamol 10 – 15 mg/Kg every 6 hours or as necessary

7.1.3.2 Admission criteria

- All infants younger than 3 months with febrile UTI (presumed pyelonephritis)
- Patients unable to tolerate adequate oral fluids or medication
- Patients with signs of urinary obstruction or underlying disease (PUV)
- Patients who are septic or toxemic

7.1.3.3 Definitive Treatment

- Infants less than 3 months should be admitted and treated with parenteral antibiotics – Gentamicin 7.5 mg/Kg once daily plus CoAmoxiclav 30 mg/Kg every 8 hours as first line. Use Cefotaxime 50 mg/Kg every 8 hours (or Ceftriaxone 50 mg/Kg once daily) alone if Gentamicin is contraindicated
- For infants aged above 3 months and children with acute pyelonephritis who are not toxic but tolerating oral intake, should be treated with oral CoAmoxiclav 30 mg/Kg every 12 hours or Cefuroxime 15 mg/Kg every 12 hours for 7 - 10 days.

- For infants aged above 3 months and children with acute cystitis, should be treated with oral CoAmoxiclav 30 mg/Kg every 12 hours or Cefuroxime 15 mg/Kg every 12 hours for 3 - 5 days.
- Asymptomatic bacteriuria should not be treated with antibiotics

Repeat urine culture 5 - 7 days after completion of the antibiotic course

Refer all infants and children with recurrent UTI or obstructive uropathy to the Nephrology clinic.

For recurrent UTI consider evaluation and management of constipation.

7.2 MINOR PENILE INFLAMMATION

Minor redness or soreness of the tip of the foreskin is very common.

Contributing factors include:

- Chemical irritation-urinary trapping, soiled nappies, soap residue, bubble bath
- Physical trauma-inappropriate attempts at retracting the foreskin for cleansing
- Candida nappy rash in infants

7.2.1 Management

- Avoiding these factors
- Reassurance
- Application of a napkin barrier cream to the tip of the foreskin will help.

7.3 BALANITIS

A severe inflammation of the glans penis. When involving the foreskin and prepuce it is referred to as balanoposthitis. Can be caused by bacteria, fungus or chemicals.

7.3.1 Diagnosis

- Pain, swelling, redness, fluid leaking from the glans, difficulty passing urine.
- Swabbing the discharge is unhelpful because the normal foreskin is usually colonized with multiple organisms.

7.3.2 Management

- Soaking in a warm salty water bath with the foreskin retracted (if retractile and not too painful) will help with cleaning and urination may be easier in the bath
- Topical antifungals - Nystatin, Clotrimazole, and Miconazole are indicated
- Topical antibiotics mupirocin or bacitracin
- If there is significant cellulitis of the whole of the foreskin or the skin of the penile shaft then bacterial infection is likely oral antibiotics should be given to cover Streptococci (including Group A), staphylococci, and gram-negative organisms. Most cases respond to oral antibiotics (Amoxicillin 25 mg/Kg/dose 8 hourly)
- Pain management with Paracetamol 10 - 15 mg/Kg every 8 hours
- Significant recurrent Balanitis may be an indication for circumcision and thus should be referred to the surgeons

7.4 ACUTE URINE RETENTION

7.4.1 Clinical presentation

Urinary retention of acute onset can be characterized by lower abdominal pain, hesitancy, and incomplete voiding for several days before presentation.

Commonest cause in children is UTI. Occasionally caused by neurological problems, PUV, ureterocele, strictures and urinary stones.

7.4.2 Treatment

- Control pain with paracetamol 15mg/kg
- Consult Paediatric Urologist
- Attempt urethral catheterization with small Foleys catheter first. Do suprapubic aspiration or in older child do cystofix to relieve bladder pressure if not successful
- Urine to be sent for urinalysis, culture and sensitivity
- Initiate antibiotic treatment if indicated
- Once acute retention is relieved, underlying problem should be appropriately evaluated.

7.5 ZIPPER INJURY

The tip of the foreskin or other skin like scrotal skin may become entrapped in the teeth of a zipper during the zipping or unzipping process.

Localized edema, superficial tissue abrasions, bruising, and pain are the most common complications of zipper injuries. Skin loss or necrosis are unusual.

7.5.1 Management

- Prior to these procedures, adequate analgesia with or without sedation should be given
- Local infiltration may be necessary (never use local anaesthetic agents with adrenaline on the penis)

If trapped between teeth below the slider:

- Cutting the median bar of the zipper with wire cutters. The median bar is the part at the top of the slider which joins the front and back plates of the slider. Once cut, the slider falls off and the zipper can be separated
- Cutting through the material either side of the zipper below the entrapped skin and then cutting across the zipper with wire

cutters/strong scissors. Then the zipper can then be separated from below

If trapped between slider and teeth of zipper:

- Liberal application of topical anaesthetic cream, then ease slider down
- Always check for injury to urethral meatus

7.6 PHIMOSIS

Phimosis is defined as a tight foreskin that cannot be retracted to expose the glans penis.

True Phimosis is when scar tissue is present in the distal foreskin and this prevents retraction.

In young children, phimosis is normal (physiologic phimosis). In older patients, infections such as balanoposthitis or other inflammatory conditions result in scarring and pathologic phimosis that requires urologic referral.

7.6.1 Indicators of true Phimosis (rather than simple non-retractile foreskin)

- Foreskin not retractile by the time of established puberty.
- Previously retractile foreskin become non-retractile
- Obvious ring of scar tissue visible at foreskin opening
- Inability to visualize urethral meatus when foreskin opening is lifted away from glans
- Ballooning of foreskin on micturition, with pinhole foreskin opening, and very narrow urinary stream (minor ballooning may occur in normal non-retractile foreskin)

7.6.2 Management

Refer to the Paediatric surgeons

7.7 PARAPHIMOSIS

This occurs when the foreskin is left in the retracted position in uncircumcised or partially circumcised males that cannot be returned to normal position.

7.7.1 Diagnosis

The glans and the foreskin distal to the tight area become oedematous. Pain and swelling make it difficult to return the foreskin to the non-retracted position.

The penile shaft appears flaccid and unaffected.

Paraphimosis is also an important cause of irritability in the preverbal infant.

7.7.2 Management

- Adequate analgesia with or without sedation should be given then referral to the Paediatric surgeons
- Surgical options include gentle compression on penile shaft to disperse edema under analgesia or incision of the tight band of the foreskin
- Once reduced, a single episode of paraphimosis is not an indication for circumcision

7.8 ACUTE SCROTAL PAIN

A condition characterized by abrupt onset of scrotal pain.

Any acute scrotal swelling requires immediate surgical assessment for torsion of the testis or strangulated inguinal hernia, which are surgical emergencies.

The most common causes of acute scrotal pain in children and adolescents include testicular torsion, torsion of the appendix testis, and epididymitis, orchitis, testicular abscess, varicoceles, trauma, insect bites, dermatologic lesions, inflammatory vasculitis (HSP), microlithiasis.

Testicular torsion has two peak incidences: a small one in the neonatal period and a large one during puberty, but it can occur at any age. May follow minor trauma in sports.

7.8.1 Clinical features

- The scrotum may be edematous, indurated and erythematous
- The affected testis usually is tender, swollen, and slightly elevated because of shortening of the cord from twisting.
- The testis may be lying horizontally, displacing the epididymis from its normal posterolateral position.
- A reactive hydrocele may also be present.
- Absent cremasteric reflex in nearly all cases of torsion, but it also may be absent in boys without torsion, particularly if they are younger than six months

7.8.2 Diagnosis

- Clinical diagnosis
- Scrotal ultrasound- Doppler ultrasound can discern testicular and epididymal size, scrotal fluid, scrotal wall thickening, enlarged appendix testis, and arterial flow in the testis and epididymis. Decreased testicular perfusion also can be seen in some patients with a large hydrocele, abscess, hematoma, or scrotal hernia

7.8.3 Management

- Early surgical consultation is vital, as delay in scrotal exploration and detorsion of a torqued testis will result in testicular infarction within 6 hours. Keep the child fasted
- Adequate analgesia (paracetamol 10-15mg/kg, ibuprofen 10mg/kg)

7.8.4 Specific management of other causes depends on the diagnosis:

- Suspected torsion of the appendix testis usually requires surgical exploration
- Incarcerated inguinal hernia must be reduced or the contents of the hernia may become gangrenous
- Epididymoorchitis should be managed with antibiotics once a suitable urine sample has been sent.
- Young infants or systemically unwell children should be admitted for IV antibiotics (e.g., Amoxicillin and Gentamicin). Adolescent with Epididymoorchitis should have a first-pass urine sample (ideally first morning urine) for Chlamydia and gonococcus
- Idiopathic scrotal oedema usually resolves spontaneously over a couple of days. No intervention is required
- Hydroceles will often resorb and the tunica vaginalis close spontaneously in the first year. If still present at 2 years, surgical referral should be made for consideration of repair
- Henoch Schonlein purpura. Check urinalysis and blood pressure. These children need close paediatric surveillance as abdominal pathology can be quite severe acutely and nephritis may develop in the convalescent period

7.9 GENITOURINARY TRAUMA

- If urine is clear do bedrest, monitor vitals, imaging studies
- If urine is bloody and patient is stable monitor and refer to Paediatric surgeon. If unstable, stabilize patient first as per ABCDE protocol then consult Paediatric surgeon

7.10 COMMON CONGENITAL GENITOURINARY ANOMALIES

- Patent urachus - urine passage through umbilicus
- Prune belly syndrome- anterior abdominal wall abnormality with urinary complications
- Exstrophy of the bladder- anterior abdominal wall defect with an open bladder
- Hypospadias - ventral urethral opening with bent penile shaft (mild or severe)

Refer all the above anomalies to a Paediatric surgeon

- Polycystic kidneys- may present with abdominal swelling in the flanks- *refer to pediatric nephrologist*
- DSD - disorders of sexual differentiation (ambiguous genitalia)- *refer to Paediatric endocrinologist*

8. CENTRAL NERVOUS SYSTEM DISORDERS

8.1 STATUS EPILEPTICUS

Status epilepticus is a condition resulting either from the failure of the mechanisms responsible for seizure termination or from the initiation of mechanisms, which lead to abnormally, prolonged seizures over 5 minutes and can have long-term consequences after 30 minutes, including neuronal death, neuronal injury, and alteration of neuronal networks, depending on the type and duration of seizures.

8.1.1 Management

8.1.1.1 *General measures*

- Manage airway, breathing and circulation (ABCs). Monitor pulse oximetry, blood pressure, and temperature. Give Oxygen. Intubate and hyperventilate if hypoventilation, hypoxia and/or hypercarbia occur or if aspiration is a concern.
- Observe and confirm the fit is Status Epilepticus.
- Pursue available drug history.
- Establish intravenous or intra osseous line.
- Obtain blood samples for initial laboratory studies.

8.1.1.2 *Medication*

For neonates:

- Phenobarbitone 20 mg/kg IV or IO at 1 mg/kg/min (infuse over 20-30 minutes)
- With control of seizures, maintenance dose is 3 – 5 mg/Kg/24 hours divided into 2 equal doses.

For older children start with:

- Midazolam 0.1 mg/kg slow IV or IO (maximum 10 mg) - May repeat every 10 minutes;
- OR IV diazepam 0.1 - 0.3 mg/Kg over 3 - 5 minutes (direct IV at maximum rate 2 mg/min), or rectal diazepam 0.2 - 0.5 mg/Kg a maximum of three doses at 15-minute intervals;
- OR Lorazepam: 0.1 mg/Kg IV at 2 mg/min to maximum of 4 mg. May repeat at 10-minute intervals twice.

If the first-line medication fails or once the convulsive activity ceases a longer acting anticonvulsant is given immediately

- Phenytoin 20 mg/kg IV or IO at 1 mg/kg/min - Maximum dosage rate 50 mg/min
- Fosphenytoin is dosed in phenytoin equivalents (PE) - 20 mg PE/kg IM, IV or IO at 3 mg PE/kg/min; - Maximum dosage rate 150 mg PE/min, - Fosphenytoin may be used IM.
- Sodium valproate 20 to 40 mg/kg IV over 5 to 10 minutes; May give additional 20 mg/kg IV.
- Levetiracetam 20 to 30 mg/kg IV at a rate of 2 to 5 mg/kg/minute, Maximum dose 3 g.

Note:

- Adjust the doses of antiepileptic drugs if child is already receiving these drugs.
- Be prepared to ventilate and give continuous infusion of anti-epileptic drugs such as midazolam, sodium valproate, thiopental, ketamine or propofol.
- For patients aged < 1 year, Consider trial of pyridoxine IV 100 mg/kg.
- Transfer to Paediatric Critical Care Unit (PCCU).
- Serum phenytoin and phenobarbital levels should be monitored during long term therapy.

8.2 FEBRILE SEIZURES

A febrile convulsion is a seizure occurring with fever in children aged between the ages of 6months and 5 years without evidence of other underlying cause.

Refer to section 3.2 for management.

8.3 BACTERIAL MENINGITIS

This is an inflammation in response to bacterial infection of the pia-arachnoid mater. Meningitis is always cerebrospinal.

8.3.1 Management

8.3.1.1 General measures

- If diagnosis is suspected, lumbar puncture should be done immediately with antibiotic therapy being begun empirically.

- If signs of increased intracranial pressure (altered level of consciousness, bradycardia, Hypertension) or focal neurological findings present, antibiotics to started without lumbar puncture and request brain imaging (CT scan or MRI).
- Increased ICP should be treated simultaneously (IV Mannitol, Hyperventilation).
- Antibiotic choice for empirical therapy is determined by susceptibility to streptococcus pneumonia.

8.3.1.2 Uncomplicated Meningitis

Empirical therapy

- Ceftriaxone 80 - 100 mg/Kg/24 hours or divided into two doses for 10 – 14 days or Cefotaxime 200 mg/Kg/24 hours in four divided doses
- If not sensitive to betalactams, use IV Chloramphenicol 100 mg/Kg/24 hours in 4 divided doses for 10 – 14 days

In infants 1-2 months old, with suspected listeria infection:

- IV Ampicillin 200 mg/Kg/24 hours given 6hourly with IV Ceftriaxone or IV Cefotaxime
- IV Cotrimoxazole is an alternative

Dexamethasone 0.15 mg/Kg/dose every six hours should be given for two days, one to two hours before initiation of antibiotics.

8.3.1.3 Supportive care

- Manage airway, breathing and circulation (ABCs). Monitor pulse oximetry, blood pressure, and temperature.

- Daily neurological assessment.
- Laboratory investigation including urea, electrolytes, creatinine, calcium, magnesium, bicarbonate levels, urine output and specific gravity, haemogram, coagulation profile.
- Closely monitor the hydration status and serum Sodium levels. Restrict IV fluids to between half and two-thirds maintenance or 800 – 1000 mL/m²/24hrs only if SIADH is confirmed. Revert to normal fluid allowances when serum sodium level is normal.

8.3.2 Complications

- If complications present, admit in Paediatric Critical Care Unit
- If in shock, aggressive treatment indicated.
- If features of raised ICP, intubate and give IV Frusemide at 1 mg/Kg or mannitol 0.5 – 1 g/Kg.
- In case of seizures – manage as appropriate.

8.3.3 Indications for repeat lumbar puncture

- Gram negative bacillary meningitis in neonates.
- Beta lactam resistant *Streptococcus pneumoniae* infection.
- Gram negative bacillary meningitis should be treated for 21 days or two weeks after CSF sterilization.

8.4 COMA

Coma is a state of altered consciousness from which a person cannot be aroused. It is reliably graded using the Glasgow coma scale.

- Coma is a symptom, not a diagnosis.
- The aim of immediate management is to minimize any ongoing neurological damage whilst making a definitive diagnosis.

- Elements of the history, examination, investigation and treatment will therefore occur simultaneously.

8.4.1 Immediate management

- Attend to airway, breathing and circulation – (see Resuscitation guidelines).
- If traumatic cause is possible immobilize cervical spine and arrange urgent neurosurgery involvement.
- Insert intravenous access.
- Perform blood glucose; if Glucometer glucose < 2.5mmol/L in a non-diabetic, send specific bloods tests and administer IV 10% dextrose 10 mL/Kg stat.
- Consider Naloxone 0.1 mg/Kg (maximum 2 mg) IV can be repeated.
- Assess and monitor pulse, respiratory rate, BP, temperature, oximetry ± ECG monitoring and conscious state.
- Look carefully for subtle signs of a continuing convulsion.

8.4.2 Ongoing care

- Will be determined by the diagnosis, level of consciousness and degree of ventilatory and circulatory support needed.
- See specific guidelines if diagnosis becomes clear.
- Consider Lumbar puncture, imaging and antibiotics.

8.5 CEREBRAL PALSY

Cerebral palsy is a persistent but not unchanging disorder of movement and posture due to a defect or lesion of the developing brain. It is accepted that children up to five years, who acquire permanent motor impairment due to non-progressive neurological insults, have cerebral palsy.

There are many causes, a wide range of manifestations of the motor disorder and various associated problems.

If the cause is clear e.g. massive antepartum haemorrhage followed by neonatal encephalopathy, no further investigations are necessary.

If the cause is not clear, investigate with urine or plasma metabolic screening, consider congenital infections, chromosomal abnormalities, and radiological investigations. MRI Scanning may indicate the cause e.g. vascular lesion, malformation or periventricular leukomalacia.

Cerebral palsy is not a single disorder but a group of disorders with diverse implications for children and their families.

8.5.1 Common presentations to the Emergency Department

- Respiratory problems particularly pneumonia
- Uncontrolled seizures/status epilepticus
- Unexplained irritability – consider acute infections, oesophagitis, dental disease, hip subluxation, pathological fracture. Review medications.

8.5.2 Management

Management involves a team approach with health professionals and teachers. Input from the family is important.

- Accurate diagnosis and genetic counselling.
- Establish the cause of cerebral palsy if possible.
- Management of the associated disabilities, health problems and consequences of the motor disorder.

8.5.2.1 Associated disabilities

- All children require **hearing** and **visual** assessments
- Assess and review anticonvulsants for **epilepsy**
- Formal **cognitive** assessment is beneficial. Children often need help with their educational program

8.5.2.2 Health problems

- Monitor **growth** and provide dietary advice. Failure to thrive is frequent. Consider nasogastric or gastrostomy feeds if there is difficulty in achieving satisfactory weight gains, or there are major feeding problems. Obesity interferes with progress in motor skills
- **Gastro-oesophageal reflux** can result in oesophagitis or gastritis, causing pain, poor appetite and aspiration
- **Constipation**
- Some children with severe cerebral palsy develop **chronic lung disease**, due to aspiration from oromotor dysfunction or severe gastro-oesophageal reflux. Coughing or choking during mealtimes or wheeze during or after meals may signal aspiration. It can also occur silently
- Monitor **ventriculo-peritoneal shunts**
- **Osteoporosis** and pathological fractures occur in severe cerebral palsy
- Monitor **dental health**
- **Emotional problems** can be responsible for suboptimal performance either with academic tasks or self-care

8.5.2.3 Consequences of the motor disorder

- **Drooling** (poor saliva control). Speech therapists assist with behavioural approaches. Medication (anticholinergics) and surgery may be helpful
- **Incontinence**. Children may be late in achieving bowel and bladder control because of cognitive

deficits or lack of opportunity to access toileting facilities because of physical disability or inability to communicate. Some children have detrusor over activity causing urgency, frequency and incontinence

- **Orthopaedic problems.** Contractures may develop and require orthopaedic intervention. Surgery is more commonly undertaken on the lower limb
- The hip. Non-walkers and those partially ambulant are at risk for hip subluxation and dislocation. Perform hip x-rays at yearly intervals. Refer children to an orthopaedic surgeon if there is evidence of subluxation or dislocation. Ambulant children occasionally develop hip problems
- The knee. Flexion contractures at the knee may require hamstring surgery
- The ankle. Equinus deformity is the commonest orthopaedic problem in children with cerebral palsy. Toe walking is treated conservatively in young children with orthoses, inhibitory casts, and Botulinum toxin A therapy. Older children benefit from surgery
- Children may require multi-level surgery (for example, hip, knee and ankle), usually between 8 and 12 years. The aims of surgery are to correct deformities and to improve both the appearance and efficiency of walking
- Several procedures are available for the upper limb
- Correction of scoliosis is sometimes necessary
- **Spasticity** management aims to improve function, comfort and care and requires a team approach. Options include:
 - Oral medications, for example, diazepam, dantrolene sodium and baclofen
 - Inhibitory casts, for example, below knee casts increase joint range and facilitate improved quality of movement.
 - Botulinum toxin A reduces localised spasticity

- Intrathecal baclofen is suitable for a small number of children with severe spasticity and may enhance quality of life
- **Selective dorsal rhizotomy** is a neurosurgical procedure whereby anterior spinal roots are sectioned to reduce spasticity
- **Assessment of the child's capabilities and referral**

9. CARDIOVASCULAR DISEASES

9.1 RHEUMATIC FEVER

The diagnosis of acute rheumatic fever can be established by the Jones criteria.

9.1.1 Modified Jones Criteria for Rheumatic Fever:

9.1.1.1 *Major Criteria*

- Migratory polyarthritis.
- Carditis.
- Subcutaneous nodules.
- Erythema marginatum.
- Sydenham's chorea.

9.1.1.2 *Minor Criteria*

- Fever.
- Arthralgia (only if no arthritis).
- High ESR, High white blood cell count, High CRP.
- Abdominal pain.
- Epistaxis.
- First degree heart block.

9.1.1.3 Criteria for diagnosis:

- Two major criteria or one major criterion and two minor criteria. In either option, evidence of streptococcal infection (High ASO Titre) is required for diagnosis.
- Chorea may occur as the only manifestation.
- Indolent carditis may be the only manifestation in patients who first come to medical attention months after the onset of acute rheumatic fever.

9.1.2 Management

All patients with acute rheumatic fever should be placed on **bed rest** and monitored closely for evidence of carditis. They can be allowed to ambulate as soon as signs of acute inflammation have subsided

9.1.2.1 Antibiotic therapy

Once diagnosis is established and regardless of throat culture results, the patient should receive:

- Oral Penicillin V 7.5 – 15 mg/Kg/dose PO 6hourly or Erythromycin 10 mg/Kg 6hourly for 10 days. Or
- Single IM injection of Benzathine Penicillin

After this initial course, long-term antibiotic prophylaxis should be initiated

The regimen of choice for secondary prevention is one of:

- IM Benzathine penicillin G every 3 - 4 weeks
 - Penicillin V 7.5 - 15 mg/Kg/dose PO 12 hourly
- In penicillin allergy:
- Sulfadiazine 500 mg - 1g once daily or
 - Erythromycin 10 mg/Kg 12 hourly

- Patients who did not have carditis with their initial episode of acute rheumatic fever have a relatively low risk of recurrences
- Antibiotic prophylaxis may be discontinued in these patients when they reach their early 20s and after at least 5 years have elapsed since their last episode of acute rheumatic fever
- The decision to discontinue prophylactic antibiotics should be made only after careful consideration of potential risks and benefits and of epidemiological factors such as the risk of exposure to group A streptococcal infections

9.1.2.2 Anti-inflammatory therapy:

- Should be withheld if arthritis or atypical arthritis is the only clinical manifestation – premature treatment interferes with development of characteristic migratory polyarthritis and obscures diagnosis
- Patients with typical migratory polyarthritis and those with carditis without cardiomegaly or CHF should be treated with oral Salicylates: Aspirin 100 mg/Kg/24hrs divided 6hourly PO for 3-5 days, followed by 75 mg/Kg/24hrs divided 6hourly PO for 4 weeks
- Patients with carditis and cardiomegaly or CHF should receive corticosteroids: Prednisone 2 mg/Kg/24hrs in 4 divided doses for 2-3 wks. followed by a tapering of the dose that reduces the dose by 5 mg/24hrs every 2-3 days. At the beginning of the tapering of the dose, aspirin should be started at 75 mg/Kg/24hrs in 4 divided doses for 6 weeks.
- Supportive therapies in mod-severe carditis include Digoxin, fluid and salt restriction, diuretics and Oxygen.

Termination of anti-inflammatory therapy may be followed by reappearance of clinical manifestations or of laboratory abnormalities. These 'rebounds' are best left untreated unless clinical manifestations are severe.

9.1.2.3 Chorea

Anti-inflammatory agents not indicated if it is an isolated manifestation. Sedatives are helpful early in the course of chorea.

Oral Haloperidol 0.01 – 0.03 mg/Kg/day in two divided doses.

9.2 INFECTIVE ENDOCARDITIS

This is often a complication of congenital or rheumatic heart disease but can also occur in children without any abnormal valves or cardiac malformations

Patients with the following conditions are likely to develop infective endocarditis and require pre procedure antibiotic prophylaxis: Prosthetic valve or material, Previous, relapse, or recurrent infective endocarditis and congenital heart disease

The procedures that require prophylaxis of infective endocarditis include dental work, respiratory incisions or biopsies, and skin and soft tissue procedures.

Viridians type streptococci (α -haemolytic) and *Staphylococcus aureus* are the leading causative organisms

Cephazolin is the prophylactic medication of choice and Azithromycin for those who may have allergies to it.

Patients on concurrent antibiotics for other indications at the time of procedure, an alternate antibiotic of a different class is often chosen

9.2.1 Management of Infective Endocarditis

- Specific antibiotic therapy guided by culture and sensitivity.
- If culture results are negative, empirical therapy with antibiotics to cover common causative organisms – use Benzyl Penicillin and Gentamicin.

9.3 CONGESTIVE HEART FAILURE

As the demands on the heart outstrip the normal range of physiologic compensatory mechanisms, signs of CHF occur. These signs include tachycardia; venous congestion; high catecholamine levels; and, ultimately, insufficient cardiac output.

9.3.1 Specific treatment

- Acute CHF in the neonate or infant - The evaluation and treatment of these patients should be in the Neonatal or Paediatric Critical Care Unit.
- Acute CHF in the older child - Intensive care for diuresis with IV Frusemide and IV dopamine infusion at a rate of 5 – 10mcg/Kg/min are appropriate until stabilization is achieved. Older children may require the placement of a central venous catheter to monitor venous pressure and cardiac output during stabilization.

9.3.1.1 *Electrolyte management*

- Supplemental potassium chloride may be required when high doses of diuretics are used.
- Because ACE inhibitors cause potassium retention, Spironolactone and supplemental potassium should be avoided except in the presence of documented hypokalemia in patients taking these medications.
- Sodium supplementation is almost never indicated in infants or children with CHF except in emergency situations. Severe hyponatremia is generally best managed by reducing the dose of diuretics or by restricting fluid intake, although the latter has little utility in small children.

9.3.1.2 Supportive care

Nutrition is crucial in the management of chronic CHF. Enhanced caloric content feedings and, in some cases, nasogastric or gastrostomy feedings may be necessary to maintain the patient's growth.

- Anemia aggravates CHF. Careful attention to iron stores or the administration of red cell transfusions results in a significant improvement. The success of medical therapy of CHF in infants and small children is judged according to the child's growth. A failure to thrive is an indication for increased medical management or, when the option exists, surgical repair of structural heart disease

Table 24: Congestive Heart Failure Medications

Agent	Paediatric Dose	Comment
Preload reduction		
Frusemide	1 mg/Kg/dose PO or IV	May increase to 6 hourly
Hydrochlorthiazide	2 mg/Kg/d PO in 2 divided doses	May increase to 6 hourly
Metolazone	0.2 mg/Kg/dose PO	Used with loop diuretic, may increase to 12 hourly
Inotropic		
Digoxin	Loading Dose Preterm infants <1.5Kg: 0.025 mg/Kg/d PO in 2 divided doses Preterm infants 1.5 - 2.5Kg: 0.03 mg/Kg PO in 2 divided doses Term Neonates to 2years: 0.045 mg/Kg PO in 1 or 2 divided doses Child aged 2 to 5years: 0.035 mg/Kg PO in 1 or 2 divided doses Child aged 5 to 10years: 0.025 mg/Kg PO in 1 or 2 doses	For IV use 75% of the oral dose Maintenance dose: 20 – 25% of the Loading dose given once a day
Dopamine	5 – 20 mcg/Kg/min IV	Gradually titrate upward to desired effect
Dobutamine	5 – 20 mcg/Kg/min IV	Gradually titrate upward to desired effect
Amrinone	5 – 10 mcg/Kg/min IV	Load: 1 mg/Kg IV over 2-3 min
Milrinone	0.5 – 1 mcg/Kg/min IV	Load: 50 – 75 mcg/Kg IV slowly over 15min
Afterload reduction		
Captopril	0.5 – 1 mg/Kg/day PO in divided doses 8 hourly	-
Enalapril	0.1 mg/Kg/day PO in 1 or 2 doses, not to exceed 0.5 mg/Kg/day	Adults: 2.5-5 mg/d PO in 1 or 2 doses, not to exceed 40 mg/day
Lisinopril	Not established	Adults: 10 mg PO once daily

Nitroprusside	0.5-10mcg/Kg/min IV	May need to monitor cyanide level
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9.4 HYPERTENSION

Hypertension is defined as systolic or diastolic systemic arterial blood pressure above normal for age.

Note: BP measurements repeated on several different occasions are required to diagnose hypertension. Formula for calculating BP is found in the appendices.

9.4.1 Classification of Hypertension

Hypertension in children is usually secondary. Look for the cause.

9.4.1.1 Asymptomatic hypertension

- No treatment required acutely. Investigate and manage as out-patient.

9.4.1.2 Acute Severe Hypertension

- These patients require admission to HDU for urgent treatment.
- Hypertensive encephalopathy presents as severe headache, visual disturbance and vomiting, progressing to focal neurological deficits, seizures and impaired conscious state, with grossly elevated BP, papilloedema and retinal haemorrhages.
- These patients almost always have chronic renal disease and are on dialysis.
- The differential diagnosis includes uremic encephalopathy and metabolic disturbance.
- BP should be lowered in a controlled fashion, with anticonvulsants given for seizures.

9.4.2 Antihypertensives

Choices include:

Intravenous Hydrallazine

- IV Hydrallazine at a dose of 0.15 mg/Kg (maximum 10 mg) diluted in normal saline given over 30 minutes is indicated for severe hypertension
- Continue with 4 - 6 micrograms/Kg/min (max 300 micrograms/min)
- Hydrallazine is contraindicated in Systemic Lupus Erythematosus due to the risk of convulsions.
- Hydrallazine may cause tachycardia, nausea and fluid retention

Intravenous Labetalol

- 0.2 mg/Kg initially; later 0.4 mg/Kg by slow push every 10min up to 3-4 mg/Kg (max. 100 mg) total dose.
- Avoid if there is heart failure, asthma or bradycardia.

Oral Captopril

- 0.1 mg/Kg initially, increasing to a maximum of 1 mg/Kg (max. 50 mg), thereafter 0.11.0 mg/Kg/dose 8-hourly.
- Captopril is usually effective within 30 - 60min.

10. BURNS

A burn is an injury to body tissues caused by very high temperatures. Causes include thermal (fire, flame, hot objects, hot fluids, friction), electricity, chemical and radiation. Injury to body tissues like burns can be caused by very low temperatures (frost bites). Burns can result in both local and systemic effects.

10.1 FIRST AID

- Inhalational burns – maintain airway and give oxygen.
- Surface burns cool the area with running water for 20 minutes but avoid hypothermia (useful up to 3 hours after burn).
- Cold water compresses (changed frequently) useful for analgesia for localised burns.
- Do not apply ice or ice slush.
- For chemical and eye burns, irrigate with copious volumes of water.
- Plastic (cling) wrap useful after cooling (limits evaporation and heat loss).
- Burns to the eyes require early copious irrigation with Normal Saline or water and ophthalmologic opinion.

10.2 ASSESSMENT

10.2.1 Airway and Breathing

- Assess for presence of stridor, hoarseness, black sputum or respiratory distress, burnt nasal hairs or facial swelling.
- Oropharyngeal burns and significant neck burn usually require immediate intubation even if the airway is not yet compromised; Admit to CCU early.

- Immobilise cervical spine if associated trauma.

10.2.2 Circulation

- Early hypovolaemia is rarely related to the burn injury and other sources of bleeding should be sought.
- For circumferential burns, check for signs of circulatory obstruction and the need for an escharotomy; elevate the limb.
- For electrical burns, monitor ECG continuously. If high voltage limb burn, early fasciotomy might be required.

10.2.3 Estimation of Surface Area

- Use a burn diagram to accurately calculate the burnt surface area, however, do not count skin with isolated erythema (no blistering).
- As a rough measure, the child's palm represents about 1% of total body surface area.
- For each area note if shallow, indeterminate or deep.

Diagram 7: The Lund and Browder Chart

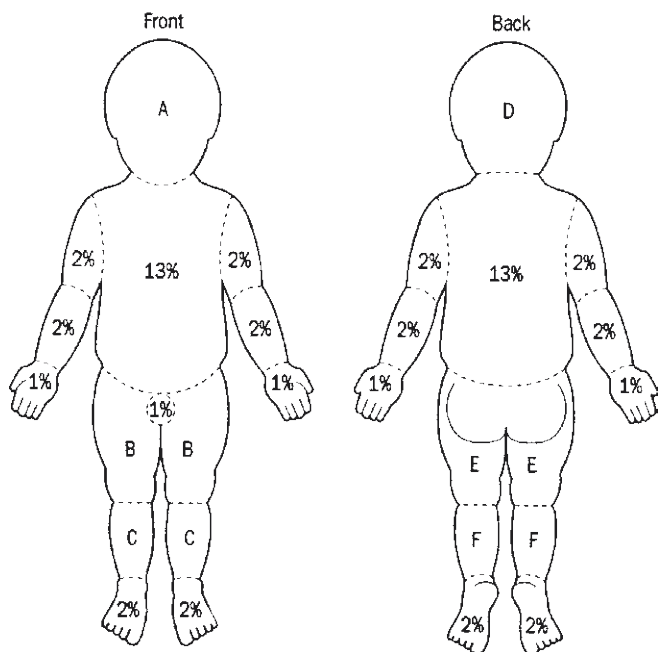


Table 25: The Lund and Browder Table

	1yr	1-4yr	5-9yr	10-14yr	15-18yr	Adult	Shallow	Indeterminate or deep
Head	19	17	13	11	9	7		
Neck	2	2	2	2	2	2		
Ant. Trunk	13	13	13	13	13	13		
Post. Trunk	13	13	13	13	13	13		
R. Buttock	2.5	2.5	2.5	2.5	2.5	2.5		
L. Buttock	2.5	2.5	2.5	2.5	2.5	2.5		
Genitalia	1	1	1	1	1	1		
R.U. Arm	4	4	4	4	4	4		
L.U. Arm	4	4	4	4	4	4		
R.L. Arm	3	3	3	3	3	3		
L.L. Arm	3	3	3	3	3	3		
R. Hand	2.5	2.5	2.5	2.5	2.5	2.5		
L. Hand	2.5	2.5	2.5	2.5	2.5	2.5		
R. Thigh	5.5	6.5	8	8.5	9	9.5		
L. Thigh	5.5	6.5	8	8.5	9	9.5		
R. Leg	5	5	5.5	6	6.5	7		
L. Leg	5	5	3.5	6	6.5	7		
R. Foot	3.5	3.5	3.5	3.5	3.5	3.5		
L. Foot	3.5	3.5	3.5	3.5	3.5	3.5		
Total								

Table 26: Depth of Burn

Depth	Surface/colour	Pain sensation
Superficial	Dry, minor blisters, erythema, brisk capillary return	Painful
Partial thickness-superficial (superficial dermal)	Moist, reddened with broken blisters, brisk capillary return	Painful
Partial thickness-deep (deep dermal)	Moist white slough, red mottled, sluggish capillary return	Painless

Full thickness	Dry, charred whitish. Absent capillary return	Painless
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Table 27: Classification of Burns

Extent of burn	Partial thickness – First and second-degree burns)	Full thickness – third- and fourth-degree burns)
Minor	<10%	<2%
Moderate	10-20%	2-5%
Major	>20%	>5%

10.3 MANAGEMENT

Document the following:

- Time of burn
- Extent – burns diagram
- Depth
- First aid
- Tetanus immunization status

10.3.1 Pain relief

- PO Paracetamol 10 - 15 mg/Kg every 6 hours.
- Morphine:
 - **Neonate:** Initially 100 micrograms/Kg IV every 6 hours, adjusted according to response or as an infusion; initially by intravenous injection (over at least 5 minutes) 50 micrograms/kg then by continuous intravenous infusion 5 - 20 micrograms/Kg/hour adjusted according to response.

- **Child:** Initially 100 - 200 micrograms/Kg every 6 hours, adjusted according to response or as an infusion; initially by intravenous injection (over at least 5 minutes) 100 micrograms/kg then by continuous intravenous infusion 10 - 30 micrograms/kg/hour adjusted according to response.

10.3.2 Management of Minor Burns

- Analgesia should be adequate; children may require morphine initially before assessment and initial dressings.
- Immobilisation with sling and splinting is suggested for upper limb burns as well as early occupational therapy consultation.
- Check tetanus status.
- Closed dressings are recommended for partial thickness burns. The wound exudate determines the number of dressing changes.
- The depth of a partial thickness burn may only be declared after 7 – 10 days.

10.3.2.1 Blisters

- Have a protective function and reduce pain if they are left intact for a few days.
- If blisters are small, not near a joint and not obstructing the dressing, leave alone.
- Large blisters and those overlying a joint should be de-roofed.
- If patient is for admission, apply topical antibiotics, dress and let debridement (de-roofing) be done in the operating theatre
- Opaque blister fluid occurring after a few days suggests infection. The burn should then be de-roofed and dressed.

10.3.2.2 Superficial burns with erythema only:

Can be treated by exposure. In infants who show a tendency to blister or scratch, a protective, low-adherent dressing with crepe bandage may be helpful.

10.3.2.3 Partial thickness burns

- Cleanse the burn and surrounding surface with saline and pat dry. If treatment is delayed or wound is dirty, use aqueous Chlorhexidine 0.1% then saline.
- For small, superficial partial thickness burns, a low adherent dressing then crepe bandage or adhesive paper.
- For more extensive or deeper partial thickness burns, a low-adherent silver sulphadiazine dressing should be applied.

10.3.2.4 Follow up

- If a superficial burn has not healed in 7 - 10 days, it has either become infected or is deeper than anticipated. If in doubt, consult the plastic surgeon.

10.3.3 Management of Major Burns

These require admission to a Burns Unit

10.3.3.1 Airway and Breathing

- For signs of airway burn or lung injury, arrange intubation as soon as possible and before airway swelling occurs.

10.3.3.2 Fluids

- If < 10% of body surface involved, commence burns fluid resuscitation and calculate fluid requirements from the time

of injury. Preferably insert an IV line through uninvolved skin.

- Initial fluid resuscitation in first 24 hours from the time of the burn
 - Ringer's lactate or 0.9% normal saline 3 – 4 mL/Kg/1% BSA over 24 hours. Give the first half within first 8 hours. Remainder within the next 16 hours.
 - Add the maintenance fluid requirements if not able to take orally as 5% dextrose in Ringer's lactate.
- Insert urinary catheter if burn < 15% BSA or if significant perineal burn.
- Monitor fluid resuscitation by urine output - target 1 to 2 mL/Kg/hr.
- Insert an NG Tube if < 10% partial thickness or full thickness burns; Start feeding within 6 – 18 hours.

10.3.3.3 Investigations

- Haemoglobin, electrolytes, blood glucose, blood group and cross match.
- Carboxyhaemoglobin and cyanide levels (if fire in confined space).

10.3.3.4 Full thickness burns

- All require admission and early debridement.

10.3.4 Burns requiring admission

- Have low threshold for admission, in-hospital team can discharge after further evaluation and admission in the ward
- Partial thickness (superficial) burns with a surface area greater than 10%, except very superficial burns.
- All full thickness burns, except those that are extremely small.

- All burns to face, ears, eyes, hands, feet, genitalia, perineum or a major joint, even if less than 5-10%.
- Admit patients with major burns to a Burns Unit as soon as possible
- Circumferential burns.
- Chemical burns.
- Electrical burns (including lightening). Extensive tissue damage can occur to underlying structures.
- Burns associated with significant fractures or other major injury.
- All inhalational burns.
- Burns in children under the age of 12 months.
- Minor burns in patients with social problems e.g., suspected child abuse, anxious parents, dysfunctional family set up

10.3.5 Transfer of patients from other hospitals for assessment

If time from burn to arrival is <6 hours and the burn area is clean:

- Wash with saline, cover with plastic cling wrap for transfer, allowing for easy assessment without undue discomfort from removal of dressings.

10.3.6 If transfer is likely < 6 hours or burn is dirty

- Dirty or charred burns should be washed with aqueous Chlorhexidine 0.1% and dressed.
- Clean burns should be dressed with a low-adherent dressing.
- The patient should receive analgesia and then ideally proceed without delay to the ward.
- Notify the Plastic Surgeon.

11. INFECTIOUS DISEASES

11.1 VIRAL INFECTIONS

11.1.1 Aseptic Meningitis

This refers to inflammation of the meninges other than that caused by pus producing organisms. It includes viral infections of the meninges. The usual cause is acute and may be relapsing.

11.1.1.1 Diagnosis

If viral -sudden febrile illness and rash, other symptoms include headache and signs of meningeal involvement.

11.1.1.2 Investigation

Main test: Lumbar puncture for CSF biochemistry, microscopy, culture and sensitivity, blood serology and CSF PCR for viral agents.

11.1.1.3 Management

- Paracetamol 10 – 15 mg/Kg/dose every 4 – 6 hours.
- Antiemetic - dexamethasone 0.25 mg/Kg/6hours and ondansetron 0.15 mg/Kg/6hours may be used.
- Start empirical IV acyclovir infusion acyclovir 10 - 20 mg/Kg every 8 hours and a broad-spectrum antibiotic with good CSF penetration as you await laboratory results.

11.1.2 Herpes Simplex

Viral disease usually seen as painful vesicles that often occur in clusters on skin, cornea, or mucous membranes; may occur as encephalitis, pneumonia, or disseminated infection.

Usual course of primary disease is 10 – 21 days

Neonates or individuals with immune compromise are at risk for major morbidity or mortality due to severe disseminated infections.

11.1.2.1 Management

Primary herpes gingivostomatitis – See Section 4.9

Herpes Simplex Encephalitis (HSE) and recurrent herpes labialis and other HSV skin infections:

- 3months to 11 years: IV acyclovir infusion 10 - 20 mg/Kg every 8 hours for 21 days
- ≥ 12 years: IV acyclovir 10 mg/kg/day 8 hourly for 14-21 days

11.1.3 Measles

An acute epidemic viral exanthem which classically presents as a confluent erythematous maculopapular rash which begins over the head and spreads inferiorly to involve the trunk and extremities.

The rash is preceded by the triad of cough, coryza, and conjunctivitis plus a pathognomonic enanthem.

11.1.3.1 Diagnosis

- Measles specific IgM antibodies or 4-fold rise in IgG been acute and convalescent sera.
- Viral isolation by RT-PCR in urine, blood or nasopharyngeal mucous in a referral laboratory.
- WHO case definition: Any person in whom a clinician suspects measles infection, **or** any person with fever **and** maculopapular rash **and** cough, coryza or conjunctivitis.

11.1.3.2 Management

Keep out of school for 4 days after rash appearance. Children with severe complicated measles require treatment in hospital.

11.1.3.3 Vitamin A therapy

Give oral vitamin A once a day for two days, to all children with measles

- 50 000 IU for a child aged <6 months
- 100 000 IU 6 – 11 months
- 200 000 IU 12 months up to 5 years

If the child shows any eye signs of vitamin A deficiency or is severely malnourished, a third dose must be given 2 - 4 weeks after the second dose to be given when the child comes for follow up.

11.1.3.4 Supportive care

- Manage fever: If the temperature is $\geq 39^{\circ}\text{C}$, give Paracetamol 10 – 15 mg/ Kg PO every 4 – 6 hours.
- **Nutritional support**
 - Assess the nutritional status
 - Encourage continued breastfeeding.
 - Encourage the child to take frequent small meals.

- Check for mouth ulcers and treat them if present

11.1.3.5 post-exposure prophylaxis

- Vaccine within 72 hours of exposure in those with no contraindication.
- Pregnant women, infants less than 6 months, immunocompromised IVIG within 6 days of exposure (0.25 mL/Kg IM in immunocompetent, 0.5 mL/Kg IM, max doses 15 mL/Kg) and vaccine within 5 - 6 months later if no contraindication.

11.1.4 Mumps

Acute generalized paramyxovirus infection usually presenting with unilateral or bilateral parotitis.

11.1.4.1 Diagnosis

Mainly clinical, serology IgM or acute rise in IgG (4-fold), RT-PCR of clinical specimens if available.

11.1.4.2 General management

- Supportive and symptomatic care, hydration, fever and pain control.
- Exclude from school for 5 days following onset of rash.
- Patients with orchitis, need surgical referral, ice packs to scrotum can help relieve pain.
- Scrotal support with adhesive bridge while recumbent and/or athletic supporter while ambulatory.
- Use IV fluids if severe nausea or vomiting accompanies pancreatitis.

11.1.4.3 Medication

- Ibuprofen 4 - 10 mg/Kg every 6 - 8 hours for pain and swelling in acute orchitis and arthritis mumps
- Paracetamol 10 - 15 mg/Kg every 4 - 6 hours for fever and/or pain.

11.2 FUNGAL INFECTIONS

11.2.1 Candidiasis

A mycosis usually confined to superficial layers of the skin (e.g. oral thrush or ulcerations, vulvovaginitis, erythematous intertriginous rash with satellite lesions, paronychia, onychomycosis).

Fungaemia related to intravascular devices.

Immunosuppressed individuals may present with systemic infections (renal, hepatic, splenic, pulmonary, or cerebral abscesses), cotton- wool retinal lesions and cutaneous nodules.

11.2.1.1 Diagnosis

Microscopic demonstration of budding yeast and pseudohyphae seen in biopsy specimens, fluid, or scrapings of lesions; positive culture

11.2.1.2 Management

Oral Candidiasis:

- First line: In infants, oral nystatin suspension 100,000 units 4-6 times a day in the buccal fold usually after feeding until resolution. Treat mother concurrently with topical Nystatin on nipples. Refer to oral candidiasis guidelines.

- Second line: Oral fluconazole 6 mg/Kg stat then 3 mg/Kg/day as a single dose for 7 - 14 days (neonates <14days 72 hourly < 14 days 48 hourly)
- Discontinuation of antibiotics or corticosteroids is advised when possible.

Cutaneous infection:

- First line: clotrimazole cream/ointment applied two times a day for 7 – 14 days
- Second line: miconazole cream/ointment applied two times a day for 7 – 14 days.
- Associated inflammation, such as severe diaper dermatitis, is also helped by concurrent use of a topical mild corticosteroid cream, such as 1% hydrocortisone.

Vaginal infection:

- Treat with clotrimazole, miconazole, triazoles, or nystatin suppositories or creams, applied once nightly for 3 - 7 days.
Avoid pessaries or intravaginal application in pre pubertal girls and those who are not sexually active.
- Oral azole therapy is equally effective

Systemic Infection

- First line: Fluconazole loading dose 12 mg/Kg followed by 6 mg/Kg IV daily (neonates <14 days 72 hourly < 14 days 48 hourly) for 7 - 14 days or continue according to response.
- In invasive fungal infection unresponsive to Fluconazole therapy, Voriconazole:
 - 2-14 years <50 Kg: 9 mg/kg/dose for 2 doses 12 hourly then 8 mg/Kg 12 hourly.
 - 14 years and above < 50kg: 6 mg/Kg 12 hourly for 2 doses then 4 mg/Kg 12 hourly IV for 7-14 days.

11.2.2 Cryptococcosis

Cryptococcus neoformans infection.

Cryptococcal meningitis is one of the most common AIDS-defining infections in HIV seropositive persons.

11.2.2.1 Diagnosis

- CSF demonstration of organism India ink positive capsular halo or budding forms and culture (CT scan before LP if increased ICP).
- Cryptococcal Antigen tests on serum and CSF.

11.2.2.2 Treatment

Cryptococcal meningitis:

- Induction: Amphotericin B deoxycholate 1 mg/Kg IV daily) or liposomal Amphotericin B 5 mg/kg/day plus Flucytosine (100 mg/Kg orally per day in four divided doses) for two weeks
- Consolidation: Fluconazole (10 to 12 mg/kg orally per day) for eight weeks.
- Maintenance: Fluconazole for 12 months at 6 mg/kg per day orally daily.

For elevated ICP, perform therapeutic lumbar punctures.

11.3 PROTOZOAL INFECTIONS,

11.3.1 Malaria

11.3.1.1 Uncomplicated Malaria (non-severe Malaria)

Patients will present with symptoms of malaria and a positive microbiological test (MPS or RDT) but with no features of severe malaria.

Fever is the cardinal symptom of malaria. It can be intermittent with or without periodicity or continuous. Many cases have chills and rigors. The fever is often accompanied by headache, myalgia, arthralgia, anorexia, nausea and vomiting.

Malaria should be suspected in patients residing in or have recently visited an endemic area and presenting with above symptoms. The diagnosis is confirmed by a high-quality blood slide.

Treatment

First line treatment: Treat as outpatient

Artemether/Lumefantrine:

Table 28: Dosage of Artemether-Lumefantrine (20mg/120mg)

5 - 14Kg	1 tablet two times a day for 3 days
15 - 24Kg	2 tablets two times a day for 3 days
25 – 34 Kg	3 tablets two times a day for 3 days
> 34Kg	4 tablets two times a day for 3 days

- Malaria patients with HIV/AIDS should be managed according to the same regimen above.
- In children below 5 Kg (under 2 months of age), malaria is not a common cause of fever. Evaluation of other causes

should be undertaken. Where malaria is diagnosed, the recommended treatment is oral quinine (WHO recommends ACT at same dose as for children weighing over 5 Kg).

Treatment failure

- Treatment failure can be defined as a failure to achieve the desired therapeutic response after initiation of treatment.
- Development of symptoms 14 days after initiation of therapy where there has been prior clearance of symptoms should be considered as a new infection and be treated with the first line drug.
- Treatment failures should be suspected if a patient deteriorates clinically at any time or symptoms persist 3 to 14 days after initiation of treatment.
- Malaria microscopy should be used to assess suspected treatment failures.
- Other potential differential diagnosis should be sought for and adequately managed.
- In cases of non-adherence with or non-completion of medications repeat a full course of the same medication is required.
- Confirmed cases of treatment failure should be treated with the second line medications.

Treatment of Non falciparum Malaria

- Where there is treatment failure, one should consider the possibility of non-falciparum malaria which may not clear due to the liver hypnozoites
- Good laboratory examination of the blood slides is required to specify the parasite type
- P.vivax and P. ovale requires Primaquine to clear the liver hypnozoites
- Primaquine causes abdominal discomfort when taken on

an empty stomach: it should always be taken with food.

- Primaquine may also cause haemolysis in patients with glucose-6-phosphatase dehydrogenase (G6PD) deficiency
- Primaquine dose of 0.25 - 0.5mg/Kg/day once a day for 14 days
- This may be used for up to 8 weeks to eradicate the liver hypnozoites to prevent relapse at a dose of 0.75 mg/Kg once a week with close medical supervision for potential primaquine-induced adverse haematological effects.

Second Line treatment

- Treat as severe malaria

11.3.1.2 Severe complicated malaria

This is characterized by Fever and **any one of**:

- AVPU less than A (either V, P or U)
- Unable to drink.
- Respiratory distress with severe anemia or acidotic breathing.
- Hypoglycaemia (glucose < 2.2mmol/L).

If a high-quality blood slide is negative, then only children with severe disease or those with severe anemia should get presumptive treatment.

Severe malaria is a medical emergency.

Other features may include:

- Prostration i.e., generalized weakness so that the patient is unable walk or sit up without assistance.
- Multiple convulsions – 2 or more episodes in 24hr.
- Circulatory collapse or shock, systolic blood pressure <50 mmHg in children.

- Clinical jaundice plus evidence of another vital organ dysfunction.
- Haemoglobinuria.
- Abnormal spontaneous bleeding.
- Radiological Pulmonary oedema

Laboratory findings

- Hypoglycaemia (blood glucose < 2.2mmol/L)
- Metabolic acidosis (plasma bicarbonate <15 mmol/l)
- Severe normocytic anemia (Hb <5 g/dL, packed cell volume <15%)
- Haemoglobinuria
- Hyperparasitaemia defined as > 2% or 100,000/ μ L in low intensity transmission areas OR > 5% or 250,000/ μ L in areas of high stable malaria transmission intensity
- Hyperlactataemia (lactate > 5mmol/L).
- Renal impairment (high serum creatinine) > 100 μ mol /L

Management

Severe malaria is a medical emergency. After rapid clinical assessment and confirmation of the diagnosis, full doses of parenteral antimalarial treatment should be started.

Artesunate

- In children $\leq$ 20 Kg give 3 mg/Kg/dose of injectable Artesunate (IV/IM) at time 0, 12 and 24 hours and continued once daily until oral administration is feasible or maximum of 7 days.
- If weight > 20 Kg give 2.4 mg/Kg/dose injectable Artesunate at time 0, 12 and 24 hours and continued once daily until oral administration is feasible or maximum of 7 days.

Quinine is an acceptable alternative if Artesunate is not available:

- Quinine 20 mg/Kg at the beginning (IV infusion **in 5% dextrose** to run over 4 hours or IM injection divided on 2 sites), then 10 mg/Kg 8 hours after the first dose, every 8 hours; infusion rate should not exceed 5 mg/Kg/hr.
- Omit loading dose if patient has received quinine in the previous 24 hours or mefloquine in the previous 7 days.
- The intramuscular route can be used as an alternative where intravenous route is not feasible. In this case the Quinine should be diluted to 1:5 ratio.
- Monitor for side effects including hypoglycaemia. In hepatic insufficiency, reduce dose by 25%.

Artemether

- 3.2 mg/Kg IM given at the start, then 1.6 mg/Kg daily
- Artemether should only be used if quinine and artesunate are not available due to its erratic absorption.
- Use only the intramuscular route for artemether administration.

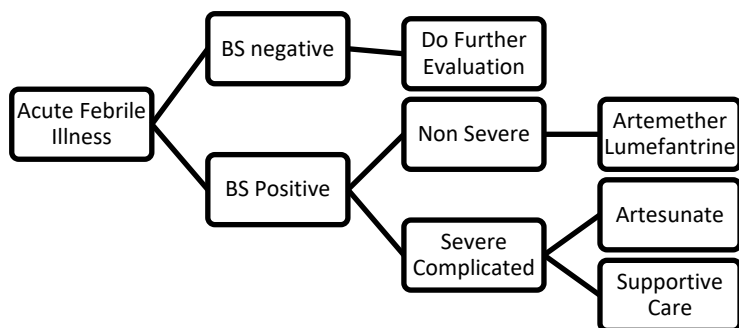
Give parenteral antimalarials in the treatment of severe malaria for a minimum of 24 hours, once started (irrespective of the patient's ability to tolerate oral medication earlier), and, thereafter, complete treatment by giving a complete course of Artemether plus Lumefantrine

Manage any complications appropriately.

Follow up

- Monitor haemoglobin levels and give haematinics
- Monitor and rehabilitate patients with neurological sequelae
- Threshold for antibiotic use should be low in severe malaria due to the association with septicaemia, give broad spectrum antibiotics in areas of moderate and high transmission until infection excluded

Diagram 8: Diagnosis and treatment of Malaria



11.3.1.3 Malaria prophylaxis

Atovaquone-Proguanil: is administered as a daily dose commencing 1 day before departure to a malaria endemic area, throughout the stay and continuing 7 days after departure from malaria endemic area.

Table 29: Atovaquone-Proguanil dosing

WEIGHT (Kg)	TABLETS	STRENGTH
<11	-	Not recommended
11 – 20	1	62.5 mg A + 25 mg P
21 – 30	2	125 mg A + 50 mg P
31 – 40	3	187.5 mg A + 75 mg P

The drug should be taken with food or milk at the same time each day.

Proguanil: is administered at a daily dose of 3 mg/Kg beginning two days before travel, continuing daily throughout and then continued for 4 weeks after departure from malaria endemic area.

Table 30: Proguanil dosing

WEIGHT	AGE	No. OF TABLETS
5 – 8 Kg	< 8 Months	¼
9 – 16 Kg	8 months – 3 year	½
17 – 24 Kg	4 – 7yrs	¾
25 – 35 Kg	8 – 10yr	1
36 – 50 Kg	11 – 13yr	1 1/2
50 + Kg	14+ yr	2

Mefloquine: is administered as a weekly dose of 5 mg/Kg body weight for children below 36 Kg and 250 mg for children above 36 Kg. Mefloquine prophylaxis is started 2 – 3 weeks before departure, throughout the stay and continued for 4 weeks after departure.

Table 31: Mefloquine dosing using 250 mg tablet

WEIGHT (Kg)	AGE	No. OF 250 mg TABLETS
< 5	<3 months	Not recommended
5 – 12	3 – 23 months	¼
13 – 24	2 – 7 yr	½
25 – 35	8 – 10 yr	¾
36 and above	11 yr and above	1

Doxycycline: should not be used in children under 8 years of age.

Table 32: Doxycycline dosing

Weight in Kg	No. of tablets
<25	Contraindicated

25 – 35	½ a tablet
36 – 50	¾ a tablet
< 50	1 tablet

11.3.2 Amoebiasis

Infection by the protozoon *Entamoeba histolytica* characterized by:

- Acute dysentery: diarrhoea with blood and mucus, abdominal pain, tenesmus.
- Chronic nondysenteric diarrhea.
- Hepatic abscess.

11.3.2.1 Laboratory detection

- Amoebas or cysts in stool (3 specimens over 3 days) or abscesses.
- Amoebic antigen in stool.
- Serologic evidence of amoebic infection.

Can present as

- Intestinal Amoebiasis
- Extraintestinal Amoebiasis

11.3.2.2 Management

- Oral Metronidazole 7.5 mg/Kg every 8 hours for 7 - 10 days combined with Diloxanide furoate 20 mg/kg per day in three divided doses maximum 500 mg per dose for 10 days only if stool microscopy reveals trophozoites of *Entamoeba histolytica*. Diloxanide not recommended for patients weighing <25 Kg.

11.3.3 Giardiasis

Infection by the protozoa *Giardia lamblia* characterized by:

- Chronic relapsing diarrhoea, flatulence, bloating, anorexia, poor weight gain.
- Acute diarrhoea that is self-limiting.
- Absence of fever and haematochezia.
- Detection of trophozoites, cysts, or Giardia antigens in stool.

11.3.3.1 Treatment

If cysts or trophozoites of *Giardia lamblia* are seen in the faeces,

- Tinidazole single dose
 - Preferred for < 3 years of age
 - 1 month – 11 years 50-75 mg/kg single dose
 - 12-17 years 2g single dose, may be repeated once if necessary
- Oral Metronidazole for 3 days (preferred in <1 year)
 - 1 – 2yr: 500 mg once daily
 - 3– 6yr: 600-800 mg once daily
 - 7 – 9yr: 1g once daily
 - 10-17yr: 2g once daily
- Oral Nitazoxanide for 3 days (approved in age < 1year)
 - 1 – 4yr 100 mg 12hourly
 - 4 – 11yr 200 mg 12hourly
 - Above 12yr 500 mg 12hourly

11.3.4 Trichomoniasis

11.3.4.1 Diagnosis

Microscopy presence of motile trichomonads on wet mounts

11.3.4.2 Treatment

- Metronidazole
 - Child 1–2 years: 50 mg 3 times a day for 7 days
 - Child 3–6 years: 100 mg twice daily for 7 days
 - Child 7–9 years: 100 mg 3 times a day for 7 days
 - Child 10–17 years: 200 mg 3 times a day for 7 days
- Alternative Tinidazole
 - Child 1 month–11 years: 50–75 mg/kg (max. per dose 2 g) for 1 single dose
 - Child 12–17 years: 2 g for 1 single dose, dose may be repeated once if necessary

11.3.5 Leishmaniasis

An infective condition caused by several species of the protozoan *Leishmania*. Can present as either:

- Visceral Leishmaniasis
- Cutaneous Leishmaniasis
- Mucocutaneous Leishmaniasis
- Diffuse cutaneous Leishmaniasis

11.3.5.1 Treatment

First line:

Visceral leishmaniasis

- Sodium antimony gluconate 20 mg/Kg IM or IV once daily maximum 850 mg for 28 days to 30 days.
- Monitor electrocardiogram (EKG), complete blood count, creatinine, transaminase, lipase, and amylase levels should be performed weekly.

Second line

- Amphotericin B IV: 0.5-1 mg/Kg on alternate days.
- Child: 1 - 3 mg/kg daily for 10–21days to a cumulative dose of 21 - 30 mg/kg, alternatively 3 mg/kg for 5 consecutive days, followed by 3 mg/kg after 6days for 1 dose.
- Pentamidine
 - Visceral leishmaniasis: Child 1 - 17years: 3 - 4 mg/kg once daily on alternate days, maximum total of 10 injections, course maybe repeated if necessary).
 - Cutaneous leishmaniasis (specialist use only) BY DEEP INTRAMUSCULAR INJECTION Child 1–17years: 3 - 4 mg/kg 1 - 2 times a week until condition resolves.

11.3.6 Cryptosporidiosis

Infection by the protozoon *Cryptosporidium parvum*.

Its characteristics depends on the immune competence of the patient:

Immunocompetent persons:

- Self-limited diarrhoea with or without abdominal cramps
- Low grade fever, nausea, vomiting, loss of appetite, and malaise

Immunocompromised patients

- Prolonged disease
- Cholecystitis, pancreatitis, hepatitis, and respiratory symptoms.
- Subsides only after the immunodeficiency is corrected

11.3.6.1 Diagnosis

Detection of oocysts in faecal smears or parasite life cycle stages in intestinal biopsy specimens

11.3.6.2 Management

Treatment and Prevention

- Exclude infected from work or day care/school environments until asymptomatic.

Immunocompetent patients:

- Antidiarrheal agents and hydration.

Immunocompromised patients:

- parenteral nutrition in addition to hydration and nonspecific antidiarrheal agents.

Medications:

- Oral Nitazoxanide for 3 days
 - 1 - 4 yr 100 mg 12hourly
 - 4 - 11 yr 200 mg 12hourly
 - Above 12 yr 500 mg 12hourly
- For patients with AIDS, institution of effective antiretroviral therapy eliminates symptomatic cryptosporidiosis

11.4 NEMATODE INFECTIONS

11.4.1 Enterobiasis (Pinworms)

Helminth infection by pinworms (*Enterobius vermicularis*), often characterized by anal pruritus.

11.4.1.1 Diagnosis

- Tape swab in the morning done 3 times (if negative) on the perianal area demonstrates or eggs on perianal skin.

- Worms or eggs in the stool.

11.4.1.2 Management

- Give treatment to all household members at the same time to prevent re-infections.
- Therapy should be repeated after 2 weeks to kill the recently hatched adults.

First line therapy

- Pyrantel Pamoate as a single dose of 11 mg/Kg, maximum 1g.

Second line therapy

- Mebendazole 100 mg PO Stat
- Albendazole 400 mg PO Stat

11.4.1.3 General Measures

Personal hygiene must be emphasized. Nails should be kept short and clean. Children should wear undergarments to bed to diminish contamination of fingers. Bedclothes should be laundered frequently.

11.4.2 Ascariasis

A helminthic infection by the worm *Ascaris lumbricoides* (roundworms), often characterized by abdominal cramps and discomfort.

11.4.2.1 Diagnosis

Large, white or reddish, round worms or ova in the faeces.

11.4.2.2 Management

- Albendazole (400 mg in a single dose)

- Mebendazole (500 mg single dose or 100 mg twice daily for 3 days),
- Pyrantel Pamoate (a single dose of 11 mg/Kg, maximum 1g)- (for pregnant patients)

11.4.3 Trichuriasis (Whipworm)

A helminthic infection by the whipworm (*Trichuris trichura*), characterized by pain, diarrhoea, and mild abdominal distention.

Symptoms are not present unless the infection is severe.

Massive infections may also cause rectal prolapse and dysentery.

11.4.3.1 Diagnosis

- Eggs in stool.
- Proctoscopy or sigmoidoscopy may show adult worms.

11.4.3.2 Management

- Mebendazole (100 mg orally twice daily for 3 days) or
- Albendazole (400 mg once daily for 3 days- second line)

11.4.4 Hookworm

Helminthic hookworm infection often associated with:

- Iron deficiency anemia
- Abdominal discomfort, weight loss

11.4.4.1 Diagnosis

- Microscopic demonstration of ova or larvae in the faeces or adult worms.

11.4.4.2 Management

- Mebendazole (100 mg orally twice daily for 3 days) or
- Pyrantel Pamoate (11 mg/Kg, maximum 1g, daily for 3 days) are the medications of choice
- Albendazole 400 mg stat dose and is also useful for creeping eruption
- Follow up stool exam after 2 weeks and repeat treatment if heavy worm burden
- Iron supplementation as needed

11.4.5 Strongyloidiasis

Helminthic infection by the worm *Strongyloides stercoralis*.

11.4.5.1 Diagnosis

- Microscopic identification of motile larvae in freshly passed stool, cultures or duodenal aspirates.
- May require repeated examinations.

11.4.5.2 Management

First line

- Ivermectin (200 mcg/Kg/d for 2 days).

Alternative (lower efficacy)

- Albendazole 400 mg bd for 3 - 7 days.

In the hyperinfection syndrome, 5 - 7 days of therapy may be necessary with Ivermectin or until clinical response and daily stool exam negative for 2 weeks.

11.4.6 Trichinosis

Infection by *Trichinella spiralis* initially characterised by muscle soreness and pain with periorbital or facial oedema with fever.

This is followed by subconjunctival, subungual, retinal haemorrhages, pain, photophobia and thirst. Profuse sweating, chills, weakness prostration and rising eosinophilia also occur.

Cardiac and neurological complications may occur in the 3rd to 4th week.

11.4.6.1 Diagnosis

- Serology
- Haemogram –marked eosinophilia
- Biopsy of muscle taken < 10 days after infection demonstrate parasite cyst.

11.4.6.2 Management

Clinical course for most *Trichinella* infections is uncomplicated and self-limited. For mild infection, antiparasitic therapy is not required; symptomatic treatment with analgesia and antipyretics is appropriate.

- Mebendazole at a dose of 200-400 mg three times daily for 3 days followed by 400 – 500 mg three times daily for 10 days
- Albendazole 400 mg orally with fatty meal twice daily for 10 to 14 days

In general, treatment of children under 2 years with above medications not advised.

- Concurrent corticosteroids to prevent the Jarisch-Herxheimer reaction associated with treatment.

11.4.7 Taeniasis

Taeniasis is caused by infection by either the beef tapeworm (*Taenia saginata*) or the pork tapeworm (*Taenia solium*).

11.4.7.1 Diagnosis

Demonstration of eggs in faeces or adult worm.

11.4.7.2 Management

Praziquantel (5-10 mg/Kg once) is the drug of choice.

Stools should be rechecked for eggs at three months (taeniasis) to document cure.

11.4.8 Cysticercosis

Cysticercosis is a tissue infection caused by larval cysts of the tapeworm *Taenia solium*. It may infect brain, muscle or other tissues.

Specific treatment is reserved for patients with meningitis or parenchymal cysts.

11.4.8.1 Treatment

The goal of treatment in neurocysticercosis is to control seizures, intracranial pressure, edema or hydrocephalus.

Those with inactive disease require only symptomatic treatment.

Antiparasitic treatment is warranted in those with viable or degenerating cysts on neuroimaging.

- 1 - 2 cysts: albendazole (15 mg/Kg divided into two doses daily for 10 - 14 days) cause disappearance of cysts.
- If more than two cysts: albendazole as above in combination with praziquantel 50 mg/kg/day in two divided doses.
- Not given to those with untreated hydrocephalus, high cyst burden, calcified cysts.
- A short course of Dexamethasone 0.15 mg/Kg/6hours to manage inflammatory oedema due to larval death.

11.4.9 Echinococcosis (Hydatid disease)

Infection due to *Echinococcus granulosus*. The infection may remain silent for years; symptoms occur due to enlargement of the cysts and will depend on the affected organs.

Rupture of the cysts should be avoided as it may lead to anaphylactic shock in addition to cyst dissemination.

11.4.9.1 Management

- PAIR (percutaneous aspiration, injection of chemicals and reaspiration) for certain cysts.
- Definitive therapy of *E. multilocularis* requires meticulous surgical removal of the cysts.
- This is preceded by careful injection of the cyst with formalin, iodine, or 95% alcohol solution to sterilize infectious protoscoleces.
- Freezing the cyst wall and injecting silver nitrate prior to removal is another technique.

11.4.9.2 Medication

- Albendazole (15 mg/Kg/d bd for 28 days with repeat courses as necessary following a 14-day rest period) 1-3 months up to 6 may be required.
- Adjunctive treatment with surgery: start medications 4 days prior to surgery (the WHO suggests 4 to 30 days preoperatively) and should be continued for at least 1 month (albendazole) or 3 months (mebendazole- 40-50 mg/kg/day 3 divided doses) following surgery.
- If the cyst leaks or ruptures, the allergic symptoms must be managed immediately and praziquantel should be used to prevent secondary cysts.

11.5 TREMATODE INFECTIONS

11.5.1 Schistosomiasis

Schistosomiasis is caused by several species of *Schistosoma* flukes.

11.5.1.1 Management

- Katayama syndrome (acute schistosomiasis): treatment with prednisolone 20 to 40 mg daily then praziquantel for 4-6 weeks following exposure.
- Praziquantel at a dosage of 20 mg/Kg three times a day for 1 day
- A repeat dose of praziquantel given two weeks after the initial dose may improve cure rate and improve the intensity of *S. mansoni* infection (not standard practice).
- Alternative Oxamniquine (20 mg/Kg/day in two doses once per day) is an alternative regimen for treatment of *S. mansoni* infection.

11.6 BACTERIAL INFECTIONS

See elsewhere in the text under systemic conditions

12. NEONATAL CONDITIONS

12.1 NEONATAL SEPSIS

12.1.1 Description:

Clinical syndrome in infant ≤ 90 days. Systemic signs of infection/isolation of bacterial pathogen from bloodstream.

Early-onset sepsis: before 7 days

Late-onset sepsis: ≥ 7 days.

Early onset: usually vertical transmission.

Late onset: Vertical or horizontal (from caregivers/environment)

Etiology: GBS, E. coli most frequent causes, other gram-negative infections, Coagulase Negative Staphylococci (CONS) in hospital-associated infections.

12.1.2 Clinical Manifestations:

- Subtle to profound septic shock
- Nonspecific: Temperature instability (usually fever)
- Respiratory symptoms (tachypnoea, grunting, hypoxia)
- Lethargy
- Poor feeding
- Metabolic signs: hypo/hyperglycemia, metabolic acidosis, Jaundice
- Neurologic signs: Irritability, seizures, bulging Anterior Fontanelle
- Cardiac signs: tachycardia, bradycardia, pallor, prolonged CRT, hypotension- late signs of shock indicate higher chances of mortality

12.1.3 Diagnosis:

All symptomatic infants should undergo full diagnostic evaluation and receive empiric antibiotic treatment

Well- appearing infants with identified risk factors (maternal GBS, maternal PROM $\geq$ 18 hours, maternal chorioamnionitis, maternal UTI) should be observed 48 hours minimum.

12.1.4 Investigations for Early-Onset Sepsis

- Blood culture
- Lumbar Puncture (if clinically stable)
- haemogram
- C-Reactive Proteins/ Procalcitonin
- Chest x-ray (if respiratory symptoms present)

12.1.5 Late-onset Sepsis:

In addition to above tests also do urine culture (obtained using catheterization or suprapubic tap - do NOT use collectors). Evaluate any other potential foci e.g., pustules, eye discharge.

12.1.6 Treatment:

12.1.6.1 Antibiotic therapy

Because sepsis may manifest with nonspecific clinical signs and its effects may be devastating, rapid empiric antibiotic therapy is recommended. Drugs are later adjusted according to sensitivities and the site of infection.

Generally, if no source of infection is identified clinically, the infant appears well, and cultures are negative, antibiotics can be stopped after 48 h (up to 72 h in small preterm infants).

Early Onset Sepsis

IV Crystalline penicillin plus aminoglycoside. Cefotaxime may be added to or substituted for aminoglycoside in suspected meningitis.

Late Onset Sepsis

IV Crystalline penicillin plus aminoglycoside, or crystalline penicillin and cefotaxime

In suspected meningitis, cefotaxime and aminoglycoside.

Monitor drug levels 48 hour after starting aminoglycoside (potential nephron and ototoxicity) and audiology screen before discharge.

12.1.7 Supportive therapy

General supportive measures, including respiratory and hemodynamic management, nutritional support (parenteral or enteral) and thermoregulatory support (radiant warmer/incubator) for infants with temperature instability are combined with antibiotic treatment.

12.2 NEONATAL JAUNDICE

All babies develop elevated serum bilirubin (SBR) levels to a greater or lesser degree in the first week of life. Due to an increased production (accelerated red cell breakdown), decreased removal (transient liver enzyme insufficiency) and increased resorption (enterohepatic circulation).

12.2.1 Classification of neonatal jaundice

Classify jaundice according to the age of the baby when becomes visibly jaundiced. Jaundice starts on the head and extends towards the feet as the level rises.

12.2.1.1 *Early jaundice (Day 1-2)*

ALWAYS pathological and usually due to the following:

Haemolysis

- Rhesus isoimmunization - ABO and other blood group incompatibilities
- Rarer causes of haemolysis: RBC membrane defects, RBC enzyme defects
- Sepsis
- Hepatitis

12.2.1.2 *Normal jaundice (Day 3 - 10)*

Very common occurrence and can be divided into:

- Physiological (uncomplicated)
- Physiological (complicated)

12.2.1.3 *Late jaundice (>Day 14):*

- Breast milk (common occurrence)
- Conjugated jaundice (uncommon)
- Inherited deficiency of glucuronyl transferase enzymes (very rare)

12.2.2 Diagnosis

Detailed history highlighting the following points:

- Maternal blood group
- Baby's blood group if available
- Feeding history
- Colour of stools/urine
- Neonatal vitamin K administration
- Onset of neonatal jaundice
- Antenatal history
- Any risks for antenatally acquired infections
- Family history (any history of severe jaundice)

Any baby having jaundice at less than 24 hours of age should be investigated promptly.

12.2.3 Investigations

- Blood Group and DAT (Direct Coomb's test)
- Full blood count (FBC) and Blood Film
- Reticulocytes Count
- Liver function test (LFT) with split serum bilirubin
- Urea and electrolytes
- Investigations for sepsis in unwell babies if clinically indicated (microbiological cultures of blood, urine and cerebrospinal fluid (CSF))

12.2.4 Treatment

Treatment where indicated is determined by bilirubin levels.

- Phototherapy
- Exchange transfusion

Table 33: Nice Guidelines for the Management of Neonatal Hyperbilirubinemia

Estimated Gestational Age						
Age in hours		28wk	30wk	32wk	34wk	36wk
		All values in micromole per litre				
Start Phototherapy	12	Any value above normal range				
	24	80	90	100	110	110
	36	110	120	130	140	150
	48	140	150	160	170	180
	60	160	170	190	200	220
	72+	180	200	220	240	260
Exchange Transfusion	12	120	120	120	120	120
	24	150	150	160	160	170
	36	180	180	200	210	220
	48	210	220	240	250	260
	60	240	260	280	290	310
	72+	280	300	320	340	360

Follow up mild to case in the OPD with monitoring of bilirubin levels

12.2.5 Prolonged Jaundice

Prolonged jaundice is defined as jaundice lasting over 14 days in a term neonate and over 21 days in a preterm neonate. Prolonged jaundice can be due to unconjugated or conjugated hyperbilirubinaemia. This must be investigated as some causes may need further treatment.

The following steps must be taken where prolonged jaundice occurs:

- a full history and clinical examination should be performed
- weight should be plotted on an appropriate weight centile chart
- stool and urine should be examined: pale, chalky stools and dark urine- suggests cholestasis
- routine metabolic screening must be performed including screening of congenital hypothyroidism
- serum bilirubin should be taken to establish whether the bilirubin is conjugated or unconjugated. Conjugated bilirubinaemia is defined as a conjugated bilirubin $>17\text{mmol/L}$ if the total bilirubin is $<85\text{mmol/L}$ or more than 20% of the total bilirubin if the total is $>85\text{mmol/L}$.
- If the bilirubin is conjugated, consultation with a specialist should take place to determine further investigation and management
- If the bilirubin is unconjugated a full blood count, liver function tests, thyroid function test and urine culture should be taken.
- Most commonly the diagnosis is unconjugated hyperbilirubinaemia due to breastmilk jaundice, but this is a diagnosis of exclusion.

12.3 OPHTHALMIA NEONATORUM

Eye most frequent site of gonococcal infection in newborn.

12.3.1 Clinical features

Purulent conjunctivitis with profuse exudate and swelling of eyelids. Without treatment, infection can extend into cornea leading to ulceration, scarring, visual impairment.

Usually manifests between 2 - 5 days after birth.

12.3.2 Diagnosis

A gram stain of conjunctival exudate: gram-negative kidney bean-shaped diplococcic. Culture using selective media.

Should also be evaluated for *Chlamydia trachomatis*. Co-infection is common.

Mother and partner should be evaluated and treated for gonococcal infection, other STIs, including HIV.

12.3.3 Treatment

- Hospitalize patient to observe response to therapy and for disseminated disease.
- Presumptive treatment after obtaining samples for diagnosis.: Single dose of IV Ceftriaxone: 25 - 50 mg/kg IM OR IV
- For infants with hyperbilirubinemia a single dose of IV/IM Cefotaxime: 100 mg/kg.
- Irrigate eyes frequently with saline until discharge clears.
- Asymptomatic infants of untreated mothers: Treat as above
- Neonatal prophylaxis: up to one hour after birth. Erythromycin or Tetracycline eye ointments.

12.4 INFANTILE COLIC

No standard definition, little consensus on how much crying is excessive.

For clinical purposes: broadly defined as crying for no apparent reason for ≥ 3 hours /day, ≥ 3 days per week, in otherwise healthy infant < 3 months of age.

12.4.1 Clinical features

Paroxysms: cry/fuss behavior- clear beginning and end, cluster often in evening hours.

Difficulty consoling

Qualitative differences- often described as more urgent, piercing, distressing: “infant in pain”

12.4.2 Evaluation

- History and examination for identifiable causes of crying/fussiness. Thoroughness of these is reassuring to parents, may strengthen clinician-caregiver bond.
- History: infant's feeding, stool and urine voiding patterns, sleeping patterns. Ask about regurgitation and vomiting.
- Prenatal and perinatal history including risk factors for sepsis.
- Psychosocial history: parent-infant interaction, influence of extended family in parenting style, soothing techniques.
- Specific questions: timing, association with feeds (air swallowing/gastroesophageal reflux: may respond to changes in feeding technique). How and what do you feed baby?

- What do you do when baby cries? Information on soothing techniques; helpful vs unhelpful, possibly harmful.
- How do you feel when baby cries? How has colic affected family?
- Examination: infant-parent interaction during crying helpful, assessment of baby's temperament: sensitivity and responsive to stimuli: does baby cry or quieten down?
- Full physical examination: Other things to look for: tongue-tie, oral thrush, ear exam for otitis media, abdominal exam: tenderness and bowel sounds, perineum for diaper rash, anal fissure, inguinal hernia.

12.4.3 Management

- Improves spontaneously with time
- Parental support is the mainstay of management. Reassurance that baby is not sick.

12.4.3.1 First-line

- Address feeding problems (under-feeding, over-feeding, inadequate burping).
- Soothing techniques, decreasing environmental stimuli. Encourage parents to experiment- see what works: using a pacifier, taking baby for a ride in car/stroller, rocking baby, rubbing abdomen, hip-healthy swaddling (room for free movement of legs), providing 'white noise.'

12.4.3.2 Second- line

- Dietary changes: time-limited trial may be warranted in babies who don't respond to first-line interventions.

- Breastfed infants- decrease in maternal milk product consumption, hypoallergenic diet (wheat, eggs). A subgroup of infants may have cow's milk allergy.
- Formula fed infants- trial of extensive hydrolysate infant formula: usually respond in 48 hrs. Discontinue if no change.
- Probiotics: *Lactobacillus reuteri* in breastfed babies may be an option. Discuss risk versus benefits with parents.
- Of unproven benefit: simethicone, herbal remedies including gripe water, homeopathic remedies.

12.5 NEONATAL FEEDING

Positive fluid balance in the first few postnatal days is unnecessary and is associated with more severe RDS and higher risk of PDA, Congestive Heart Failure, Pulmonary Oedema, Necrotising Enterocolitis and Chronic Lung Disease (BPD). Term newborns have a urine output of 1 mL/Kg/hour for the first few days of life

Feeding should start at 60 – 100 mL/Kg/day depending on the maturity and respiratory status. This should increase gradually and as the baby's fluid input and output are monitored. As a guideline, use the table below:

Table 34: Nasogastric 3-hourly feed volumes for babies on full volume feeds

Weight (Kg)	1.2	1.4	1.6	1.8	2.0	2.2	2.4	2.6	2.8	3.0	3.2	3.4	3.6	3.8	4.0
Day 1	3	4	4	5	5	6	6	7	7	8	8	9	9	10	10
Day 2	4	5	5	6	7	7	8	9	9	10	11	11	12	13	13
Day 3	5	6	7	8	8	9	10	11	12	13	13	14	15	16	17

Day 4	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Day 5	7	8	9	11	12	13	14	15	16	18	19	20	21	22	23
Day 6	8	9	11	12	13	15	16	17	19	20	21	23	24	25	27
Day 7	9	11	12	14	15	17	18	20	21	23	24	26	27	29	30

When the baby is not feeding, the same fluid volume is calculated and given as an IV infusion proving dextrose at a minimum of 6-8 mg/Kg/min and only add electrolytes from day 2 if passing urine

$$\text{Volume of feeds} = \frac{(40 + 20 \times \text{Age in days})}{8}$$

up to 8 days only, thereafter a maximum of 180 – 200 mL/Kg/day

The very low birthweight infants can be fed 2 hourly; in this case the formula will be:

$$\text{Volume of feeds} = \frac{(40 + 20 \times \text{Age in days})}{12}$$

up to 8 days only, thereafter a maximum of 180 – 200 mL/Kg/day

13. EYE DISORDERS

13.1 ACUTE EYE INJURIES

The following traumatic conditions threaten vision:

- Ruptured globe
- Foreign body either intraocular or deep corneal
- Large hyphaemas (causing acute glaucoma)
- Retinal detachment
- Corneal burns, either chemical or thermal - alkalis penetrate deeper and have greater potential for serious and delayed burns
- Contact lens - related corneal infections (bacterial keratinise)

13.1.1 Management

If an adequate examination is not possible due either to the child's age or cooperation level, specialist assistance should be sought. Adequate analgesia will aid assessment. Topical anaesthetic such as amethocaine 1% (one drop) may be used once in the emergency department (local anaesthetic causes direct epithelial toxicity and should not be used repeatedly)

13.1.2 Corneal abrasions/Foreign bodies

- If very large or deep defect seen, presume full thickness
- Exclude foreign body, including under eyelid
- Avoid removing large, deep or central corneal foreign bodies, refer these to the Ophthalmologist

- Gently use moistened cotton bud for small superficial foreign body. A small gauge needle may be used in older, cooperative children, using slit lamp with approach to the patient from the temporal aspect of the eye
- Chloramphenicol ointment/drops or Tobramycin ointment/drops
- Pad the eye for four hours (to prevent accidental further eye injury due to anaesthetic effect). Occasionally, prolonged eye pad may help ease pain, but be aware not to apply too tightly as this can impair epithelial healing
- Cycloplegic eye drops (Cyclopentolate 1% or Homatropine 2%) can be used for relief of a very painful eye
- Patients need daily review until corneal ulceration healed
- If not able to examine the child due to age or uncooperative child refer to ophthalmologist.

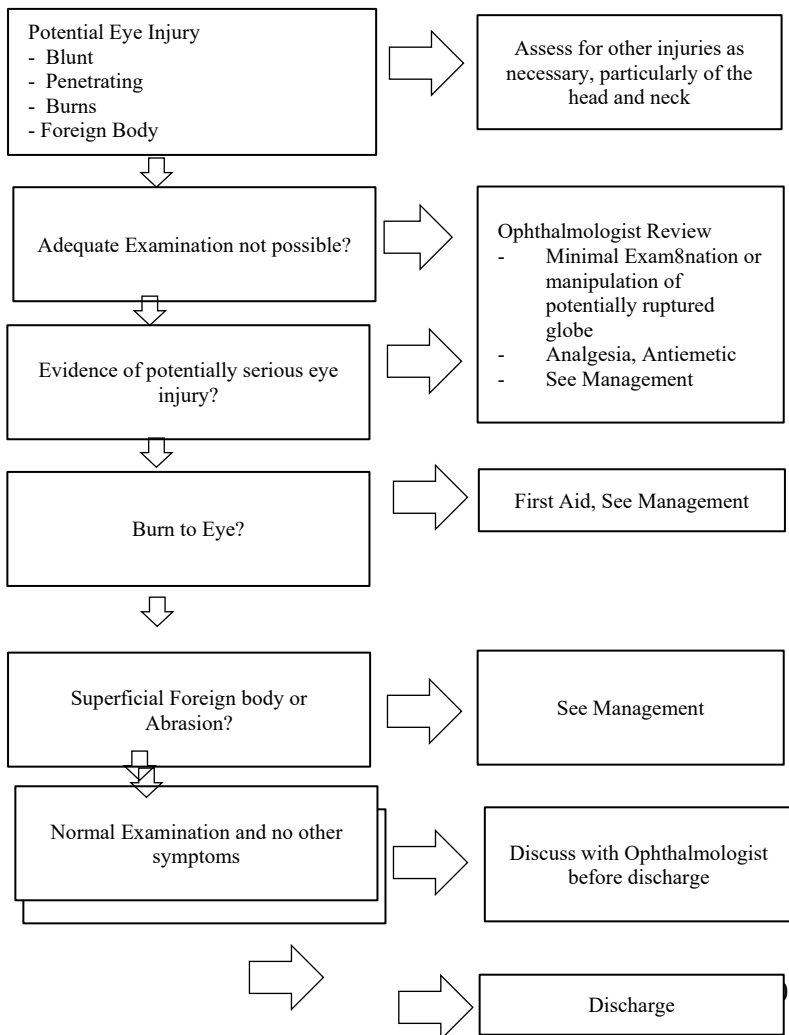
13.1.3 Penetrating eye injuries

When a penetrating eye injury is suspected:

- **Do not force eyelids open.** Pressure on the lids may cause extrusion of ocular contents.
- **Do not attempt to remove a protruding foreign body from the globe.**
- Fast the patient from the time they are seen.
- Use appropriate analgesia. Consider NSAIDs. If opiates are required, consider concurrent antiemetic as vomiting increases intraocular pressure and may cause expulsion of ocular contents. Use ondansetron rather than agents which may precipitate dystonic reactions.
- Notify ophthalmology for all suspected penetrating eye injuries.

- After discussion with ophthalmology, image the orbit (x-ray or CT scan) in cases where an intra-ocular foreign body is suspected.

Diagram 9: Flow Chart of Acute Eye injuries in Children



13.1.4 Ocular burns

Discuss all eye burns with the Ophthalmologist

13.1.4.1 Chemical (Strong Acid/ Alkali) Burns

- Extensive immediate irrigation: Urgent copious irrigation after local anaesthetic drops including under the upper eyelid is to be done. Use sterile high osmotic solutions like Ringers Lactate through a fluid giving set over 15 – 30 minutes or until the PH normalises (PH 6 – 8)
- Sedation or urgent GA may be required. Particulate alkaline matter needs urgent, total removal
- Immediate treatment is required to reduce pain, inflammation and risk of infection

13.1.4.2 Management

- Systemic Analgesics
- Course of steroids drops 2 – 4 hourly for 10 days
- Cycloplegic – Cyclopentolate eye drops or Atropine eye drops
- Prophylactic topical antibiotic – Ofloxacin eye drops 4hourly for 1week or Tetracycline eye ointment to be applied 8hourly

13.1.4.3 Thermal Burns

If corneal damage present, manage as for other corneal abrasions

13.1.5 Contact Lens Associated Red Eye

- Anaesthetise eye and remove contact lenses if possible
- Stain eye and check for corneal abrasions or foreign body
- Swab if discharge present- Gram-negative bacteria including *P. aeruginosa* are frequently the cause, thus requiring broader-spectrum antibiotics like Ofloxacin
- If ulcer identified refer urgently to ophthalmologist
- Start topical antibiotics and arrange for follow up

13.2 THE ACUTE RED EYE

- Common causes of a red eye include conjunctivitis (viral, bacterial, allergic or chemical), foreign body, corneal ulceration and subconjunctival haemorrhage
- Uncommon causes include iritis, scleritis, episcleritis and glaucoma
- Discharging non-red eye in infants is most likely due to nasolacrimal duct obstruction

13.2.1 Conjunctivitis (Non-Neonatal)

- May be difficult to clinically differentiate bacterial, viral and allergic conjunctivitis
- Suspect bacterial conjunctivitis if discharge is purulent. Treat with eye toilet with warm clean water every 2 – 4 hours and topical Chloramphenicol

- Suspect herpes simplex infection if there are lid vesicles. If a dendritic ulcer is present treat with acyclovir eye ointment and contact Ophthalmologist
- Suspect allergic conjunctivitis if bilateral watery discharge with burning or foreign body sensation and eyelid swelling, itching especially in an atopic child. Consider treating with mast cell stabilisers, antihistamines (oral or topical) and artificial tears
- **DO NOT PRESCRIBE STEROID EYE DROPS UNLESS ADVISED BY AN OPHTHALMOLOGIST**

13.2.2 Conjunctivitis (Neonatal)

- Pathogens include staphylococcus, haemophilus, Chlamydia, streptococcus, gonococcus and herpes simplex
- Obtain conjunctival scrapings for gram stain, Giemsa stain and cultures
- Use Chlamydia kit for immunofluorescence and treat Chlamydia conjunctivitis with eye toilet with warm clean water and oral erythromycin for 3 days
- Consider Gonococcal conjunctivitis if severe purulent discharge with conjunctival and lid oedema. Perform an urgent gram stain and contact Ophthalmologist. May need septic work up and systemic Cefotaxime or Ceftriaxone
- Treat other organisms with topical Chloramphenicol

13.2.3 Foreign Body

- Anaesthetise conjunctiva with Amethocaine 1%
- Examine surface of eye and under lids, using slit lamp if possible
- Stain with fluroscein to assess corneal injury

- Remove foreign body with moistened cotton bud, apply Chloramphenicol ointment and pad for four hours
- Need daily review until epithelium healed
- Refer embedded or metallic foreign bodies to ophthalmologist

13.2.4 Corneal Ulceration

- May be caused by trauma (including foreign body) or herpes simplex infection
- Visible after fluorescein staining
- If traumatic apply Chloramphenicol ointment and review in 24 hours
- If any ulcer associated with hypopyon or is very large, contact Ophthalmologist

13.2.5 Iritis

- Presents with pain, photophobia, blepharospasm and lacrimation
- Pupil may be small and poorly reactive
- May occur in association with juvenile chronic arthritis
- Consult the ophthalmologist

13.2.6 Scleritis and Episcleritis

- Present with localised areas of inflammation with tenderness and lacrimation
- Strongly associated with systemic disease
- Consult the ophthalmologist

13.2.7 Glaucoma

- Presents with an enlarged, hazy cornea, photophobia and lacrimation

- Contact ophthalmologist

13.2.8 Trauma

- Consider if hyphaema or focal conjunctival injection
- Consider penetrating injury
- Contact ophthalmologist

13.2.9 Sty (Hordeolum)

- Infection of the eyelash follicle (external hordeolum) or meibomian glands (internal hordeolum) at the eyelash root.
- Usually caused by staphylococcus aureas. Presents as an
- acute painful lid margin swelling
- Supportive treatment is by use of warm compresses
- Use antibiotic eye ointment or drops
(Tobramycin, Chloramphenicol 8 hourly for 1 week)

13.2.10 Eyelid Cellulitis

- This is infection of the eyelid secondary to trauma, insect bites, upper respiratory tract infections or spread from a sty. Presents with a tender periorbital swelling. The rest of the eye examination is normal
- Treatment is by use of systemic broad-spectrum antibiotics for 1 week
- Oral anti-inflammatory analgesics should be used for pain and swelling

13.3 CONGENITAL NASOLACRIMAL DUCT OBSTRUCTION

Usually caused by the persistence of a thin mucous membrane at the lower end of the nasolacrimal duct at distal valve of Hasner.

13.3.1 Symptoms

- Presents by the age of 1 month with tearing and sticky mucoid or mucopurulent discharge on the lid margins and eyelashes
- Digital pressure over the lachrymal sac may produce reflux of mucoid material.
- Affected infant is usually normal if any photophobia or corneal irritation suspect glaucoma or corneal disease.

13.3.2 Treatment

- Treatment is by use of Broad-spectrum antibiotic eye drops (Ofloxacin, Chloramphenicol) 6 hourly for weeks
- Digital massage with the pulp of the index fingers on the medial part (canthus) of the eye of the lachrymal sac – 10 strokes 6 hourly for 3 months
- If any swelling at the lachrymal sac, persistent tearing or discharge, refer to the Ophthalmologist

14. AEMATOLOGICAL CONDITIONS

14.1 ANEMIA

Anemia is defined as a decrease in red blood cell (RBC) mass, or Haemoglobin (Hb) less than the lower limit of the reference range for age. It is a symptom of disease that requires investigation to determine the underlying aetiology.

Table 35: Lower Normal Limit of Haemoglobin for Age

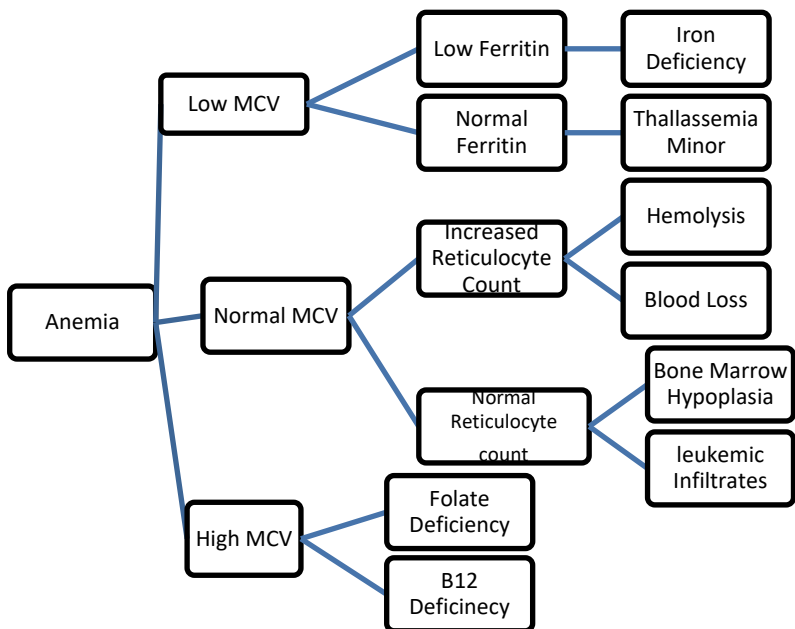
Age	Lower Limit of Normal Range g/dL
< 2 months	9.0
2 – ≤ 6 months	9.5
< 6 months – 2 years	10.5
< 2 years – 12 years	11.5
< 12 years	Girls – 12.0, Boys – 13.0

14.1.1 Causes

- Genetic; Haemoglobinopathies, Thalasseмии, Abetalipoproteinemia, Fanconi anemia, etc
- Nutritional e.g., Starvation and generalized malnutrition
- Haemorrhage
- Immunologic - Antibody-mediated abnormalities
- Physical effects e.g., Trauma, Burns, Prosthetic valves and surfaces
- Drugs and chemicals, Aplastic anemia, Megaloblastic anemia

- Chronic diseases and malignancies, e.g., Celiac disease, Chronic Renal Failure, Leukaemia
- Infections e.g., malaria, hookworm infestation,
- Thrombotic thrombocytopenic purpura and haemolytic uremic syndrome

Diagram 10: Types of Anemia



14.1.2 Iron Deficiency Anemia

14.1.2.1 Causes

- Blood loss, either from disease or injury
- Not getting enough iron in the diet

Not being able to absorb the iron in the diet

Note:

Iron deficiency that lasts more than 2 months while being treated should be considered chronic anemia and other explanations must be sought

14.1.2.2 Basic laboratory Tests

- Haematocrit and PBF for reticulocyte count
- Serum ferritin reveals the amount of iron stored in body
- Serum iron shows how much iron is in blood
- Total iron binding capacity (TIBC)

14.1.2.3 Sources

The best sources of iron include:

Breast milk (the iron is very easily used by the child), Infant cereals and other iron-fortified cereals, Baby formula with iron, Liver, meats including poultry, Dried beans and lentils, Eggs, Fish, Molasses, Peanut butter, Soybeans.

14.1.2.4 Treatment

- Stop blood losses e.g., acute haemorrhage, GIT bleeding, infestations, infections,
- Transfusion with packed cells if severe and symptomatic e.g., CCF
- Iron infusions if malabsorption of iron is the cause
- Oral iron supplements if not severe and asymptomatic.
- Iron (elemental) at 6 mg/Kg/day for at least 3 months after correction of haemoglobin (to replace stores)
- Counsel on diet

14.1.2.5 Follow up

- Refer all children with anemia for Paediatrician follow up
- If not responding as expected in 2 months on adequate treatment, then re-evaluate and refer to haematology clinic
- Monitor growth and development and refer as needed.

14.2 Haemophilia

- Haemophilia A is Factor VIII deficiency
- Haemophilia B is Factor IX deficiency

14.2.1 Management

- Most bleeds will require factor replacement except for bruises and minor soft tissue injuries that do not impact on function and mobility
- Invasive procedures such as arterial puncture, lumbar puncture must only be performed after clotting factor replacement
- Do not give intramuscular injections

14.2.1.1 Clotting Factor Replacement

- Recombinant clotting factor concentrates are the product of choice. Doses should be rounded up to the nearest vial size
- Do not discard any product (use whole vial). e.g., 20 Kg child requiring 30 units/Kg factor VIII, give 750 units
- Specify product type and name e.g., 750 units Recombinant Factor VIII (Recombinate). See table below for list of products

Note that there are three recombinant Factor VIII products available. Whilst it is recommended that patients maintain treatment with their established brand of FVIII product, **in an emergency administration of any brand of recombinant FVIII is acceptable**

Table 36: Clotting factor Replacement

Product type	Comment
Recombinant FVIII	ED alert system or card will indicate patient's treatment product. Should plasma FVIII level be required, indicate product name on pathology request form
Recombinant FIX	
Plasma derived FVIII	Also contains von Willebrand Factor
Plasma derived FIX	
Recombinant VIIa	Once reconstituted must be used immediately

14.2.1.2 Recommended clotting factor dosage

- The dosage of the clotting factor depends on the haemophilia type and the site and amount of bleed
- Consult haematologist
- The general guideline on the dosage is as follows:
- Haemophilia A: 30 – 75 Units/Kg on day 1 and may repeat prophylactic dose day 2 and 5
- Haemophilia B: 50 – 125 Units/Kg on day 1 and may repeat prophylactic dose day 2 and 5
- Antifibrinolytic therapy is critical in oral and dental bleeds
- Watch for compartment syndrome in major muscle bleeds

14.2.1.3 Bleeds requiring admission

- Suspected intracranial haemorrhage
- Bleeding into neck/throat
- Forearm/calf bleed with suspicion or evidence of compartment syndrome
- Bleeding into hip or inguinal area, suspected ileopsoas haemorrhage
- Undiagnosed abdominal pain
- Persistent haematuria
- Bleeds causing severe pain

14.2.1.4 General Measures - Joint and Muscle Bleeds

For muscle and joint bleeds R.I.C.E.S will limit bleeding and reduce pain. Initiate on arrival

- R = Rest (in position of comfort)
- I = Ice (Cold pack to reduce bleeding and pain)
- C = gentle compression bandage
- E = Elevation
- S = splint (severe/recurrent bleeds)

14.2.1.5 Venous Access

- Venous access is often the major fear and stressor for children with haemophilia and their parents. Treat veins with care; they are the lifeline for patients with haemophilia. Apply pressure for 3 minutes post venepuncture

14.2.1.6 Analgesia

- Paracetamol/codeine is enough in most cases, however morphine, Tramadol can be used for severe pain
- Splinting/immobilization is an effective adjunct for reducing severe pain

14.2.1.7 DDAVP (*Desmopressin*)

- Releases stored FVIII (and von Willebrand factor) into the circulation
- Used in patients with mild haemophilia A where there is documented evidence in the medical record of safe and satisfactory response (DDAVP challenge). DDAVP challenge can be performed from around 5 years of age
- Not adequate for haemostasis in major bleeding
- Generally, not recommended in young children (< 3 years) due to documented reports of hyponatremia and seizures
Relatively contraindicated in children with previous seizure disorders
- **DDAVP Dose:** 0.3 microgram/Kg (max 20microgram) in 50 mL 0.9% Normal Saline given by intravenous infusion over at least 30 minutes – available as 4 mcg injection

14.2.1.8 Antifibrinolytic Therapy (*Tranexamic Acid*)

- Reduces breakdown of blood clots and is effective for preventing recurrence of mouth bleeds and epistaxis
- Contraindicated for treatment of haematuria
- Dose of Tranexamic acid 25 mg/Kg/dose 8 hourly max. 1.5 g for 5 – 7 days

Table 37: Tranexamic Acid Dose (500 mg tablets)

Weight (Kg)	Tranexamic Acid 8-hourly (mg)
<20	250
20 - 30	500
30 - 40	750
> 40	1000

Notes: Regular follow up

14.2.1.9 Physiotherapy

Referral to physiotherapist for joint and muscle bleeds is essential. Following severe joint/muscle bleeds, the affected joint should be rested. For other joint bleeds, physiotherapy should commence as soon as pain has subsided, and bleed has been controlled.

14.2.1.10 Fresh Frozen Plasma or Cryoprecipitate

Use of Fresh Frozen Plasma or Cryoprecipitate is restricted to emergency situations where there is no clotting factor available.

14.3 SICKLE CELL DISEASE

Sickle cell disease is caused by structurally abnormal haemoglobin (Hb S) that causes a rigid distorted red blood cell (sickle cell). There are 3 common types

- Sickle cell anemia (SS disease) is the most common
- Sickle β Thalassemia
- Sickle haemoglobin C disease

14.3.1 Fever

Patients are functionally asplenic and thus at greater risk for invasive disease particularly by encapsulated organisms

14.3.1.1 Management

- Give high flow Oxygen
- Do a haemogram including reticulocyte count
- Do a blood culture, Urine culture and culture other sites, as indicated
- Obtain chest IM if respiratory signs or symptoms, or high fever with no focus of infection
- Consider parenteral antibiotic(s) (Flucloxacillin 50 mg/Kg 6 hourly + Gentamicin 7.5 mg/Kg (6 mg/Kg if < 10yrs) daily; do not delay for FBC or urine culture results
- Consult a Haematologist
- All children with fever > 38.5⁰ C should be admitted

14.3.2 Painful Vaso-Occlusive Crisis

All episodes of pain should be treated initially as vasoocclusive disease. Other diagnoses may need to be considered later. Pain may be on a limb, back, chest or abdomen

NB: Chest pain should be treated as an acute chest syndrome and not simply as a vasoocclusive disorder

14.3.2.1 Management

Start analgesics promptly to provide effective relief of pain.

Mild pain

- Paracetamol 10 – 15 mg/Kg + codeine 1 mg/Kg every 6 hours
- Encourage plenty oral fluids
- If not settling, then add Ibuprofen 10 mg/Kg every 6 hours
- If still not settling, manage as severe pain

Moderate to severe pain

- Give high flow Oxygen
- Take blood for FBC, Reticulocyte count, group/cross match and bilirubin
- Give bolus of 10 mL/Kg 0.9% Normal Saline followed by maintenance rate 6 mL/Kg/hour (including oral intake) of 5% Dextrose in 0.45% Saline.

Recommended analgesics

- Morphine 0.05 mg/Kg/dose IV Repeat as necessary (see analgesia and sedation guideline)
- Some patients may require higher individual doses, based
- on prior history or may benefit from continuous infusion via
- Patient Controlled Analgesia (PCA)
- May need leucocyte filtered blood transfusion
- Discuss with haematologist

14.3.3 Acute Chest Syndrome

Sickle cell disease can produce an acute illness related to infarction of the lung tissue. Usually associated with lower respiratory symptoms, hypoxaemia and a new infiltrate on CXR. Chest pain and hypoxaemia may be the only signs

Chest pain should be treated as an acute chest syndrome and not simply as a vasoocclusive disorder

NB: This is a life-threatening illness and patients may deteriorate quickly

14.3.3.1 Management

- Give high flow Oxygen
- Maintain and treat airway, breathing and circulation problems first
- Maintain hydration with IV + oral intake at maintenance rate
- Treat pain aggressively by giving adequate analgesia to control pain
- Obtain CXR
- All patients with chest pain should be admitted, irrespective of CXR findings
- Cross match for possible exchange or simple transfusion
- Consider broad-spectrum antibiotics (see fever and sickle cell guideline)
- Consult Haematology urgently and prior to transfusion or admission

14.3.4 Acute Splenic Sequestration

This is defined as a haemoglobin drop of at least 2 g/dL below patient's baseline level with an acutely enlarged spleen. Mild to moderate thrombocytopenia is often present

Reticulocyte count is equal to or greater than patient's usual baseline. Consider co-existent aplastic anemia if reticulocyte count is low

It is most common in infants and young children. It has a high mortality rate

14.3.4.1 Investigations

- Group and Cross match Blood
- FBC including platelet count
- Reticulocyte count
- Blood and urine cultures if febrile; Chest x-ray if febrile and respiratory symptoms

14.3.4.2 Management

- While waiting for blood, give 0.9% Normal Saline 10 mL/Kg to treat hypovolaemia. Be careful in patients who have pre-existing cardiomyopathy
- Initiate transfusion of 10mL/Kg of packed red cells for patients with haemoglobin <4 g/dL or signs of Shock.
- Do not raise Hb above baseline since the spleen will shrink and autotransfusion will occur. This will result in an increase in the percentage of Hb S and risk of stroke (due to hyperviscosity)
- Treat with antibiotics if febrile (see fever), and analgesics for pain.

14.3.5 Aplastic Crisis

Acute illness associated with Hb below baseline for that patient and associated with a substantially decreased reticulocyte count (usually <1%). Usually associated with acute infection particularly parvovirus infection

May be associated with enlarged spleen as well

14.3.5.1 Management

- Give high flow Oxygen
- Maintain and treat airway, breathing and circulation problems first
- IV + oral fluids at maintenance rates – do not add potassium to IV fluid
- Transfuse patient if symptomatic anemia or Hb < 5g/dL (usually 5 mL/Kg over 4 hours). The blood product used should be WBC filtered and, if the patient is on Hydroxyurea should be irradiated also. Close observation for fluid overload. Transfusion may need to be repeated.
- Treat fever and pain as required (see fever and vasoocclusive crisis guidelines)
- Discuss with Haematologist

14.3.6 Stroke

Acute neurological events occur in about 10% of patients with Hb SS

These can present as

- Hemiparesis
- Monoparesis
- Aphasia or dysphasia
- Seizures
- Cranial nerve palsies
- Coma
- Can occur suddenly or as a complication of acute chest syndrome or aplastic crisis

NB. Consider other causes as well (see coma and seizure guidelines)

14.3.6.1 Management

- Give high flow Oxygen
- Maintain and treat airway, breathing and circulation problems first
- Investigations:
 - Haemogram
 - Cross match
- Arrange for partial exchange transfusion but remember not to over transfuse (e.g., Hb <10g/dL)
- Do a head CT without contrast to exclude intracranial haemorrhage
- Notify Haematologist urgently
- May need admission to PCCU for immediate exchange transfusion

14.3.7 Priapism

Priapism is prolonged painful erection of the penis often starting in the early hours of the morning and not due to sexual stimulation

Occurs in 2 forms

- Stuttering episodes which last 2 - 4 hours but are often recurrent and may precede a severe episode
- An attack lasting longer than 4 hours and can result in impotence.

Recurrent episodes should be evaluated by haematologist and Paediatric surgeon

14.3.7.1 Assessment

- Length of current episode
- Associated symptoms – fever, dysuria, dehydration or pain at other locations

- History of prior episodes and the previous treatments and effectiveness
- Symptoms of obstructive sleep apnoea

14.3.7.2 Investigations

One should not wait for results before treating patient:

- Check haemogram and reticulocyte count
- Group and Cross match Blood
- Cultures and chest x-ray as required

14.3.7.3 Management

- Administer intravenous fluids at maintenance rate
- Give Morphine for analgesia – use the Morphine infusion chart to titrate dose to effect
- Encourage emptying of the bladder; catheterize, if unable to empty bladder
- Do not use ice or ice packs
- Consult Urologist and Haematologist if priapism has lasted more than 3 - 4 hours (may require aspiration and drainage)
- If no response to treatment, arrange for partial exchange transfusion
- Admit under Haematology Unit

Notes:

- Simple measures should be tried first at home, particularly if less than 3 - 4 hrs since onset:
 - Encourage plenty oral fluids
 - Analgesia
 - Urination
 - Moderate exercise

- Take a bath or shower

14.4 BLOOD AND BLOOD PRODUCT TRANSFUSION

Blood and blood product transfusions may be required for acute blood loss, or for failure of production such as bone marrow suppression.

Blood product therapy should only be given when the expected benefits to the patient are likely to outweigh the potential hazards.

Blood products

- Packed Red Blood Cells
- Platelet Concentrate
- Fresh Frozen Plasma
- Cryoprecipitate

14.4.1 Indications for Packed Red Blood Cells

- Hb < 5g/dL
- Haemoglobin levels between 5 and 7g/dL can be managed by medications as indicated e.g., Iron
- **Transfusion may be indicated at higher thresholds for specific situations:**
 - Hb < 7 - 10g/dL during surgery associated with major blood loss or if evidence of impaired Oxygen transport

- Hb < 8g/dL; patients on a chronic transfusion regimen or during marrow suppressive therapy (for symptom control and appropriate growth)
- Hb < 10g/dL; only for very select populations (e.g., neonates)

Table 38: Indications for Platelet Concentrates

Clinical Situation	Indication for Platelet Transfusion
Bone marrow failure	Platelet <10x10 ⁹ /L if no other risk factors for bleeding (see below) Platelet <10x10 ⁹ /L if risk factors present (fever, antibiotics, haemostatic failure, risk of intracranial haemorrhage)
Surgery/invasive procedure	Platelet <50x10 ⁹ /L. However, higher counts may be needed in surgery with high risk of bleeding e.g. neurosurgery
Platelet function Defects	Transfuse if there is bleeding or high risk of bleeding, regardless of actual platelet count
Bleeding/Massive transfusion	Maintain Platelet <10x10 ⁹ /L if thrombocytopenia likely contributing to bleeding Maintain Platelet <100x10 ⁹ /L in the presence of diffuse microvascular bleeding (DIC) or CNS trauma

14.4.2 Indications for Fresh Frozen Plasma, FFP

FFP is appropriate for bleeding with coagulation in the following:

- Warfarin effect, in addition to the use of vitamin K and vitamin-K dependent clotting factor
- Liver disease
- Acute disseminated intravascular coagulation, DIC

- Following massive transfusion
- Cardiac bypass

14.4.3 Indications for Cryoprecipitate

Fibrinogen deficiency, in the setting of clinical bleeding, an invasive procedure, trauma or DIC

14.4.4 Indications for Leukocyte depleted blood products:

- Immunocompromised patients (oncology, transplant recipients, CCU patients, and other congenital and acquired immune deficiencies)
- Patients requiring chronic transfusions
- Infants under 12 months
- Intrauterine or exchange transfusions

14.4.5 Indications for Irradiated blood products:

- All immunocompromised patients, including all oncology patients, cardiac neonates and all patients in ICU, to prevent graft-versus host disease.
- CMV negative products:
- Leukocyte depleted blood products, are considered an acceptable alternative to CMV seronegative products

14.4.6 Pre-transfusion Assessment

- The indication to transfuse (see above)

- Discuss re the reason for transfusion and its consent (also see hospital guideline: Blood transfusion, consent and documentation)
- Collect pre-transfusion sample and document
- A sample for cross matching must be collected in a 1.4 mL Red EDTA Tube. Patients known to have red cell antibodies or haemolytic anemia will require a larger sample
- Correctly identify the patient during the collection of the pre-transfusion sample. Identification must include 2 unique identifiers: full name, date of birth and Registration number. This, together with completing the bedside check prior to blood administration are the most vital steps in preventing serious transfusion errors.
- Request the appropriate blood product component and special requirements
- Calculate and prescribe the transfusion volume with consideration to pack sizes
- Prescribe the blood product and rate of administration on the fluid order chart
- All transfusions must be completed within 4 hours of spiking a pack

14.4.7 Management of Transfusion

The key steps include:

- A formal checking process prior to commencement of transfusion
- The use of correct equipment (filters, pump, consideration of blood warmer)
- Correct transfusion documentation including patient observations, start and finish times

14.4.8 Complications during transfusion

- The most common immediate adverse reactions to transfusion are fever, chills and Urticaria
- The most potentially significant reactions include acute haemolytic transfusion reactions, bacterial contamination of blood products and transfusion related acute lung injury
- During the early stages of a reaction, it may be difficult to ascertain the cause
- **All suspected transfusion reactions must be reported to the issuing blood bank immediately.** Consult the haematologist provide advice regarding investigation and ongoing transfusion support

14.4.9 After transfusion

- Document the effect of transfusion on the patient's condition including haemoglobin level if repeated

Table 39: Blood product Transfusion Volumes and Rates of Infusion

Blood product	Formula for calculating transfusion volume	Pack sizes	Rate
Packed Cells	Packed Red Cells (mL) = Wt (Kg) x Hb deficit (g/ dL) x 4	250 – 300 mL Pack 50 – 60 mL Pedipack	Start at no more than 5 mL/min for the first 15 minutes, unless of an emergency
Platelets	5 - 20 mL/Kg (5 – 10 mL/Kg will raise the platelet count by 50 – 100x10 ⁹ /L)	Neonatal/ paediatric <40Kg patients: • Paediatric (single donor) – 40 - 60 mL • Apheresis (single donor) split into 4 - 8 x 40 - 60 mL packs Paediatric < 40Kg or adult: • Apheresis (single donor) < 200 mL or split into 2 x <100 mL packs • Pooled from 4 - 5 donors - < 160 mL	3 mL/Kg/hr over 2-3 hours. (Occasionally platelets are given over 30 minutes, but this may contribute to an increased risk of some reactions (fever/chills) and fluid overload)
Fresh Frozen Plasma	10 - 20 mL/Kg	300 mL/pack 50 mL/pack for neonates	Start at no more than 5 mL/min. for 15 minutes

Cryoprecipitate	5-10 mL/Kg	30 – 40 mL Pack	Start at no more than 5 mL/min. for 15 minutes
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15. ONCOLOGICAL CONDITIONS

There are approximately 300,000 new cancer cases diagnosed each year in those younger than 19 years of age worldwide. It is estimated that approximately 80,000 children will die of cancer each year around the world and most of these are in low- and middle-income countries (LMIC). Late diagnosis and a lack of knowledge and awareness of childhood cancer amongst health care workers are among the contributing factors.

The following early warning signs and symptoms that can easily be confused with other commoner childhood illnesses have been designed to help health care workers identify early warning signs and symptoms of childhood cancer.

Diagram 11: early warning signs of childhood cancer

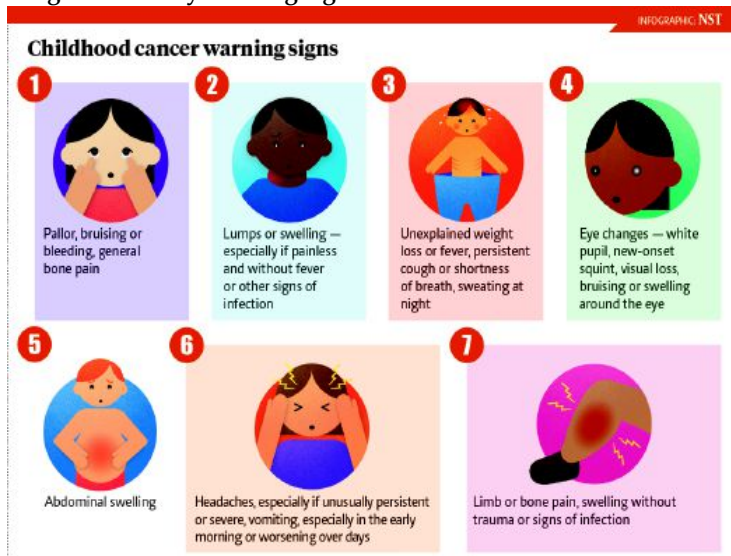


Table 3: Common differential diagnosis for some of the symptoms and signs of Childhood cancer

Presenting signs and symptoms	Common Differential diagnosis (Nonmalignant conditions)	Suspected Cancer
Low blood counts	Infection	Leukemia
Bleeding symptoms	Coagulation disorders Platelet disorders Bone marrow failure Syndromes	Leukemia
Lymph node swelling	Infection like tuberculosis	Leukemia Burkitt's Lymphoma Other Non-Hodgkin's Lymphoma (NHL) Hodgkin's Lymphoma (HL)
Abdominal swelling	Kidney cyst Hepatosplenomegaly Constipation Ovarian cyst	Wilms tumor Neuroblastoma Germ Cell Tumor Rhabdomyosarcoma
Mediastinal mass with shortness of breath	Infection	T cell Leukemia HL NHL Neuroblastoma Germ Cell tumor
White pupil	Congenital Cataract	Retinoblastoma
Persistent headache/Early morning vomiting/paresis/Ataxia	Meningitis Migraine Sinusitis	Brain tumor
Bone pain/swelling on arm/leg without trauma/infection	Bone cyst Osteomyelitis	Bone tumor
Soft tissue swelling	Lipoma	Sarcoma

15.1 ACUTE LEUKEMIA

Acute Leukemia is a cancer involving the cells in the bone marrow that form part of the hematopoietic system. It constitutes of Acute lymphoblastic leukemia (80%) and Acute Myeloid leukemia (15%). It accounts for approximately 35% of all childhood cancers; approximately 1 in 1000 children will be diagnosed with leukemia by the age of 19, although it is more common in children under the age of 10.

15.1.1 Diagnostic tests

- Complete Blood Count with differential white cell count: white blood cells may be low, normal or raised, neutropenia, anemia, thrombocytopenia.
- Peripheral morphology: blasts (immature red blood cells may be seen).
- Bone marrow Aspirate/Trephine biopsy for cytology/Immunophenotyping and cytogenetics: diagnosis of ALL > 25% blasts and AML > 20% blasts
- CSF cytology
- Serum chemistries: Liver function tests, Renal function tests, Lactate dehydrogenase, Serum Calcium/Phosphorus/Uric acid
- Coagulation screen (especially for Acute Promyelocytic Leukemia (APL) where patients are at risk of bleeding)
- Group and Cross match
- Chest radiograph: to check for mediastinal mass
- Echocardiography (pretreatment assessment)
- Blood culture (if fever present)

15.1.2 Treatment recommendations

15.1.2.1 Definitive

Acute leukemias are mainly treated with combination chemotherapy in several phases guided by the oncologist

15.1.2.2 Supportive care

1. Tumor lysis prevention/treatment: Start hydration at 3000 mL/m²/day, Allopurinol 100mg/m² TID, Strict Input /Output monitoring (Target urine output 2 – 3 mL/Kg/hour), Monitor for hyperkalemia/Hyperphosphatemia and treat appropriately, do not treat hypocalcemia unless having seizures
2. Treat infection urgently if suspected
3. Transfuse with appropriate blood products especially in APL which require early initiation and use of fresh frozen plasma (FFP) and Platelets. Be careful with transfusion of packed red blood cells especially if WBC count > 100,000/ μ L
4. Patients with suspected mediastinal mass should **ALWAYS** be **PROPPED UP!**
5. Inform pediatric oncologist as soon as diagnosis is suspected.

15.2 LYMPHOMA

Lymphomas are malignant cell infiltrations of the lymphatic system. Lymphomas are broadly classified as Hodgkin's and Non-Hodgkin's. The two are distinguished by cell type.

15.2.1 Signs and Symptoms

- swollen lymph node, especially in the neck, armpit or groin

- swelling of the face
- weakness, tiredness
- sweating, especially at night
- unexplained fever
- unexplained weight loss
- abdominal pain or swelling
- pain
- breathing difficulties, occasional cough, sometimes difficulty in swallowing

15.2.2 Diagnostic tests

- Complete Blood Count with differential white cell count
- Bone marrow Aspirate/Trephine biopsy for cytology (for Non-Hodgkin's Lymphoma)
- CSF cytology (Burkitt's lymphoma mainly)
- Serum chemistries: Liver function tests, Renal function tests, Lactate dehydrogenase, Serum Calcium/Phosphorus/Uric acid
- Lymph node biopsy and cytology
- Chest radiograph: to check for mediastinal mass
- Echocardiography (pretreatment assessment)

15.2.3 Treatment recommendations

Lymphomas are usually treated by a combination of chemotherapy, radiation, and/or bone marrow transplants.

Supportive care: Check Acute Leukemia

15.3 WILMS TUMOR

Wilms tumor is a cancerous tumor on the kidney, although it is totally unrelated to adult kidney cancer. It accounts for 6-7% of childhood cancer cases.

15.3.1 Signs and Symptoms

- abdominal swelling and/or pain - the pain can be intermittent rather than constant
- nausea, vomiting, constipation, loss of appetite
- fever of unknown origin
- night sweats
- abnormal urine color or blood in the urine
- malaise

15.3.2 Diagnostic tests

- Complete blood count
- Serum chemistries: renal and liver function tests
- Urinalysis
- Abdominal Ultrasound/CT scan
- Chest CT scan

15.3.3 Treatment recommendations

Wilms tumor is treated with surgery, chemotherapy and radiotherapy

15.4 RETINOBLASTOMA

Retinoblastoma is a malignancy of the retinal cell layer of the eye and is the most common eye tumor in children. It usually occurs before the age of five and can occur in one or in both eyes and is hereditary in some cases. Retinoblastoma accounts for 3-4% of all childhood cancers.

15.4.1 Signs and Symptoms

- Whitish color behind the pupil
- Problems with eye movements (crossed eyes)
- A red irritation that persists

15.4.2 Diagnostic tests

- Ophthalmologist review under anaesthesia of both eyes
- MRI scan of the orbit
- Histologic examination of the orbit after enucleation
- CSF/Bone marrow aspirate (only for patients with high-risk pathology)

15.4.3 Treatment recommendations

Retinoblastoma is treated by surgery, chemotherapy, and radiation.

15.5 NEUROBLASTOMA

Neuroblastoma is a cancer of the sympathetic nervous system. It accounts for 5%-7% of all childhood malignancies

15.5.1 Signs and Symptoms

- lump or mass in the abdomen, chest, neck, or pelvis
- loss of appetite, nausea, weight loss, stomach pain, constipation, difficulty urinating
- changes in the eyes: black eyes, a droopy eyelid, a pupil that doesn't constrict, vision problems
- pain in the chest, difficulty breathing, persistent cough

- pain or numbness in the lower extremities, limping, inability to stand, stumbling
- bone pain, fever, irritability, listlessness
- backaches (backaches in children are not usual)

15.5.2 Diagnosis/Investigations

- complete blood count
- blood tests of liver and kidney function,
- bone marrow aspiration/trephine biopsy
- Urine catecholamine (Vanillylmandelic Acid/Homovanillic Acid)
- Ultrasound/CT scans of the neck, chest or abdomen depending on presentation
- Radionuclide scans: MIBG scan

15.5.3 Treatment recommendations

Neuroblastoma is usually treated by a combination of surgery, chemotherapy, radiation, and/or bone marrow transplants.

15.6 RHABDOMYOSARCOMA

Rhabdomyosarcoma is a fast-growing, highly malignant soft-tissue sarcoma which arises in undifferentiated striated muscle cells. It can occur in a variety of places in the body: the head, neck, and around the eyes; the extremities (shoulders, arms, and legs); in the pelvic region and genitourinary tract; and in the chest and lungs.

Rhabdomyosarcoma accounts for 5-8% of childhood cancers and usually affects children the ages of 2 to 6 and 15 to 19.

15.6.1 Diagnosis/Investigations

- Complete blood count
- Serum chemistries: liver and renal function tests
- Biopsy/Gross resection of the mass for histology
- MRI scan of the mass
 - Chest CT scan: look for metastasis
 - Bone marrow biopsy (look for metastasis)

15.6.2 Treatment recommendations

Treatment for rhabdomyosarcoma usually includes surgery to remove the tumor, radiation, and/or chemotherapy.

15.7 BRAIN TUMORS

Brain cancers account for 15% of pediatric cancers. The symptoms depend on the location of the tumor.

15.7.1 Signs and Symptoms

- a seizure not related to high fever
- staring, repetitive automatic movements
- persistent vomiting without any known cause (projectile vomiting), nausea
- progressive weakness or clumsiness; neck tilt, squint
- walking, balance problems
- precocious puberty; growth retardation
- sleep apnea
- vision problems

- headache, especially that wakes the child up at night or is early in the morning
- pain, especially back pain, which should be taken seriously in a child
- changes in personality, irritability, listlessness
- excessive thirst and excessive urination (rare, if the tumor is pressing against the pituitary)

15.7.2 Diagnostic tests

- MRI scans of Brain and /or spine

15.7.3 Treatment recommendations

Brain tumors are treated with surgery, radiation, and chemotherapy

15.8 BONE SARCOMA

Osteosarcoma and Ewing's sarcoma are the most common malignancies of bone tissues in children. Osteosarcoma, the more common of the two types, usually presents in bones around the knee; Ewing's sarcoma may affect bones of the pelvis, thigh, upper arm, or ribs. Bone cancers are most common in ages 10 - 20 and they account for about 6% of all childhood cancers.

15.8.1 Signs and Symptoms

- pain in a bone
- swelling or tenderness around a bone or joint
- interference with normal movements
- weak bones, leading to fractures
- fatigue, fever, weight loss, anemia

15.8.2 Diagnostic tests

- MRI scan of the limb with joint above and below
- Biopsy of the mass
- Chest CT scan (look for metastasis)

15.8.3 Treatment recommendations

Bone cancer is treated by surgery to remove the tumor combined with chemotherapy

16. DERMATOLOGIC CONDITIONS

15.1 BACTERIAL SKIN INFECTIONS (PYODERMA)

- Bacterial skin infections can be primary or secondary (to damaged skin from injury or other skin diseases).
- Some bacteria like *Staphylococcus aureus* act as super- antigens triggering off eczematous lesions in atopic dermatitis.
- The presence of bacterial infection in an existing skin condition may not always be obvious to the untrained eye.
- The extent and severity of infection and other factors will determine whether a topical antibiotic is enough or systemic antibiotics are also indicated.

15.1.1 Descriptions of Pyoderma

- **Cellulitis** is a spreading infection of the skin extending to involve the subcutaneous tissues. The most common causes are group A β -haemolytic streptococci (GABHS) and *Staphylococcus aureus*. Predisposing factors include skin abrasions, lacerations, burns, eczematous skin, etc., although the portal of entry of organisms is often not seen.
- **NOTE:** Allergic reactions/contact dermatitis (e.g. to insect bites, immunisations, plants, etc.) are frequently misdiagnosed as cellulitis. If there is itchiness and no tenderness, cellulitis is unlikely.

- **Erysipelas** is a specific superficial form of cellulitis usually caused by GABHS. There may be lymphatic involvement.
- **Impetigo** (commonly called "school sores") is a highly contagious infection of the epidermis, particularly common in young children. Causative organisms are GABHS and *S. aureus*.
- **Staphylococcal scalded skin syndrome (SSSS)** is a blistering skin disorder induced by the exfoliative (epidermolytic) toxins of *S. aureus*. It primarily affects neonates and young children.
- **Necrotising fasciitis** is a rapidly progressive soft tissue infection characterised by necrosis of subcutaneous tissue. Aetiology is often polymicrobial. Causative organisms include GABHS, *S. aureus* and anaerobes. It can cause severe illness with a high mortality rate (25%).
- **Folliculitis** is inflammation of one or more hair follicles. Infection of a hair follicle that goes deeper into the skin layer and is associated with pus accumulation is a **furuncle** or boil. A **carbuncle** is larger and deeper and is formed by a group of furuncles. These may be caused by *Staphylococcus aureus* or mixed infections.
- Consider herpetic infection when vesicles are present and send appropriate specimens for immunofluorescence and viral culture.
- There are many other forms of skin infection that are not covered in this guideline.

15.1.2 Management of Pyoderma

15.1.2.1 Cellulitis

Involves subcutaneous lesions; several organisms are implicated.

- Systemic therapy and inpatient treatment required. Several other bacterial infections of the skin with a mixture of gram-positive gram-negative bacteria and anaerobes may be seen.
- Broad spectrum antibiotics may be necessary, and therapy should be directed towards culture and sensitivity of specimens taken.
- Widespread, recurrent and non-responding infections should be managed in conjunction with the surgical team.
- Initial treatment with flucloxacillin 25 mg/Kg (max 500 mg) PO 6 hourly for 7 days.
- If severe/extensive, systemically unwell or not responding to oral treatment; admit and put on IV Flucloxacillin 50 mg/Kg (max 2 g) 6 hourly (consider adding Clindamycin if rapidly progressive to inhibit toxin production).
- For facial/periorbital cellulitis: consider adding Cefotaxime if the child is <5 years and non-Haemophilus influenza type b immunised, or not responding to Flucloxacillin alone.
- For suspected or confirmed MRSA infection use IV Vancomycin 15 mg/Kg every 8 hours.

15.1.2.2 Folliculitis

- If mild and localized – topical antibiotics may be enough.
- When widespread or with systemic symptoms – systemic antibiotics are added.

15.1.2.3 Carbuncles and Furuncles

- Flucloxacillin (dosage by age/weight) or Clindamycin
- Carbuncles may require surgical incision and drainage

15.1.2.4 Erysipelas

- As for cellulitis

15.1.2.5 Impetigo

- Uncomplicated localized: wash crusts off - topical Mupirocin 2% ointment or Fucidin 12hourly
- If extensive multiple lesions present not responding to topical treatment, treat as for cellulitis
- Impetigo is very contagious - the child should be excluded from childcare/kindergarten/school until treatment has started and the sores are completely covered with watertight dressings

Staphylococcal Scalded Skin Syndrome

- As for cellulitis

Necrotising fasciitis

- Admit
- IV Flucloxacillin 50 mg/Kg (max 2 g) IV 6hourly plus IV Clindamycin 10 mg/Kg (max 600 mg) IV 6hourly.
- Refer to surgical team if pus collection is present.

15.2 NAPPY RASH (DIAPER DERMATITIS)

- Nappy rash is a dermatitis confined to the area covered by the nappy
- It is most commonly characterised by confluent erythema of the convex surfaces of the buttocks, the areas of skin in closest contact with the nappy and it spares the groin folds
- Nappy rash is not one distinct diagnosis, but is a multifactorial problem

15.2.1 Management

- Use disposable nappies.
- Increase the frequency of nappy changing and cleansing the skin.
- Use disposable towels or face washers soaked in water or olive oil to cleanse the area.
- Application of a barrier cream like Zinc in castor paste at every change. Apply extremely thick layer and should not be removed completely after each nappy change, rather apply another layer over the top.
- Letting the child be without a nappy on for as long as possible, lying on a soft absorbent sheet that is changed as soon as it is wet. Sunlight plays a role.
- If there is associated Candida infection, leading to erythema in the folds and satellite pustules then topical anti-candida therapy (Clotrimazole or Nystatin) should be applied. This therapy is often combined with 1% Hydrocortisone to reduce the associated inflammation.
- Antibiotic creams used when bacterial infection is present.

15.3 ATOPIC ECZEMA

Contact (irritant or allergic) dermatitis, Seborrhoic dermatitis

- Neutralizing balancing creams, moisturizers that restore and maintain optimal conditions of moisture lipid and PH of the skin should be used regularly.
- Emollients-moisturize the skin frequently and should be used up to four times, avoid aqueous cream.
- Avoid allergens or irritants in contact with the skin. Clothing should be of soft fabric (cotton) and loose fitting.
- Mild potency steroid cream like 1 %hydrocortisone twice a day
- Topical or systemic antibiotics should be given when bacterial infection is present.
- Antifungal creams should be used where indicated.
- Tacrolimus 0.03% cream is used on affected eczematous areas for short periods and monitored.

15.3.1 Important

- Avoid topical combinations with corticosteroids.
- Cold water or lukewarm compresses are good antipruritics.
- Caution with systemic antihistamines, many are not to be given under 2 years of age.
- Topical antihistamines are not to be used.

15.4 SCABIES

Caused by the mite; *Sarcoptes scabiei var hominis*

- Itchy condition with papules on fingers, soles, sides of hands, feet, axillary areas, wrists and can be all over the body, often excoriated from scratching
- Mites or eggs spread the infection by close contact with infected persons, or infected articles.

15.4.1 Treatment

- All clothing and infected articles should be properly laundered.
- Crothamiton cream or lotion is mild and relatively safe in children and relieves the pruritus.
- Permethrin 5% cream is the drug of choice for those over 2 months of age. It is applied all over the body avoiding the centre of the face. It is washed off after 6 hours in those aged 6 months or less and after 12 hours for the older ones. This is repeated the following day and the 2-day cycle repeated the following week.
- For children aged 2 months and below, use 10% Sulphur in soft white paraffin applied topically four times a day for 2 days
- Use of oral ivermectin and benzyl benzoate to be discussed with the dermatologist

15.5 FUNGAL SKIN INFECTIONS

The most common fungal infection in children is Tinea Capitis (dermatophyte infection of scalp).

- Other dermatophyte infections are Tinea corporis (body) Tinea cruris (groin) and Tinea pedis (feet).
- Dermatophytes respond best to Terbinafine cream.
- In more complicated or deeper infections Terbinafine tablets may be given once daily for 2 – 4 weeks dosage by weight.
- If Terbinafine is given for more than 4 weeks, liver function tests must be done.
- In mixed fungal skin infections, yeasts, moulds like Candida and Pityriasis versicolor, topical preparations of

Nystatin, Clotrimazole, Isoconazole, Bifonazole, Miconazole can be used for treatment

15.5.1 Duration of treatment

Duration of treatment for superficial dematophytes (tinea) depends on region affected. Broadly tinea corporis is curable by topical antifungal for 4 – 6 weeks. Tinea capitis (hair) for 4 – 6 weeks, onychomycosis (nail) for 2 to 3 months

15.5.2 Classes of antifungals

- Topical agents – shampoo and topical antifungal creams (azoles)
- Systemic antifungals
 - Antifungal antibiotics
 - Azoles
 - Allylamine
 - Newer ones – Benzylamine

15.6 VIRAL SKIN INFECTIONS

15.6.1 Herpes simplex

- Herpes simplex is characterized by grouped vesicles around oral or genital area, recurrent infection in common

15.6.1.1 Treatment

- Acyclovir 5% cream for primary infection 2 or 3 times a day for 7 days
- Oral Acyclovir for recurrent infections – *See infectious diseases section for further discussion*

15.6.2 Chicken pox

This is varicella infection with widespread vesicles at different stages of development and a positive history of contact

15.6.2.1 Prevention

Immunization at the age of 1 year.

15.6.6.2 Management:

- Isolation of skin contact, soothing agents like calamine lotion, Crotamiton, stop as soon as lesions crust.
- Widespread disease is treated with systemic Acyclovir 100 - 200 mg/Kg PO 5 times a day for 7 days. Oral antihistamines and antipyretics with paracetamol are desirable.
 - Child < 2 years 100 mg PO 5 times a day
 - Child $\geq$ 2 years 200 mg PO 5 times a day

15.6.3 Molluscum Contagiosum

Umbilicated skin lesions

- Refer if lesions multiple.
- Advise against manipulation.

15.6.4 Common warts

Condylomata acuminata: <3 lesions or persistent lesions.

- Advise against prurging and manipulation
- Treat with topical duofilm solution or podophyllin to be applied 3 times weekly for 4 weeks.

16. BONE AND CONNECTIVE TISSUE DISEASE

16.1 THE ACUTELY SWOLLEN JOINT

- Acutely swollen joints may reflect local pathology (e.g., trauma, sepsis) or generalized pathology localized to a joint/s (vasculitis) or post infective arthritis.
- Often the diagnosis only becomes apparent with time and initial treatment is on a presumptive basis.
- For a significant number of patients this involves symptomatic measures only.

16.1.1 Management

In the acute phase the most important tasks are to identify those conditions requiring more than just symptomatic treatment, and to ensure that those being treated symptomatically have appropriate follow up.

Consider referral to Rheumatologist if:

- Symptoms for more than 4 weeks.
- A significant joint effusion.
- Significant limitation of activity.
- Multiple joint involvement.
- Evidence of joint contractures.
- Vasculitis other than HSP.

In many cases there may not be a clear diagnosis by the end of the child's assessment in the emergency department - the

results of some investigations may not be available for days, and others may help only in ruling out certain conditions.

For such children, symptomatic outpatient treatment with Non-Steroidal Anti-inflammatory Medications (e.g. Ibuprofen) with careful follow up is appropriate.

These children should be followed closely until their symptoms resolve, or the diagnosis becomes clear.

Any child with symptoms not resolved after four weeks or any child in whom NSAIDs do not provide adequate relief of symptoms should be re-evaluated.

17. ENDOCRINE DISORDERS

17.1 HYPOTHYROIDISM

Hypothyroidism is a common endocrine disorder resulting from deficiency of thyroid hormones.

Cretinism refers to untreated congenital hypothyroidism, which affects about 1 per 4000 newborns.

Subclinical hypothyroidism, also referred to as mild hypothyroidism or compensated hypothyroidism, is defined as normal serum free T4 levels with raised serum TSH concentration.

17.1.1 Investigations

- Thyroid function test
 - TSH assays
 - Free T4
 - TRH stimulation tests (to be done by an endocrinologist)
- Imaging studies
 - Thyroid ultrasound scan
 - Iodine Isotope scans
- Fine needle biopsy

17.1.2 Treatment

Treatment is thyroid hormone replacement.

Effects are seen in 4-5 days and maximal effects in 6 weeks and the TSH corrects in about 4 weeks.

Give a starting dose, then refer to endocrinology clinic.

Levothyroxine given preferably in the morning (Titrate the dose to individual requirements)

- Neonate to 6 months: 12 – 18 microgram/Kg PO once daily
- 6 – 12 months: 8 – 12 microgram/Kg once daily
- 1 – 5 years: 6 – 8 microgram/Kg once daily
- 6 – 12 years: 4 – 6 microgram/Kg once daily
- < 12 years: 2 – 3 microgram/Kg once daily

17.2 HYPERTHYROIDISM

Hyperthyroidism refers to overactivity of the thyroid gland leading to excessive synthesis of thyroid hormones and accelerated metabolism in the peripheral tissues.

Thyrotoxicosis, on the other hand, refers to the clinical effects of an unbound thyroid hormone, whether the thyroid gland is the primary source.

17.2.1 Investigations

- Free T4, Free T3, and TSH measurements.
- Diagnostic radioiodine ^{131}I uptake is performed rarely.
- Either technetium ^{99}Tc m or ^{123}I scan may be useful if the gland does not have a uniform consistency.

17.2.2 Medical Treatment

Give a starting dose; then refer to Endocrinology clinic

- Methimazole 0.5-0.7 mg/Kg/day given every 8-12 hours
- Carbimazole 0.25 – 0.7 mg/Kg/day given every 8-12 hours
- Lugols Iodine Solution 1 – 5 drops PO 8hourly
- Radioiodine (^{131}I , Iodotope), 4-10 microCi, one to 2 doses is enough. Patient will need thyroxine replacement

- Propranolol, 80 mg/m²/d, or 2-4 mg/Kg/d PO in 2 divided doses in patients with marked cardiac manifestations
- Hydrocortisone 100 – 200 mg/m²/day PO/IV (in very severe conditions)

17.2.3 Treatment Options

17.2.3.1 Medical Management

- Control: Dosage of methimazole is titrated to maintain T4 concentration within the normal range. As the disease comes under control and TSH levels rise, the dose is decreased and eventually discontinued.
- Block and replace: High doses of methimazole are used to cause hypothyroidism and then thyroxine replacement is started.

Note: Methimazole and Carbimazole may cause agranulocytosis. Check TBC before and regularly thereafter. Treatment must be stopped if this happens.

17.2.3.2 Surgical Treatment

Total thyroidectomy can be performed. Surgical complications can include hypoparathyroidism and damage to the recurrent laryngeal nerve.

17.3 DIABETES MELLITUS

This is a disorder of metabolism resulting from impaired utilization of glucose from the bloodstream as a result of impaired insulin activity.

17.3.1. New presentation, mildly ill

17.3.1.1 Assessment

<3% dehydration, no acidosis and not vomiting.

Investigations include:

- Blood glucose, urea, and electrolytes.
- Arterial or capillary acid/base status.
- Urine - ketones, microscopy, culture.
- Check for precipitating cause e.g. infection (urine, full haemogram, blood cultures; consider Chest x-ray).
- Islet cell antibodies, and/or GAD antibodies.
- Total IgA, antiendomysial IgA antibody and thyroid function test for all newly diagnosed patients

17.3.1.2 Management

Initial Treatment:

- 0.25 units/Kg of quick-acting insulin SC. stat. If within 2 hours of a meal give mealtime dose only. Halve dose if child is less than 4 years old.
- Use anaesthetic cream for initial doses of insulin in a newly diagnosed child.
- Before breakfast and lunch (7.30a.m., 11.30a.m.) give 0.25 units/Kg of quick acting insulin. Before the evening meal (5.30p.m.) give 0.25 units/Kg of quick-acting insulin and 0.25 units/Kg of intermediate-acting insulin. If this is the first insulin

dose, give 0.25 units/Kg quick-acting insulin only, followed by further 0.25 units/Kg quick-acting insulin at midnight followed by a snack.

Ongoing Treatment:

- Once normoglycaemia is achieved and ketonuria disappears, change insulin to twice daily mixture of short and intermediate insulins.
- Give insulin as twice daily mixture of short and intermediate insulin, usually at 1 unit/Kg but may need modification. Given as two-thirds in the morning, a third at night, two-thirds of each dose intermediate-acting, a third as short-acting. Occasionally older adolescents go onto a basal bolus regimen: 30 - 40% intermediate acting insulin given at 2200hr, rest given as short-acting insulin in 3 equal doses before meals.
- Consult an Endocrinologist.

17.3.2 Hyperglycemic, mildly ill diabetic patients (already on insulin)

- Usually advised to take 10% of total daily dose as rapid-acting insulin every 2 hours until normoglycemic (in addition to usual insulin).
- Notify endocrinologist if there are any management issues that you want to discuss.

17.4 HYPOGLYCAEMIA

Beyond the neonatal period, hypoglycaemia is defined as blood glucose less than 2.5mmol/L.

There should be a low threshold for performing a Random Blood Sugar (Glucometer) in the acutely unwell child.

17.4.1 Assessment

17.4.1.1 *Effects of hypoglycaemia:*

- CNS Effects - irritability, coma, depressed level of consciousness, convulsions.
- Adrenergic overdrive - tremor, jitteriness, pallor, sweating.

17.4.1.2 *Causes of Hypoglycemia*

- Hypopituitarism (Measure height and weight. Look for midline defects, micropenis, optic nerve hypoplasia).
- Galactosaemia (Cataracts).
- Glycogen storage disease (Hepatomegaly).
- Adrenal insufficiency (Hyperpigmentation).
- Beckwith-Wiedemann syndrome (Hemihypertrophy, exomphalos, macroglossia and transverse ear creases).
- Hyperinsulinism is the commonest cause of hypoglycaemia in the under two years old, except preterms and small for gestational age newborns. This diagnosis is excluded by presence of ketonuria.
- "Accelerated starvation" (ketotic hypoglycaemia) classically manifests itself between the ages of 18 months and 5 years, and generally remits spontaneously before the age 8 or 9 years. A presumptive diagnosis is made by documenting a low blood sugar in association with ketonuria, ketonaemia and typical symptoms of hypoglycaemia. The definitive diagnosis is established by demonstrating an inability to tolerate a provocative ketogenic diet, or a fast. Susceptible or affected children develop severe hypoglycaemia and ketosis on this diet within 24 hours.

17.4.1.3 Investigations

Blood tests include:

- Glucose and lactate
- Insulin, cortisol, growth hormone
- Ammonia
- Ketones and free fatty acids
- Amino acids, electrolytes
- Acid-base
- Acyl-carnitine profile

Urine tests include:

- Ward test for ketones, glucose, reducing substances.
- 10 – 20 mL for amino acids and organic acids.

17.4.2 Management

Symptomatic hypoglycemia should be treated with an IV bolus of 5 mL/Kg 10% Dextrose (0.5 g/Kg Dextrose). The expected maintenance infusion rate is 3 – 5 mL/Kg/hr of 10% Dextrose (6 – 8 mg/Kg/min).

A required infusion rate above 10 mg/Kg/minute is consistent with hyperinsulinism.

All patients with hypoglycaemia presenting to the emergency department require admission.

17.5 DIABETES INSIPIDUS

Diabetes insipidus (DI) is an uncommon condition with either relative or absolute lack of anti-diuretic hormone (ADH). This causes inability to concentrate the urine and subsequent polyuria/ polydypsia with potentially fluid and electrolyte imbalance.

This can be seen in a variety of conditions in the paediatric population, most commonly in patients post neurosurgery or with cerebral malformations. It may be central or nephrogenic.

17.5.1 Assessment

Consideration should be given to:

- Hydration status/fluid balance/urine output.
- Presence of intercurrent illness like urinary tract infection, pneumonia, meningitis.
- Causes of excess fluid loss like gastroenteritis and surgical drains.
- History of diabetes insipidus with similar episode.
- Change in weight as marker of fluid status.

Baseline investigations include:

- Urea and electrolytes
- Urinalysis
- Paired serum and urine Osmolality

Diabetes insipidus is suspected if the serum Osmolality is raised (< 295 mOsmol/Kg water) with inappropriately dilute urine (urine Osmolality < 700 mOsmol/Kg of water). The serum sodium is often elevated due to excess free water losses.

17.5.2 Management

17.5.2.1 Rehydration

After assessment of level of dehydration and ongoing losses, adequate rehydration therapy should be commenced. If the serum Sodium is <150 mmol/L, rehydration should occur over 48 hours, if >170 mmol/L, admit to the Paediatric Critical Care Unit.

17.5.2.2 DDAVP (1-deamino-8-Arginine Vasopressin) administration

All patients should be discussed with the endocrinologist prior to the commencement of DDAVP therapy

DDAVP acts on the distal tubules and collecting ducts of the kidney to increase water reabsorption, as a long-acting analogue of ADH

There are several formulations available:

- Intranasal solution - 100 mcg/mL
- Intranasal spray (10 mcg/spray)
- Parenteral (IV/IM) – 4 mcg/mL - used rarely
- Oral - 200 mcg tablets

(Roughly 10 mcg intranasal $\equiv$ 200 mcg oral)

Administration principles:

- For infants discuss with an endocrinologist.
- Under 2 yr, dose is usually 2 - 5 mcg intranasal.
- From 2 yr on, dose use adult dose (5 - 10 mcg/day).
- Oral dose 50 mcg to 2 mg per dose, given 2 or 3 times a day.
- Dosage effect is all or nothing - in general, the dose determines the duration of action NOT the degree of response.

- Nasal administration is operator dependent - also need to consider effectiveness if problems with nasal mucosa like intercurrent URTI, hay fever, or post operatively.
- Careful fluid balance needs to be maintained to prevent fluid overload/hyponatremia.

17.5.2.3 Ongoing management of patients on DDAVP - General principles

- At a minimum, daily serum electrolytes and Osmolality and daily urine osmolality are required until stable - consider more frequent electrolytes if hypernatraemic or concerns about fluid state.
- Ensure most recent serum sodium result is above 135mmol/L prior to administration of DDAVP.
- 1 - 2 hrs of diuresis needed prior to administration of next dose to allow free water clearance and avoid hyponatremia.
- Specific gravity of all urine checked and documented.
- Strict fluid balance chart.
- Patient should be weighed daily.

Complications of management

- Hyponatremia
- Hypernatremia
- Fluid overload

Inform the Endocrinologist if:

- Urine output <4 mL/Kg/hour for two consecutive hours - may need repeat serum sodium.
- If DDAVP due and there has been no urine output for previous 12 hours, or urine output less than < 1.5 mL/Kg/hr may need to reduce or omit the dose.
- If serum electrolytes are not within normal range.

- If child is exhibiting signs and symptoms of dehydration or fluid overload.

17.6 RICKETS

Deficient mineralization of the primary spongiosum at the growth plate may result in impaired growth and skeletal deformity. Hypophosphatemia resulting from phosphate, vitamin D, and/or calcium deficiencies results in impaired bone development. Maximal effect is at the metaphyses where bone is growing rapidly.

17.6.1 Risk factors and risk populations

- Premature infants (especially if on parenteral nutrition)
- Unbalanced infant nutrition
- Protracted exclusive breast feeding
- Non-vitamin D supplemented formula fed infants
- Vegetarian diets
- Maternal vitamin D deficiency
- Lack of sun exposure and dark skin in sun-poor countries
- Malabsorption syndromes

17.6.2 Clinical presentation

Severe cases usually present with skeletal changes and bone pain in the second year of life. Signs and symptoms include:

- Developmental delay
 - Delayed walking, abnormal gait
 - Hypotonia
- Limb deformity usually in weight-bearing limbs
 - Swelling of wrists, ankles $\pm$ knees
 - Forearm deformities in crawling child

- Wind-swept deformity in walking child
 - Limb pain and fracture
- Short stature
- Chest deformity
 - Rachitic rosary (enlarged costochondral joints felt lateral to the nipple line)
 - Harrison sulcus - horizontal indentation along lower border of ribs at level of diaphragm
- Skull deformity
 - Frontal bossing
 - Craniotabes - softening of skull bones, usually evident on palpation of cranial sutures
 - Delayed fontanelle closure
- Spinal deformity
 - Lordosis
 - Kyphosis
- Dental abnormality
 - Delayed tooth eruption
 - Enamel hypoplasia
- Hypocalcemia
 - Seizures, tetany, carpopedal spasm, laryngospasm
 - Irritability
- Brown tumor secondary to hyperparathyroidism

17.6.3 Diagnostic Studies

17.6.3.1 Radiological appearance

- Cupping, splaying and fraying of the metaphysis of the ulna, radius and costochondral junction
- Course trabecular pattern of metaphysis
- Osteopenia

- Fractures

17.6.3.2 Laboratory features

- Serum calcium, phosphate, and alkaline phosphatase, BUN, creatinine, and 25-hydroxyvitamin D3 (25 OHD)
- Typical rickets biochemistry
 - Decreased phosphate
 - Increased alkaline phosphatase
 - Normal or decreased calcium

17.6.4 Calcipenic Rickets

Vitamin D deficiency most commonly due to

- Reduced intake or synthesis of vitamin D3
 - Vitamin D deficient mother (often dark-skinned women or those who actively avoid exposure to sunlight)
 - Prolonged breast feeding without vitamin D supplementation
 - Dark skin color
 - Reduced sun exposure e.g., veiled, or modest clothing, chronic illness/hospitalization, or excessive sunscreen use
 - Low intake of vitamin D-containing foods
- Abnormal gut function or malabsorption
 - Small bowel disorders e.g., celiac disease
 - Pancreatic insufficiency e.g. cystic fibrosis
 - Biliary obstruction e.g., biliary atresia
- Reduced synthesis or increased degradation of 25-OHD or 1,25-DOHD
 - Chronic liver disease or chronic renal disease
 - Drugs: rifampin, isoniazid and anticonvulsants

17.6.5 Complications

- Severe lower limb deformity
- Fractures
- Growth abnormalities

17.6.6 Prevention of Vitamin D Deficiency

The daily Vitamin D requirement for all infants is 400 IU (with or without calcium) to prevent nutritional rickets. This may be conveniently given as single oral or intramuscular doses of 50,000 IU at ages 3, 6, 9 and 12 months.

- In the absence of a positive family history, all rickets can be assumed to be due to vitamin D deficiency until proven otherwise.

17.6.6 Treatment of Nutritional Rickets with Vitamin D3 or D2

Table 41: Vitamin D dosing

Age	Daily dose for 3 months	Single dose	Maintenance dose
< 3 months	2,000 IU	N/A	400 IU
3 – 12 months	2,000 IU	50,000 IU	400 IU
1 – 12 years	3,000 - 6,000 IU	150,000 IU	600 IU
> 12 years	6,000 IU	300,000 IU	600 IU

The preferred route of administration of the single dose or oral, the parenteral route can be used if the oral route is contraindicated by disease process.

Ensure a Calcium intake of at least 40 mg/Kg/day.

Reassess response to treatment after 3 months as further treatment may be required

17.6.8 Other forms of Rickets

Other forms or causes of Rickets not discussed here include:

- Dietary Calcium Deficiency
- Phosphopenic Rickets
- X-Linked Hypophosphatemic Rickets
- Vitamin D Dependent Rickets
 - Type I Vitamin D Dependent Rickets
 - Type II Vitamin D Dependent Rickets
- Hepatic Rickets
- Hypophosphatasia
- Osteopenia of Prematurity

17.7 WEIGHT MANAGEMENT

Childhood obesity and overweight are emerging as common problems in children attending Gertrude's Children Hospital. However, the prevention and management of childhood obesity is not adequately addressed despite a rapid rise in its prevalence.

This guideline is to be used by medical, nursing and allied health staff at the GCH to assist them in assessing and addressing issues of weight and obesity in patients. It provides a simple, but evidence based clinical approach for the identification of childhood overweight and obesity and an

opportunistic approach to addressing these issues with the family and patient.

17.7.1 Definition of terms

Body Mass Index: Body Mass Index (BMI) is currently seen by health professionals as the most appropriate measure of adiposity in children.

Calculating Body Mass Index: BMI is calculated by dividing the weight (Kg) by the squared of the height (M^2). However calculated BMI values need to be compared with age and sex reference standards due to BMI changes that occur in normal growth.

Overweight: Overweight is defined as a BMI greater than the 85th percentile.

Obesity: Obesity is defined as a BMI greater than the 95th percentile for age.

17.7.2 Consequences of childhood overweight

- Much greater risk of adult obesity (a 5yr old who is obese has 8 times increased risk of adult obesity).
- Psychosocial morbidity (bullying, teasing, lower self-esteem, poorer socioeconomic prospects).
- Range of physical morbidities: type 2 diabetes, hypertension, dyslipidaemia, non-alcoholic steatohepatitis, orthopaedic disease, obstructive sleep apnoea, infertility.

17.7.3 Aetiology of childhood overweight

The large increase in the prevalence of obesity in the last three decades points to

- Widespread environmental factors
- Lifestyle changes

- Genetic predisposition
- Rarely, specific chromosomal or genes defects

17.7.4 Risk factors

- Genetics: At least 5 single gene defects have been found but these are all extremely rare and all are associated with severe and very early onset obesity and should prompt referral for further assessment.
- Environment/Lifestyle: Physical inactivity, increase in sedentary behaviour (especially screen-based activities), increased consumption of high fat foods and sugar sweetened drinks, children's food advertising.
- Other risk factors for obesity: Early infant feeding, parental obesity, parental encouragement of children to eat, higher socioeconomic status.
- Underlying medical conditions: (very rare) secondary obesity may occur due to hypothyroidism, hypercortisolism, growth hormone deficiency and hypothalamic damage.
- Drugs: Steroids, antipsychotic drugs (Risperidone) and some antiepileptic medications.

17.7.5 Assessment

The Body Mass Index (BMI) is recommended as a practical measure of overweight and obesity in children

- Rapid changes in BMI can occur in normal growth
- BMI varies with age and sex
- BMI rises in the first year of life, then falls during the preschool year, before rising again into adolescence (adiposity rebound)

Therefore, calculated BMI values need to be compared with age and sex reference standards (use BMI charts)

1. Calculate BMI using the formula $BMI = \text{Weight (Kg)} \div \text{the square of the height or length in metres (m}^2\text{)}$

Compare and chart BMI on percentile chart

2. Calculate Body Surface Area (m^2) using the formula:

$BSA = \text{The product of the Weight (m) and Height (cm)} \div 3600$

7.7.6 Addressing the issue of Obesity

The issue of obesity as an important health concern is often not addressed in the clinical setting. Once you have noted that the child is overweight you may address the issue using the following approach:

- Do no harm! Approach with respect but address the issue!
- Assess child and parent's perceptions of the issue - do they even see it as a concern? - Do they have differing awareness/concerns?
- Highlight the issue of weight in the context of health, show the growth chart to the family and explain what the healthiest weight for their child would be; explain they are still growing in height, so probably do not need to lose weight rather they need to grow into their weight
- Ask what they think they could do and possibly suggest some behaviour change ideas
- Organise follow up - the options are local doctor, dietician, paediatrician or a weight management clinic

7.7.7 Parental Perception

We know that approximately 50% of parents of obese children do not perceive that their child is overweight. It is therefore useful to gauge their opinion and experience of this issue as

this can shape the ensuing discussion e.g. *'What do you think about your daughter's weight?'*

- They (and the child) may respond that they are aware and concerned about the issue. This is when you can discuss specific behaviour changes and arrange referral.
- They may instead state that they think her growth is fine. Then your goal is to raise their awareness of the health issues, not necessarily to solve the problem!
- Some behaviour change ideas you can discuss with families.
- Physical activity: any increase in activity is an improvement!
- Aim for 'lifestyle' exercise: using the stairs, walking to school, walking the dog.
- Involve the whole family (everyone can benefit regardless of weight status).
- Use after school time to get outdoors and be active.
- Decrease screen-based activities (TV, Computer, Play station).
- Have bikes, helmets and balls ready to go, by the door!

7.7.8 Nutrition, and do not forget drinks!

- Water is the best drink for kids: cut out cordial, soft drink, fruit juice.
- Better to eat the fruit rather than drink fruit juice.
- Low fat (2% fat) milk (500 mL/day) is preferred for children over 2 years of age.
- Importance of breakfast, regular meals and healthy snacks.
- Basic food label reading, and awareness of the 'traps' i.e., 'no fat' might mean large amounts of sugar and therefore the same number of calories.
 - Serving sizes (does the 5-year-old get served as much as Mum or Dad?)

- Planning, avoiding regular take-away fast food.

7.7.9 Management

Management of obesity is complex and ongoing.

The aim of this guideline is to provide clinicians with an approach for identifying and addressing issues of overweight and obesity in the clinical setting. Weight management is most successful when addressed in the context of the whole family.

7.7.10 Follow up Review and Referral

Obesity is a chronic disease that may require comprehensive multidisciplinary assessment; management and long term follow up. Health care professionals involved should include:

- Maternal child health nurse
- Paediatrician
- Local doctor
- Dietician

Include a copy of the growth chart and BMI calculation with referral letter.

18. MALNUTRITION

The cellular imbalance between the supply of nutrients and energy and the body's demand for them to ensure growth, maintenance and specific functions

18.1 MANAGEMENT OF MALNUTRITION

Feeding children with severe malnutrition – use EBM and/or infant Formula, if aged < 6months

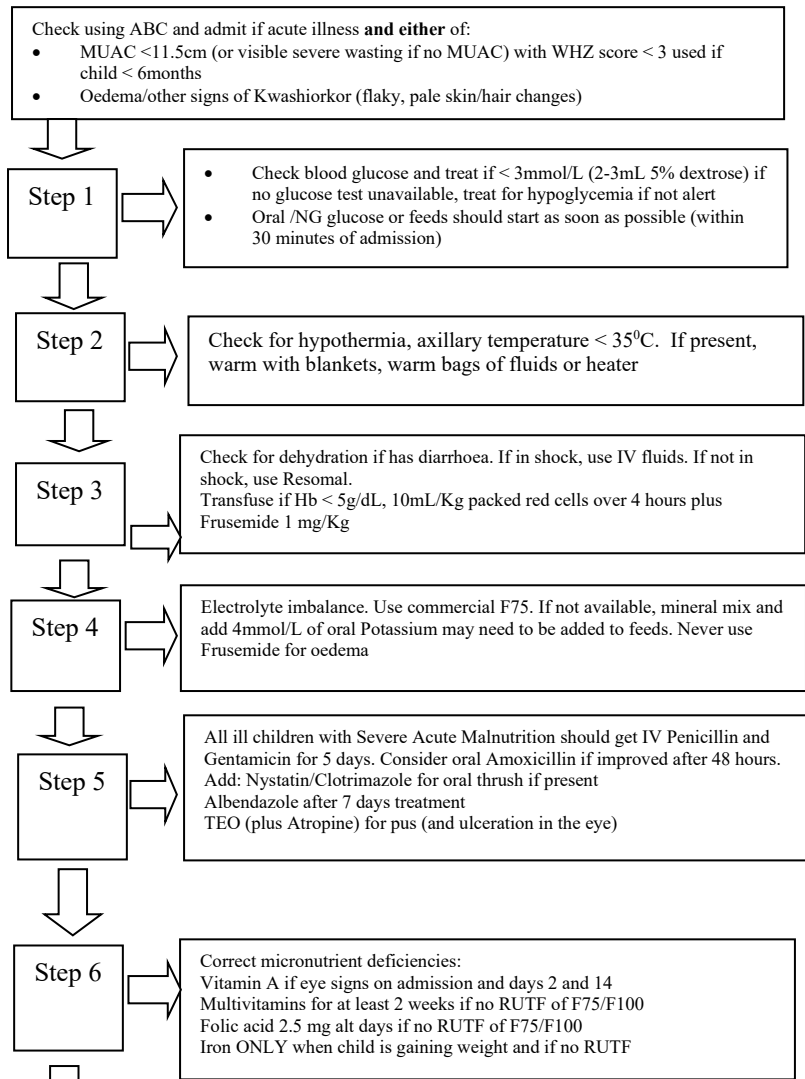
- If respiratory distress or oedema get worse or the jugular veins are engorged reduce feed volumes
- When appetite returns and oedema improves change from F75 to F100
- For the first 2 days of F100, use the same feed volumes as for F75 and continue increasing feeds by 10 mL per feed stopping when the child is not finishing the feeds or if the maximum is reached

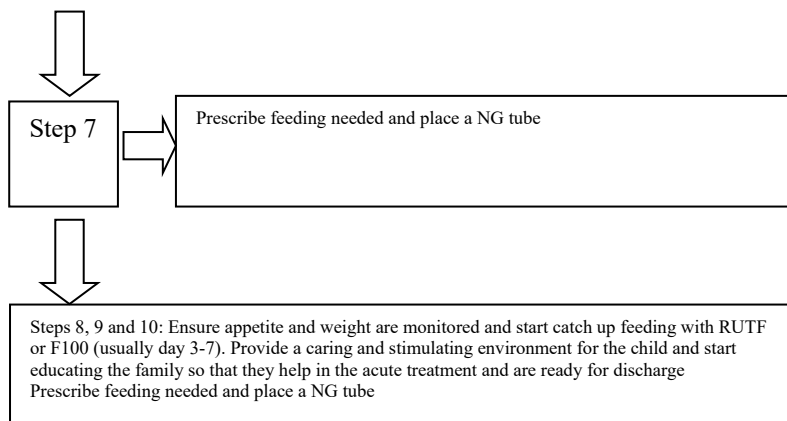
Admit to hospital if patient has

- Visible severe wasting (buttocks)
- Oedema/other signs of Kwashiorkor (flaky, pale skin/hair changes)
- Poor appetite/failed appetite test
- Malnutrition with illness
 - Respiratory - Inadequate breathing, difficulty breathing
 - Circulatory - circulatory compromise, hypothermia
 - CNS - Altered consciousness, Convulsions
 - Metabolic - Hypoglycaemia
 - Infections - Severe infections
 - Haematology- Severe anemia

- GIT - Chronic diarrhoea, vomiting with inability to retain

Diagram 12: The 10 steps in the Management of Severe Malnutrition





18.2 DIARRHEA IN SEVERE MALNUTRITION

18.2.1 Shock

AVPU < A, plus absent, or weak pulse plus prolonged capillary refill (>3s) plus cold peripheries with temperature gradient, give **10 mL/Kg in 2 hrs of Ringers lactate with 5% dextrose** (add 50 mL 50% dextrose to 450 mL Ringer's lactate)

If severe anaemia, start urgent blood transfusion not Ringers.

18.2.2 Dehydration

If not in shock or after treating shock

- If unable to give oral / NGT fluid because of very poor medical condition use / continue with iv fluids at maintenance regimen of 4 mL/Kg/hr.

- If able to introduce oral or ng fluids/feeds, give ReSoMal at 10 mL/Kg/hour for 2 hours then ReSoMal at 7.5 mL/Kg over 1 hour then introduce first feed between 5 - 10 mL/Kg/hr.

At 12 hours switch to 3 hourly oral / NG feeds with F75 (see table on *Feeding Children with Severe Malnutrition*)

Table 42: Fluid Management in Shock Complicating Malnutrition

Weight (Kg)	Fluids for Shock complicating malnutrition	Oral/NGT First 12 hours	Maintenance
	10 mL/Kg/hr. for 2 hours	7.5mL/Kg/hr for 10 hr	4 mL/kg/hr.
	Ringers in 5% dextrose	ReSoMal/F75 (10 mL/Kg first 2 hr.	Ringers in 5% dextrose
	Intravenous	Oral/ NGT	Intravenous
4	80	30	16
5	100	37	20
6	120	45	24
7	140	52	28
8	160	60	32
9	180	67	36
10	200	75	40
11	220	82	44
12	240	90	48
13	260	97	52
14	280	115	56
15	300	122	60

Table 43: Feeding Children with Severe Malnutrition

Weight (Kg)	F75 – Acute Feeding				F100 - Catch up Feeding		
	No or Moderate oedema 130 mL/Kg/day		Severe oedema, even face 100 mL/Kg/day		150 mL/Kg/day	3 hourly Feed volume	
	Total Feeds/24hr	3hrly Feed volume	Total Feeds/24hr	3hrly Feed volume	Total Feeds/24hr	Minimum 150mL/Kg/day	Maximum 220mL/Kg/day
3.0	390	50	300	40	450	55	80
3.5	455	60	350	45	525	65	95
4.0	520	65	400	50	600	75	110
4.5	585	75	450	60	675	85	120
5.0	650	80	500	65	750	95	135
5.5	715	90	550	70	825	105	150
6.0	780	100	600	75	900	115	165
6.5	845	105	650	85	975	125	175
7.0	910	115	700	90	1050	135	190
7.5	975	120	750	95	1125	140	205
8.0	1040	130	800	100	1200	150	220
8.5	1105	140	850	110	1275	160	230
9.0	1170	145	900	115	1350	170	245
9.5	1235	155	950	120	1425	180	260
10.0	1300	160	1000	125	1500	190	275
10.5	1365	170	1050	135	1575	200	285

11.0	1430	180	1100	140	1650	210	300
11.5	1495	185	1150	145	1725	215	315
12.0	1560	195	1200	150	1800	225	330

19. APPROACH TO THE ADOLESCENT PATIENT

According to the WHO an adolescent is a person aged 10-19 years.

19.1 APPROACH

1. When an adolescent enters your room, first establish rapport. Inquire if they are more comfortable with a male or female doctor, depending on the issue that has brought them. If under the age of 18 years, they should be accompanied by a parent or guardian (refer to the hospital policy on the management of unaccompanied minors)
2. If accompanied, inquire if the patient is comfortable continuing with the examination with the parent or caregiver in the room, or if they may step out for examination if it's the opposite sex have a chaperon (male or female medical staff)
3. All adolescents and their families, including caregivers should be told about confidentiality at the beginning of the interview. Document that you have informed them concerning confidentiality and that you will only break confidentiality if there is mandatory reporting needed in the best interest of the patient or with the permission of the patient to disclose the information shared to the parents/caregivers.
4. Assess the social, cognitive and biological stages of development
5. Biological staging - Tanner's staging for male and females (use diagrams as reference) If the patient is not comfortable with getting

undressed, kindly ask them to point out from the diagram at which stage they are at.

19.2 ASSESSMENT

It is important to do a **HEADSSS** assessment for every adolescent who presents to the hospital and document each response. Interview of the adolescent should be in this format:

- **H** - Home and environment (Where do you live and with whom? Siblings? Anyone who has recently moved in or out of the home? Is there any adult at home with whom you talk to or ask for advice in the home?)
- **E** - Education and Employment (Are you in school? How are you performing in school? Relationship with peers and teachers? Any part time or full-time work?)
- **E- Eating** (how many meals do you eat in a day? Are there any meals or foods you skip taking? Are you on a specific meal plan or diet?)
- **A** - Activities (Do you have any activities in or out of school? What do you do for fun?)
- **D** - Drugs (Do you drink alcohol or smoke cigarettes or take any other drug? If so, which ones? How frequently do you take them? How do you pay for them?)
- **S** - Sexuality (Are you or have you been involved in a relationship? Have you ever been sexually active? When was your last sexual encounter? Did you use any protection? Have your periods started? When was your first period? When was your last period? For boys-when did you have your first nocturnal emission or wet dream? When did you last have a nocturnal emission?)
- **S** – Suicide, Self-Image, Depression (What time do you sleep? What time do you wake up? Are there any challenges with sleeping or waking up? how do you feel about yourself? what have others said

about your appearance? Do you ever feel low, how many days in a week do you feel this way? Have you ever thought of harming yourself or those around you? Have you ever encountered a problem that had you wanting to harm yourself?)

19.3 VACCINATION

- Hepatitis B vaccination can be done for those aged 11-15 years who did not receive these vaccinations in infancy in two adult doses four weeks apart.
- For patients aged 6 - 18 years with chronic heart disease, chronic lung disease, chronic renal failure, diabetes mellitus, alcoholism, congenital or acquired immunodeficiency, anatomic or functional asplenia, malignancies should receive a dose of pneumococcal vaccine PCV 13.
- Ensure all school going adolescents have received two doses of MMR vaccine four weeks apart
- Varicella vaccine – for those aged 7-12 years of age should receive two doses three months apart, those over the age of 13 years, two doses 4 weeks apart.
- HPV vaccines: 3 dose series at 0, 1 - 2 and 6 months to all patients aged from 9 years and above.
- Meningococcal conjugate vaccine: single dose at age 11-12 years with a booster dose to be given at 16 years of age. If first dose is given at when over 16 years of age, no booster is required. In those patients aged 11-18 years with HIV infection two dose series of the vaccine is required 8 weeks apart.
- Tetanus and diphtheria toxoids and acellular pertussis (Tdap vaccine) - Administer one dose to adolescents aged 11-18 years

19.4 CONSENT FOR ASSESSMENT OR EXAMINATION

In cases of abuse:

- Age 0 – 5 years, obtain written consent from the non-offending care giver
- Age 6 – 11 years, obtain oral assent from child and written consent from the non-offending care giver
- Age 12 - 14 years, obtain written assent from the child and/or written consent of non-offending care giver or written consent from an emancipated minor
- Age 15 - 18 years, obtain written consent from the child

20. POST RAPE/DEFILEMENT PROTOCOL

It is critical to appropriately define and document your findings when a person presents to the hospital with history of rape or defilement

20.1 DEFINITIONS

Age

A child is any person who is under the age of 18 years.

An adult is any person who is aged 18 years and above.

Consent

To agree to and/or to give permission for an act to be done to oneself and the same can manifest in free will to participate in an act.

It is important to note that children under the age of 18 years and persons with mental incapacity cannot consent to acts under this law.

Indecent Act

This makes specific reference to any unlawful intentional act done on purpose with the result that it causes any contact between the *genital organs* of a person, his or her breasts and buttocks with that of another person but does not include an act that causes penetration.

It may also include exposure or display of any pornographic material to any person against his or her will.

Rape

A person commits the offence of Rape if, he or she intentionally and unlawfully commits an act which causes penetration with his or her genital organs. The other person does not consent to the penetration; or the consent is obtained by force or by means of threats or intimidation of any kind.

The penalty for rape ranges from ten (10) years to life imprisonment.

Attempted rape

This is described as trying to rape someone.

The offence carries a penalty of five (5) years to life imprisonment.

It is notable that the Act envisages that men can also be victims of rape and that rape can also occur among people who are having a mutual relationship where it is termed date rape.

Sexual Assault

Sexual assault is described as a case whereby a person penetrates the genital organs of another person or any of their body part or of another person or any object which may be capable of causing penetration including but not limited to a stick, a bottle among others.

The offence is punishable by imprisonment ranging from ten (10) years to imprisonment for life.

Compelled or Induced Indecent Act

A person, who intentionally and unlawfully forces or causes another person to take part in an Indecent Act with him/herself, or another person, or an object, including any body part of an animal, is guilty of this offence which carries a penalty of not less than five (5) years imprisonment. The exception to this section is in cases where an act which may be perceived

as being indecent is done to save a life or in the course of a medical procedure.

Acts which cause penetration or indecent acts committed within the view of a family member, a child or person with mental disabilities

A person who intentionally commits rape or an indecent act with another within the view of a family member, a child or a person with mental disabilities is guilty of an offence and is liable upon conviction to imprisonment for a term which shall not be less than ten years

Defilement

Defilement is defined as the act of having sexual intercourse with a child below the age of 18 or a minor. The key factor in this offence is the fact that there can be no consent by a child to a sexual act as the child is deemed to be incapable of giving such consent. The offence is punishable in varying degrees depending on the age of the child.

The sentence for defilement depends on the age of the child:

- Children below 11 years – Life imprisonment.
- Children between 12 – 15 - Twenty (20) years in jail or more.
- Children between 16-18 years - fifteen (15) years in jail or more.

There are some allowable defenses to this offence including if the child misled the perpetrator that they were over 18 years old and/or that the said perpetrators believed. These circumstances will only amount to an allowable defense if the perpetrator proves in evidence that he took all reasonable measures to ascertain the child's age. Notably this defense will not be allowed should the accused be related to the child

The main question with respect to this section has been whether children can commit this offence against other children. The simple response is that it would be hard to qualify two children engaging in a sexual act as defilement as the matter may better attended an act of delinquency that can be handled through counseling

On the other hand, if the act of sexual intercourse between children is accompanied by use of force by one child as against another child, then the child using force may be charged with an offence under this section and will be treated as a child sex offender. The legal question will be what level of punishment will be meted and just considering the fact the perpetrator himself or herself may not have truly appreciated the offence. There is a high likelihood that the child may be sent to a correction institution like the Borstal Institutions

Attempted Defilement

This carries a sentence of not less than ten (10) years in jail

Gang Rape

Gang rape envisages a situation whereby two or more people rape an adult or defile a child and the penalty ranges from fifteen (15) years to imprisonment for life

Notably any individual will be *individually* charged with the offence of rape as well as being charged *together* with the rest of the offenders for the offence of gang rape

20.2 MEDICATION REQUIRED POST RAPE OR DEFILEMENT

20.2.1 Post Exposure Prophylaxis (PEP) for HIV

ARV agents for PEP for 28 days, to be started no later than 72 hours after the sexual assault or defilement

Based on age:

>15 years old or > 35 Kg body weight

TDF +3TC+ DTG (or TDF+3TC+ATV/r for women and adolescent girl of childbearing age)

0 -14 years of age or < 35 Kg of body weight

ABC+3TC+LPV/r

To review the patient at 7, 14 and 28 days at Sunshine clinic and repeat HIV Elisa at 3 and 6 months after incident

20.2.2 Emergency contraceptive

For females aged 8 years and above, either of:

- Postinor 2 tablets in a single dose
- Microgynon or Nordette 4 tablets stat then repeat after 12 hours

20.2.3 Sexual reproductive tract infection treatment

Azithromycin 20 mg/Kg once daily for 3 days

20.2.4 Vaccinations

Hepatitis B vaccination if not previously immunized

Diphtheria-Tetanus toxoid for any physical injury of skin or mucous membranes

Table 44: Post Rape Care Algorithm

Outpatient Emergency Management

- Escort patient to a private room and alert doctor on call for urgent review *do not refer to another health facility*
- Clinical Evaluation and Documentation, fill PRC 1(Post Rape Case 1) form in triplicate (white to survivor, yellow attach to P3 form and green file as hospital copy)
- Post Exposure prophylaxis, STI treatment
- Emergency Contraceptive, vaccinations

Counseling at First Contact

- Alert Clinical Psychologist/nurse counsellor on call
- Trauma and crisis counseling required at first contact at time of reporting
- Clear documentation of counseling done, and persons present at time of reporting

Laboratory

- Post rape Kit - gloves, swabs, plastic bags, glass slides for preparation of specimen mounts, sanitary pads and a speculum, tape measure
- HIV Elisa test to be done at first contact, to repeat at 6, 12 and 24 weeks after first test
- Investigations - LFTs, urinalysis, VDRL, Hepatitis BsAg, FHG, UECs

Report to Police

1. Record to be made in occurrence book and survivor given an OB number
2. Record statement and sign it
3. Survivor will be provided with P3 form to be filled and signed by the Police Surgeon/doctor

Ongoing Follow up at 2 Weeks and 4 Weeks at Sunshine Smiles Clinic

Follow up with post exposure prophylaxis adherence and post exposure HIV test results and lab results. Fill PRC 2 form. Counseling

21. REFERRAL TO PSYCHIATRY CLINIC

1. Declining or persistently poor academic performance
2. Persistent boredom, irritability or low mood (more than 2 weeks)
3. Frequent symptoms that cannot be medically explained especially headaches and stomach-aches.
4. Sleep disturbances including too much sleep; very little sleep, persistent nightmares, sleep walking and night terrors.
5. Regression of milestones especially bedwetting, clingy behaviour or tantrums
6. Aggressive behaviour or persistent defiance to authority (Teachers, parents, Nannies etc.)
7. Self-harm including deliberate cutting and burning self. (Always examine teenager's arms and thighs)
8. Eating disorders with significant weight or behavioural changes
9. Relationship and socialization challenges including persistent fighting with siblings, vulgar language, isolating self, school expulsions and bullying.
10. Substance Use in anyone below 18 years of age including alcohol, cigarettes, cannabis, cough syrups etc.
11. Suicidal thoughts, intents, plans or attempts.
12. Extreme hyperactive behaviour

13. Conversion Disorder may warrant admission for thorough investigation to rule out neurological conditions but NOT factitious disorder/ malingering

Note:

- Do not ignore any vocalised suicide threat
- Admission criteria includes: - A patient with a potential risk of harm to themselves or to others after your preliminary assessment, Suspected sexual, physical, emotional abuse in home environment or for a medical reason.
- Refer for outpatient evaluation (specialist clinic) if you are unable to explain to the parent your findings.
- Teenagers undergoing urine toxicology testing for substances of abuse should void under direct observation of a staff member of the same gender and positive results referred for further management.

22. ANIMAL BITES

Animal bites are common with children being bitten more often than adults. Annually, most animal bites are caused by snakes (4.5 million), dogs (2.4 million), with the remainder caused by cats and rodents.

The most feared complications of animal bite are envenomation, rabies, and skin infection. Some bite wounds can be serious, causing injury and permanent disability and even amputations. Bite wounds to the hand carry an especially high risk for serious complications because the skin surface is so close to the underlying bones and joints.

22.1 OVERVIEW OF MANAGEMENT

22.1.1 ASSESSING FOR RABIES EXPOSURE

Anyone who is bitten by an animal needs immediate medical attention, even if the bite is small and does not appear to be infected. The animals can be carriers of rabies, and post-bite rabies medications should be started as soon as possible. The offending cat or dog should be quarantined and observed for 10 days for any sign of illness or unusual behaviour.

22.1.2 ANIMAL BITE TREATMENT

After being bitten by an animal or human, it is important to immediately clean the wound thoroughly with soap and a large

amount of water to reduce the risk of infection. If there is bleeding, a clean towel or gauze should be pressed to the wound to slow or stop the bleeding.

It is best to be evaluated and treated as soon as possible after being bitten to reduce the chance of developing an infection. This is particularly true if the bite occurs in developing countries where animal vaccination programs to prevent rabies are not prevalent.

It is not necessary evaluated anyone after a scratch from a domestic cat, unless there are signs or symptoms of infection. A scratch from a stray cat that attacked without provocation, should be considered a high risk for rabies.

22.1.2.1 Antibiotics

The most common complication of an animal bite is infection. Antibiotics are generally recommended to prevent infection in people with high-risk wounds, such as deep puncture wounds, wounds involving a bone or joint, and those with immunosuppression.

22.1.2.2 Diphtheria-tetanus immunization

Tetanus is a serious, potentially life-threatening infection that can be transmitted by an animal or human bite. The vaccination status should be updated – receive a diphtheria-tetanus vaccine if less than three doses of the vaccine have been received, or the last dose was given more than 5 years before.

If the wound is contaminated, tetanus immune globulin is recommended to provide additional protection against tetanus infection.

22.1.2.3 Rabies immunization

This is recommended for a victim bitten by an animal that could be infected with rabies, either stray, wild or unprovoked. This is given days 0, 3, 7 14 and 28. The last two doses are omitted if the animal is proven not to be rabid or remains healthy after 10 days from the time of the bite.

22.1.2.3 Wound Repair

Some wounds can be sutured within several hours of the injury, after the wound is thoroughly cleaned. Wounds to the face are usually closed immediately to avoid developing a scar.

However, due to the risk of infection, some bite wounds may not be sutured immediately. These wounds may be left open and sutured 72 hours after the injury. When wounds are left open, they are flushed with saline, left open to drain, covered with a dressing, and examined daily to look for signs of infection. These include crush injuries, puncture wounds, hands, face, presentation several hours after the bite, or for immunosuppressed patients.

An x-ray may be done to ensure that there is no bone fracture and that tooth fragments are not buried in the wound.

22.2 TYPES OF ANIMAL BITES

22.2.1 *Snake bites*

Although snakes bite humans, envenoming cause considerable morbidity and mortality. Poisoning lead to many deaths with many survivors getting amputations and other severe health consequences, such as infection, tetanus, scarring, contractures, and psychological sequelae. Poor access to health care and scarcity of antivenom increases the severity of the injuries and their outcomes.

Most snake bites occur in Africa and South-East Asia among people living in rural, resource-poor settings, who subsist on low-cost, non-mechanical farming and other field occupations.

Approximately 600 species of snake are venomous and approximately 50-70% of bites by this cause envenomation. At the time of a bite, the cornerstone of care is complete immobilization of the affected body part and prompt transfer to a medical facility. Tourniquets and cutting wounds can worsen the effects of the venom and should not be used as first aid.

Victims of bites from poisonous snakes require treatment with antivenom specific to the snake species. Where this is not known, a multivalent antivenom is to be used to avoid wasting valuable time getting the specific antivenom. Additional measures include wound cleansing to decrease infection risk, supportive therapy such as airway support, and updating of the diphtheria-tetanus vaccination.

Prevention of snake bites involves informing communities about snake bite risks and prevention techniques, such as to:

- avoid tall grassy areas;
- wear protective shoes/boots;
- keep storage areas clear of rodents;
- remove rubbish, woodpiles and low brush from around the home;
- store food in rodent-proof containers, raise beds above floor level and tuck mosquito nets securely under sleeping mats within the home.

To prevent or limit the serious health consequences of snake bites, health-care providers should be educated on snake-bite management, including the proper use and administration of antivenom.

Prehospital management of snake bite

- Reassure the victim (70% of all snakebites are by nonvenomous snakes and 50% of bites by venomous species are dry bites)
- Immobilize the affected limb (by bandage or clothes to hold splint, but tight arterial compression is not recommended)
- Promptly transfer of victim to hospital

Hospital management of snake bites

- An assessment of the airway, breathing, circulatory status, and consciousness is made with interventions as may be found necessary.
- Emergency resuscitation will be needed in those in shock (cardiovascular toxicity), those with respiratory failure

(neurotoxin), and in those who have had cardiac arrest (due to hypoxia, cardiac toxicity, or hyperkalemia from rhabdomyolysis).

- Try confirming if the victim was bitten, and if so confirm if it is by a snake or another animal. Establish the time from the bite
- Obtain history on tetanus vaccination status, any other systemic disease and allergy history.
- Examine the local area as well as pulses distal to the bite area and complete the systemic examination.

Anti(-snake) venom

Antivenom should be ideally administered within 4 hours of the bite but is effective even if given within 24 h.

The recommended initial dose is 20 mL by intravenous slow push or as an infusion in 250 mL of normal saline over 1 hour.

The dose can be repeated in 1 hour if symptoms persist or worsen

22.2.2 Dog bites

Victims of dog bites frequently know the dog that attacked them. Most dog bites occur in children, with the highest number seen in boys between the ages of five and nine years old. The head and neck are the most common site of bites in children up to age 10 years, probably because a child's head is close to the level of a large dog's mouth. The arms and legs, particularly the right hand, are the most frequent site of injury for older children. A dog bite can lead to a range of injuries, including scratches, deep open cuts, puncture wounds, crush injuries, and tearing away of a body part. Dog bites rarely cause death.

Management of dog bites include early irrigation and cleaning of the wound, primary closure for a low-risk wound, prophylactic antibiotics, rabies post exposure treatment and updating of the diphtheria-tetanus vaccination.

22.2.3 Cat bites

Cats can cause wounds with their teeth or claws. Two-thirds of cat bites involve the upper extremities (arms and hands). Scratches typically occur on the upper extremities or face.

Deep puncture wounds are of particular concern because cats have long, slender, sharp teeth. When the hand is bitten, bacteria can get into the tissue that surrounds the bones or into a joint and result in osteomyelitis or septic arthritis.

If infection occurs, it generally progresses rapidly, causing skin redness, swelling, and intense pain as quickly as 12 to 24 hours after the bite.

22.2.4 Rodent bites

Rats cause most rodent bites. Most bites occur at night on the face or hands in children five years old or younger.

22.2.5 Monkey bites

These are an important risk to travelers to areas where monkeys are present like animal parks, zoos and forests. The management of monkey bites is like that of dog bites including the risk of rabies.

22.2.6 Human bites

Children are the most common victims of human bites, usually because of aggressive play with another child. Human bites can

cause a semicircular or oval area of skin redness or bruising, and the skin may be punctured. Human bites are typically located on the face, upper extremities, or trunk.

22.2.7 Other types of bites

The bites of most other animals, such as squirrels, rabbits, and guinea pigs, are generally treated the same way as cat bites

23. PROCEDURES

23.1 ANALGESIA AND SEDATION

Analgesia and behavioural strategies should be used in all children prior to painful procedures, with sedative drugs added where necessary.

23.1.1 Analgesic options

23.1.1.1 *Paracetamol*

- Neonate under 32 weeks postmenstrual age: 15 mg/Kg 12hourly (max. 30 mg/Kg/day)
- Under 3months: 15 mg/Kg 6 – 8hourly orally or PR (max. 60 mg/Kg)
- Children above 3 months: 15 mg/Kg 4 -6hourly orally or PR (max. 90 mg/Kg/day)

23.1.1.2 *Non-Steroidal Anti-Inflammatory Drugs*

- Ibuprofen 2.5 - 10 mg/Kg 6-8hrly orally (max 30 mg/Kg/day)

23.1.1.3 *Opioids*

- Codeine 0.5 - 1 mg/Kg/dose 6hourly orally
- Pethidine 0.5–2 mg/Kg/IM or 0.25 - 0.5 mg/Kg IV
- Morphine 0.1 mg/Kg IM or 0.05 - 0.1 mg/Kg IV
- Cautions:
 - Use IV if older child/adolescent with easy intravenous access.

- Titrate boluses (i.e., give half of the dose first to see effect, then repeat at 5 - 10-minute intervals as required up to maximum total dose).
- Respiratory depression risk - reduce doses if combined with sedatives. May get delayed respiratory depression after treating cause of pain.
- Multiple trauma - do not withhold but give with caution, particularly if hypovolemic.

23.1.1.4 Sucrose

- Oral sucrose reduces pain in infants less than three months of age during minor procedures
- Oral sucrose may be given to infants during procedures such as blood collection, IV insertion, eye examination and lumbar puncture
- Sucrose may be more effective if given with a dummy as the dummy promotes non-nutritive sucking which contributes to calming
- Other strategies, which assist in calming infants and can be used as an adjunct to sucrose, include feeding (if allowed), cuddling, and wrapping
- Other appropriate local or systemic analgesic agents should be administered as required
- Note that:
 - Sucrose is only effective if given orally. There is no effect if given via an oral or nasogastric tube
 - The addition of non-nutritive sucking enhances the analgesic effect of sucrose
- Dosage is as follows:
 - Maximum 2 mL Sucrose 25 – 33% (0.5 mL for infants below 1500 grams) administered orally for each procedure

- Two minutes prior to a painful procedure, administer a small amount (around 0.25 mL) of sucrose onto the infant's tongue. Offer a dummy if this is part of the infant's care
- Continue giving remainder of sucrose slowly during the procedure for a total dose of 2 mL, until the procedure is completed

23.1.2 Specific uses of Analgesia

23.1.2.1 IV insertion

Topical anaesthetic cream for 30 - 45 min; distraction e.g., calico dolls

Oral sucrose combined with non-nutritive sucking for infants under three months

23.1.2.2 Lumbar puncture

Topical anaesthetic cream for 30 - 45mins, 1% Lignocaine with 25 G needle

Oral sucrose combined with non-nutritive sucking for infants under three months

23.1.2.3 Removal of nasal/pharyngeal foreign body

Topical Phenylephrine (0.5%) and Lignocaine spray or nebulised Lignocaine (1.0%) 1 mL in 3 mL 0.9% Saline

23.1.2.4 Earache

Topical 1% Lignocaine 2 drops

23.1.2.5 Eye

Topical Amethocaine 0.5% to examine; patch +/- atropine (to relieve iris spasm)

Oral sucrose combined with non-nutritive sucking for infants under three months

23.1.3 Midazolam Sedation

Indications for sedation with midazolam include procedures such as suturing, removal of foreign body among others.

Remember that midazolam is a sedative and has no analgesic effect, therefore for painful procedures, add an analgesic.

23.1.3.1 Dosage

Intravenous Midazolam by IV injection over 2 - 3 minutes given 5 - 10 minutes before procedure

- Child 1 month 6 years, initially 25 - 50micrograms/Kg, increased in small steps (max. total dose 6 mg)
- Child 6 - 12 years initially 25 - 50 micrograms/Kg, increased, if necessary, in small steps (max. total dose 10 mg)
- Child 12 - 18 years initially 25 - 50 micrograms/Kg, increased, if necessary, in small steps (max. total dose 7.5 mg)

Oral /buccal Midazolam 0.5 - 1.0 mg/Kg. (max. 20 mg). Draw up solution (5 mg/mL) in a 1- or 2-mL syringe. May be mixed with small volume of cordial to disguise acid taste.

Intranasal Midazolam 0.6 mg/Kg (max 10 mg) - may have more rapid onset

23.1.3.2 Cautions

- Observe the child in Emergency until fully alert (usually recover by 60 minutes).
- Midazolam will potentiate the effects of other sedative drugs e.g., opiates.
- Midazolam should not be given to children with pre-existing respiratory insufficiency, or neuromuscular problems such as myasthenia gravis.

23.1.4 Other options

- Regional anaesthesia
- Systemic analgesics – nitrous oxide

Involve pain management team.

23.2 LUMBAR PUNCTURE

Lumbar puncture may be performed as part of the initial work up of a sick child, or later in the course of an illness once the child has stabilised if there were initial contraindications

It is preferable to obtain a CSF specimen prior to antibiotic administration; however antibiotic administration should not be unduly delayed in a child with signs of meningitis or sepsis

23.2.1 Indications

- Suspected meningitis or encephalitis
- Suspected sub-arachnoid haemorrhage with a normal head CT scan

23.2.2 Contraindications

Do not do a lumbar puncture (LP) if the child is so sick that you would give antibiotics for meningitis even if the CSF is normal

The following clinical findings suggest that you should give dexamethasone and antibiotics immediately, and delay lumbar puncture for 1 - 2 days until the child is improving:

- Coma: absent or non-purposeful response to painful stimulus - squeeze earlobe hard for up to one minute. A child over 3 months of age should push you away and seek a parent
- Signs of raised intracranial pressure like abnormal pupillary responses, unilateral or bilateral motor posturing or papilloedema (NB papilloedema is an unreliable and late sign of raised ICP in meningitis; a bulging fontanelle in the absence of other signs of raised ICP, is not a contraindication)
- Cardiovascular compromise/shock
- Respiratory compromise
- Focal neurological signs or seizures.
- Recent seizures (within 30 minutes)
- Coagulopathy/thrombocytopenia
- Local infection (in the area where an LP would be performed).
- The febrile child with purpura where meningococcal infection is suspected

23.2.3 Assessment prior to lumbar puncture

- A thorough physical and neurological examination is needed
- Head CT scans should be done if there are focal neurological signs
- CT Scans are not helpful in most children with meningitis.
- A normal CT scan does not tell you that the patient does not have raised intracranial pressure
- Herniation may occur even in the presence of a normal scan

23.2.4 Complications

Informed verbal consent should be obtained. Complications of LP may include:

- Failure to obtain a specimen/need to repeat LP/Traumatic tap (common).
- Post dural puncture headache (common) - up to 5 - 15%.
- Transient/persistent paraesthesia/numbness (very uncommon).
- Respiratory arrest from positioning (rare).
- Spinal haematoma or abscess (very rare).
- Tonsillar herniation (extremely rare in the absence of contraindications above).

23.2.5 Analgesia, anaesthesia and sedation for Lumbar Puncture

Non-pharmacological techniques should be used where possible, including explanation (in an older child), distraction, and the presence of a parent

All children should have some form of local anaesthetic for lumbar puncture

- Use topical anaesthetic cream except where specimens are required urgently
- Subcutaneous Lignocaine may be used in addition to or instead of topical anaesthetic
- Use up to 0.4 mL/Kg of 1% Lignocaine (4 mg/Kg)
- Oral sucrose 25 – 33% should be used for infants < 3 months
- Sedation, including nitrous oxide, should be considered for children older than 6 months with normal conscious state

23.2.6 Monitoring:

Monitor all sedated or seriously ill children with continuous pulse oximetry. All vital signs should be checked quarter hourly till child is fully awake.

23.2.7 Equipment

- At least one trained assistant to hold the child
- Sterile gloves
- Sterile drapes and procedure tray
- Skin preparation: Povidone iodine solution or Chlorhexidine
- Local anaesthetic Lignocaine, 2-mL syringe, 25G needle
- CSF tubes (2)
- Spinal needle

23.2.7.1 Spinal Needles

- 22 G bevelled spinal needles with stylet (the use of needles without a stylet has an associated risk of spinal epidermoid tumours)
- Consider 25 G pencil point needles for older children or adolescents
- Pencil point (Blunt) needles) reduce the risk of headache in adults, however the evidence is not convincing in children. Their use may be appropriate in adolescents

23.2.8 Procedure

The most important determinant of a successful lumbar puncture is a strong, calm, experienced assistant to hold the patient. Appropriate positioning of the patient is critical for successful lumbar puncture

23.2.8.1 Position

- Lumbar puncture may be performed with the child lying on their side or sitting up
- Aim for maximum flexion of the spine (curl into fetal position), but avoid over flexing the neck, especially in infants as this may cause respiratory compromise. Ask an adolescent to slouch rather than bend from their hips
- Ensure that the plane of the back is exactly at 90 degrees to the bed (i.e. not leaning towards or away from you). Make sure the hips and shoulders are in line
- Draw an imaginary line between the top of the iliac crests. This intersects the spine at approximately the L3-4 interspace (mark this if necessary)
- The conus medullaris finishes near L3 at birth, but at L1-2 by adulthood
- Aim for the L3 - 4 or L4 - 5 interspace

23.2.8.2 Preparation

- Wash hands and aseptically put on sterile gloves
- Prepare the skin with Povidone-iodine or chlorhexidine and set up sterile drapes
- Allow adequate time for the skin preparation to dry
- Take the tops off the tubes, ensuring that they remain sterile.
- Infiltrate the skin with 1% Lignocaine using a 25 G needle

23.2.8.3 Lumbar Puncture

- Position the needle in the midline with the bevel pointing towards the ceiling (lateral decubitus position) or to the side (sitting)
- Pierce the skin with the needle and pause. Wait for the child to stop wriggling if there is movement

- Re-orientate (ensure that back is vertical, needle is parallel to the bed and perpendicular to the back). Aim for the umbilicus (i.e. slightly cephalad)
- Advance the needle into the spinous ligament (increased resistance). Continue to advance the needle within the ligament until there is a fall in resistance. Stop when you feel a slight give. Remove the stylet.
- If CSF is not obtained replace the stylet and advance the needle slightly then recheck for CSF
- An alternative technique is to remove the stylet once the needle is in the ligament and advance very slowly without stylet watching for CSF to flow back. This has the advantage of making it harder to go unintentionally past the subarachnoid space
- If the needle meets resistance, withdraw the needle slowly whilst watching for CSF. If none is obtained, replace the stylet, re-orient the needle and try again
- If blood-stained fluid is obtained collect some for culture. If it clears it can be used for a cell count. If it fails to clear another attempt at a different level may be required
- If CSF is flowing, collect into 2 numbered sterile tubes (5-10 drops each is usually adequate)
- Replace the stylet (this may reduce risk of headache) and remove the needle and stylet
- Apply brief pressure to the puncture site
- Send specimens urgently to the lab for microscopy, protein, glucose, culture, bacterial antigen tests, and viral tests if encephalitis is suspected

23.2.9 Post-Procedure Care

- Cover the puncture site with a band-aid or occlusive dressing. The dressing should be removed, and the LP site checked after 24 hours.
- Bed rest following lumbar puncture is of no benefit in preventing headache in children or adults

23.3 SUPRAPUBIC ASPIRATE

Suprapubic aspiration is the gold standard for obtaining urine specimens for culture. Any growth of pathogenic bacteria in a suprapubic aspirate specimen is considered significant

23.3.1 Indications

Any child (regardless of age) who is unable to void on request, who requires a urine specimen for the diagnosis or exclusion of urinary tract infection

23.3.2 Contraindications:

- Bleeding diathesis
- Abdominal distension
- Massive organomegaly

23.3.3 Complications

Complications may include:

- Macroscopic haematuria (infrequent — not usually clinically significant)
- Bladder haematoma (rare)
- Bladder haemorrhage (very rare)
- Intestinal perforation (rare - not usually clinically significant)

- Anaerobic bacteraemia or abscess formation (very rare)

23.3.4 Equipment

- One assistant to hold the infant (not parent)
- Specimen jar for urine
- 23G needle (25G for premature infants)
- 5-mL syringe

23.3.5 Analgesia, Anaesthesia, Sedation

- Topical anaesthetic cream should be used except where specimens are required urgently (e.g., prior to starting antibiotic treatment in a septic infant)
- Oral Sucrose should be used for infants <3 months
- Sedation should be considered for children older than 6 months especially where several procedures are required (e.g., lumbar puncture, IV cannulation)
- Nonpharmacological methods should be used where possible, including explanation (in an older child), distraction, and the presence of a parent

23.3.6 Procedure Rules

- Never undo the nappy until you have a urine jar handy and someone ready to catch
- Do the suprapubic aspirate before collecting blood or CSF as the child may void while having venepuncture/lumbar puncture
- The use of ultrasound increases the chance of success. Ultrasound does not tell you where to put the needle - only whether there is likely to be enough urine present
- If no bladder ultrasound available:
 - History of no voiding in the past 30 minutes, and the presence of a dry nappy increases the chance of a successful tap

- Pre-hydration increases the chance of a successful blind tap.
- If bladder is dull to percussion, there is a higher chance of successful aspiration

23.3.7 Procedure

- Ask assistant to hold infant supine with legs extended
- Ask parent to be ready to catch urine if the patient voids
- Wipe the skin with an alcohol swab
- Identify the puncture point
 - Midline
 - Lowest abdominal crease
- Insert needle perpendicular to the skin, aspirating gently as you advance the needle
- If no success, withdraw the needle to just under the skin, and advance at an angle with the needle aimed more away from the pelvis
- If urine is obtained, remove needle and squirt urine into sterile urine jar

23.3.8 Post-Procedure Care

- Place a band aid over the puncture site (optional)
- Warn parents that there may be a small amount of blood in the urine in the next day, but that they should return if there are large amounts or if they are concerned

23.4 NEEDLE THORACOCENTESIS

23.4.1 Indications

- Primary spontaneous pneumothorax
- Tension pneumothorax

23.4.2 Relative contraindications

Thoracocentesis should be considered only in consultation with a senior emergency physician in the following:

- Spontaneous pneumothorax in patients with underlying lung disease
- Traumatic pneumothorax without tension

23.4.3 Equipment

- Dressing pack
- Aspiration device
- Large bore cannula (12 or 14 gauge)
- Central venous catheter (CVC) or Pigtail catheter are alternatives
- 20-mL or 50-mL syringe
- 3-way tap
- Antiseptic solution
- 1% Lignocaine

23.4.4 Analgesia, Anaesthesia, Sedation

Analgesia and local anaesthesia are mandatory except with tension pneumothorax, which is immediately life-threatening.

- Use topical local anaesthetic
- Consider oral or parenteral analgesia pre- and post-procedure

23.4.5 Procedure

- Place patient on continuous cardiac monitoring and pulse oximetry
- Place trauma patient in a head-up, supine position
- All other patients should be placed in 45-degree, sitting position

- Palpate landmark (the upper border of the 3rd rib in the midclavicular line) and antiseptically prepare the area
- Attach a 5-mL syringe to the catheter device
- Puncture the skin at the level of above landmark
- Carefully insert the needle at a slightly downwards angle into the pleural space while aspirating the syringe
- In tension pneumothorax, often you will feel a pop or a change in resistance
- Withdraw the needle while gently advancing the cannula downwards into position
- Secure cannula/CVC with tape
- Attach 3 way tap and 50-mL syringe
- Drain until no further drainage or to a maximum of 30 mL/Kg (maximum 2.5 L)

23.4.6 Post-Procedure Care

- Reassess ABCs
- Consider need for further analgesia
- Plan for chest drain (intercostal catheter insertion) in patients with tension pneumothorax
- Organise appropriate patient placement post procedure

APPENDICES

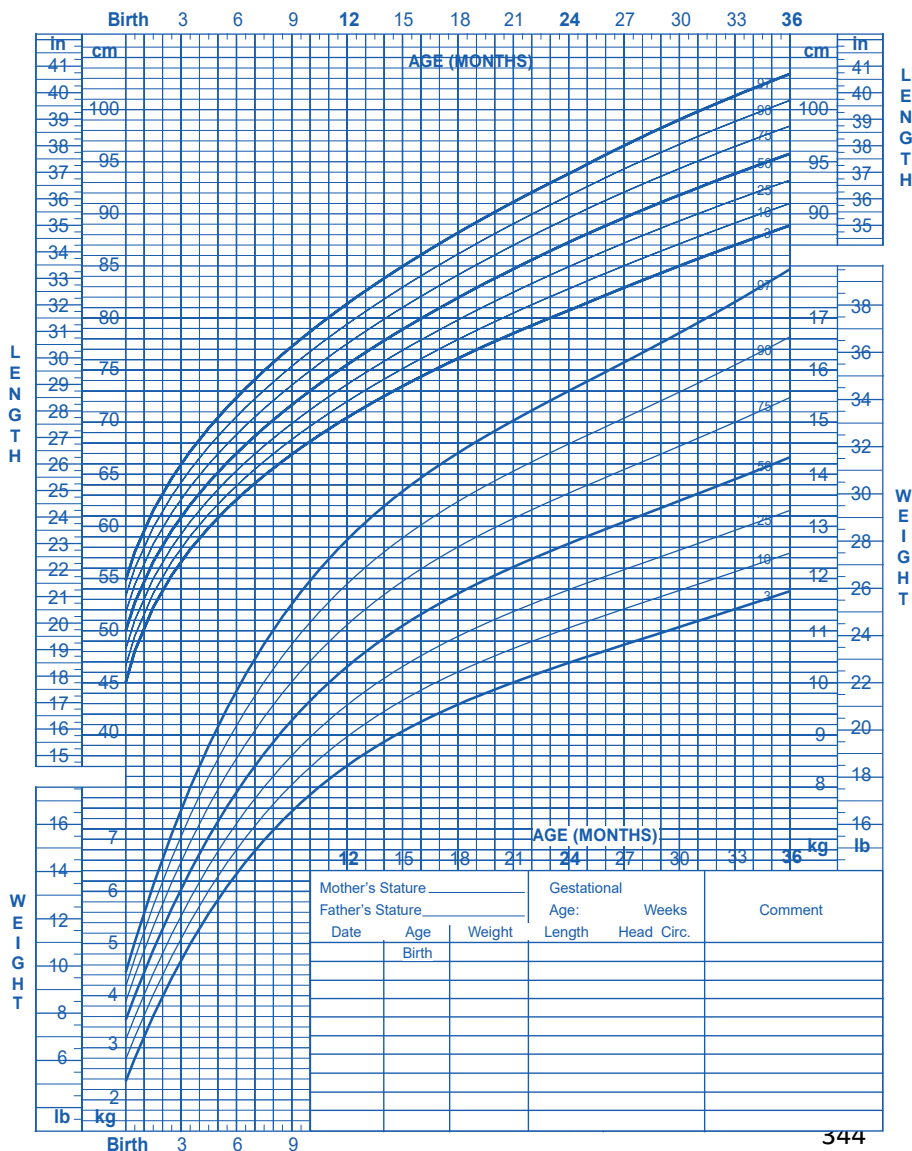
APPENDIX I: GROWTH CHARTS

Birth to 36 months: Boys

Length-for-age and Weight-for-age percentiles

NAME _____

RECORD # _____



Published May 30, 2000 (modified 4/20/01).

SOURCE: Developed by the National Center for Health Statistics in collaboration with the National Center for Chronic Disease Prevention and Health Promotion (2000).
<http://www.cdc.gov/growthcharts>



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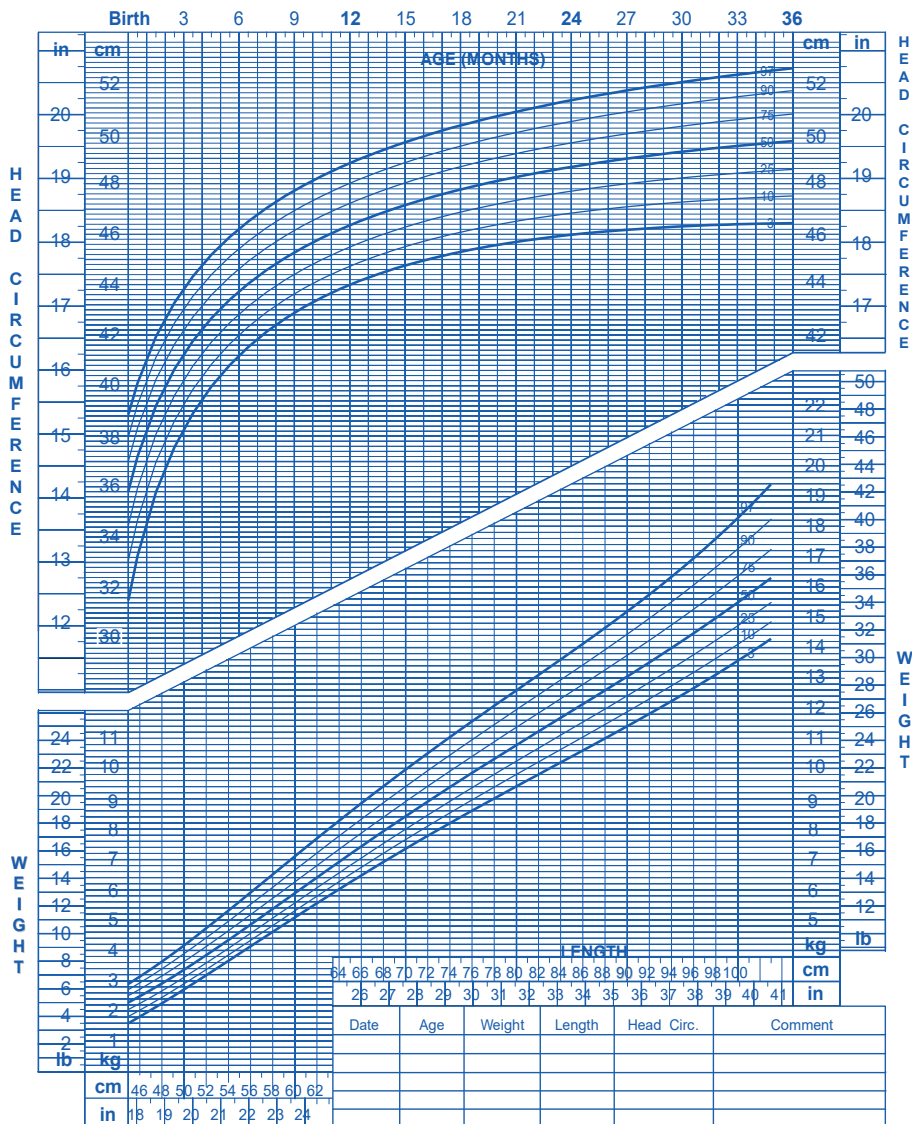
RECORD # _____



Birth to 36 months: Boys
Head circumference-for-age and
Weight-for-length percentiles

NAME _____

RECORD # _____



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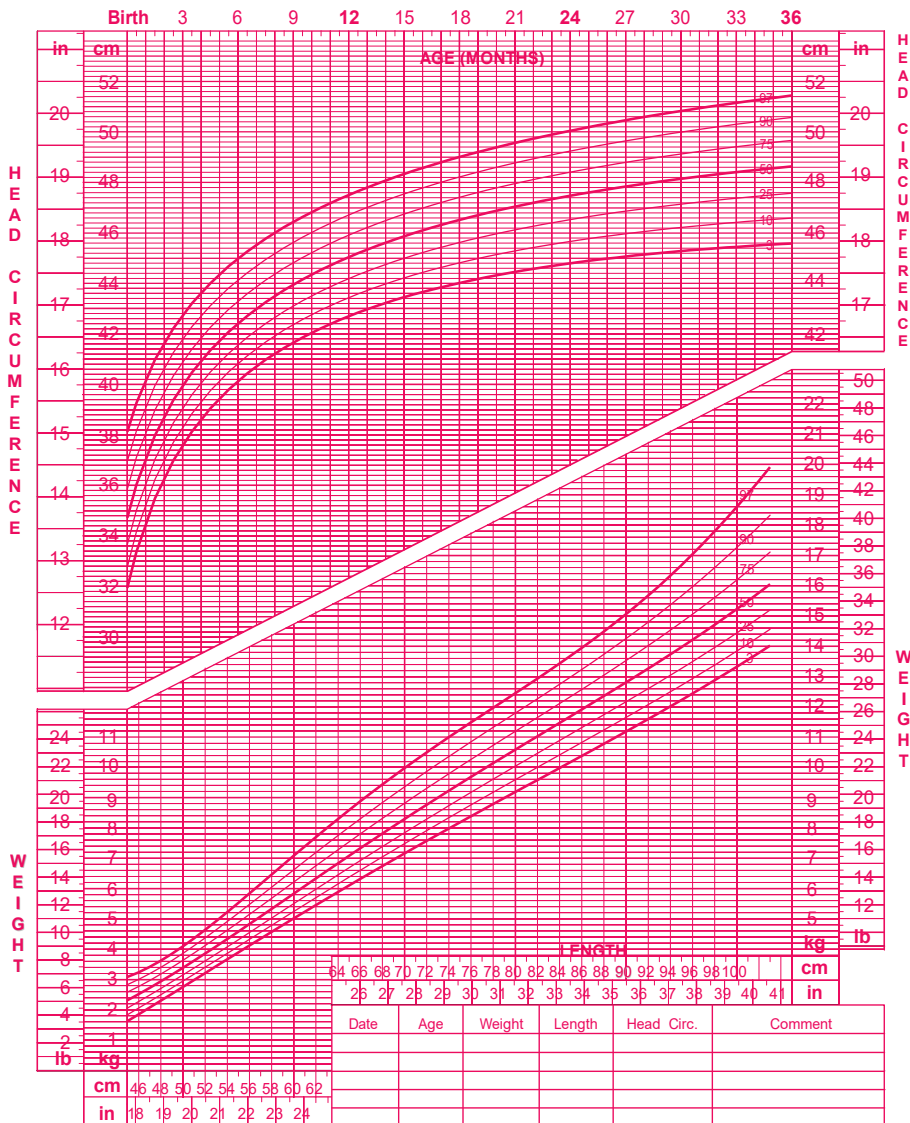
Birth to 36 months: Girls

Head circumference-for-age and

Weight-for-length percentiles

NAME _____

RECORD # _____



Published May 30, 2000 (modified 10/16/00).

SOURCE: Developed by the National Center for Health Statistics in collaboration with the National Center for Chronic Disease Prevention and Health Promotion (2000).
<http://www.cdc.gov/growthcharts>



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Stature-for-age and Weight-for-age percentiles

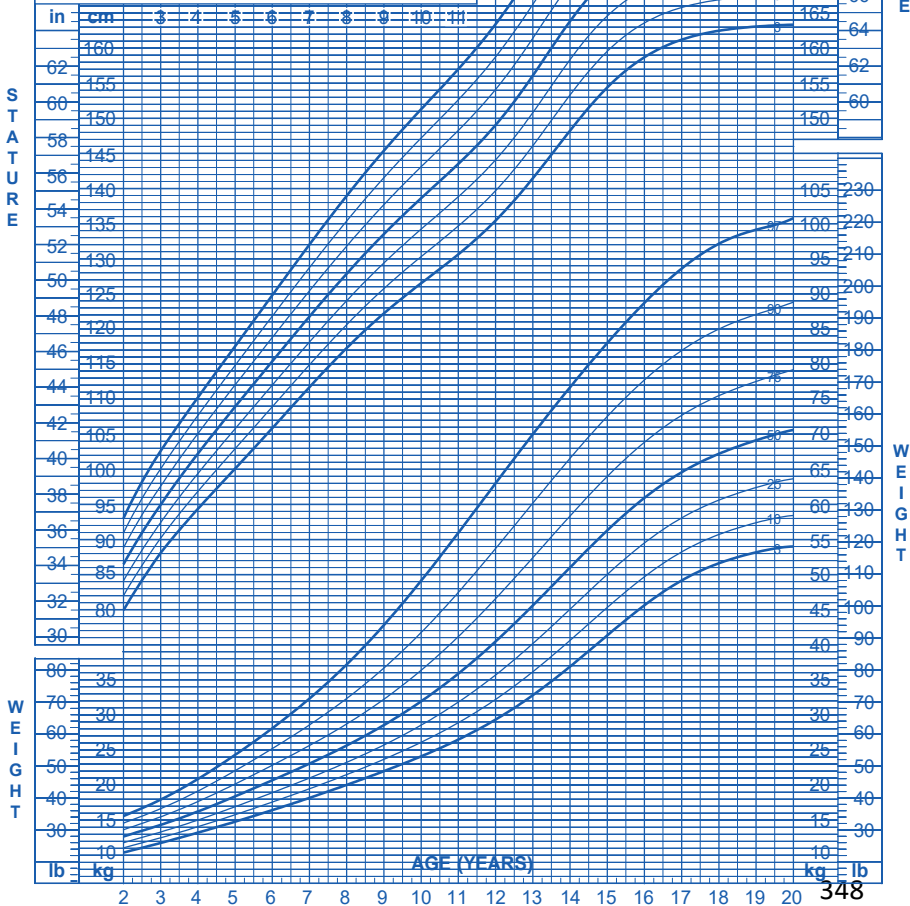
NAME _____

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12 13 14 15 16 17 18 19 20

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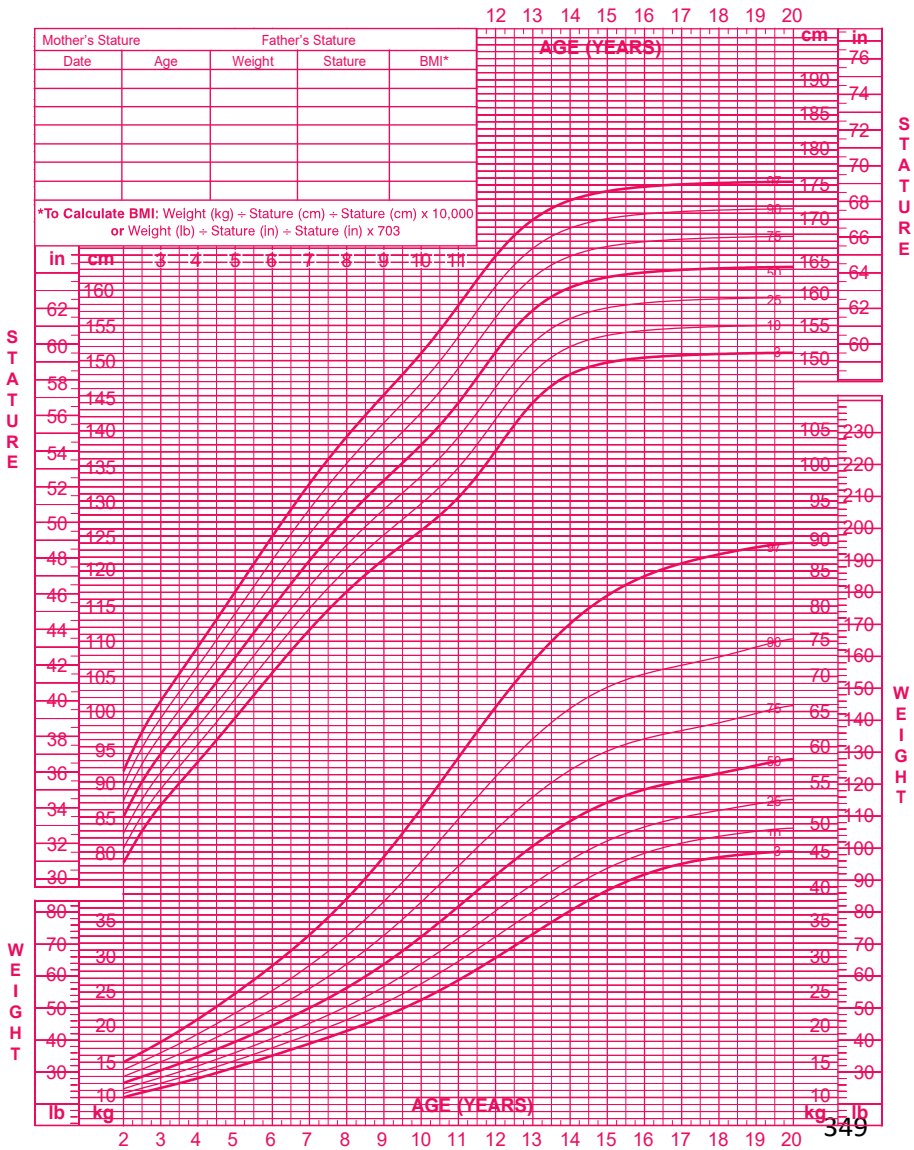
***To Calculate BMI:** Weight (kg) ÷ Stature (cm) ÷ Stature (cm) x 10,000
or Weight (lb) ÷ Stature (in) ÷ Stature (in) x 703



Stature-for-age and Weight-for-age percentiles

NAME _____

RECORD # _____



Published May 30, 2000 (modified 11/21/00).

SOURCE: Developed by the National Center for Health Statistics in collaboration with the National Center for Chronic Disease Prevention and Health Promotion (2000).

<http://www.cdc.gov/growthcharts>



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RECORD # _____

***To Calculate BMI:** Weight (kg) ÷ Stature (cm) ÷ Stature (cm) x 10,000
or Weight (lb) ÷ Stature (in) ÷ Stature (in) x 703



RECORD # _____

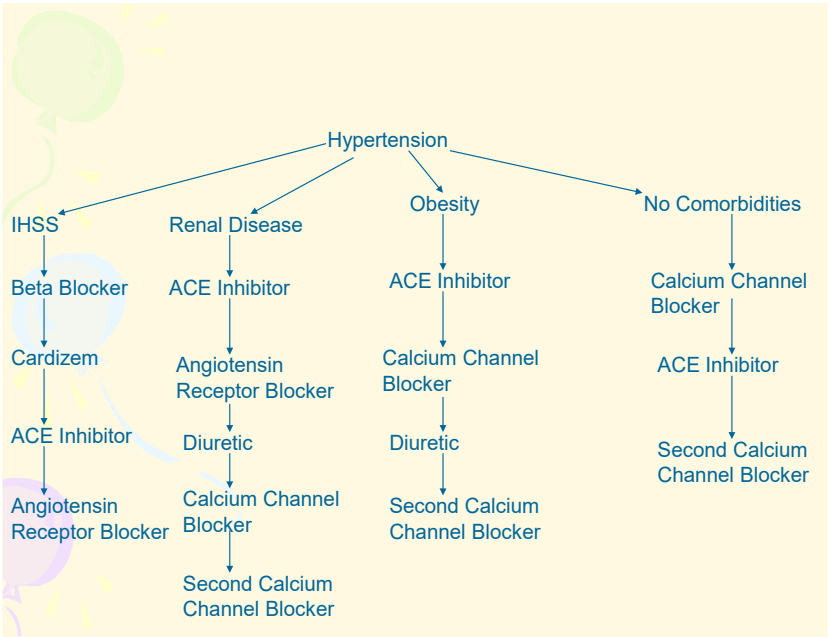
Published May 30, 2000 (modified 10/16/00).
SOURCE: Developed by the National Center for Health Statistics in collaboration with the National Center for Chronic Disease Prevention and Health Promotion (2000).
<http://www.cdc.gov/growthcharts>

APPENDIX II: BLOOD PRESSURE REFERENCES

- Formulas for high blood pressure 95th centile for age 2 – 6 years
 - Systolic = 100 mmHg + (3 x age in years)
 - Diastolic = Calculated systolic minus 20mmHg
- Formulas for normal blood pressure 50th centile for age 1-10 years
 - Systolic = 90 mmHg + (2 x age in years)
 - Diastolic = 70 mmHg + (2 x age in years)

Age	HR/min	RR/min	BP mmHg
Preterm	100 -180	40-60	50-60/35-45
Term newborn	100 -180	40-60	60-90/40-45
0 -1 mo.	93 -182	26 -65	45 – 80/33 – 52
1 – 3 mo.	120 – 178	28 – 55	65 – 85/35 – 55
3 – 6 mo.	107- 197	22 – 52	70 – 90/35 – 65
6 – 12 mo.	108 – 178	22 – 52	80 – 100/40 – 65
1 - 2 yrs.	90 – 152	20 – 50	80 – 100/40 – 70
2 – 3yrs.	90 – 152	20 – 40	80 – 110/40 – 80
3 – 5 yrs.	74 – 138	20 – 30	80 – 115/40 – 80
5 – 7 yrs.	65 – 138	20 – 26	80 – 115/40 – 80
8 – 10 yrs.	62 – 130	14 – 26	85 – 125/45 – 85
11 – 13 yrs.	65 – 130	14 – 22	95 – 135/45 – 85
14 – 18 yrs.	62 – 130	12 – 22	100 – 145/50 - 90

Hypertension Treatment Algorithm



APPENDIX III: ADMISSION CRITERIA

SUGGESTED GENERAL ADMISSION CRITERIA

- Serious bacterial infection requiring IV antibiotics – Meningitis, Pyelonephritis, Neonatal Sepsis, Severe Pneumonia, Osteomyelitis
- Oral feeding not desired or possible
- Intravenous fluid therapy
- A new infiltrate on chest x-ray in a patient with Sickle Cell Disease
- Oxygen requirement of FIO₂ more than 40% to maintain saturations of ≥95%
- Continuous Salbutamol nebulization in acute bronchospasm
- Worsening Asthma Or severe acute asthma
- Burns – more than 10%BSA, Perineum, hands, feet, face, joints. 3rd degree, Electrical, Chemical, Inhalational, co-morbidities,
- Questionable social support and unlikely to follow up
- Concerns about the parent's ability to monitor their child's condition and to provide appropriate care, consider admission to hospital
- Suspicion of child abuse
- High fever < 38.3 °C in a child less than 3months old and < 38.9 °C in an older child
- Altered mental status AVPU <A
- History of Apnoea

Suggested Admission Criteria for Neurological disease

- Status Epilepticus
- Any child having a **first convulsion** needs observation and exclusion of meningitis
- Children having **prolonged convulsions** exceeding 30 minutes
- **Recurrent seizures.** Children who have had more than one convulsion during an episode of illness
- **Focal neurological deficits** following a seizure
- Children with a **persistently altered level of consciousness** exceeding two hours after termination of the seizure
- Any Child who was **drowsy before** the seizure
- Children who have received **multiple doses of diazepam, intravenous Phenobarbitone or Phenytoin** for control of seizures
- Convulsion associated with **trauma.**
- Parents who are not reassured despite an apparent recovery after seizure
- Children with seizures living in **“inadequate / at risk”** home environments
- Convulsions with suspected **Non-accidental injury**

- Children known to have **poorly controlled** convulsive disorders
- Convulsions associated with **illnesses** that require management such as severe dehydration, Otitis media, Bronchopneumonia, and severe malnutrition etc.
- Recurrent vomiting with dehydration
- Head injury

Suggested Admission Criteria for Malaria

- Prostration (inability or difficulty to sit upright, stand or walk without support in a child normally able to do so, or inability to drink in children too young to sit)
- Alteration in the level of consciousness – AVPU less than A
- Cerebral malaria (unarousable coma not attributable to any other cause in patients with falciparum malaria)
- Respiratory distress (acidotic breathing)
- Multiple generalized convulsions (2 or more episodes within a 24-hour period)
- Circulatory collapse (shock, septicemia)
- Pulmonary oedema
- Abnormal bleeding (Disseminated Intravascular Coagulopathy)
- Jaundice
- Haemoglobinuria (black water fever)
- Acute renal failure – presenting as oliguria or anuria

- Severe anaemia (Hb <5g/dl or Hct < 15%)
- Hypoglycemia (blood glucose level < 2.2 mmol/l)
- Hyperparasitemia (parasitemia of < 3+ or < 1%)
- Hyperlactatemia
- For the patients sent home on antimalarials but on subsequent review found to be having:
Persistent vomiting or persistent fever

ICU AND HDU ADMISSION CRITERIA

Respiratory System

1. Endotracheal intubation (**PCCU**)
2. Severe Respiratory distress
3. Rapidly progressive respiratory failure (**PCCU**)
4. High supplemental oxygen requirement ($\text{FIO}_2 \geq 0.5$)
5. Newly placed tracheostomy
6. Acute barotrauma (**PCCU**)

Cardiovascular System

1. Shock (**PCCU**)
2. Post cardiopulmonary resuscitation
3. Life-threatening dysarrhythmias (**PCCU**)
4. Unstable congestive heart failure
5. Congenital heart disease with unstable cardio-respiratory status

6. After high-risk cardiovascular and intra-thoracic procedures **(PCCU)**
7. Need for monitoring of arterial, central venous, or pulmonary artery pressures
8. Need for temporary cardiac pacing

Central Nervous System

1. Seizures requiring continuous infusion of anticonvulsive agents **(PCCU)**
2. Acutely deterioration of consciousness **(PCCU)**
3. After neurosurgical procedures requiring invasive monitoring or close observation
4. Head trauma with increased intracranial pressure
5. Preoperative neurosurgical conditions with neurologic deterioration
- 6.
7. Progressive neuromuscular dysfunction **(PCCU)**
8. Spinal cord compression or impending compression
9. Placement of external ventricular drainage device

Hematology/Oncology

1. Exchange transfusions, Plasmapheresis or leukapheresis
2. Severe coagulopathy
3. Severe anemia resulting in hemodynamic and/or respiratory compromise
4. Severe complications of sickle cell crisis **(PCCU)**
5. Initiation of chemotherapy with anticipated tumor lysis syndrome
6. Tumors or masses compressing or threatening to compress vital vessels or airway **(PCCU)**

Endocrine/Metabolic

1. Severe diabetic ketoacidosis (**PCCU**)
2. Hyperkalaemia (**PCCU**)
3. Severe hypo- or hypernatraemia (**PCCU**)
4. Hypo- or hypercalcaemia (**PCCU**)
5. Hypo- or hyperglycemia requiring intensive monitoring
6. Severe metabolic acidosis (**PCCU**)
7. Complex intervention required to maintain fluid balance
8. Inborn errors of metabolism

Gastrointestinal

1. Severe acute gastrointestinal bleeding
2. After emergency endoscopy for removal of foreign bodies
3. Acute hepatic failure leading to coma, hemodynamic, or respiratory instability

Surgical

1. Cardiovascular surgery
2. Thoracic surgery
3. Neurosurgical procedures (**PCCU**)
4. Otolaryngologic surgery
5. Craniofacial surgery
6. Orthopedic and spine surgery
7. General surgery with hemodynamic or respiratory instability
8. Organ transplantation
9. Multiple trauma with or without cardiovascular instability (**PCCU**)

10. Major blood loss, either during surgery or during the postoperative period

Renal System

1. Requirement for acute dialysis, ARI
2. Acute rhabdomyolysis with renal insufficiency

Others

1. Toxic ingestions and drug overdose with potential acute decompensation of major organ systems
2. Multiple organ dysfunction syndrome (**PCCU**)
3. Suspected or documented malignant hyperthermia
4. Electrical or other household or environmental (e.g., lightning) injuries
5. Mild to moderate Burns covering >10% of body surface area. **Major Burns (see main text) require admission to a Burns Unit**

APPENDIX IV: RELIABILITY OF CORE MEASURES

ASTHMA

- History – classic asthma symptoms (any of recurrent or **persistent cough, wheeze, difficulties in breathing, tight chest**); **triggers** (weather, nocturnal/diurnal pattern, exercise, inhaled triggers)
- Examination – **SPO₂, RR** and interpretation. Chest auscultation findings (**wheeze, air entry vs silent chest**).
- Classification – **level of Control** (well controlled, partial control or uncontrolled), asthma exacerbation.
- Management – **Controller (ICS +/- LTRA OR LABA) AND Reliever med (SABA)**
- **Follow-up booking at chest clinic.** (newly diagnosed cases and poorly controlled cases have written plan)

PNEUMONIA

- History of **cough, fever and/or rapid breathing, or difficulty in breathing** ascertained and documented
- **Respiratory rate, SPO₂ Oxygenation assessment**/comment
- Given **appropriate antibiotics** (in line with pneumonia classification)

- **Supportive treatment given** (oxygen as needed, antipyretics) as appropriate
- Documented **Follow up within 48 hours**

MALARIA

- History of **Fever**
- History **Travel** to malaria endemic area documented
- **Severity** of disease with its criteria and complications documented
- **Appropriate Treatment** Given (AL for Non severe, Artesunate for Severe)
- **Supportive treatment** prescribed (Antibiotic in severe Malaria, Antipyretics)

ACUTE GASTROENTERITIS

- History of **diarrhoea and/or vomiting** documented
- Presence or absence of **fever or other systemic symptoms** documented
- **Degree of dehydration** documented with associated criteria if dehydrated
- Rational use of Fluids - ORS and/or IV Fluids
- Use of **Zinc Sulphate** documented if diarrhea present

OTITIS MEDIA

- **Pain** in the ear or **irritability or tugging on the ear** documented if appropriate

- Presence or absence of **fever or other systemic symptoms** documented
- Examination of the ear drum documented e.g., **erythema, bulging, opacity, dullness, discharge**
- **Appropriate antibiotics** and analgesics prescribed
- Documented **follow up review**

ACUTE TONSILOPHARYNGITIS

- History of **fever, painful swallowing or sore throat, foul breath, drooling**
- On examination: **fever, inflamed tonsils and/or inflamed pharynx presence or absence of exudates, with or without enlarged cervical nodes**
- Management: appropriate antibiotics and **analgesia**
- **Patient education** on medication compliance, when to return.
- **Follow-up** review planned and documented

ACUTE NASOPHARYNGITIS (COMMON COLD)

- History of cough, **runny nose, fever.**
- Examination: fever, **rhinorrhea**, pharyngeal inflammation
- Management: Symptomatic relief e.g., **antipyretic, nasal saline; Lots of fluids.**
- **No antibiotic use, No cough syrup/expectorants, No antihistamine**
- **Patient education:** duration of symptoms and danger signs

ACUTE GINGIVOSTOMATITIS

- **Ulcerations** around the gums, tongue and mouth, **reluctance to feed**
- Fever, drooling
- Foul odour from mouth
- Use of Topical remedies – antiseptic analgesic **oral gel**
- **Pain relief - Paracetamol**
- **Nutritional support** – Consider intravenous Fluids, eating soft, bland foods, avoid salty or hot food
- Use of **Aciclovir**
- Secondary Bacterial Infection to be treated with **CoAmoxiclav**

ATOPIC DERMATITIS

- History of **rash, dry skin, pruritus**, recurrence, and history of atopy in the patient/family.
- On examination: **distribution and characteristics of rash** e.g., weeping, crusting, lichenification
- etc.
- Management: **moisturize** the skin, first-line: avoid steroids
- Education: **avoidance of triggers**
- **Appropriate referral/follow up** of recurrent or persistent/severe cases

GASTROESOPHAGEAL REFLUX

- **Vomiting** or **bringing up feeds**
- **Poor weight gain** or **weight loss** or **failure to thrive**

- **Irritability or fussiness**
- management: **lifestyle modifications** (thickening of feeds, positioning, weight reduction for obese, smaller more frequent feeds)
- **Follow up** within 2 weeks/referral to gastroenterology

URINARY TRACT INFECTION

- Non-specific symptoms for the child less than 2 years old (**fever, vomiting, irritability, failure to thrive**)
- Urinary symptom for the older child (**dysuria, frequency, urgency, haematuria**)
- Abdominal examination (presence or absence of **suprapubic tenderness**, phimosis (males), costovertebral tenderness)
- **Urinalysis and culture** requested
- **Appropriate antibiotic** Prescribed
- **Follow up** after treatment (repeat urine after treatment)

CROUP (ACUTE

LARYNGOTRACHEOBRONCHITIS, LTB)

- History of **harsh or barking cough, stridor or hoarse voice**
- **Severity** of croup with grading
- Presence or absence of **drooling or toxic appearance**
- Rational use of **steroids and adrenaline**
- **Supportive treatment:** antipyretics

ADENOID "HYPERTROPHY" (ADENOIDS)

- History of **snoring, nasal blockage or mouth breathing**
- **Disturbed Sleep** (frequent awakening, sleep apnoea)
- Presence or absence of **complications** (hepatomegaly, Cardiac examination)
- **Supportive treatment:** Intranasal decongestants/Steroids, Antipyretics
- **Follow up** (referral or review)

ALLERGIC RHINITIS

- History of **stuffy/runny nose**, throat clearing, Atopy
- **Exacerbating factors** e.g., cold, dust, pollen
- Examination: **rhinorrhea**, hypertrophy of nasal turbinates, erythema of nasal mucosa
- Management: **saline drops**, antihistamine, intra-nasal steroids
- **Follow up** (referral or review)

CONSTIPATION

- History of **reduced frequency of bowel movements, hard consistency, encopresis, feeling of inadequate evacuation, pain on passing stool.**
- **Dietary history:** inadequate fibre intake, inadequate fluid intake, high intake of processed foods
- Examination of the abdomen for **fecal loading, rectal impaction (DRE)** and **examination of perianal area.**
- **Patient Education:** causes, dietary modifications, adjustment of bowel habits

- Management: **evacuation of bowel, stool softener for two weeks, follow up.**

MENINGITIS

- History: At least two of: **Fever, Reduced level of consciousness, Stiff neck, Bulging fontanelle, Convulsions, Vomiting**
- Examination: **Fever, Coma or AVPU < V, inability to drink / feed, Stiff neck, Positive Kernig's sign, Bulging fontanelle**
- Plan of Care: **Admit, Lumbar Puncture +/- Head Imaging (MRI or CT scan), Haemogram, Urea, Electrolytes and Creatinine**
- Management: **Appropriate antibiotics [Ceftriaxone]**
- Supportive treatment: **Analgesic/Antipyretic**

APPENDIX X: INFECTION CONTROL GUIDELINES FOR ALL INFECTIONS

Infection control precautionary measures are divided into two:

1. **Standard precautions:** applied in management of all patients
 2. **Transmission based precautions:** specific to a given diagnosis.
-
1. **Standard precautions:** Consider the following measures when managing **all medical** conditions:
 - 1.1 Hand hygiene
 - 1.2 Respiratory Hygiene/Cough Etiquette
 - 1.3 Use of personal protective equipment-PPE
 - 1.4 Linen and Laundry management
 - 1.5 Safe injection practices
 - 1.6 Cleaning and disinfection of the environment & equipment
 2. **Transmission based precautions:** consider the following specific precautions when managing patients in these 3 categories
 - 2.1 Contact precautions
 - 2.2 Droplet precautions
 - 2.3 Airborne precautions
-
- 2.1 **Contact based precautions:** Patients requiring Protection against infections transmitted by contact.

In addition to Standard Precautions:

- 2.2.1 Use non-sterile, clean, disposable gloves, gown/coverall, apron
- 2.1.2 Use disposable or dedicated reusable equipment (which must be cleaned and disinfected before use on other patients)
- 2.1.3 Limit patient contact with non-infected persons
- 2.1.4 Isolate patient in a single room or cohort with others

2.2 Droplet precautions: Protects against respiratory pathogens transmitted by large droplets ($>5\mu\text{m}$)

In addition to Standard Precautions:

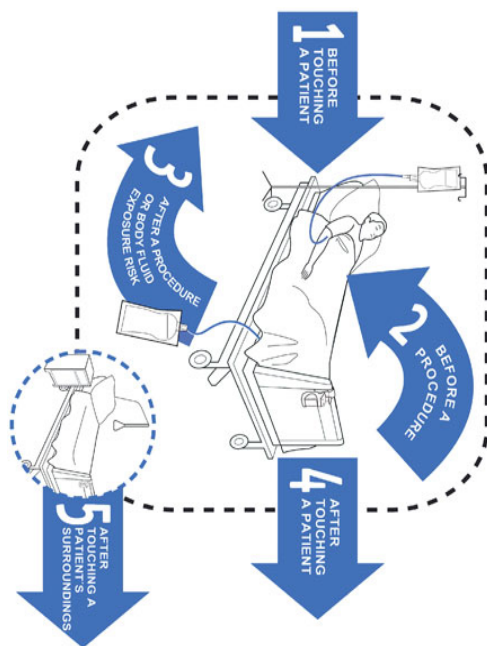
- 2.2.1 Use a face mask when within less than 1 m of patient
- 2.2.2 Maintain a distance of above 1 meter between infectious patient and others
- 2.2.3 Place patient in isolation or cohort with similar patients
- 2.2.4 Limit patient movement and bring toys to the room if necessary

2.3 Airborne precautions: Protects against inhalation of tiny infectious droplet nuclei eg tuberculosis, measles, chickenpox

In addition to Standard Precautions:

- 2.3.1 Use particulate respirator (e.g., N95)
- 2.3.2 Isolate patient in adequately ventilated room (negative pressure room)
- 2.3.3 Limit patient movement and bring toys to room if necessary
- 2.3.4 Limit number of visitors

OBSERVE THE 5 MOMENTS FOR HAND HYGIENE



Procedure for hand Hygiene (wash/rub)

How to handrub?

WITH A LCOHO L-BASED FORMULATION



Apply a small amount of the product in a cupped hand and cover all surfaces.



rub hands palm to palm



backs of fingers to opposing palms with fingers interlocked



right palm over left dorsum with interlocked fingers and vice versa



rotational rubbing of left thumb clasped in right palm and vice versa



palms to palms with fingers interlocked



rotational rubbing, backwards and forwards with clasped fingers of right hand in left palm and vice versa



20-30 sec



...once dry, your hands are safe.

How to handwash?

WITH SOAP AND WATER



Wet hands with water



apply enough soap to cover all hand surfaces.



rinse hands with water



dry thoroughly with a single use towel



use towel to turn off faucet



40-60 sec



...and your hands are safe.



Department of Health
Government of Kenya

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APPENDIX V: LABORATORY REFERENCE NORMAL VALUES

Biochemistry

ANALYTE	SPECIMEN	AGE	REFERENCE INTERVALS
Adrenocorticotrophic hormone (ACTH)	Plasma (EDTA)	Cord	11-125 pmol/L
		Newborn	2.2 - 41 pmol/L
		Child	< 26pmol/L
		Adult	< 26pmol/L
Alanine aminotransferase (ALT, SGPT)	Serum	Males	0 - 40 U/L
		Females	0 - 35 U/L
Albumin	Serum	0 - 4 days	28 - 44 g/L
		≥1 year	35 - 50g/L
	Urine (24hr)		3.9 - 24 mg/dL
Aldolase	Serum	10months - 2yrs	10-40U/L
		2- 16yrs	5-20U/L
		Adult	2.5-10
Alkaline phosphatase	Serum	4-15yr (Male and female)	54 -369U/L
		Male:20-50yr	53-128U/L
		Female:20-50yr	42 -98U/L
Amylase	Serum	1-13 years:	8-79U/L
		≥14 years:	21-110U/L

		Adult	30-110U/L
Aspartate transaminase (AST, SGOT)	Serum	Males: 1-13 years:	0- 60 U/L
		≥14 years	0 - 48U/L
		Female: 1-13 years	0 - 50 U/L
		≥14 years	0 - 43 U/L
Bilirubin Total	Serum	Cord	< 34.2μmol/L
		Premature: 0 - 1 day	17 - 187μmol/L
		Premature; 1 - 2 days	103 - 205μmol/L
		Premature: 3 - 5 days	171 - 240 μmol/L
		Full term: 0 -1 day	34 - 103 μmol/L
		Full term: 1 -2 days	103 - 171μmol/L
		Full term: 3 - 5 days	68 - 137μmol/L
		Adult	3 - 17μmol/L
Bilirubin Conjugated	Serum		0 - 3.4μmol/L
Calcium (Total)	Serum, Plasma	Neonate	1.85 - 2.75mmol/L
		Infant/child	2.35 - 2.65mmol/L
		Adult	2.10 - 2.60mmol/L
		Adjusted Calcium =	

		Total Calcium + 0.015 (40 – Albumin)	
Urine calcium	Urine (24hr)		
		Neonate	Up to 1 mmol/L
		Child and Adult	2.5 – 7.5mmol/24hr s
	Spot Urine	Calcium/creati nine ratio: in 2nd morning urine mmol/mmol	
		< 6 months	< 2.42 mmol/mmol
		6-12 months	0.09 - 2.2
		1-2 years	0.07 - 1.5
		2-3 years	0.06 - 1.4
		3-5 years	0.05 - 1.1
		5-7 years	0.04- 0.8
		< 7 years	0.04 - 0.7
		Adult	<0.75
Chloride	Serum, Plasma	Child	95 – 110mmol/L
		Adult	98 – 107mmol/L
Cholesterol (Total)	Serum	Coronary heart disease risk	

		Child: Desirable	< 4.4mmol/L
		Borderline high	4.40 - 5.15mmol/L
		High	< 5.15mmol/L
		Adult: Desirable	<5.18mmol/L
		Borderline high	5.18 - 6.19mmol/L
		High	< 6.19mmol/L
Cortisol (Total)	Serum, Plasma	Cord	138 - 469nmol/L
		Infant	53 - 304nmol/L
		1 - 16 years (AM)	83 - 580nmol/L
		Adult: AM	138 - 635 nmol/L
		Adult: PM	83 - 441nmol/L
C-Reactive protein	Serum	Child/Adult	< 10 mg/L
Creatine Kinase	Serum, Plasma	Neonatal	Up to 900 IU/L
		Infant	100 - 480 IU/L
		Child/Adult	
		Male	30 - 200IU/L
		Female	29 - 168IU/L
Creatinine	Serum, Plasma	Cord	53 - 106μmol/L
		newborn	27-88μmol/L
		Infant	18-35μmol/L

		Child	27-62 μ mol/L
		Adolescent	44-88 μ mol/L
		Adult female	53-97 μ mol/L
		Adult Male	62-115 μ mol/L
	Urine (24hr)	Child	71 - 177 μ mol/Kg/ day
		Adolescent	71 - 165 μ mol/Kg/ day
		Adult	124 - 230 μ mol/Kg/ day
Dehydroepiandrosterone sulphate	Serum, Plasma	Male: Tanner Stages	
		1	5 - 265 ug/dL
		2	15 - 380
		3	60 - 505
		4	65 - 560
		5	165 - 500
		Adult (18 - 30yrs)	125 - 619
		Females Tanner stages	
		1	5 - 125ug/dL
		2	15 -150
		3	20 - 535
		4	35 - 485
		5	75 - 530
		Adult (18 - 30 yrs)	45 - 380

Digoxin	Serum (Trough)	All ages	0.8 - 2.0ng/mL
Ferritin	Serum	Newborn	20 -200ug/L
		1 month	200-600ug/L
		2 -5 months	50 - 200ug/L
		6months - 15yrs	7 - 140ug/L
		Adult males	30 - 400ug/L
		Adult females	15 - 150ug/L
FSH	Serum	Males:	
		1 - 10yrs	0.3 - 4.6IU/L
		< 13yrs	1.4 -18.1IU/L
		Females	
		1-10yrs	0.7 - 6.7IU/L
		Follicular phase	2.5 - 10.2IU/L
		Mid-cycle	3.4 - 33.4IU/L
		Luteal phase	1.5 - 9.1IU/L
		Pregnant	<0.3IU/L
		Post menopause	19.3 - 116IU/L
Gamma Glutamyl transferase (GGT)	Serum	Neonate	Up to 215U/L
		Infant	Up to 107IU/L
		Child	Up to 50 U/L
		Adult males	< 60U/L
		Adult female	< 38 U/L
Glucose	Plasma (Fluoride)	Fasting	

		Neonate	2.5 - 5.5 mmol/L
		Child	3.5 - 6.5 mmol/L
		Adult	3.5 - 5.5 mmol/L
	CSF	Infant/child	3.3 - 4.5 mmol/L
		Adult	2.2 - 3.9 mmol/L
Glycosylated Haemoglobin (HbA1c)	Whole blood (EDTA)	Non-diabetic range	4 - 6%
		In diabetics –	
		good glycaemic control	< 6.5%
		borderline control	6.5 - 7.5%
		poor control	> 7.5%
Growth Hormone	Serum	Basal (Fasting)	2 - 5 ng/mL
HDL-Cholesterol	Serum	Desirable	< 1.2 mmol/L
Immunoglobulin A	Serum	1 month	0.1 - 3.2 g/L
		6 months - 1 yr	0.2 - 3.4 g/L
		< 6 yrs	1.1 - 6.6 g/L
Immunoglobulin G	Serum	1 month	4 - 16 g/L
		6 months - 1 yr	5 - 19 g/L
		< 15 yrs	10 - 23 g/L
Immunoglobulin M	Serum	1 month	0.05 - 2.2 g/L
		6 months - 1 yr	0.06 - 2.5 g/L
		< 6 yrs	0.05 - 2.5 g/L

Iron	Serum	Newborn	17.9 - 44.8µmol/L
		Infant	7.2 - 17.9µmol/L
		Child	9 - 21.5µmol/L
		Adult	10 - 30µmol/L
Lactate	Plasma (Fluoride)	1 - 12months	1.1 - 2.3mmol/L
		1 - 7yrs	0.8 - 1.5 mmol/L
		7 - 15yrs	0.6 - 0.9 mmol/L
		Adult	0.7 - 2.5mmol/L
Lactate dehydrogenase	Serum	Newborn	125 - 765U/L
		2 - 12yrs	125 - 220U/L
		< 12yrs	180 - 360U/L
LDL-Cholesterol	Serum	Desirable	<2.59mmol/L
		Near optimal	2.59 - 3.34mmol/L
		Borderline high	3.37 - 4.12mmol/L
		High	< 4.12mmol/L
Magnesium	Serum, Plasma	Newborn	0.62 - 0.91mmol/L
		6 - 12yrs	0.7 - 0.91mmol/L
		< 12yrs	0.8 - 1.0mmol/L

	Urine (24hr)	Adult	3 - 5mmol/24hrs
Oestradiol	Serum	6months - 10yrs	<55pmol/L
		Adult Female	
		Follicular phase	0 - 979pmol/L
		Mid-cycle	430 - 1300pmol/L
		Luteal phase	95 - 606pmol/L
		Post- menopausal	0 - 150pmol/L
		Adult Male	0 - 150pmol/L
Osmolality	Serum	Newborn	275 - 300mmol/Kg
		7days - 1 month	274 - 305mmol/Kg
		Adult	280 - 295mmol/Kg
	Urine	Random specimen	50 - 1200mmol/K g
Phenytoin	Serum	Trough sample	10 - 20ug/mL
Phosphate	Serum	Child	1.30 - 2.26mmol/L
		Adult	0.81 - 1.45mmol/L
	Urine, 24hr (in acid)	Adult, unrestricted diet	29 - 42mmol/day

Potassium	Serum, plasma	Newborn	3.7- 5.9mmol/L
		Infant	4.1- 5.3mmol/L
		Child	3.4- 4.7mmol/L
		Adult	3.5- 5.1mmol/L
Protein (Total)	Serum	Neonate	46- 86g/L
		1 month - 1 yr	48 - 76g/L
		1 - 18yrs	60 - 80g/L
		Adult	60 -85g/L
Sodium	Serum, plasma	Newborn (cord)	126- 166mmol/L
		Child/Adult	135- 146mmol/L
Testosterone	Serum	Male: 6 months - 8yrs	< 0.9 nmol/L
		Female: 0 - 8 yrs.	< 0.9 nmol/L
		Adult: Male	8.4 - 28.7 nmol/L
		Adult: Female	0 - 2.8 nmol/L
TSH	Serum	Cord	0.4-12 UIU/L
		Newborns	0.4-39 UIU/L
		1-11 months	0.8-6.3UIU/L
		< 1yr	0.25 - 4.7UIU/L
Free T3	Serum		2.6 - 5.4pg/mL

Free T4	Serum		0.8 - 2.0 ng/dL
Triglycerides	Serum, Plasma	0 - 9yrs	0.34 - 1.13mmol/L
		10 - 14 yrs.	0.36 - 1.41mmol/L
		15 - 20yrs	0.42 - 1.67mmol/L
		Adults	0.5 - 1.5mmol/L
Urea	Serum	Newborn	1.4- 4.3mmol/L
		Child	1.7 - 8.3mmol/L
		Adult	2.1- 7.1mmol/L
Uric acid	Serum	Child	0.12- 0.32mmol/L
		Adult male	0.21- 0.42mmol/L
		Adult female	0.15- 0.35mmol/L

HAEMOGRAM REFERENCE VALUES			
	<1 month	2months-1yr	< 1yr-6yrs
RBC (millions/mm ³)	3.0-5.4	3.0-4.6	3.9-5.3
HB (g/dl)	11.0 - 17.0	9.5-13.5	11.0-14.0
PCV (%)	31-55	28-42	34-40
MCV (femtolitres)	85-123	74-108	75-87
MCH (pg)	28-40	25-35	24-30
MCHC (g/dl RBC)	29-37	29-40	31-37
RETICULOCYTE (%)	0.3-1.0	0.4-1.0	0.2-2.0
WBC (x10 ⁹ /L)	4.1-15.8	4.1-15.8	4.9-13.6
NEUTROPHILS (x10 ⁹ /L)	1.3-7.5	1.3-7.5	1.3-7.5
LYMOPHOCYTES (x10 ⁹ /L)	2.0-10	2.0-10	2.5-9.0
MONOCYTES (x10 ⁹ /L)	0.2-1.8	0.2-1.8	0.2-1.0
EOSINOPHILS (x10 ⁹ /L)	0.1-1.0	0.1-1.0	0.1-1
BASOPHILS (x10 ⁹ /L)	<0.1	<0.1	<0.1

APPENDIX VI: INTERPRETATION OF BLOOD GAS ANALYSIS

Arterial Blood Gas Analysis Interpretation					
ABG Parameter			Calculation and Interpretation		
PH	> 7.45	Alkalemia	PH	PCO ₂	Interpretation
	7.35 – 7.45	Normal			
	< 7.35	Acidemia	⇓	⇓	Metabolic Acidosis
PCO ₂	> 45	High	⇑	⇑	Metabolic Alkalosis
	35 - 45	Normal	⇑	⇓	Metabolic Alkalosis
	< 35	Low	⇓	⇑	Respiratory Acidosis
HCO ₃	< 26	High	Corrected Standard for AG Albumin		
	22 - 26	Normal	Albumin/4 + 15 x Phosphate		
	< 22	Low			
Anion Gap	> 16	High	Anion Gap Calculation		
	8 - 16	Normal	[Na] – [Cl+HCO ₃] = 8 to 16		
	< 8	Low	Corrected [Na] for AG in Hyperglycemia		
Glucose	< 10	High	Corrected [Na] = [Na] + (Glucose – 5)/3		
	< 2	Low			
Gap: Calculation	$\Delta AG = \frac{AG - 12}{24 - HCO_3}$		Gap: Gap Calculation for Metabolic Acidosis		
			< 0.4		Low or Normal AG Metabolic Acidosis
			0.4 – 0.8		Normal + High AG metabolic Acidosis
Lactate	< 1.9	Normal	0.8 – 2.0		Pure High AG Metabolic Acidosis
	> 2	High	> 2.0		Metabolic Acidosis with metabolic Alkalosis/Respiratory Acidosis
PO ₂	80 - 100	Normal	PaO ₂ = [713xPIO ₂] – [PCO ₂ x1.25]		
	< 80	Hypoxemia	A-a Gradient = PAO ₂ -PaO ₂ = (Age/4) + 4		
Compensation rules for:					
Expected pCO ₂	Metabolic Acidosis		Metabolic Alkalosis		
	1.5 x [HCO ₃] + 8 (±2)		0.7 x [HCO ₃] + 20 (±5)		
Expected HCO ₃	Respiratory Acidosis		Respiratory Alkalosis		
	Acute	Chronic	Acute		Chronic
	24 + (pCO ₂ - 40)/10	24 + (pCO ₂ - 40)/2.5	24 - (40 - pCO ₂)/5		24 - (40 - pCO ₂)/2

APPENDIX IIV: MEDICATION DOSING GUIDELINES

DRUG	DOSE	ROUTE	FREQUENCY	MAX. DOSE
Acetyl cysteine	150 mg/Kg(loading) over 1 hour then, 50 mg/Kg/over 4 hours then, 100 mg/Kg over 16 hours	Slow IV Infusion		300 mg/Kg
Activated Charcoal (Reduction of absorption of poisons)	1 g/Kg (Neonate)	PO		
	1g/Kg (1month - 12 years)	PO		50g
	50g (12-18 years)	PO		
Activated Charcoal (Active Elimination of poisons)	1g/Kg (Neonate)	PO	4 Hourly	
	1g/Kg (1 month-12 years)	PO	4 Hourly	50g
	50g (12-18 years)	PO	4 Hourly	
Acyclovir Oral	100 mg (< 2 years)	PO	5 times daily	
	200 mg (≥ 2years)	PO	5 times daily	
Acyclovir Injection	10 – 20 mg/Kg	IV	8 Hourly	
Adrenaline 1:10,000	0.1mL/Kg	IV/IO	As required	
Adrenaline (1:1000)	400 mg/Kg	Nebulization	Repeat after 30 minutes if needed	5 mg
Albendazole (2Years and above) Strongyloidiasis	400 mg	PO	12 Hourly	800 mg/day
Hydatid Disease	7.5 mg/Kg	PO	12 Hourly	400 mg/dose
Hookworms Infection	400 mg for 1 dose			
Albumin	0.25 – 1 g/Kg/day	IV Infusion over 4 Hours		2g/Kg/day

Amikacin	15 mg/Kg	24 Hourly		1.5 grams
Amiodarone Neonate (First 7-10 days)	5 - 10 mg/Kg	PO	12 Hourly	200 mg/day
1 month-12 years	5 - 10 mg/Kg	PO	12 Hourly	
Maintenance dose	5-10 mg /Kg	PO	24 Hourly	
Neonate (First 7-10 days)	5 mg/Kg	IV Infusion	12-24 Hourly	
1 month-18 years	5 - 10 mg/Kg	IV Infusion		
Amoxicillin (Neonate)	25 mg/Kg/dose	PO	12 Hourly	125 mg/dose
Amoxicillin (Mild infection)	30 mg/Kg/dose	PO	8-12 Hourly	1gram/dose
Amoxicillin (Pneumonia)	45 mg/Kg/dose	PO	12 Hourly	
Amphotericin B	0.25-1.5 mg/Kg	IV Infusion	Over 2-6 hours	1.5 mg/Kg/day
Artesunate 20 Kg or below	3 mg/Kg	IV/IM	0 hours,12 hours, then 24-hourly	
Above 20Kg	2.4 mg/Kg	IV/IM		
Azithromycin	10-20 mg/Kg	PO	24 Hourly	500 mg/day
Benzathine Penicillin Children Less than 30Kg	600,000 International Units	IM	Stat or Every 3-4 weeks	2.4 million units/dose
Children above 30Kg and Adults	1,200,000 International Units	IM		
Caffeine Citrate Loading Dose	20 mg/Kg	PO/IV	24 Hourly	
Maintenance	5-10 mg/Kg	PO/IV		
Captopril Neonate– Infant(6mos)	0.01-0.3 mg/Kg/dose	PO	8-12 hourly	6 mg/Kg/day
Child	0.3-0.5 mg/Kg/dose	PO	8-12 hourly	
Carbimazole	0.75 - 1 mg/Kg /day	PO	8 – 24 Hourly	

Cefixime	4 mg/Kg /day	PO	12-hourly	400 mg /day
Cefotaxime <i>Neonate up to 7 days</i>	25 mg/Kg/dose	IV	12 Hourly	
<i>Neonate 7days-21 days</i>	25 mg/Kg/dose	IV	8 Hourly	
<i>Neonate 21-28 days</i>	25 mg /Kg/dose	IV	6-8 Hourly	
<i>Child</i>	50 mg/Kg/dose	IV	8-12 Hourly	
Cefotaxime (Meningitis) <i>Neonate up to 7 days</i>	50 mg/Kg/dose	IV	12 Hourly	
<i>Neonate 7days-21 days</i>	50 mg/Kg/dose	IV	8 Hourly	
<i>Neonate 21-28 days</i>	50 mg /Kg/dose	IV	6-8 Hourly	
<i>Child</i>	50 mg/Kg/dose	IV	6 Hourly	Max 12 g
Cefpodoxime	5 mg/Kg	PO	12 Hourly	400 mg
Ceftazidime <i>Neonate 0-7 days</i>	25 mg/Kg-50 mg/Kg	IV	24 Hourly	
<i>Neonate 7-20 days</i>	25 mg/Kg-50 mg/Kg	IV	12 Hourly	
<i>Neonate 21- 28 days</i>	25 mg/Kg-50 mg/Kg	IV	8 Hourly	
<i>Child</i>	50 mg/Kg	IV	8 Hourly	Max 6 g
Ceftriaxone	25 - 50 mg/Kg (up to 80 mg/Kg for some infections)	IV	24 Hourly	100 mg/Kg/day
Ceftriaxone <i>Meningitis</i>	80 - 100 mg/Kg	IV Infusion	24 Hourly	4 grams /day
Cefuroxime <i>3 months -2 years</i>	10 mg/Kg	PO	12 Hourly	125 mg/dose
<i>2-12 years</i>	15 mg /Kg	PO	12 Hourly	250 mg/dose
<i>12-17 years</i>	250 mg	PO	12 Hourly	500 mg/dose
<i>1 month -18 years</i>	20-60 mg/Kg	IV	8 Hourly	750 mg /dose 1.5 grams/dose - in severe cases

Cetirizine <i>From 1 Year</i>	250 micrograms	PO	12 Hourly	
<i>2-6 years</i>	2.5 mg-5 mg	PO	12 Hourly 24 Hourly	
<i>Above 6 Years</i>	5 mg-10 mg	PO	12 Hourly 24 Hourly	10 mg
Chloramphenicol	12.5 - 50 mg/Kg	PO, IV	6 – 8 hourly	100 mg/Kg/day 4 grams
Chlorpheniramine <i>1 Month- 2 Years</i>	1 mg	PO	12 Hourly	6 mg/day
<i>2 Years – 5 Years</i>	1 mg	PO	4-6 Hourly	12 mg/day
<i>6 Years -11 Years</i>	2 mg	PO	4-6 Hourly	24 mg/day
<i>12 Years -18 Years</i>	4 mg	PO	4-6 Hourly	24 mg/day
Cholestyramine	1 - 4 grams /Dose	PO	6-8 Hourly	36 grams/day
Cimetidine	2.5 – 10 mg/Kg	PO	6-12 Hourly	40 mg/Kg/day
Ciprofloxacin	10-20 mg/Kg	PO	12 Hourly	750 mg/dose
	6-10 mg/Kg	IV	8 -12 Hourly	400 mg/dose
Clarithromycin	7.5 mg/Kg/dose	PO/IV	12 Hourly	500 mg/dose
Clindamycin	3-7 mg/Kg	PO	6-8 Hourly	600 mg/dose
	3.75-10 mg/Kg	IV/IM	6-8 Hourly	1.2g/dose
Clonidine	1-2 mcg/Kg/hour	IV Infusion		2 mcg/Kg/hour
Co-trimoxazole	24 mg/Kg/Dose	PO	12 Hourly	960 mg/Dose
	18-27 mg /Kg/Dose	IV	12 Hourly	1.44 g/Dose
<i>PCP Treatment</i>	120 mg/Kg/day	PO/IV	6-12 Hourly	
<i>PCP Prophylaxis</i>	450 mg/m ² /Dose	PO	12 Hourly	960 mg/dose
Codeine Phosphate	0.5-1 mg/Kg/dose	PO	4-6 Hourly	

Crystalline Penicillin	25,000-100,000 units/Kg	IV	12 Hourly 6-8 Hourly	
Danazol	100-400 mg	PO	BD	800 mg/day
Desloratadine				
1 -2 years	1.25 mg	PO	24 hourly	
1 - 5 years	2.5 mg	PO	24 Hourly	
6-11 years	5 mg	PO	24 Hourly	
12-17 years	10 mg	PO	24 Hourly	
Dexamethasone Croup	150mcg/Kg	IV/IM	Stat	
Diazepam	0.1-0.5 mg/Kg	IV	As required	10 mg/dose
	1.25 mg-10 mg	Rectal	As required	
	250mcg- 10 mg	PO	12 Hourly	40 mg
Diclofenac	0.3-1 mg /Kg/day	PO	8 Hourly	50 mg /dose
Dobutamine	2-20 mcg/Kg/min	IV Infusion	as required	40 mcg/Kg/min
Dopamine	1-20 mcg/Kg/min	IV Infusion	as required	50 mcg/Kg/min
Doxycycline	1-4 mg/Kg/dose	PO	12 Hourly	200 mg/day
Enalapril	100mcg-1 mg/Kg/day. 2.5 mg/dose	PO	12-24 Hourly	40 mg/day
Erythromycin	12.5 mg/Kg	PO	6 Hourly	2 grams/day
Esomeprazole				
1-11 years(10-20Kg)	10 mg	PO/IV	24 Hourly	
1-11 years (< 20Kg)	10-20 mg	PO/IV	24 Hourly	40 mg/day
12 -17 Years	20-40 mg	PO/IV	24-hourly	40 mg/day
Ethambutol	15-25 mg/Kg	PO	12 Hourly	1.2 grams/day
Fentanyl	1-5 mcg/Kg/dose	IV Infusion	as required	8mcg/Kg/hr
Flucloxacillin	25-50 mg/Kg/dose	PO	6-12 Hourly	2g/day
	12.5-50 mg/Kg/hr	IV/IM	6 Hourly	2g/day
Fluconazole	3-12 mg/Kg	PO/ IV	24 Hourly	800 mg/day

Flumazenil	2-10 mcg/Kg/hr	IV		400 mcg/hour
Frusumide/Furose mide	0.5-2 mg/Kg/dose	PO	8-24 Hourly	80 mg/day
	0.5-5 mg/Kg/dose	IV	6-12 Hourly	1 g/day
Gentamicin	2.5-7.5 mg/Kg	IV/IM	8-24Hourly	7.5 mg/Kg/day
Hydralazine	0.25-0.5 mg/Kg/dose	PO	8-12 Hourly	7.5 mg/Kg/day or 200 mg/day
	0.1-0.5 mg/Kg/dose	IV	4-6 Hourly	3 mg/Kg/day
Ibuprofen	5-10 mg/Kg/dose	PO	6-8 Hourly	40 mg/Kg/day or 2.4g/day
Iron (Anemia)				
Child	3-6 mg/Kg/day	PO	8-12 Hourly	200 mg/day
Iron Prophylaxis				
Preterm/ low birth weight infants	2-4 mg/Kg/day	PO		15 mg/day
Full Term	1 mg/Kg/day	PO	24 Hourly	15 mg/day
Children < 10yrs in Puberty Male	2 mg/day	PO	24 Hourly	
Children < 10yrs in Puberty Female	5 mg/day	PO	24 Hourly	
Isoniazid	10-15 mg/Kg	PO	24 Hourly	300 mg/day
Ivermectin (Child < 5yrs)	0.15 – 0.2 mg/Kg/dose	PO	24 Hourly	
Ketamine	1-2 mg/Kg/dose	IV	Single Dose	4.5 mg/Kg/day
	4-12 mg/Kg/dose	IM	Single Dose	
Lansoprazole				
<i>Less than 30Kg</i>	0.5-1 mg/Kg/dose	PO	24 Hourly	15 mg/day
<i>More than 30Kg</i>	15-30 mg/dose	PO	24 Hourly	
Lisinopril				
6-11 Years	70mcg/Kg- 600mcg/Kg/day	PO	24 Hourly	Initial dose should not exceed 5

				mg. 40 mg/day
12-17 Years	2.5 mg-20 mg/day	PO	24 Hourly	80 mg/day
Loratidine				
Child 2-11 Years (<31Kg)	5 mg	PO	24 Hourly	
< 31KG	10 mg	PO	24 Hourly	
Lorazepam				
1 Month – 11 Years	0.1 mg/Kg/dose	PO/IV	Single dose	4 mg/dose
12 – 17 Years	1 - 4 mg	PO		
Mannitol				
1 Month – 11 Years	250 mg–1.5g/Kg/dose	IV infusion	as required	
12 – 17 Years	250 mg-2g/Kg/dose			
Mebendazole				
Enterobiasis	100 mg/dose	PO	Stat (Repeat after 2 weeks if required)	
Ascariasis/ Trichuriasis	100 mg/dose	PO	12 Hourly	
	Alternatively, 500 mg (Children < 1yr)	PO	Stat	
Trichinosis	200-500 mg/dose	PO	8 Hourly	
Methimazole	0.1-0.5 mg/Kg/day	PO	8 Hourly	1 mg/Kg/day
Metolazone				
1month-11 Years	0.1-0.2 mg/Kg/dose	PO	12-24 Hourly	
12-17 Years	5-10 mg	PO	12-24 Hourly	
Metronidazole Less than 30days old		PO	BD, TDS	1200 mg/day
	7.5 mg/Kg	PO/IV	12 Hourly	Oral: 400 mg/dose IV: 500 mg/dose
	7.5 mg/Kg	PO/IV	8 Hourly	
Midazolam				
	0.05-0.3 mg/Kg/dose	IV	Single dose	6 mg per course or not more the 0.3

				mg/Kg per dose
	0.2-0.5 mg/Kg/dose	PO	Single dose	20 mg/day
Morphine (neonate)	0.05-0.1 mg/Kg/dose	Slow IV	6 Hourly	
Morphine (infant and Child)	0.1-0.2 mg/Kg/dose	IV	4-6 Hourly	
	0.1 mg/Kg/dose	IV	4-6 Hourly	
	0.05-0.3 mg/Kg/dose	PO	4 Hourly	
Nystatin	100,000 units (1mL)	PO	6 Hourly	
Paracetamol				
<i>Neonate (28-32 Weeks)</i>	10-15 mg/Kg/dose	PO/PR	8-12 Hourly	30 mg/Kg/day
<i>Neonate (< 32 weeks)</i>	10-15 mg/Kg/dose	PO/PR	6-8 Hourly	60 mg/Kg/day
<i>Child 1-2 months</i>	10-20 mg/Kg/dose	PO/PR	8 Hourly	60 mg/Kg/day
<i>Child< 3 months</i>	10-20 mg/Kg/dose	PO/PR	4-6 Hourly	90 mg/Kg/day or 4g per day
<i>Child <10Kgs</i>	10 mg/Kg/dose	IV	4-6 Hourly	30 mg/Kg/day
<i>10-50Kg</i>	15 mg/Kg/dose	IV	4-6 Hourly	60 mg/Kg/day
<i>< 50Kg</i>	1g	IV	4-6 Hourly	4g per day
Penicillin V	12.5 mg/Kg/dose	PO	6 Hourly	4 g/day
<i>1 Year- 4 Years</i>	125 mg/dose	PO	12 Hourly	
<i>5 – 17 Years</i>	250 mg/dose	PO	12 Hourly	
Pentamidine	4 mg/Kg	IV/IM	24 Hourly	
Phenobarbitone (loading dose)	15-20 mg/Kg/dose	IV	Single dose	1g maximum per dose
Phenobarbitone (maintenance dose)	2.5-5 mg/Kg/dose	IV	12-24 Hourly	600 mg/day
Phenobarbitone (loading dose) <i>1month-11 years</i>	1-1.5 mg/Kg increase up to 2 mg/Kg	PO	Single dose	

Phenobarbitone (maintenance dose) 1month-11 years	2.5-4 mg/Kg/dose	PO	12-24 Hourly	180 mg/day
Phenytoin (loading dose)	15-20 mg/Kg/dose	IV over 20-30 minutes		
Phenytoin (maintenance dose)	2.5-5 mg/Kg/dose	PO/IV	12 Hourly	300 mg per day and PO-7.5 mg/Kg/dose
Praziquantel	5-25 mg/Kg	PO	Variable	60 mg/Kg/day
Prednisolone	1-2 mg/Kg	PO	24 Hourly	60 mg/day
Pyrazinamide	25-35 mg/Kg	PO	24 Hourly	1.6g/day
Pyridoxine (Prophylaxis)				
5-7 Kg	12.5 mg	PO	24 Hourly	200 mg
8-14 Kg	25 mg	PO	24 Hourly	200 mg
< 15Kg	50 mg	PO	24 Hourly	200 mg
Pyridoxine (Treatment)	100 mg	PO	12 Hourly	200 mg
Quinine	10 mg/Kg/dose	PO	8 Hourly	600 mg/dose
<i>Loading dose</i>	20 mg/Kg/dose	IV infusion	Stat	1.4g per dose
<i>Maintenance dose</i>	10 mg/Kg/dose After 48hours reduce to 5-7 mg/Kg/dose	IV	8 Hourly	700 mg/dose
Rifampicin	10-20 mg/Kg/day	PO	24 Hourly	600 mg/day
Salbutamol	2.5-5 mg/dose	Nebulisation	As required	
Resonium	0.5-1g/Kg/day	PR/PO	Divided doses	PO maximum 60g/day PR Maximum 30g/day

Spironolactone (child< 1 month)	1-3 mg/Kg/day	PO	12-24Hourly	9 mg/Kg/day or 400g/day
Terbinafine				
10-19Kg	62.5 mg	PO	24 Hourly	
20-39Kg	125 mg	PO	24 Hourly	600 mg/day
< 40Kg	250 mg	Nebulisation	As required	
Thyroxine (initial)	5-15 mcg/Kg	PO	24 Hourly	50 mcg/dose
(maintenance)	20-200mcg/ day	PO	24 Hourly	Neonate- 50mcg 1month-1 year-75mcg 2-11year- 100mcg < 12 years- 200mcg
Tranexamic Acid	15-25 mg/Kg/dose	PO	8-12 Hourly	1500 mg/dose
10-19Kg	10 mg/Kg/dose	IV PO	8-12 Hourly Stat	1g/dose
Vitamin A				
(0-6 months)	50,000 International units			
(6-11 months)	100,000 International units	PO	Stat	
(12months and above)	200,000 International units	PO	Stat; then every 6 months up to the age of 5 years	
Vitamin D				
(0-6 months)	75,000 units	PO/IM	Stat	
(6-12 months)	150,000 units	PO/IM	Stat	
(12 months and above)	300,000 units	PO/IM	Stat	
Zinc Sulphate				
(0-6 mo. Or <10Kg)	10 mg	PO	24 Hourly	
> 6mo. Or < 10Kg	20 mg	PO	24 Hourly	

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